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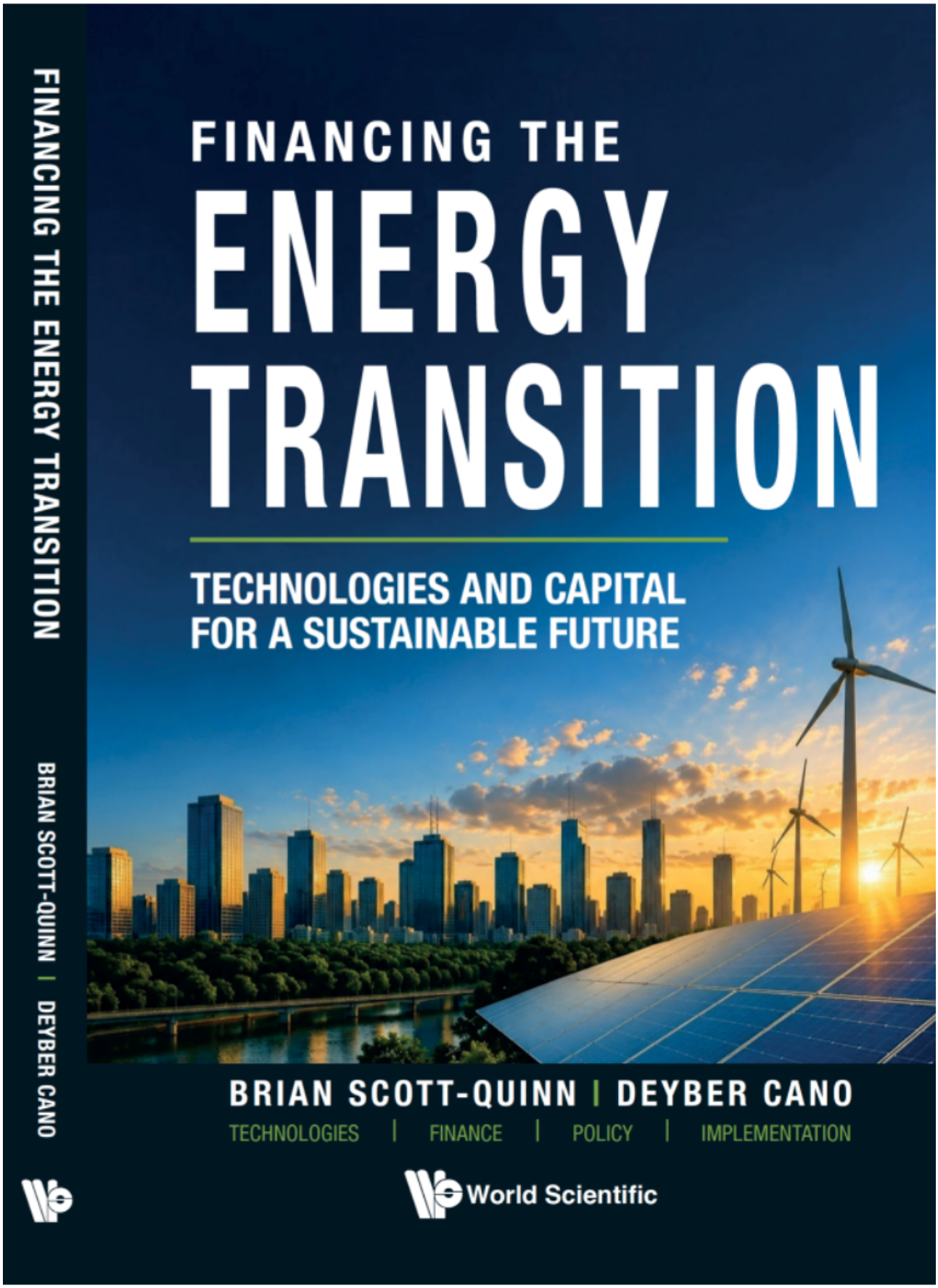

FINANCING THE ENERGY TRANSITION
TECHNOLOGIES AND CAPITAL FOR A SUSTAINABLE FUTURE
BRIAN SCOTT-QUINN | DEYBER CANO
TECHNOLOGIES | FINANCE | POLICY | IMPLEMENTATION
World Scientific
FINANCING THE ENERGY TRANSITION
BRIAN SCOTT-QUINN | DEYBER CANO

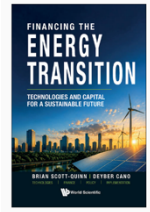
Back Cover

Infrastructure finance, capital markets and emerging energy technologies are central to delivering the global energy transition.

This book brings these disciplines together, showing how innovative financing structures and technological advances must work hand in hand to accelerate the transition to a low-carbon economy. As one of the defining infrastructure and financing challenges of the twenty-first century, the energy transition will require trillions of dollars of investment, new financial models and the successful deployment of transformative technologies.

Combining academic rigour with real-world case studies, the book examines the financial architecture behind the transition—from venture capital, project finance, green bonds and blended finance to public–private partnerships, innovative financing structures and the growing role of AI and hyperscalers in energy investment.

It also explores the technologies shaping tomorrow's energy systems, including advanced nuclear, renewable energy, concentrated solar power, hydrogen, energy storage, carbon capture and AI-enabled electricity networks. Written by authors with experience across the energy industry, investment banking and academia, this book is an essential reference for investors, infrastructure professionals, policymakers, engineers, researchers and graduate students seeking to understand—and finance—the future of global energy systems.



Financing the Energy Transition

Technologies and Capital for a Sustainable Future

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About the Authors, Acknowledgements, the International Capital Market Association (ICMA) and the ICMA Centre for financial markets

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In 2014 he became a founding director of Civilised Bank Plc (now Allica Bank), an SME lending bank. He was also Chairman of its Audit and Assurance Committee, a member of the Risk Committee and a Financial Conduct Authority SMF 11. He is a founding director of Madiston plc – a fintech which produces banking software. From 2012 to 2021 he was a member of the Advisory Board of Penultimate Power UK which was aiming to bring the manufacture of advanced technology small modular reactors to the UK. He was a consultant to US SPAC, Growth for Good (G4G) undertaking search and due diligence on clean energy acquisition targets. Along with Deyber Cano (see below) he provided financial advisory services on project financing as part of a consortium of Eversheds, IDP, CEG and IPSL in respect of a proposed oil pipeline and storage facility for Petroci, the Côte d'Ivoire state oil and gas company. He advised an overseas investor on a proposed multimillion pound sterling project to manufacture sustainable aircraft fuel (SAF) in the UK. He has provided input for a proposed air taxi service project for vertical take-off and landing (VTOL) electric, battery or hydrogen planes to replace many of the uses of helicopters. In February 2026 he was invited to present to the UK Parliamentary Liaison Group (PLG) on issues arising from achieving net zero in aviation and those related to the length of the proposed 3rd Heathrow runway.

He is the invited academic member of two tripartite (industrialists, policy makers and an academic) Brussels-based High-Level Groups (HLGs) on Financing the Sustainable Industrial Transition and on Clean Energy and Supply Security. He has been involved with the Bank of China (London) for some 10 years as an Executive Education provider on debt capital markets, portfolio construction and sustainable finance. He is currently a member of its International Advisory Group (IAG) on Strategic Issues in Globalisation. His previous book is entitled "*Commercial and Investment Banking and the International Credit and Capital Markets* "**.** He always welcomes comments on a book's content and would be pleased to have any suggestions emailed to him at bsq@icmacentre.ac.uk

Deyber Cano. My particular thanks to Deyber Cano for his major input to many of the chapters and to the structuring of this book. He is an economist with a strong track record in infrastructure investment, energy systems, oil and gas, and real estate development. He holds an MSc with Distinction in Infrastructure Investment and Finance (IIF) from University College London (UCL). His work sits at the intersection of project finance and public policy, with a focus on real estate development and structuring and financing carbon-free energy. His co-authored publications with Brian Scott-Quinn include:
https://www.pbookshop.com/media/filetype/s/p/1511312227.pdf
https://www.afme.eu/globalassets/downloads/publications/afme_guide_to_infrastructure_financing2.pdf

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Dr Michael Lai was a PhD student in the Dept. of Meteorology, University of Reading whose research interest is climate variability. His personal interest is in climate finance and climate justice. He now works for the Met Office in Exeter.

Dr Lisa Schopohl is Associate Professor of Finance focussed on socially responsible investing. Along with a colleague, Dr Ivan Sangiorgi, she contributed to data analysis for the for the Climate Bonds Green Bond Treasurer Survey.

Aditya Sharma is a civil engineer with an MBA from Henley Business School. At Rising Sun he managed a 140 MW solar power project then moved to ACME, India's largest solar power producer and is now a Senior Associate in the Climate and Energy sector at PwC

John Chessher was originally an investment banker in Europe and Asia and subsequently a fund manager in the City of London. He went on to found Cenkos Asia. He now teaches banking and sustainable finance at the ICMA Centre.

Morgan Qian Cheng was an ICMA Centre & Meteorology Department PhD student researching the impact of weather on power output, risk and investment return of intermittent generation. He was jointly supervised by Brian Scott-Quinn, Nadia Kappou & David Brayshaw (Meteorology Department). He now works for Alpha ESS, a global Chinese energy storage provider.

ICMA Centre for financial markets, Henley Business School, The University of Reading

We would like to acknowledge the huge contribution made by ICMA at the time of the Centre's founding in 1991/2, its board of directors, its CEO (John Langton) its Chairman (Jan Ekman) and its deputy chairman (Osama Elansari) which included providing over £8m of financing to enable the construction of the two buildings shown on the left, below. Also, the contribution made by Reuters/Refinitiv/LSEG which provided twenty data terminals for the first trading room in a European university. In addition, its founding was enabled by senior colleagues in the University of Reading who supported a new venture of a nature which, 30+ years ago, was an innovation which many other universities would have found difficult to support. The Centre is part of the world-renowned Henley Business School, one of the few schools in the world with accreditation from all three accrediting agencies, the US, the European and the UK.

The University of Reading has been recognised as *The Times and The Sunday Times Sustainable University of the Year 2025*, a well-deserved celebration of its outstanding commitment to tackling climate change and driving sustainability. Since 2009, the university has slashed its carbon emissions by nearly 60% and reduced its waste by 35%. Whether its cutting emissions, saving energy, or finding smarter ways to manage resources, Reading has consistently shown that sustainability isn't just a goal, it's a way of life.

The ICMA Centre buildings [the two on the left] which are part of Henley Business School were gifted by the International Capital Market Association Picture credit *https://www.icmacentre.ac.uk/about*

International Capital Market Association (ICMA) www.icmagroup.org
ICMA, with a head office in Zurich, promotes well-functioning cross-border capital markets, which are essential to fund sustainable economic growth. It is a not-for-profit membership association with other offices in London, Paris, Brussels and Hong Kong, serving over 620 member firms in 70 jurisdictions. Among its members are private and official sector issuers, banks, broker-dealers, asset managers, pension funds, insurance companies, market infrastructure providers, central banks and law firms. It provides industry-driven standards and recommendations, prioritising four core fixed income market areas: primary, secondary, repo and collateral and sustainable finance. ICMA works with regulatory and governmental authorities, helping to ensure that financial regulation supports stable and efficient capital markets.

ICMAs Role in Sustainable Finance
ICMA plays a recognised leading role in supporting the development of sustainable finance in the bond and wider debt capital markets. It provides the Secretariat for the Green Bond and other Principles (GBP), collectively known as "the Principles" – which are the de facto global issuance standard for the international sustainable bond market.

The ICMA Centre and the University of Reading Meteorology Department
As well as undergraduate and MSc courses in sustainable finance, the Centre offers an **MSc Climate Change, Sustainable Business and Green Finance** jointly with the **University's Meteorology Department** which is the top-rated meteorology department in the UK and fifth highest rated in the world. The Meteorology Department also offers an **MSc Climate Change and Artificial Intelligence[1].** In addition, the European Centre for Medium-Range Weather Forecasts (ECMFW) will shortly be re-locating to the University of Reading campus.

[1]Brochure available at https://www.reading.ac.uk/ready-to-study/study/2025/meteorology-and-climate-pg/msc-climate-change-and-artificial-intelligence

Preface

Without an understanding of the new underlying technologies of the energy transition, it is quite risky to undertake financial due diligence on any of the new types of asset which will be the backbone of the transition. For that reason this book is almost equally divided between financing issues and infrastructure asset technology issues. The need for finance professionals to invest on a much greater scale into carbon free infrastructure assets has become clearer every year of this decade. To enable this, a 'techno-economic' understanding of the new asset types is critical.

2024 was the first calendar year when global temperatures reached 1.5°C above the pre-industrial level. 2025 looks like it will break 2024's record particularly in relation to air and sea temperatures particularly in Europe. On 25 October 2024, the UN reported that the world is on course for a temperature rise of more than 3°C – twice the 1.5°C Paris Agreement limit. In many regions a warming world is supercharging the impact of wildfires, drought and flooding and causing falling crop yields and falling labour productivity in agriculture. **Thus increasing the speed of clean asset buildout and the consequent fall in emissions is a key requirement if global society is to minimise climate change impact on us and future generations**. Speed may require more regulation rather than 'voluntarism'. On 23 July 2025, the Lex column in the Financial Times noted that "that companies that rushed to sign up to net zero initiatives [a few years ago] are stampeding for the exit." Similarly, large banks are "deserting the Net-Zero Banking Alliance." The FT goes on to say that the obvious alternative is for "governments to spearhead decarbonisation via legislation or capital allocation" as an alternative to voluntary coalitions that tend to fall apart over time.

There are other issues beyond the voluntarism/regulatory issue which must be overcome to speed up the transition. Scaling up the necessary technologies and the volume of financing while providing investors with a reasonable certainty of their project generating an appropriate risk adjusted return, will be key. The whole energy **system** needs a huge level of investment – it's not just more solar, wind and nuclear generation that is required. The spokesperson for the Dutch power grid operator has said that it is not only Holland that is having difficulties with transmission grid system, but that "Belgium is in trouble. The UK is in trouble. In Germany there's lots of trouble because in Germany all the wind is in the north and demand is in the south." Holland is also short of 28,000 technicians that are necessary for the transition.

We focus on heat and heat networks as much as on electricity systems as decarbonisation of an economy will require decarbonisation of both industrial and space heating (buildings). While the energy transition in most countries will require around twice as much electricity as is being used at present, heat is currently a larger component of the total energy system than electricity and thus we must also consider how heat can be decarbonised to eliminate the use of natural gas or coal. Because of our ''whole energy system' focus in this book, our most detailed chapter is on **nuclear** as this is one of the few types of infrastructure asset which provides both **decarbonised *heat* and *power*** and in addition offers many advantages for desalination – something which will be increasingly required in Europe as well as globally (covered in Chapter 13.3). Will Advanced Nuclear Technology (ANT) small modular reactors push out the use of natural gas in the next decade? We consider this issue in a key part of the text which is Part V: ***Nuclear – A New Approach to De-Carbonisation: Innovation in Technology, Co-Location of Supply & Demand and Nuclear Financing.***

The New Energy System requires a more complex network with many more types of distributed asset connected to it than before. AI technology is necessary to integrate all these assets into the 'system' to optimise individual projects and whole system performance. Such AI systems are needed to minimise the risk of brownouts or blackouts such as when around 30GW of power disconnected from the Iberian peninsula grid on 28th April 2025 causing a massive blackout. The digitalisation of electricity systems is covered in Chapter 12 which examines 'the other transition' – the **digital transition**. But the transition to **Generative AI** and its use by customers has changed all forecasts of when global emissions will decline. It has led to a massive increase in the use of natural gas to power the new Gigascale datacentres of the so-called 'hyperscalers'. One such, being constructed in Pennsylvania, is expected to require 4.5 gigawatts of power (from natural gas) which is enough to power several large cities. In Malaysia, datacentres will account for 30% of total power demand by 2030 with gas-fired power capacity expected to rise by around 50% by 2030

In the past, geopolitical risk has not been much considered by analysts. But what is now termed 'geoeconomics' has become critical. A European example of potential risk is that the cables, known as Offshore Transmission Owner (OFTO) electricity cables bringing offshore wind power to the shore can be severed by countries such as China, Russia or Iran causing severe disruptions to European power supplies.

We have aimed for a wide readership, not only finance industry professionals worldwide but government ministers and their staff, energy regulators and those employed in the power, energy and transport sectors who need to understand more about how their company is viewed by financiers when undertaking due diligence on a proposed project. In addition it will be useful in business schools and executive education programmes in corporates and financial firms. We hope we have made a contribution to speeding the transition which becomes ever more critical as each year goes by.

Introduction to Part 1: Financing Established and Innovative Clean Energy Technologies

Introduction *to the Pivotal Roles of Technologies & Financing in the Transition to Carbon-Free Electric Power, Heat, Hydrogen & Desalination*

Capital and loan markets provide the function of matching corporate **real asset** projects such as wind and solar farms, nuclear power stations, geothermal energy sources, hydropower, energy storage, electricity and transmission and distribution and many other real tangible asset types with the buyers of the **financial assets** needed to finance them. But investors and lenders will only provide financing if the potential returns are at least equal to those of other financial assets of comparable risk which are available to them **globally**. It is the function of financial intermediaries, sometimes with the assistance of government agencies, to structure financial instruments in such a way that investors and lenders, through undertaking the necessary due diligence (DD) on a project, will be convinced that the expected returns will indeed be likely to offer them their required return on investment for the level of risk they are taking. Only then will they supply the funds required for such projects.

However, neither side of the equation – the financing side or corporate decision making on clean energy infrastructure asset investment – is working as well as it should. The problem appears to be that equity and debt investors are not used to projects which are not only large in terms of the capital required but require many more years to see the initial investment returned – they are long-lived projects. In the case of nuclear, the expected life of a power station may be 60 years or more. In addition, many investors and lenders have not previously invested in this type of asset and are not willing to undertake the necessary work to enable them to assess the level of risk involved and whether or not the expected return is at the required level.

Lack of understanding of new technologies as well as the uncertainties inherent in the new business models of potential energy transition projects creates **market failures in scaling-up transition financing**. Overcoming this requires, inter-alia, innovative public sector risk mitigation mechanisms and more co-financing by both public and private sectors. The US (until the second Trump administration), the European Union and China have all intervened in markets to support climate finance – the EU with the 'Green Deal' and China with support for solar, wind, energy storage and automotive technologies including batteries. However, much of the world including those countries, has not come close to meeting the requirements for a 'pathway' to 'net zero' emissions by 2050. Indeed, at the Federal level US the new administration in 2025 seems to have chosen to withdraw US participation in the global energy transition by abandoning the so-called "Inflation Reduction Act" which, despite its name, had been designed to greatly speed up the energy transition and also to withdraw from the Paris Agreement.

In all regions except China, problems in scaling up the production and commissioning of the necessary assets has been much slower than might have been expected. A key issue is regulatory and permitting problems. It can take ten years to obtain permission for a project and as long to obtain the needed transmission cables to transport the power from where it is produced to where it is needed. But the financing issues referred to above have also been critical in some cases. Many of these issues can be considered to be market failures

Capital flows into clean energy technologies in developing economies have been particularly anaemic. Indeed, much of the benefit from reductions in global carbon emissions that has been achieved by the actions of the US, EU and China is being dissipated by the growth of emissions in developing economies. As it is the **global** concentration of greenhouse gases in the atmosphere that determines the extent of global warming, the logic is that the large economies would gain a net benefit by paying for a high proportion of the investment required by such countries to achieve their own energy transitions. Without this aid, it is unlikely that the world will achieve the necessary transition before temperatures become unbearable for human and animal life in many parts of the globe. While superficially this is a financing problem it is as much, or more, a political and social issue that can only be resolved through political will across the world.

The need to scale-up and speed-up the energy transition worldwide is a key, indeed the key existential issue that mankind faces at present. Much greater financing of established technologies such as wind and solar is a critical need but so also is the scaling-up of innovative and existing technologies including energy storage, synthetic fuels, carbon capture and advanced nuclear. In the case of new technologies it is proving even harder than for existing technologies. While much of the problem arises from slow permitting of potential projects, many projects have

been cancelled even after construction has commenced because of the problem of political uncertainty (such as the outcome of the November 2024 US election). This has made firms hesitate to invest.

One example of this is the cancellation in August 2024, by the large wind power developer, Ørsted, of a US project to make so-called e-fuels. These are manufactured using power from windfarms. Without the certainty of tax incentives from governments and mandatory offtake (purchases) by users through regulation of, for example, the proportion of clean fuel in aircraft fuel, investment has become just too risky and too large for many investors. In 2024 and early 2025, share prices of many 'clean asset manufacturers' such as wind and electrolysers for the manufacture of hydrogen, have declined sharply due to regulatory issues and lack of offtaker demand. This and cutting back of clean project investment particularly into assets such as wind farms to which the US President seems to have an emotional objection, has held back the transition.

There is no absolute global shortage of funds available for investment in financial assets of many different types which are available to investors worldwide but in the case of 'clean tech', often the needs of investors and lenders and the objectives of entrepreneurs, developers and corporates do not match resulting in too slow a growth in clean energy assets and hence too slow an energy transition. Entrepreneurs and developers say that investors won't look at projects which have even a relatively small degree of uncertainty. This problem is often a **result of a lender's or investor's lack of knowledge and understanding of the new technologies and hence of their risks and potential returns**. They therefore shy away from such investments in favour of asset types that they understand better and which have quicker returns. Competition for funds comes from AI and datacentre investments and also from the health sector in companies such as Novo Nordisk.

There is a symbiosis between financing and the technologies that require to be scaled. It is for this reason that we have covered both sides of this equation in some detail. Without knowledge of technologies, many investors and lenders will continue to shun new types of asset and miss out on many potentially successful projects. This suggests on the one side, the need for a better understanding of the techno-economics of new types of real asset by providers of capital, regulatory agencies, planning authorities, official bodies and government. On the other side, there is a need for a better understanding by developers of the approach taken by lenders and investors to due diligence (DD) on potential projects and on structuring the financing of such projects.

Just as the new technologies have to 'work' in the sense of achieving what they were designed to achieve technologically, so also the economics of the transition have to 'work' to ensure that investors and lenders can expect comparable returns for the risks they take relative to other potential equity or debt investments that are available to them. This is what we mean by "techno-economics". Historically, energy transitions such as those we cover in Chapter 1.3 have been driven by favourable economics such as when animal power, water power, steam power and electricity were introduced and supplanted earlier technologies. But even though the price of solar & wind power and battery storage have all declined dramatically over the last twenty years, the economics is not yet in their favour. Despite the cost of solar and falling to record low levels, the costs of integrating these variable output technologies into a grid system that requires continuous generation 24/7/365 is high. In fact, power companies revert to using natural gas turbines on a large scale when power from renewables is in short supply.

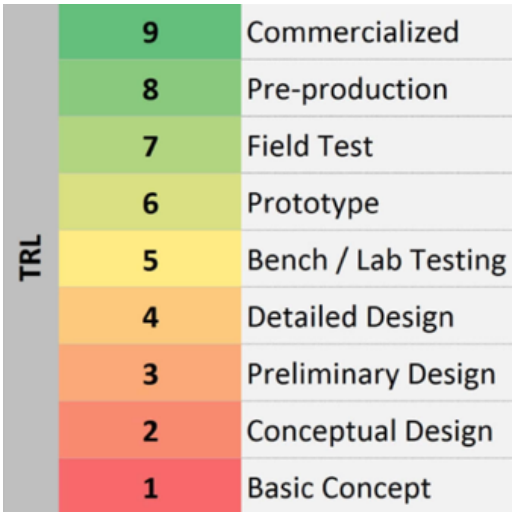
In practice there is a bifurcation in capital markets between '**infrastructure capital**' on the one hand and '**venture financing'** on the other hand which leaves a gap in the market, the so-called 'Missing Middle' which is a **market failure**. Perhaps only government interventions can fill this gap though many in the private sector have tried. This bifurcation misses out on the need for capital focused on companies that have matured out of the venture stage but have not yet scaled-up and de-risked sufficiently to access the 'fully commercial asset' capital market.

In 2024 a number of companies and projects that looked as if they would soon scale-up output to a mass-production level using capital market finance were in practice unable to do so. One reason is that investors have become more used to 'big tech' projects as we have noted above which are very different in nature than long-lived asset heavy projects. There is a big difference between digital projects where human capital is central and returns may increase relatively rapidly and 'hardware' projects where the construction period can be long as can the asset's operational life. In practice is it clear that it is difficult for many potentially highly successful companies to access the capital necessary to bridge the gap between **venture capital** and **infrastructure investment capital**.

The problem of the Missing Middle arises both in obtaining growth equity capital and also the capital required for the first-of-a-kind (FOAK) project capital. Both are critical to enable the technical de-risking of a project and the

scale-up of the technology. This is necessary to enable companies to advance from the venture stage to being bankable by infrastructure debt providers.

In effect, the climate tech market is two different markets. The first is in relatively early stage technologies and the second is in well-established technologies. A successful and timely energy transition will require the use of both and their respective financing techniques. The first of these financing techniques is required to bring new technologies to the point where they can be commercialised and then scaled. The second is required not only to enable new technologies to move to a commercial scale but also to massively scale-up existing technologies which are already in use. We cover both sectors in the following chapters. Where the problem still lies is in the 'missing middle' that we referred to above.



When considering the 'technology stage of development', the generally accepted approach to defining the stage a technology has reached is one developed by the US National Aeronautics and Space Administration (NASA) – the Technology Readiness Levels (TRLs) shown in Figure 1.1.

Figure 1.1 NASA Technology Readiness Levels (TRLs)

TRL 9 is the stage at which all companies are likely to be aiming at which point they should certainly be able to access infrastructure finance often by using project finance techniques which we cover shortly. While some companies start life at TRL 1 mostly they would start at a later stage as they are likely to be building on fundamental research (TRL1) frequently carried out by others including universities, research institutes and government laboratories.

The winners in the finance industry of the future are likely to be those firms which develop a deep understanding of the science of climate change and relevant technological innovations to mitigate and adapt to climate change impact and thus have a better insight into which technologies are likely to be the most rewarding for corporates and thereby for asset owners, asset managers and lenders. That insight will be provided by analysts who understand the new technologies. It is for this reason that we cover innovative technologies extensively as well as covering technologies.

Although the cost of solar and wind has fallen dramatically over time (though wind has risen in cost very substantially in the years since Covid) it has been government regulations, taxes and incentives – in particular guarantees and revenue subsidies from government for projects – that has driven forward solar, wind and storage projects. But much investment is required beyond just solar and wind farms. The cost of the necessary upgrade of, for example, the UK National Grid is estimated at £58bn ($74bn) This is due to an expected 64% increase in electricity demand by 2035 driven by sectors such as transportation and heating transitioning to electric power. The investment programme is also likely to involve the expansion of the offshore power grid and the construction of new north-to-south electrical "spine" in a number of countries. Despite much support for new technologies the UK government, in July 2025, declined to support the Xlinks project (see Chapter 10.2) on the grounds that it did not provide power generated domestically and also on 'security' grounds'. The Xlinks project cost was expected to require $22bn of investment.

For lenders and investors to release their capital for investment in the energy transition and thereby divert it from being applied to other opportunities, requires clear signalling from governments and an absence of frequent changes in policy. It also requires actions to reduce risks to investors in both new and existing technologies. One risk arises from the possibility of long delays because of planning rules and resistance from local communities. These issues occur in most countries but some countries are trying to tackle what is known as the NIMBY problem or the 'not in my back yard' problem which often delays project by some years at the planning stage.

In the UK, the energy company SSE has noted that by the time communities have been consulted, ground conditions mapped, seabird flocks assessed and grid connections secured ten years may have passed between the

granting of a seabed lease by the Crown Estate and construction commencing. As a result, developers may abandon projects. Since the Covid emergency ended, costs of wind turbines and interest rates have risen so much as to invalidate the assumptions behind their original techno-economic assessment.

As well as covering projects which are at the stage of the Final Investment Decision (FID) we cover innovation and financing of start-ups given the importance of research and development (R&D) in achieving the transition in time to avoid the ever increasing hazards to life on Earth with each fractional rise in the Earth's temperature.

While at one time, hydropower was a reliable source of dispatchable (continuous, always available) power, today in many regions it no longer so reliable and alternative sources of clean continuous electricity supply must be found. The reason is that droughts have caused very large reductions in power generation from hydro in many African countries, in Ecuador, in the US and elsewhere. The Kariba Dam, constructed in the 1950's with a total capacity of 2 gigawatts used to provide 70% of the electricity consumed in Zimbabwe. But its huge reservoir was only 3% full in 2022 due to repeated droughts. It continues to have a low or zero output due to drought which may be a function of the changing climate of Africa because of global warming. In the US, the Hoover Dam on the Colorado River has seen its usable power capacity plunge by up to **70%**, reducing its normal capacity of 1,302 megawatts down to 382 megawatts when the reservoir crosses the critical 1,035-foot threshold. In China, a severe drought started in 2022, forcing the authorities to revert to the use of coal to meet demand for electricity. In China as a whole, hydro capacity rose 18% between 2019 and 2023 but the amount of hydro electricity generated fell 1%.

Two alternative non-fossil sources of continuous electricity generation are geothermal and nuclear. Geothermal energy comes from heat generated during the formation of planet Earth and the nuclear decay of materials of which the earth is composed and will be able to continue to provide 24/7 energy for thousands of years. It is likely that with more investment, geothermal could become a significant source of power in **some** geographical regions. Increasingly, technology that comes from the oil and gas industry is being used such as hydraulic fracturing of rock. Such techniques enable the construction of what are known as 'Enhanced Geothermal Systems (EGS)'.

Advanced nuclear technology (ANT) reactors which we cover in some detail in Part V on Nuclear and which will quite soon be available on the market have few similarities to the megawatt reactors currently being installed worldwide. They introduce the concept of 'distributed' power i.e. small (and also micro) power generation systems similar in concept to solar and wind farms. But unlike wind and solar, they can also provide sources of heat and can be installed close to, or within, built-up areas close to sources of demand. They can provide high temperature heat which can substitute for fossil generated heat in many industrial applications. Such a distributed power approach with 'micro' reactors installed very close to sources of demand would also greatly reduce the high cost of strengthening national grids which is the investment urgently required to bring solar and wind power from often far distant locations to sources of demand.

At this point we must emphasise that when we write about the 'energy transition' we are not writing about a transition to renewables. **While renewables are critical for the energy transition, on their own they may not achieve the best balance between cost, security of power supply and a fully decarbonised electricity system**. We use the term 'clean energy' or 'carbon-free energy' (CFE) or carbon-free electricity (CFE) or carbon free heat (CFH) as the objective of the energy transition is to transition from a fossil energy based economy to a carbon-free energy system not a renewable system although renewables are likely to remain a key input in the transition.

Around the world, trade associations have an important role in helping companies to contribute to a speedier climate transition. Internationally, The International Capital Market Association (ICMA) which sponsored the ICMA Centre buildings, runs the 'Green Bond Principles'. Green bonds provide capital-raising for new and existing projects with environmental benefits.

In the UK, a relevant trade association for financial companies focussed on sustainability is the **UK Sustainable Investment and Finance Association (UKSIF)**. It has some 300 members managing over £19trn in assets under management (AUM), including investment managers, pension funds, banks, financial advisers, research providers, NGOs, among others. In turn, UKSIF is a member of Eurosif. Eurosif works as a partnership of Europe-based national Sustainable Investment Fora (SIFs) with the direct support of its network which spans over 400 Europe-based organisations drawn from the sustainable investment industry value chain. These organisations include institutional investors, asset managers, financial services, index providers and ESG research and analysis firms.

Chapter 1: Clean Energy Infrastructure and the Energy Transition

1.1 *Infrastructure as a Financial Asset Class*

It is not so long ago that infrastructure was not a capital market asset class as most types of infrastructure in most countries such as electricity and gas production were state owned. But the late 1970's, in the global drive to privatise state-owned industries and companies, the shares of many such companies worldwide (after the conversion of the entity to a corporate form) were transferred from state ownership to private ownership. The industries in which this happened includes companies providing telecoms services, electric power generation, water supply, coal gas, natural gas, railways, oil companies, airlines, bus companies, ports, iron and steel manufacturers, post offices and mail delivery services. Institutional investors were keen buyers of the new financial assets being created often simply because they believed they were being sold off too cheaply. They saw the opportunity to increase efficiency by reducing costs and so generate substantial increases in profitability and market value. Until twenty or thirty years ago, infrastructure investment referred mainly to investment in utilities such as telecoms, water, gas and electric companies and energy pipelines or 'economic' infrastructure as it is known. Today we also have renewables, LNG terminals, public transport, toll roads, airports, ports, energy storage assets, transmission cables, mobile phone masts, datacentres, 5G wireless for mobiles and by the early 2030s possibly small modular reactors.

In more recent times, also, infrastructure has transitioned from a relatively minor asset class in institutional investor portfolios to being increasingly important. Infrastructure equity is generally a 'private' asset class i.e. not traded on a stock exchange. It has different performance characteristics to those of listed equities which are traded on a stock exchange. Infrastructure project finance debt also has characteristics which differ from those of corporate debt. These differences can make infrastructure equity and debt attractive as diversifying components of an equity, bond or loan portfolio. As a result, investors have perceived considerable diversification and other benefits from adding these asset classes to their existing equity and debt portfolios.

Infrastructure assets are a component of the 'real' asset class which is assets which are tangible such as real estate and commodities (gold, silver, copper, etc.). Retail investors cannot in most instances invest in a private infrastructure project for their own portfolio. However there are many options for investing in **funds** which hold only infrastructure assets (including fossil-free funds) and are traded on markets such as the London Stock Exchange giving both institutional and retail investors the ability to obtain the benefits that infrastructure equity and debt can provide. By spreading an investor's liquidity over a number of projects such funds enable investors to overcome the lack of access to and illiquidity of, individual assets thus enabling them to benefit from stock market liquidity. We provide a case study of one of these funds in Chapter 9.4.

A fund manager may invest in **already existing** projects by purchasing a proportion of the total shares or debt of a project company in a bilateral transaction with the current holder of such financial assets. Alternatively, they may provide 100% of the necessary funding or a part thereof to a **developer** to enable a **new project** to be constructed. This latter type of investment creates **'additionality'** i.e. it increases the stock of such assets whereas investment in existing shares does not. Infrastructure assets generally have a long expected lifetime – often up to 50 or more years. In any investment today, environmental, social and governance (ESG) issues are important for ethical reasons but also for practical and regulatory reasons. It should be noted, however, that many 'clean energy' assets have low carbon emissions when operating but have a high carbon input in their manufacture. This includes the carbon content of the concrete in the foundations, the steel in the tower and the mining of the copper required for the turbine itself. These embedded carbon costs of manufacture should be amortised over the expected lifetime of the asset when comparing GHG emissions of any two technologies. Thus most generation technologies are, in practice, low carbon assets rather than being truly 'carbon free' assets.

We referred above to "asset managers" who manage portfolios of financial assets. But there are also "physical asset managers". This refers to the function of sustaining the physical assets so they continue to perform optimally over their expected life. It also includes optimising the use of such assets to generate the maximum revenue without this causing a shortening of their economic lifespan. This is facilitated by the internet of things (IoT) which enables information on the often geographically scattered assets to be transmitted to artificial intelligence (AI) systems which can alter parameters to increase the efficiency of the utilisation of such assets. The other type of asset manager is the financial or portfolio or fund asset manager. While a good physical asset manager is critical

to the success of physical assets in a portfolio, we will not be covering this function in any detail but dealing mainly with management of financial assets in a portfolio.

Renewable energy is just one of many categories of economic infrastructure. The fact that renewable energy and other "new energy assets" must compete with all the other options that investors have available to them in the wider arena of economic infrastructure means that project developers of clean energy assets must consider how they can offer equity and debt which provide the characteristics investors are seeking from private infrastructure assets and are competitive on the features that investors seeking. These include:

1. a low correlation with other asset classes;
2. relatively low volatility;
3. in many cases long "offtake" contracts which guarantee revenue levels for many years;
4. returns which to some extent are linked to inflation, and
5. higher expected risk-adjusted returns than listed equities.

The fifth feature above, if manifested in practice, is certainly an attraction. It arises because private assets such as infrastructure have relatively low liquidity (cannot be sold quickly through a stock exchange). This means that investors will **expect** to receive an 'illiquidity premium' and thus achieve a higher **expected** risk-adjusted return than from listed equities, though this premium may not materialise in practice.

While any asset, including listed and highly liquid equity, may be held for long periods, infrastructure is an asset class that is particularly appropriate for 'long-term' investors. In practice what his means is that the transaction costs involved in buying and selling are higher than for most other assets and hence, to achieve the expected return on the asset requires that the higher transaction cost be amortised over a lengthy period. Thus whereas investors in listed assets may consider trading their assets on a regular basis in the belief that this may improve performance (which in practice it often does not) or to rebalance a portfolio, assets with high transaction costs cannot be traded frequently without substantially reducing expected return. Thus these are assets which are more relevant to investor classes which do not trade actively.

Infrastructure assets often have long-term contracts with entities with a high credit rating such as governments, local authorities and major companies. Contracts such as power purchase agreements (PPAs) 'guarantee' future revenues for many years ahead. As a result, such assets can be considered relatively low volatility. For some types of investor, the long duration of infrastructure assets **is** attractive as it provides them with long periods of revenue generation and thus certainty. The type of investor to whom such assets should be attractive are, for example, pension funds and insurance companies that have long-term liabilities. In addition, infrastructure assets have a relatively low correlation to traditional asset classes in part because their value does not rise or fall during business cycles as much as traditional assets because their revenue is not as volatile as that of other assets classes and is often inflation-linked. However, sometimes this low correlation is illusory as it results from the absence of market prices for private assets and hence difficulty in valuing them in particular when quoted market prices are falling.

In addition to sources of energy generation, the energy transition will also require a large investment in energy storage (energy which can be quickly converted to electricity) to "balance" supply and demand at every moment as renewable energy has a highly variable output. Such storage needs to cope with short term variations in supply and demand and also seasonal variation (such as winter when there is much less sun) and times when there is a "wind drought" as has happened in Europe for periods of weeks at a time. Also required will be facilities that manufacture hydrogen from water and electric power using electrolysers for use in so-called 'hard to decarbonise sectors' and the manufacture of synthetic liquid hydrocarbon fuels from hydrogen and captured carbon.

The total amount of financing that will be required to achieve the Paris Agreement goals will also need a much wider range of assets than just those we have referred to above, **not all of which will be held in the financial portfolios of infrastructure funds**. Bloomberg New Energy Finance (BNEF) produces a report published annually entitled "New Energy Outlook (NEO)". In its NEO21 and subsequent issues, BNEF has used a new, broad measure of **'energy transition investment'** which, as well as including wind and solar generation, now includes investment in a wider scope of transition areas and end-uses **including the purchase of battery electric vehicles by consumers and new heating systems by households as well as energy storage, hydrogen production, and carbon capture and storage**. They estimate "the investment required globally in energy supply and infrastructure between 2020 and 2050 to achieve net zero by 2050, is between $92 trillion and $173 trillion." If this range is correct, then

between $3.1 trillion and $5.8 trillion per year on average will need to be invested over the next three decades which is greater than twice today's investment of around $1.7 trillion per year as we noted earlier.

The European commission estimates that the "twin transitions", the green energy transition and the digital transition (which is also highly relevant to the energy transition) require €620bn and €125bn each year respectively to 2030 if Europe is to achieve its committed contribution to the global climate mitigation effort. For other countries and regions similar sums are, proportionately, likely to be necessary. Unfortunately, in Europe, unlike the United States, the capital markets do not seem to be structured appropriately to enable them to channel these amounts of funding into these two types of infrastructure – energy and digital. The grand plan to create a "Capital Market Union" in the European Union has come to nothing in the last decade making large scale financing of projects either impossible or only at a considerably higher expected rate of return for equity investors and higher interest returns to lenders. The regulatory requirements of what is known as Solvency II also seem to inhibit capital investment in infrastructure by institutional investors – though there are considerable pressures for these rules to be modified. In addition, Europe's venture capital sector is unlikely to be able to fund sufficient early stage ventures to enable them to reach commercialisation. This is due in part due to 'the missing middle' as it has been called to which we referred in the last section meaning finance for growth-stage climate tech companies, particularly those at Series B+ investment stage having achieved initial venture financing from 'seed' finance and Series A equity issues (we cover this in Chapter 2.3)

Other countries have managed to resolve such issues such rather better than those in Europe and are also more able to mobilise savings for investment in the transitions (both green energy and digital) – particularly the United States with a single national capital market on Wall St. and its so-called Inflation Reduction Act (IRA) designed to subsidise green infrastructure (though President Trump has eliminated this). China, it should be noted, has created more clean energy assets than any other country and transitioned to electric vehicles more quickly than most though its use of fossil fuels has also increased substantially.

1.2 What is Energy, What is the Energy Transition, Sources of Energy, Decarbonisation

Clearly "the Energy Transition" is about changing the energy system from one that has been in use for many decades and is based on low cost oil, gas and coal to one based on 'clean assets'. To create a carbon-free energy (CFE) economy, project developers and innovators must understand how lenders and equity investors think and ensure that their projects are likely to meet their needs. On the other side of the equation, lenders and investors need to have an almost equally deep understanding of the new Carbon Free Energy/Electricity (CFE) assets and other relevant assets in order to undertake the necessary due diligence (DD) on potential opportunities. The first step in this process is understanding the energy system as a whole and the different ways in which power (by which we mean electricity) and green hydrogen can be produced as these two energy carriers (along with derivatives of hydrogen and clean heat energy) will be the basis for almost all de-carbonisation.

Energy is the potential to do work and comes in many forms – chemical, atomic, thermal, human power, animal power etc. It has been the availability of fossil fuels since the start of the Industrial Revolution which commenced in the United Kingdom around 1760 that has driven economic growth and productivity dramatically upwards in the major economies. Figure 1.2[2] shows energy in two forms. First kinetic energy and second potential energy that has yet to be released).

[2] University of Central Arkansas https://faculty.uca.edu/njaustin/PHYS1400/LabSims/sim05Energy.html

Under the Kinetic Energy heading, the first type is energy is in the form of motion such as cycling on a bicycle (human energy). Electrical Energy is the energy that can be carried through a cable. Thermal Energy arises from combusting wood, coal or natural gas and can then be utilised by, for example, using the heat that has been generated from combustion to turn water into steam and passing the steam through a turbine to create mechanical energy. This energy can then be used to drive an electric generator which converts mechanical energy into electrical power.

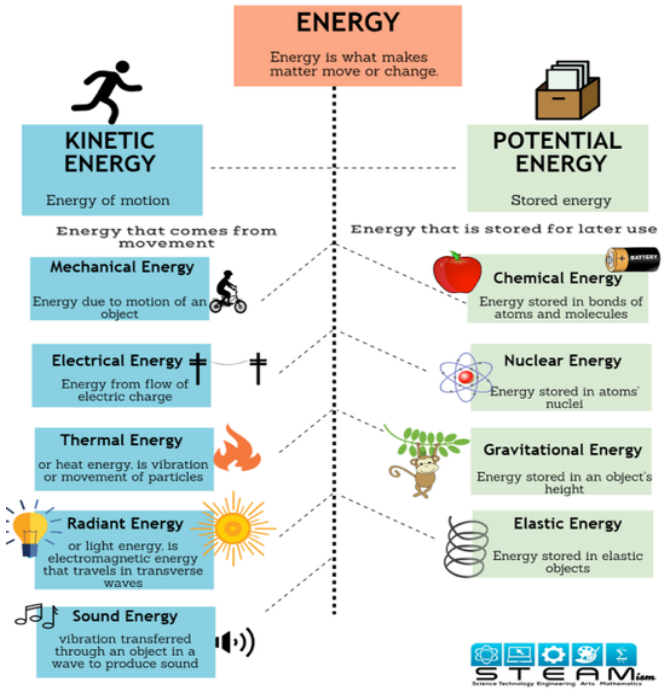


Figure 1.2 Types of Energy

Potential Energy (on the right hand side of the diagram) is not in itself energy but is energy stored in some form. Gasoline/petrol/essence is chemical energy in the form of a liquid. But as soon as it is oxidised by igniting it in air, it explodes releasing a huge amount of heat energy. In an internal combustion engine (ICE), this explosive force is controlled and turned into mechanical energy. Likewise, hydrogen which is chemical energy, is not energy in itself but when oxidised, releases its stored energy as heat.

Nuclear energy requires a fission process to release its potential energy content. This is a process whereby an atom splits into parts which releases a large amount of energy also in the form of heat. This heat can then be used to generate steam to drive an electrical (power) generator which converts heat energy to electrical energy (power).

Sources of Energy: *Conversion of Primary Energy Sources to Secondary Energy (energy carriers) & Decarbonisation of the Economy*

To understand the transition better and why it offers opportunities to developers and financiers, it is helpful to start an explanation with the original source of energy (primary energy) as shown in Figure 1.3[3]. For example, primary energy in the form of crude oil cannot be used directly but has to be converted to a usable form such as petrol, diesel, kerosene (for aircraft) or bunker fuel (for shipping). Nuclear energy is also primary energy but produced initially in the form of heat which, if not required directly as heat, can be converted to electricity.

Photovoltaic solar (solar PV) produces electricity **directly** (but not 24/7) by converting the light component of the energy in sunlight (primary energy) to electricity (an energy carrier). Concentrating solar power (CSP) which we cover later, converts the heat energy of the sun (infra red rays) into usable heat at a high temperature. Wind turbines convert the kinetic energy of wind (produced by the sun's heat energy generating air currents) directly to electricity by means of an electric generator. Hydro power is kinetic energy driven by the heat energy of the sun (infra-red) raising water up into the sky as moisture which then falls on higher ground as rain. Reservoirs are therefore storing kinetic energy which can be used to drive a generator to produce power.

In Figure 1.3 fossil fuels (at the top left of the graphic under the heading 'Primary Energy') have been the mainstay of the global economy for many decades. Initially coal, then oil, then natural gas.

[3] Source: Masao Hori *Application of nuclear produced hydrogen for energy and industrial use* https://slideplayer.com/slide/14403122/

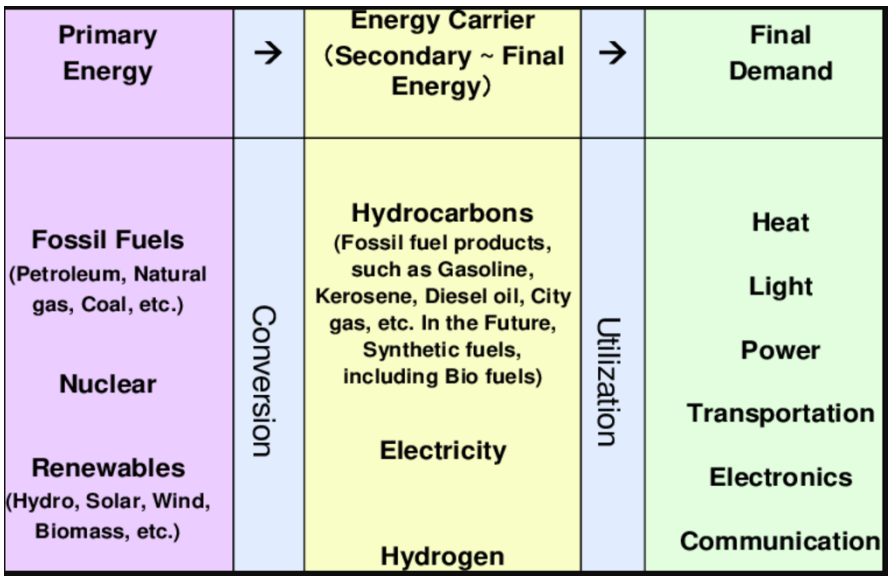


Figure 1.3 From primary energy through energy carrier to final demand

A major part of the transition involves ceasing to use these sources of primary energy and moving to nuclear, hydro, geothermal and renewables (note that batteries do not **produce** energy or electric power and only store energy). Primary energy is energy in a form found in nature e.g. crude oil, wind energy, solar energy and nuclear energy from Uranium or Thorium. It then goes through a conversion process. In contrast, primary energy from concentrating solar power provides heat energy directly for its many final demand uses.

An energy carrier (secondary or useful energy) is a means of providing energy in a more useful form. The uses of energy, namely Final Demand are shown in the last column. Examples of final energy use as heat can be found in industries such as steel, aluminium and glass manufacture and of course domestic and commercial space heating. Transport is one of the largest uses of energy for achieving motion. Domestic lighting was originally created by heating a metal filament by passing electricity through it which then glowed a warm white colour. It provided light but at the cost of also mainly producing a high (wasteful) output of heat energy. Today most light is produced by passing electricity through light emitting diodes (LEDs) which produces much less waste heat and so is much lower cost to run. It also means that less generating capacity is required for this particular use of power.

Another way of looking at energy carriers is to consider the two forms in which they exist – **electrons or molecules**. Electrons are what provide electric power. Molecules are gases or liquids (usually hydrocarbons). Examples of molecules are hydrogen itself and its derivatives (methane, methanol, ammonia, synthetic liquid hydrocarbon fuels). In the first instance molecules produce heat energy through oxidation (combining with oxygen normally from the air) otherwise known as combustion. This energy is then converted to either mechanical energy (gasoline/ petrol energy to propel a motor car) or into electrical power (by using the heat when oxidised to generate steam to turn a generator). The problem of the energy transition is how to generate electrical power, motive power for transport and heat for industry and space heating (homes and offices) without using fossil fuels and in the most economical way.

The key objection to fossil fuels is that their production and use creates emissions of carbon dioxide (CO_2), methane (CH_4) and nitrous oxide (N_2O) which are, collectively, the principal greenhouse gases (GHGs). While we tend to consider mainly CO_2, in fact methane is 30 – 80 times more potent per kilogram in trapping heat in the atmosphere than CO_2. In addition to the global heating impact of GHGs, the local impact of the fumes emitted by combusting fossil fuels in cities and in homes can also have severe impacts on human health including through the emission of very small particles called PM2 from the combustion of diesel fuel in trucks and cars.

While the key intention of the energy transition is to convert as many end uses of energy as possible to electrical power, some uses of fossil fuel are difficult to convert. Some industrial processes which use coal, oil or natural gas as sources of heat and air travel which cannot yet (and may never be able to) use electrical power to replace the high energy density of aviation fuel are examples. This is where hydrogen (as a molecule) may have a role at least for short-haul planes. We cover hydrogen in some detail in Chapter 16 and also derivatives of hydrogen such as methanol, methane and ammonia and also e-fuels such as petrol, kerosine and diesel (synthetic fuels in the above graphic). Nuclear energy, although not referred to above, is another important source of carbon-free heat and this heat energy enables the production of the carbon free electric power that is required by society which we cover in Chapter 13

1.3 Energy Transitions Through Time: Innovation in Energy Technologies

In respect of the successive industrial transitions that the world has experienced, Figure 1.4[4] shows the radical changes in the technologies employed since the 1st wave of innovation in the world which started with the UK's Industrial Revolution. The Industrial Revolution is said by some historians to have been the most important event in the history of humanity since the earlier era when mankind started to domesticate animals (horses, cows etc) and plants (the agricultural revolution). It is these industrial technologies along with the massive increase in farming in particular of beef cows which emit methane and whose pasture land requires large amounts of nitrogenous fertiliser made from natural gas which has resulted in huge volumes of greenhouse gases being emitted into the atmosphere. Growth of beef cow farming has also resulted in large areas of land being de-forested to provide pasture land for cows. Such land use change is a particular issue in the Amazon.

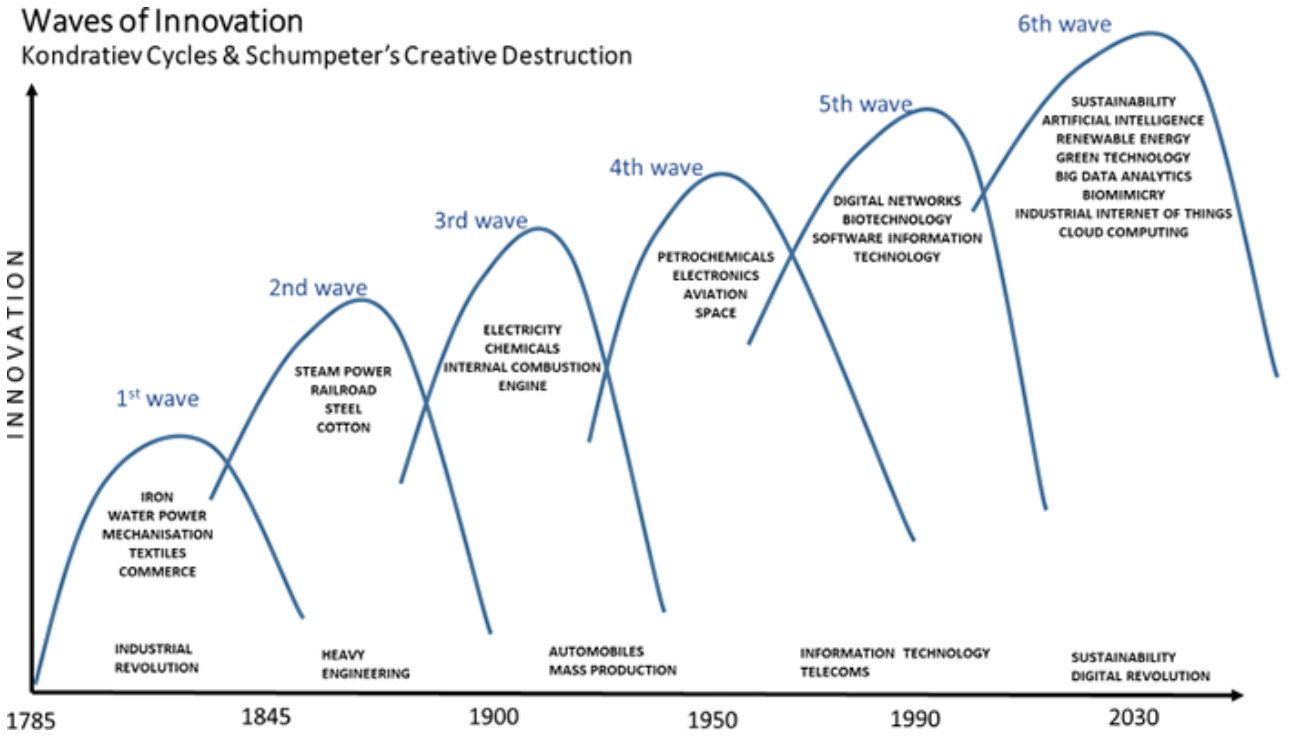


Figure 1.4 The five waves of innovation over 250 years

The 1st wave of the Industrial Revolution was a transition from the use of energy in the form of human power to animal power (horse and oxen) to drive the economy and subsequently to the use of energy from flowing rivers (waterpower). Waterpower technology enabled the development of a large-scale textile industry in the UK (mainly cotton mills) and water-powered flour mills to make flour for bread. This source of power was limited by the need to be near a river with a sufficient flow of water to provide the necessary energy. So just as solar and wind power are location (and weather) dependent, so too was water power. Wind power has also been used over the ages and in particular in Holland (Netherlands) where it was, and still is, used to pump water to prevent reclaimed land from flooding. Indeed, it is believed that windmills were used in Holland as early as 1,200 CE (AD) for grinding grain.

In the 2nd wave, fossil fuel (coal) was be used to create heat (to make iron and subsequently steel) and to provide mechanical power through the invention of the steam engine. This development enabled the creation of the iron industry and subsequently the steel industry, the development of steam powered iron ships (rather than wooden ships with sails) and railways which required iron and steel for the tracks, engines and coaches and used coal as the source of energy for propulsion. It was in this second wave that global warming emissions started to increase dramatically. Later in the industrial revolution, coal was used to produce steam which was fed to a steam engine (an external combustion engine) used to drive an electric generator.

[4] Right to reproduce this graphic kindly provided by Professor Clark of University of Technology, Sydney, Australia

The 3rd wave involved the use of fossil fuels other than coal. Crude oil, refined into vehicle fuels, could be used in internal combustion engines (ICEs) which enabled a massive increase in efficiency compared with the external combustion engine (the steam engine). The 4th wave was electronics and aviation. The 5th wave was digitisation. At the present point in the 21st century, mankind is in the 6th wave and we are seeing yet again 'creative destruction' which refers to the incessant technological (product, process and business model) innovation by which new production units replace outdated ones. In particular, as shown in the graphic, this wave includes most of what we cover in this book. Today we are probably entering the 7th wave (the graphic is from 2019).which involves new clean technologies, Generative AI, the internet of things and robots and drones (including drones for weaponised attacks in war). The expression **'creative destruction'** was coined by Joseph Schumpeter, a political economist and a one-time finance minister who considered it "the essential fact about capitalism" (along with the need for competition and innovation). In today's case, some of the destruction will be of formerly valuable fossil fuel assets which, in the new fossil-free world, will be valueless, so-called *stranded assets*.

Heat energy, supplied almost completely by fossil fuels, is as important as electrical energy as it fuels much of industry as well as its use for heating domestic and commercial premises. Transport is a huge user of (fossil) energy to generate heat which the internal combustion engine (ICE) converts to motion. When 'peak' oil and gas usage is reached, many of the assets in the oil and gas industry will start to become stranded assets.

A key reason for focussing mainly on electric power production and use in this book is that electrification of end-uses such as road vehicles, home heating and industrial processes will require a large increase in the supply of carbon-free electric power. In addition, almost all the green hydrogen that may become necessary for the energy transition (and its derivative products such as synthetic liquid hydrocarbon fuels) is manufactured in electrolysers which require large amounts of electric power. AI Large Language Models (LLMs) are also much more energy intensive than traditional AI and may consume 5 – 8% of total power generation by the next decade.

Chapter 2: Financing Infrastructure Projects: Corporate Finance, Project Finance, Financing Climate Tech Innovation

2.1. The Financing Choices – corporate finance or project finance

Financing the capital expenditure of *any* of company involves the exchange of a cash inflow to the company now (a loan or share investment) against an expectation by the lenders or shareholders that they will receive a greater amount of cash in the future generated by the assets financed by the cash inflow. In the case of debt financing the inflow may be in the form of a fixed or floating rate loan, a floating rate security or a fixed rate bond. In fact there is little legal difference between loans and bonds once common distinction that loans cannot be traded while bonds can, is no longer the case. But the key issues from the financing point of view are, firstly, whether the venture will be financed by using corporate finance techniques or by using project finance techniques. Following that decision, the next decision is the proportion of debt to equity that is optimal. This is determined mainly by focussing on what structure is likely to achieve the lowest cost of capital.

There are two ways of 'housing' a new project:

1. Using the **corporate structure** of an existing large company which is the corporate finance structure also known as **on-balance sheet financing**, and;

2. using the **project structure** by creating a new company just for the purpose of 'housing' the new project using project finance techniques also known as **off-balance sheet financing**

Corporate finance structures are relatively well known as corporate finance is a standard topic in a finance degree. Project finance is much less well known so we cover it in much more detail. In any financing, there are two aspects to which lenders pay attention:

> *First*, we assume that the economic unit will to be a going concern for the life of the loan, and therefore that the project will generate sufficient cash to service its debt;
>
> *Second*, if the entity ceases to be a going concern (becomes a gone concern*)* there must be a mechanism to enable the investors to recover as much principal and unpaid interest as possible.

The two financing techniques – corporate finance and project finance – are shown in Figures 2.1 & 2.2[5].

Corporate Finance (1) where we have an existing company running an ongoing business.

The average of the cost of each type of capital is its Weighted Average Cost of Capital (WACC). The company has decided to start a new project and to **house it within the existing corporate structure**. The project may be financed by existing resources obtained from retained earnings on the balance sheet, additional debt and/or the issue of additional shares by the corporate. The key point is that if the new project fails, the existing shareholders will take any loss. If the project is thought by investors to be risky, the WACC of the company will increase meaning that the share price is likely to fall and the cost of any future debt raise will be higher. To avoid this risk the company may opt instead to use project financing to minimise any direct impact on its own cost of capital and share price.

[5] Project Finance 101 Bryan C. Associate Director @ Buro Four | Data-driven Project Management & Engineering Professional | Mission Critical https://www.linkedin.com/pulse/project-finance-101-bryan-chan-ceng-mba-pmp/

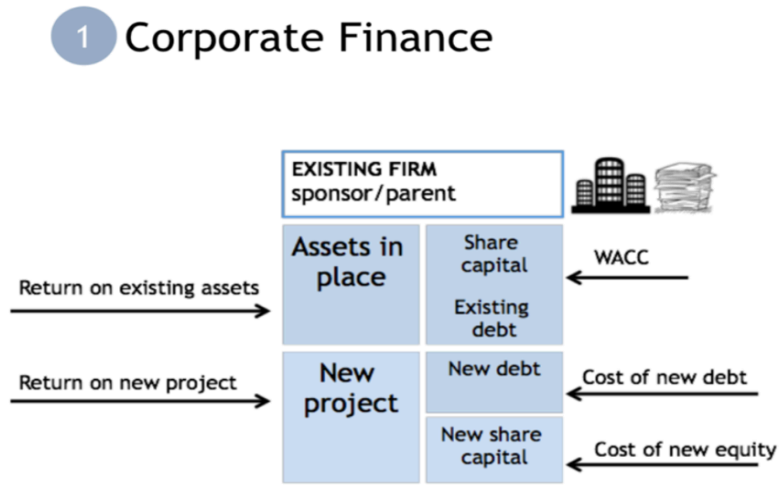


Figure 2.1 Corporate finance

Project Financing (2), we can see below that there are now **two companies** with the lower one being the project company. This company is called a Special Purpose Vehicle (SPV) or special purpose company which is "special" only in the sense that it was created only for the purpose of this one project. If the project fails and lenders are not repaid, this does not directly impact the sponsor (which may be a corporate or a developer) as the debt is not on its balance sheet. As a result it should not have much if any impact on the corporate's cost of capital. However, the drawback from the point of view of lenders is that they cannot demand repayment of the debt from the corporate sponsor or any other equity investor since the venture is quite separate legally from the sponsor and any other investors. As a result of the consequently potentially greater risk to lenders, to obtain debt funding steps must be taken to mitigate the risks of default though even with such steps the venture may still remain riskier for lenders compared with corporate financing. In consequence the spread (cost above the interest rate on loans to government of the same maturity (tenor) on any loan is likely to be higher than if corporate finance were employed and thus the WACC is likely to be higher than if financed on balance sheet.

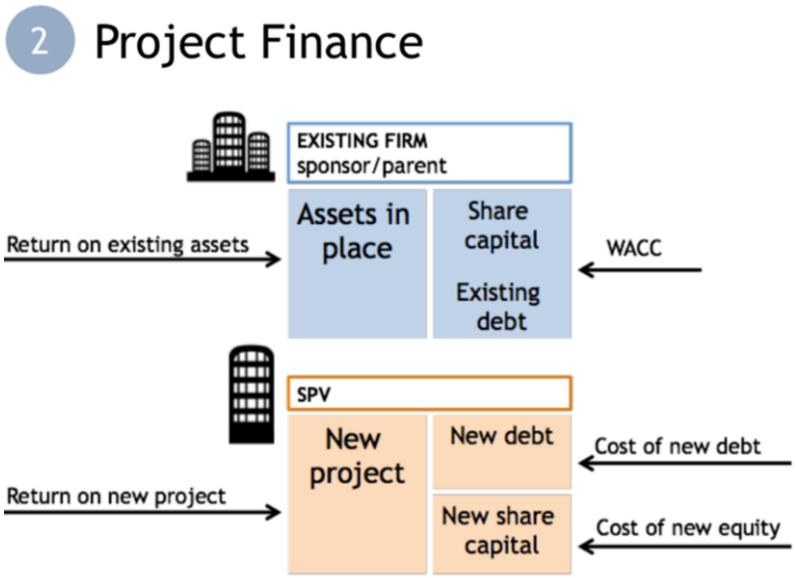


Figure 2.2 Project finance

It is frequently said that the main difference between these two structures is that in the first, lenders have recourse i.e. have a claim on all the *assets of the corporation* if the debt is not serviced i.e. interest and any capital repayments are not made on time while in the second, they only have recourse to the cash flows of the *project* and to its assets but not to the sponsor(s). That is, however, rather too simplified an explanation. In fact, most project financings involve only 'limited' recourse not non-recourse at least until a project is operational.

In addition to the fact that the spread on a project loan or bond is likely to be higher than on a corporate loan or bond leading to a higher cost of debt, the *structuring* of a project financing is much more complex than a corporate

financing and can be costly in fees. The structuring must overcome (mitigate) the problem of the potentially highly volatile nature of cash flows from a single new project i.e. undiversified idiosyncratic cash flows. The structuring also must also try to overcome the fact that if a project fails due to problems in completing it, then it is very hard to realise much value from the assets of an incomplete infrastructure project. In project finance, continuation of the project after re-structuring is almost always preferable to liquidation which might not be the case if the project was in a corporate structure.

Because project companies are almost always much more highly leveraged (high proportion of debt to equity) than corporates, any rise in the cost of debt over the life of the project (which could result if the initial financing has a relatively short term and requires re-financing) or any reduction in revenues less expenses can have potentially disastrous effects on the Cash Flow Available for Debt Service (CFADS).

Lenders also gain risk mitigation in project financing by requiring that even when it is a going concern (an accounting term for a business that is assumed will meet its financial obligations when they become due), **no dividend payments can be made to shareholders before all debt service has been covered.** If debts cannot be repaid from the project as a going concern, then the financial structuring of a project company should have been designed to ensure either that the debt holders can take steps (such as finding new managers) as they consider appropriate to allow it to revert to being a going concern or, if this is not possible, the structuring should ensure that lenders can recover the maximum percentage of money owing to them.

An additional key difference between corporate and project financing is in the structure of the loan repayments i.e. principal (P) repayments. In a corporate which is expected, at least in theory, to have an on-going infinite life, investors are willing to accept that loans will not be repaid from surplus cash generated by operations but from new loans being issued (rolling-over a loan). Thus corporate bonds will often have 'bullet' repayment terms i.e. no principal payments will be made during the life of the bond and redemption will come in one payment when the bond reaches its final maturity date. The cash required for repayment will be raised by issuing a fresh bond (often bought by the same investors that then receive the proceeds of the redemption of the maturing bond).

In contrast to this, in most project financings a bond issue, as with many bank financings, will have amortisation payments i.e. capital repayments at regular intervals. This is because a project company will generally have a fixed and pre-determined life – normally the period of any concession agreement to which it is a counterparty or a few years short of this. If a project company will be dissolved in, say, 20 years lenders would require it to have amortisation payments made to them on regular payment dates – thus taking cash out of reach of management and enhancing the probability of full repayment. Generally also, for additional security, the loan would have a final redemption date a few years prior to the project termination date. This so-called *tail* provides the borrower with the additional security of a few years more revenue flowing into the project company should be borrower default on the scheduled final redemption payment.

Some bond issues and many bank loans have structures that try to match amortisation payments with expected cash flows over the life of the project. Sometimes indeed, capital payments for redemption will be very much 'back-ended' i.e. very much biased towards the end of the project life in order that the cash requirements in the early years are minimised.

A project company has no existing projects generating revenue and no revenue will be generated until the completion of the construction phase (which will be measured in years). It will also only ever have one project which means that *without risk mitigation measures, both revenue and costs may be highly volatile*. This could result in a project company missing debt servicing payments even though in a subsequent year it may be easily able to meet such payments.

Project finance is undertaken through a special purpose vehicle (SPV) which, as we noted above, is a company with only one clearly defined objective – the development and operation of a single project. The sponsors who are promoting the project will often be those companies which will benefit from contracts for construction, equipment (such as solar installations or wind turbines) and operation and maintenance (O&M). Those who intend to buy the power (offtakers) who are intending to sign Power Purchase Agreements (PPAs) with the SPV which guarantees the purchase price of power for a period of years may also invest in the equity. Tech companies such as Microsoft and Amazon and others have an enormous, and rapidly growing, need for power particularly as large

language models (LLMs) such as Chat GPT consume much more power than traditional AI. Such companies are likely to assist the financing by investing in the equity as well as being offtakers.

The directors and officers of the SPV are constrained by **covenants** imposed by lenders which may restrict management discretion. A covenant is a formal agreement or promise between two or more parties that specifies what actions may or may not be taken by the parties. A key feature of project finance covenants is that lenders demand a lot of control over how the company is operated, particularly in terms of the use of cash flow, compared to corporate lending. Covenants lay down activities that may or may not (negative covenant) be undertaken. These are **designed to protect lenders by minimising management discretion in particular over the use of cash flow. This is to avoid cash outflows unrelated to the project.**

Four covenants which most project company SPVs will be required to consent to in the credit agreement are:

Negative pledge (the borrower agrees not to pledge any of its assets if that would reduce the security of the existing lenders;
Restrictions on additional borrowing;
Restriction on activities that the borrower may undertake;
Restrictions on capital spending and acquisitions.

The effect of these restrictions is that the SPV management may not take assets (collateral) away from the control of the lenders, may not borrow additional amounts (which would make the position of existing lenders less secure), may not diversify (which would introduce different business risks) and may not acquire another company or undertaken capital spending.

Another, somewhat different but key restriction is a *covenant to restrict the distribution of dividends*. This can be achieved by laying down uses of cash flow which must be prioritised prior to any dividends (distributions) being paid to shareholders. These are outlined below.

The revenue and cost stabilising measures that are taken should reduce the volatility of cash flow and the restrictions on management discretion should reduce opportunities for cash to leak from the company. But the lenders will also wish to try to ensure that the leverage of the company is such that it is almost certain that more cash will be generated than the minimum required to service the debt. To achieve this, lenders will set certain "cover ratios".

Cover Ratios.

Cash Flow Available for Debt Service (CFADS)
This is the starting point for calculating ratios. The lenders will require, for example, that in each period the Debt Service Coverage Ratio (DSCR) which is CFADS divided by debt service cost (P & I payment), is greater than 1. The level that they will require for the DSCR is determined by the riskiness of the project measured as the expected variability of operating revenue.

For example, if the payment mechanism is designed so that the company does not take demand risk (meaning revenue risk), lenders might be satisfied with a projected debt service cover ratio of 1.3x. But if the company does bear substantial risk (such as vehicle numbers on a toll bridge), then lenders may insist on a minimum DSCR as high as 2.0x. Lenders use detailed forward-looking financial models to estimate future cash flows and cover ratios.

There are other cash flow coverage ratios in addition to the annual measure which are intended to provide an even greater assurance of future ability to service debt such as:

Loan Life Cover Ratio (LLCR)
This is a comparable measure of ability to service debt but measured over the whole life of the loan. At any point in time, this is the discounted value of future CFADS over the life of the project plus any cash in the Debt Service Reserve Account (DSRA) discounted back to a present value divided by the outstanding amount of debt at the time the calculation is undertaken. A further calculation that is sometimes made particularly if there is some doubt about the project being able to service its debt before maturity is the Project Life Cover Ratio (PLCR). This adds cash flow after the point when the debt should have been

redeemed i.e. during the *tail* period of the project before a concession expires to test if this would enable full debt service.

Use of cash reserves to minimise risk of temporary debt service shortfalls
If despite these measures above, the actual cash flow, rather than the expected cash flow from the financial model, in a particular period is not adequate to service the senior debt perhaps because of a temporary fall in revenue, the lenders still want to receive the expected P & I payments. If the actual DSCR has fallen below one then the company will not have generated sufficient cash in that period to meet debt service. To protect themselves against this possibility there is a back-up measure included in the security documentation for the project. The lender will require that *debt service cash reserves* are built up after operating costs have been met and the senior debt has been serviced. This reserve might cover a period of 6 months or more but cannot overcome a continuation of the shortfall in cash flow once it has been used. It can also be applied to the subordinated debt.

The Cash Flow Waterfall and Distribution Covenants.
The successive uses for project revenue during the operational stage only are shown in Figure 2.3[6] (a toll road bridge project). Distribution *covenants* are designed to ensure that there is a clear priority in the use of cash to meet upcoming liabilities with senior debt service obviously having first priority and distributions to shareholders having the lowest priority. Operating expenses are paid first (since if the project does not operate there will be no revenue). Then reserve funds are built-up to enable maintenance and debt service for some months even if the bridge is closed with no revenue. What is left can be distributed to shareholders if the amount proposed does not exceed the year's profit plus any reserves from previous years (restricted under company law).

The distribution covenants are designed to ensure that there is a clear priority in the use of cash to meet upcoming liabilities with senior debt service having first priority and distributions to shareholders having the lowest priority. This is often referred to as the *cash flow waterfall*. In the waterfall graphic below, these various outflows can be seen. Operating expenses are paid first (since if the project does not operate there will be no revenue). We assume that there are no more construction costs at this point when operations start. The first call on the available cash after that is the payment of P & I on the senior debt followed by P & I on the subordinated debt. If there is cash remaining after these withdrawals then in the waterfall in the graphic, the lenders have the right to undertake a *cash sweep.*

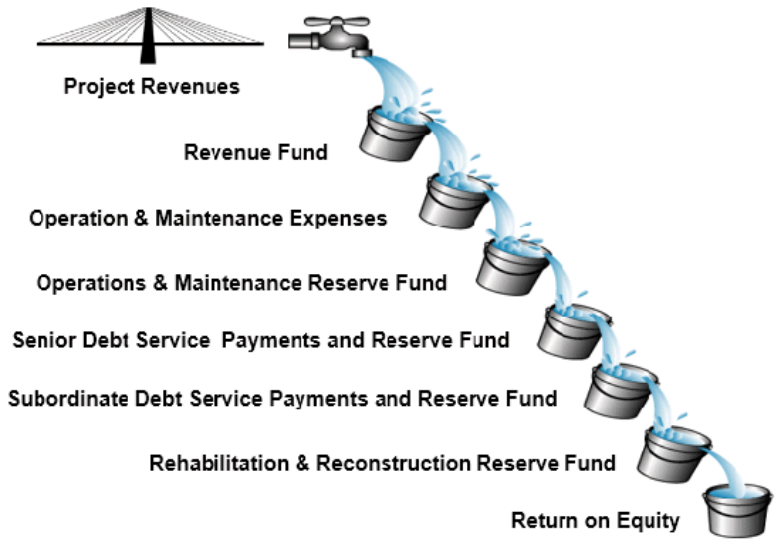


Figure 2.3 The cash flow 'waterfall' in project finance

A cash sweep refers to cash available in the company in excess of that required to provide debt service on all debt. One third to one half of the 'surplus' cash can then 'swept' out of the company and used to further pay down the senior debt beyond any mandatory amortisation payment. The remaining cash flow will then be used to top up the major maintenance reserve account (though it may already be at the required level and not require topping-

[6] US Dept. of Transportation Center for Innovative Finance Support
https://www.fhwa.dot.gov/ipd/p3/toolkit/toolkit_webinars/financing_highways/

up) and can also then be used to repay the subordinated debt if this is required by the loan agreement. Finally, what is left can be **distributed** to shareholders (**this is not a dividend payment**) provided the amount proposed to be distributed does not exceed the year's profit plus any reserves from previous years (restricted under company law). Thus the Distribution Account enables any remaining cash to be distributed to shareholders provided there have been no defaults and all financial tests area met.

After the construction period and once the project is operating satisfactorily and physical and revenue risks are low, the owners of the SPV may choose to re-finance its debt using either bank finance once again or by means of a bond issue. With risks greatly reduced at this point the credit risk spread will fall sharply. At this point, the original sponsors may also sell their shares to an infrastructure corporate or to institutional investors such as pension funds or to clean energy funds and take the capital gain resulting from the fall in the level of risk compared to the point in time at which they invested. This then enables them to "recycle" their capital into another project.

While project finance as covered above is a common way to finance projects, the 'big oil' corporates which have large cash flows when oil prices are high, will often finance a new project, whether it is oil and gas exploration and production (E&P) or a solar farm, from internal resources i.e. cash on hand (from retained earnings), corporate borrowing and if necessary a new equity issue using corporate finance i.e. *on-balance sheet financing*. Alternatively if undertaking a joint venture project, oil companies may choose to use project finance instead and create a new and separate joint venture company just to undertake a single jointly owned project using project finance i.e. *off-balance sheet financing*.

Deeply Subordinated Debt

Project equity, unlike corporate equity, is not permanent capital in the normal meaning of the term for the simple reason that project companies have a finite life. Project equity is available to the company for the duration of the project and must be repaid at the latest at the time the project company is liquidated (in the case of concessions, when the concession expires).

Ideally, project equity would be redeemed over the planned life of the company along with debt redemption. This is because over time the company needs less permanent capital simply because its assets are falling in value over time as recognised by the depreciation charge and liabilities are also likely to be falling as bonds or loans amortise. However, company law in most countries only allows dividends to be paid out of current and retained profit. Dividends cannot be financed from 'paid-in capital' i.e. the original equity capital put into the project cannot be paid out. This leaves project companies with the problem that paid-in equity (the original owners' capital contribution) cannot be paid back to them except when the project company is liquidated and all the assets sold.

In fact, 'dividends' paid out by project companies to investors are not dividends as they are understood in corporate finance. In corporate finance, a dividend is paid out as a proportion of annual profit after taxation as a result of a decision of the directors – often around a 50% pay-out. In a project company, what is paid out at intervals is a '*distribution'* of cash generated beyond that required by the project company which in practice means what the lenders agreed with the company in the covenants and in the **cover ratios** *(*see below*)*. What is 'required' by the company and what may be paid out is not at the discretion of the directors.

However, even when a given level of cash distribution is allowed under the terms of the loan agreements, this might exceed what is permitted by company law i.e. be more than the annual profit and any retained profit. This can easily happen in a project company as it is not allowed to use surplus cash to undertake new investment. As the depreciation charge on the profit and loss account is a non-cash charge, the company may have low profits for the year yet have a high level of cash generation. Thus cash can accumulate in the company on the asset side of the balance sheet as the counterpart of the original paid-in equity on the opposite side of the balance sheet. In effect the assets are being liquidated but the surplus cash is trapped within the project company.

To overcome this problem, many project companies use *deeply subordinated debt* to act as quasi equity along with a small proportion of actual ordinary share capital (common stock). As debt can be repaid to investors, it is possible to use such debt as a means of repaying this 'quasi-equity' which will normally have been provided by sponsors plus any financial investors in the project. It also has a tax advantage in that any interest paid on such debt is deductible against corporate taxation.

When considering what the word '*deeply'* means, what we are actually asking is what is the difference between equity and debt and thus the real question is whether we can we have financing instruments which are somewhere

in-between. If there is such an instrument, this would be valuable as part of the financing in the 'capital structure' of a project company. In fact there is such an instrument, if properly structured.

Deeply subordinated debt **will** act as 'equity' provided it is seen as a 'cushion' for the senior debt by having almost all the features of 'real' equity though no voting or ownership rights. Provided a rating agency accepts that the debt is truly, deeply, subordinated, the objective of providing virtually full support to the senior debt will be achieved.

Deeply subordinated debt, because it is considered virtually equity by rating agencies, allows a higher proportion of debt in the capital structure without reducing the credit rating (or credit quality if unrated) on the senior debt. However its use in the capital structure gives another benefit as we noted above. It provides a means to return to investors the initial 'equity' put into the business by the sponsors or financial investors. By paying back this debt using any surplus cash available after meeting all the requirements of the cash flow waterfall a project company can achieve a higher IRR on its equity than it could if return ***of*** equity had to be postponed to a time closer to when the project company was being wound up.

Summary of the Difference Between Corporate and Project Finance

Corporate finance, in contrast to project finance, involves financing of new projects by a medium or large company which has **an existing revenue and earnings stream** and has generated retained earnings which are now available for funding the project. In addition, funds raised from new on-balance sheet corporate borrowing or a secondary equity issue can be used. Energy majors such as the large oil and gas companies can raise finance from banks and from bond markets but raised against **the strength of their own balance sheet and cash flow (corporate finance) and not on the strength of the expected future cash flows from the project as in project finance.**

Borrowing to finance such a new project using corporate finance will often be described in the prospectus or loan documentation as being ***'for general corporate purposes'*** meaning that it is not described as being for the purpose of financing a specific new asset. Consequently the lender to the corporate has security from a call (ability to take control of) on all the corporate assets on the balance sheet in the event of default. To sum-up the differences, we have listed the main ones below:

- **Corporate finance is about financing a company which currently has multiple existing activities and new projects**
  - and is permitted to undertake future acquisition, expansion and diversification plans using retained earnings

- **Project finance is about financing a company which has no existing activities and only one new project**
  - and is not permitted to make future acquisition, expansion or diversification plans
  - will return all surplus cash to shareholders and not re-invest it
  - uses a special purpose entity (SPE)/ special purpose vehicle (SPV) and is a 'limited recourse company'
  - and the debt issued by the SPV has support only from the cash flows from the project, not from any sponsors (unlike an on-balance sheet corporate finance project)

Additional differences include that:

- **A corporation (a corporate)** is a:
  - **a multi-purpose vehicle** not a special purpose vehicle
  - generally has a very diverse range of activities and intended projects and,
  - is expected to have an **infinite life**,

- **A project company (a special purpose vehicle or SPV)** is:
  - a **single purpose vehicle**
  - has only one project and no growth projects
  - has a pre-determined **finite life**.

- **Project finance equity and debt will have been repaid by the time the SPV is liquidated. Corporates, on the other hand, are assumed (usually wrongly) to have an infinite lifespan (unlike the humans who manage them) and therefore do not repay their equity financiers (unless they choose for some reason to buy back their own shares in the stock market.**

Figure 2.4 shows a simplified graphical representation of the contractual structures involved in a project, in this case an offshore wind project[7]. At the centre is the project company (the SPV) which is sometimes known as the nexus of contracts (connection of links between things). The boxes show the main parties involved in these contracts and what the contracts are for. The construction contracts are between the SPV and the parties involved in constructing the offshore wind farm.

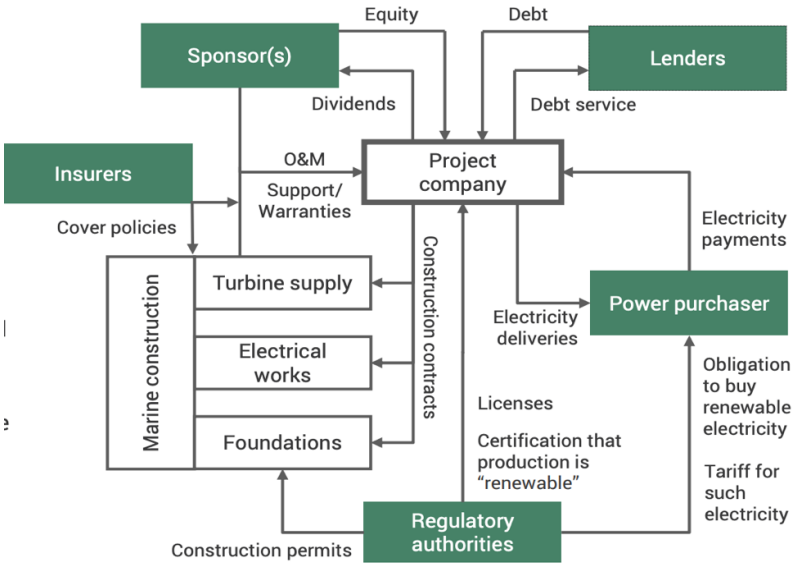


Figure 2.4 Contractual structure of a project financing for offshore wind

There are likely to be different specialist contractors for each type of work though some contractors will provide a "turnkey" service i.e. many or all aspects of the construction – engineering, procurement and construction (EPC). A link between construction and the sponsors is shown and the reason for this is that some or all of the sponsors are likely to be the firms which have bid for the contracting work and will generally also have invested in some of the equity of the SPV.

There is also likely to be an advisory firm which will work alongside project developers, contractors, investors and asset managers. There are a number of firms which provide such advice including the large consulting practices. One of the independent firms is Operis whose advisory teams specialise in infrastructure, the energy transition and digital infrastructure projects. This is a good combination of skills as the energy transition requires not just an understanding of generating and transmission infrastructure but also the digital networks and artificial intelligence systems which control projects, networks and indeed the whole electrical power system in a country. It has a particular expertise in the financial modelling of project finance transactions.

An example of the type of services an advisory firm might provide for a greenfield project i.e. the development of a new clean energy infrastructure asset, would include:

- advice on bid strategy
- contractual negotiation
- transaction model development
- running funding competitions from multiple potential sources of debt
- sensitivity analysis based on due diligence

[7] *Simplified contractual structure for an offshore wind project. Source: Green Giraffe. "Recent trends in offshore wind finance", April 2019 S7* https://green-giraffe.com/wp-content/uploads/2019/04/190411_green_giraffe_bilbao_-_trends_in_ow_finance_final.pdf

One of the objectives would be to optimise the SPVs financial structure and advise the sponsors on debt raising. Advice on contractual negotiations of project documents such as CfDs & PPA agreements can also be critical to the on-going success of a project.

There will be a number of regulatory authorities involved which will provide the necessary licenses, certifications and permits that are required. Not shown in the graphic is the agreement with the owner of the sea bed for offshore wind which in the case of the UK is the Crown Estate. If it is a land-based wind or solar project, then there will be an agreement for rental payments for the use of the land during the life of the project. Insurers are necessary for any project of this type, both 3rd party cover in case of damage to others and commercial and industrial (C&I) cover for risks to the assets. Lenders require such insurance cover so that in the event of damage to the facility or a claim by a third party, this will not affect cash flows to them.

After the design and construction stages, the operational phase requires mechanisms to stabilise *operational revenue and operational costs* over a major part of the life of the project in order to stabilise the cash flow (revenues minus cash costs) of the project company. This is to maximise the probability of having a stable cash flow and hence the highest probability of being able to meet P&I payments when they fall due. Mitigation of revenue and cost variability will reduce much of the perceived credit risk on the operational stage of a project and hence reduce the credit risk premium that a lender would charge at that point.

> *On the operational **revenue** side*, mechanisms such as a power purchase agreement (PPA) or long-term take-or-pay contracts (where the buyer of the output such as electric power must pay for an agreed amount of output even if it does not accept delivery) can be used. To reduce credit risk **on the SPV**, the counterparty to the PPA must have a strong credit rating of its own.
>
> *On the operational **cost** side*, of solar and wind are not subject to the high cost variability of fossil fuel sources of energy. Other types of clean energy asset which have electric power as an input, can clearly benefit from a PPA from the point of view of fixing input cost. An example would be fixed price long-term supply contracts between a wind farm owner and an electrolyser facility to manufacture hydrogen. These would be offtaker agreements with a generator (it would be the counterparty side of a wind farm PPA). Such contracts are designed to achieve stability of the major cost input.
>
> Operation and Maintenance (O&M) of an asset once it is operational is one of the ongoing costs in any project. An example of a contractor which provides both EPC and O&M services is Belelectric a large multinational Germany based contractor. It is a 'turnkey' provider which offers all the necessary services to enable it to hand over a completed project. Subsequent to handover it offers plant maintenance generally through multiyear contracts in order to fix the cost over that period.

Project Phases

Figure 2.5[8] (an onshore wind power project) summarises the four stages of a project's life with the final investment decision (FID) and financial close (the point at which construction can commence) in the middle.

Value creation from greenfield to brownfield – example of onshore wind development

	Greenfield		Brownfield	
	Pre-planning development	Post-planning development	Construction	Operations Ramp-up | Growth | Maturity
Duration	12–24 months	3–6 months	12–24 months	20+ years
Description	• Before permits • High risk • 1 in 3 gets approved • €0.5–1 million invest	• Permits received • Negotiation of all relevant contracts • Pre-financial close	• Post-financial close • Post all contracts • Construction risk relatively low • Pre operations	• Once constructed and connected to the grid, priority of dispatch given with guaranteed take-off agreements for energy generated • Ramp-up and growth phase not applicable
Equity*	€0–5k per MW	€150–200k per MW	€300–400k per MW	€600–800k per MW
Costs	€1–2 million per project		€15–50 million per project	

* Figures for development value increase are a generic, rough European wide onshore wind 'ndex', pre-financial crisis in 2008, not including debt.

Source: Author's image, based on Platina Partners' research.

Figure 2.5 How equity value increases at each stage of a project development

[8] Source: Javier Herrero y Saenz de Cabezon at http://jherrerosdc.typepad.com/

Before financial close (prior to the point at which contracts are signed for the debt financing) at which point the sponsors are committed to the project going ahead and construction can start, the first stage is the pre-planning development stage where, critically, the fundamental techno-economics of the project must be evaluated (conceptual design or feasibility study) and all necessary permits obtained. After that is the post-planning development including Front-End Engineering Design (FEED). The FEED focuses on the technical requirements as well as estimating an approximate investment cost for the project and expected rates of return on debt and equity capital invested after risk mitigation. Without these steps, it is not possible to put a proposal to potential funders. After this stage is the construction period which for the windfarm in this example is estimated to be about two years. During this time, the project does not generate revenue. Once construction has been completed, the turbines will be brought up to full operational output and the whole system tested and certified. Finally, the operational phase which, is estimated to be around 20 years commences and revenues start to flow.

What can be seen in the graphic above is that **an investor coming into a project at a very risky early stage** (on the left-hand side of the graphic) where cash is being spent on project planning and design with only a 1 in 3 chance of getting planning permission, **is able to 'buy into' a potential 20 year revenue stream from selling the power output at a cost of not more than €5,000 per MW**. It would normally be the project sponsor(s)/ developers who would invest at this stage. Once permits have been granted, the next stage is agreeing all the contracts necessary for construction, operation, maintenance, power purchase agreements, etc. As the project is now much less risky since the 2 in 3 probability of the project not receiving planning permission has not materialised, the price at which further equity shares in the project can be sold shoots up to € 150,000 – 200,000 per MW. Then a high-risk taking private equity fund might be interesting in investing.

Once this stage has been completed, the project company moves to financial close. At this point, an investor buying into the project will pay €300,000 to €400,000 to buy-in since risks have been very substantially reduced (the higher the price paid, the lower the subsequent return to reflect the lower risk). At this stage, other private equity funds may be interested in investing but there is likely to still be substantial 'construction risk', i.e., project over-budget and completion delay. Finally, once the project is up and running, pension funds and insurance companies may want to invest in the equity to generate relatively safe, long-term returns without any construction risk, to match their need for safe, relatively low return but long-maturity investments. These investors will pay as much as €600,000 - €800,000 per MW which means as noted above, that their returns on a successful project will be much lower than for those who bought into the equity earlier, but equally their returns are more certain as major risks have been mitigated by completing construction, testing and signing of guaranteed revenue contracts. Note that these costs per MW are hypothetical and change year by year.

On the debt side, while the project is likely to have been financed initially by a bank or syndicate of banks using a relatively short-term loan (probably 3-year maximum), once the project has been shown to be operating as designed and is generating revenue, it is likely that bond investors (relatively low risk investors which once again are mainly pension funds and insurance companies) will wish to invest. Thus at this point, the bank lending may be *re-financed* by a long-term bond issue. As bonds have a fixed rate of interest, this avoids the risk of an interest rate rise for the remainder of the project life (20 years) and thus contributes to stabilising the margin between costs and revenue.

2.2: Bond Financing of Infrastructure Projects

The transition to greater use of capital market (bond) financing requires a substantial shift in mindset from *sponsors* and *developers* seeking to finance projects using project bonds. Investment bankers with whom developers have to deal in arranging a bond issue take a transactional approach to clients rather than the relational approach with which the industry is more familiar. This arises in part from the fact that a debt capital market (DCM) banker is not providing other services to their client such as foreign exchange, treasury, transaction services, interest rate hedging, working capital lending, revolving credit, invoice discounting etc. which a commercial banker may be providing and on which the bank makes a good return. In contrast the investment banker makes a return only on a single bond issue as a project company is unlikely to ever issue another bond or equity for the same company.

In addition, a commercial banker is also going to be the investor in the debt, at least initially, and is thus both the arranger (originator) and the investor. In contrast, in the debt markets the arranging (originating) banker is only an intermediary between the developer/sponsors and investors and consequently, a second key to a successful

bond issue is an understanding of the needs of bond investors. Frequently institutional investors are seeking long-term assets to match their long-term liabilities such as pension policies, which is in direct contrast to bank lenders for whom short-term assets are more attractive and require less regulatory capital.

A bond issued at the time of financial close (rather than at a subsequent re-financing of a bank loan) offers the advantage that it will not require such subsequent re-financing because it can have as long a maturity as required by the project company. This therefore mitigates the perceived risk of a future re-financing raising the cost of debt service or not being available at all at any reasonable price.

One of the key advantages of using longer-term debt finance at the time of the initial financing if it is possible is that debt service is spread out over a longer period. This results in a smaller principal repayment each year (amortisation) and hence a greater amount of free cash flow is available to the project company in the early years which helps the project get off to a good start. In this respect, project financing which involves annual amortisation of the bond principal, is different from corporate financing where 'bullet' bonds with no, or limited, amortisation are issued and, in effect, the old bond is "rolled over" by issuing a new bond to refinance the old one at maturity.

If, after meeting all prior claims in the cash flow waterfall there is a cash surplus available to make an early equity distribution as a result of the lower amortisation payments, then the IRR of the project will be higher. This obviously makes such financing more attractive to equity investors. The abbreviation "RET" stands for Renewable Energy Technology.

Corporate bond financing and the Green Bond market
A large energy *corporate* such as Exxon Mobil is regularly in the market with new corporate bond issues (both to raise new finance and to re-finance maturing issues) in which the 'Use of Proceeds' paragraph in the prospectus would simply state 'For General Corporate Purposes'.

To the extent that such corporates choose to finance new projects on balance sheet using 'green bonds', increasingly, this *use of proceeds* clause is **not** going to satisfy investors. They will wish to know if any of the proceeds of the issue might leak into carbon-emitting investment projects. If so, many will not buy such securities or the corporate will have to offer a substantial premium. The reason is simple – many of the underlying beneficiaries of institutional investment funds will not permit their fund to make investments which increase global CO2 emissions.

One (partial) solution to the problem of the *Use of Proceeds* clause in a corporate bond is to provide the necessary disclosures on use of proceeds to satisfy the Green Bond criteria. To help with this process, and in order to make this growing asset class more appealing to institutional investors, the International Capital Market Association, ICMA (the sponsor of the ICMA Centre), developed the "Green Bond Principles." These principles are supported by over 115 institutions that participate in the Green Bond market. Traditionally Green Bonds have been issued by corporates. Increasingly they are also being issued by project companies. Indeed, it is likely that carbon-emitting projects (including projects such as combined cycle gas turbine (CCGT) power stations will find that new financing but also re-financing in the bond markets becomes increasingly difficult as investors choose to hold only Green Bonds and thus invest only in Green Project Bonds.

Project bond financing & solving potential drawbacks of such financing
While institutional investors find long bonds attractive whether corporate or project, there are a number of problems which arise when trying to fit project bonds into the capital structure of a greenfield project. The 'sticking point' for bond investors is that project bonds without risk mitigation features may have some characteristics which are not attractive to them.

> *Credit Risk Mitigation from sponsor support (construction performance support packages)*
> Institutional investors generally require a higher credit rating (at least investment grade) than bank investors and are thus likely to require some additional type of credit enhancement if they are to be attracted to investment at the start of the construction phase.
>
> One of the ways in which the investment bankers have tried to overcome this issue is by using a construction performance support package (which would include support by credit worthy sponsors)

which limits the exposure of bond investors to construction cost overrun or extended construction time and thereby may enhance the bond's rating.

If investors are persuaded that the risk of the construction phase has thereby been mitigated adequately, they may be persuaded to finance this phase. Some of these mechanisms are considered in the following sections.

Construction phase – risk mitigation by limited recourse
Project finance is not always *non*-recourse. In fact non-recourse finance is relatively unusual. There is usually some form of recourse back to sponsor. If additional funds to meet cost overruns come from the sponsors this would be provided in the form of additional equity.

The World Bank views this type of limited recourse as 'Sponsor Support'.

> *Sponsor support* may include:
>
> - shortfall guarantees, where the banks, after enforcing all other security rights, experience a shortfall;
> - buy-down undertakings: a promise to prepay project debt to ensure specified ratios, in certain circumstances;
> - price guarantees, to ensure the price of off-take;
> - market price purchase guarantees, to purchase a minimum quantity of product at market price over a set period;
> - tax loss purchases, where a shareholder agrees to purchase certain tax losses from the project company;
> - technical support, extended warranties and maintenance arrangements; and
> - contingent equity or subordinated debt commitments to cover construction or other price overruns.
>
> Source: World Bank Group PPPIRC[9]

It should be noted that during the initial periods (permitting, contracting and obtaining financing), all the cash inflow is likely to be in the form of equity from the developer and other equity investors with no dividend payments or cash outflows to investors. This is because many projects do not manage to reach the "start construction" phase and fail at an earlier stage. Thus until all permitting etc. is in place and the lenders have agreed that its conditions have been met and that the project can proceed to the point of the **final investment decision (FID)** and the **financial close**, the risk to providers of capital is much greater than after financial close. The green line in the graphic shows the level of risk and how it changes over time while the blue line shows cash flow becoming ever more negative with time and as construction starts, debt funding comes in to use and enables a very negative cash flow until production commences.

Project stages, risks and cash flow in graphical form Figure 2.6 from Green Giraffe[10], a specialist advisory firm focused on renewable energy, shows risk declining over the period up to the end of construction (as we examined above) and also the pattern of cash flow. Large amounts of capital are not required prior to the construction stage and are met by sponsor (and other investor) equity.

[9] http://ppp.worldbank.org/public-private-partnership/financing/sources
[10] https://green-giraffe.eu/sites/green-giraffe.eu/files/190411_green_giraffe_bilbao_-_trends_in_ow_finance_final.pdf

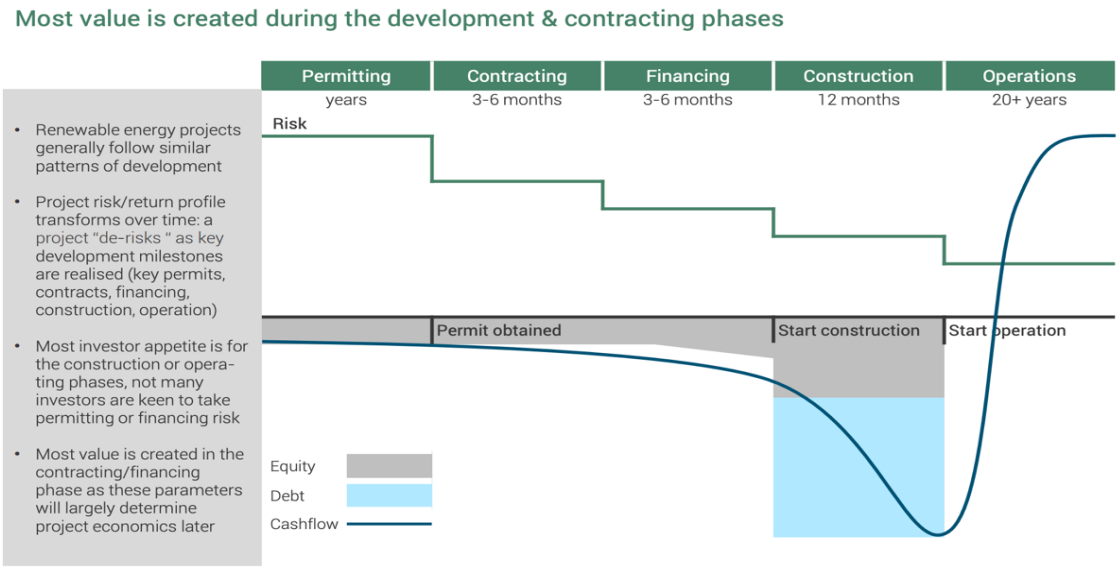


Figure 2.6 Most value created during the development and contracting phases

Examples of actual projects are shown below.

The first two are road projects. The third is for an Offshore Transmission Operator (OFTO) which means the cable which connects an offshore windfarm into the national grid. The fourth is also an OFTO while the fifth (Castor) was intended to be for natural gas storage as a reserve for use in winter in Spain. We provide a case study of Castor below. Figure 2.7 is from the AFME/ ICMA Guide to Infrastructure Financing[11] on which Brian Scott-Quinn and Deyber Cano were the principal authors with valuable input from a Committee of AFME/ ICMA Member Firms and staff members from ICMA.

	CAV Roads Project	N25 Road Project	WoDS OFTO	Greater Gabbard OFTO	Castor Gas Storage Project
Country	Italy (Baa2 stable)	Ireland (Baa1 positive)	UK (Aa1 stable)	UK (Aa1 stable)	Spain (Baa2 stable)
Project Co	Concessioni Autostradali Venete S.p.A.	New Ross N25 By-Pass Designated Activity Company	WoDS Transmission plc	Greater Gabbard OFTO Plc	Escal UGS S.L.
Issuer	Concessioni Autostradali Venete S.p.A.	New Ross N25 By-Pass Designated Activity Company	WoDS Transmission plc	Greater Gabbard OFTO Plc	Watercraft Capital S.A.
Bonds	EUR830 million 2.115% due Dec 2030	EUR145.4 million 2.684% due Jan 2042	GBP254.8 million 3.446% due Aug 2034	GBP305.1 million 4.137% due Nov 2032	EUR1,400 million 5.756% due Dec 2034
Issue Date	12-Apr-16	2-Feb-16	25-Aug-15	29-Nov-13	2-Aug-13
Project Summary	Operation and maintenance of a motorway network comprising four sections with an aggregate length of 74 km, in the region of Venice, north-east Italy	Construction, operation & maintenance of approximately 13.6km of new dual carriageway, 1.2km of new/ upgraded single carriageway and various structures including a 0.9km bridge	Operation and maintenance of electricity transmission assets that connect the 389MW West of Duddon Sands wind farm, some 14 km offshore north-west England, to the UK grid	Operation and maintenance of electricity transmission assets that connect the 504MW Greater Gabbard offshore wind farm, some 40km east of the port of Harwich, to the UK grid	Construction and operation of an underground gas storage facility utilising a depleted oil field located some 20 km offshore, near the port of Vinoraz, Spain
Project Life	16 years	25 years post completion	20 years	20 years	24 years
Term of debt	14 years	Construction + 23 years	19 years	19 years	21 years

Figure 2.7 Examples of actual project financings

[11] https://www.afme.eu/globalassets/downloads/publications/afme_guide_to_infrastructure_financing2.pdf

2.3: Venture Capital and Private Equity: financing climate tech innovation

> *For new technologies to reach TRL 9 (scaling-up and commercial production levels) they need to be financed through all the stages prior to this. Venture capital and private equity are involved in these early stages as well as financing through grants from governments and foundations*

In this chapter we examine the techniques for financing risky start-ups and financing their growth to enable them to reach TRL 9.

To achieve a much more effective and rapid energy transition from 2030 to 2050 using new technologies currently being developed, companies with well thought through concepts must find sources of risk capital, initially 100% equity (the risk capital). The technologies that are being developed, trialled and demonstrated will have to be trialled in the market and then, if successful, put into large scale production. Having covered the financing of projects at TRL9, we are now looking at those from TRL 1 – 8.

The issue in financing "post 2030 technologies" is that each stage requires a different type of investor and a different scale of capital. In addition, there are relatively few 'analysts' able to assess these new technologies and choose companies for a fund to invest in from the hundreds that may approach them. Figure 2.8[12] provides a good picture of the problem and in particular of the so-called **'valley of death'** that companies need to traverse before even potentially valuable technologies become mainstream. The valley of death has two phases but both result from the difficulty of finding sources for the additional capital that needs to be continually provided for such ventures which consume capital but at this stage have no revenue and indeed may never generate revenue. Thus cash flow remains negative for some years.

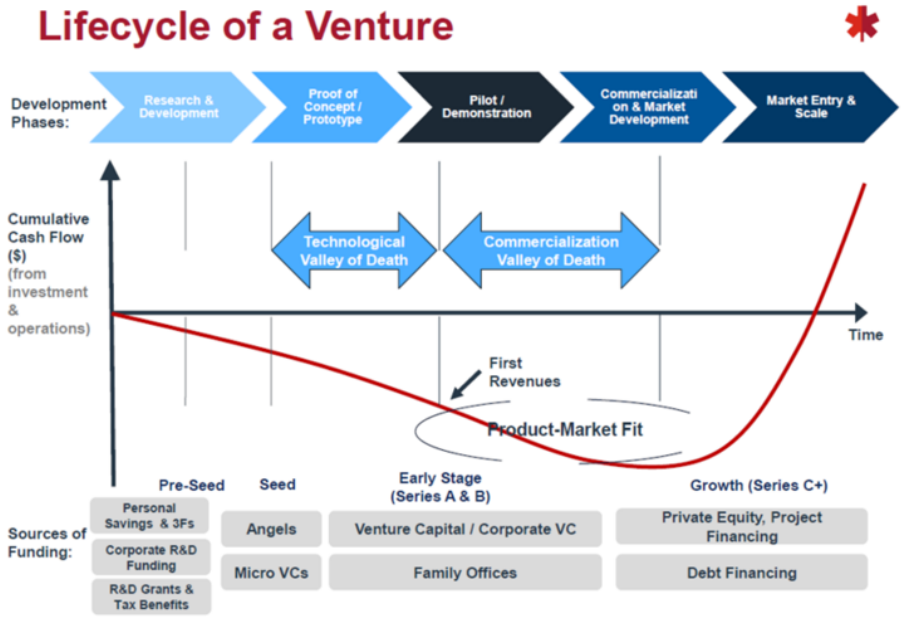


Figure 2.8 Lifecycle of a new venture

Initially many projects at the R&D stage [pre-seed capital as in the Life Cycle diagram above] are financed by research institutes, government agency grants (for example the through the US Department of Energy, universities or/and with funding from the entrepreneur's savings or borrowing or "friends, family and fools!!". That's what the 3F means in the 'sources of funding' above, described this way since putting money into a venture at this stage which has only a small chance of being successful is only for those who are foolish (hoping to make a quick buck) or are able to bear the high probability of loss.

On the other hand, venture capitalists operate by putting relatively small amounts of money into a large number of ventures. While 80% of their investments could fail to make a high return or fail completely, the 20% which succeed, may provide very high returns. Thus this diversification over many projects protects professional investors from the most serious losses on their portfolio.

[12] Graphic from Business Development Bank of Canada (BDC) *BDC's Industrial, Clean and Energy Technology (ICE) Venture Fund* https://www.bdc.ca/en/bdc-capital/venture-capital/funds/energy-cleantech-venture-fund

In the graphic, the terms Series A & B are used. While a Series A funding round is to develop the product, a Series B Funding round is when the business is being taken to the next level, past the development stage. Series A is about getting the product to the point where there actually is a product. Series B is designed to enable the business to start selling product, while Series C is about scaling-up operations and possibly preparing for an exit like an IPO or acquisition to enable earlier investors to realise their capital gain (or loss).

Entrepreneurs can also work within large corporates when they are known as **intrapreneurs**. Indeed the first personal computer, the IBM PC, was funded internally by IBM [corporate R&D funding] but in a separate facility far from head office (known as the 'skunk works). Many of those in head office thought the venture was almost certain to fail. The PC was introduced to the market in August 1981, only a year after the company gave the team of intrapreneurs the go-ahead. Initially it was thought by IBM management that there were not enough applications for this device to justify the idea of large scale production. But as we now know, it was a roaring success against all odds, in particular the odds against incubating a completely novel idea for a desktop computer being produced by the manufacturer of 'big iron' (mainframes).

After the pre-seed stage when the basic R&D has shown the possibility of a saleable product, we have the seed capital stage as the next stage on the timeline. At this point, some capital may come from 'Angel Investors' (those outside family and friends but who are willing to help what seem like a promising venture such as Henley Business Ventures). At this stage some small VC firms may also make some contribution. This funding is all in the form of equity i.e. risk capital as it is unlikely that any financial institution would lend to a venture at this stage.

The technological 'valley of death' is the period when the technology is being tested and almost certainly, there will be technology failures along the way. If investors keep faith and the entrepreneurs and scientists seem able to overcome the technology problems, investors may put in more capital. If not, the company will collapse (we give two on-going examples of this problem at the end of this chapter). Those investors would be the larger venture capital companies and the venture capital arms of corporates. In addition family offices which are the private wealth management companies of ultra-high net worth (UHNW) individuals may contribute capital. They would be subscribing to what is known as Early Stage Series A & B series funding rounds prior to the hoped-for Series C funding round. A key distinction between these successive funding rounds is the **valuation of the business which is likely to rise at each stage as risk is reduced and hence value in increases**. The different stages also reflect its maturity level and growth prospects and the greater these are, the higher the valuation.

While there may be some early sales of the product/service to 'early adopters', this does not guarantee a market large enough to enable survival. This is the 'commercialisation valley of death' point in the timeline which is partly during early stage (which may not go as well as hoped) or during the early growth stage when expectations are not met and the promised funding is discontinued due to the company not meeting certain targets. This is still a period in which cash flow is negative (the red line) and revenues are still well behind costs. But once the cash flow graph turns up as it does in the graphic, the company 'is on its way' and at that point, but generally not before, it may (as the graphic shows) be able to obtain some debt financing including project finance.

Financing a project becomes increasingly difficult as its capital intensity (in this case defined as absolute amount of capital) rises and also as technology risk rises. While venture capital firms may be able to finance a high risk project if the absolute capital requirement is not too large (and they take only a proportion), if the project also has a high capital requirement that is not within the business model of most such firms. The reason for this is that a VC firm, while only investing in firms where they have a high expectation of success, in practice, as they also know, many of these firms will fail. Thus they need to have many investments (a large, diversified portfolio as we noted before) to ensure that on average they generate their expected rate of return for the risks they take and consequently, cannot put too large an investment into any one firm.

Unfortunately most clean energy infrastructure assets involve a high absolute amount of capital and a long period of time to recover the initial investment. We are looking here at the 'bankability' of a project. Simply put, this means structuring a project such that the type of financier being approached is satisfied by the expected return in relation to risk. In practice this means designing an optimal risk-sharing protocol at the project development phase. This will frequently mean a need for government being involved in risk sharing. It is likely to apply to, for example, nuclear reactors whether these are giga scale power generation projects or small modular reactors.

British Volt and Northvolt

While we have not covered innovation in the transport transition detail, it is important to note here that the transition from fossil-fuel powered vehicles to battery electric is also proceeding much more slowly in Europe in particular than expected by some. Converting factories from internal combustion propulsion to electric (EV or, though less so, hybrid) has not been going well. This is also a result of a dramatic change in technology from traditional vehicles to so-called 'software defined vehicles' (SDVs) which has been happening at the same time. These are vehicles in which software has a key role to play in managing its operations and features, rather than, as traditionally, employing hardware. SDVs are also a key part of the automotive industry's shift toward self-driving and connected cars. Additionally, batteries are a key component (and comprise up to 40% of its cost) of such a vehicle and European companies have been less successful in making such batteries at the scale required.

Most batteries come from South Korea and China. The UK's attempt at creating a start-up battery factory called BritishVolt, ended in failure within around three years when the company filed for bankruptcy. In Continental Europe, Northvolt was a brave attempt to create an operation able to provide batteries on a 'gigawatt hour' scale to power vehicles made in Europe. But by September 2024 it was in danger of going the same way as British Volt and has now filed for Chapter 11 reorganisation. Northvolt raised more capital than any other private group in Europe — more than $15bn in equity, debt, and government support. Its problems signify **the difficulty in 'scaling-up' production from initial prototypes or small scale production to the huge volumes of battery cells required by road vehicle manufacturers**. That then leads to difficulties in obtaining scaling-up financing and breaking through to mass production. Without this breakthrough, sales are low, revenues are low, costs are high and investors are very cautious. Indeed this stage of financing as we noted above, is known as 'the missing-middle' – between venture capital/private equity and infrastructure finance. Meantime, such a company is 'burning cash' and investors become increasingly nervous. This is a fate that governments need to ensure that Gen IV AMR manufacturers do not approach.

Chapter 3: Green & Sustainable Finance

3.1 Financing the Climate Transition: ICMA Green, Social & Sustainability-Linked Bond Principles & Sustainability Bond Guidelines (collectively GSS bonds), Transition Finance Market

For many investors there is a requirement, either from a regulatory perspective or from the conditions in an investor mandate, to invest in 'sustainable' financial assets. Here we cover mainly investment in bonds as infrastructure assets are generally financed with a high proportion of debt relative to equity. However, there is often confusion in terminology in relation to the word 'sustainable'. There are many terms in use making it necessary to define these terms more precisely. ICMA, the Zurich-based body which originally financially sponsored the ICMA Centre has taken a particular interest in sustainability. ICMA serves as Secretariat to the Green Bond Principles (GBP), the Social Bond Principles (SBP), the Sustainability Bond Guidelines (SBG) and the Sustainability-Linked Bond Principles (SLBP) - collectively known as "the Principles"[13]. ICMA has also produced a document entitled *Sustainable Finance – high-level definitions*[14] which provides a good graphical presentation in Figure 3.1 of how various terms – sustainable, green, social and ESG – relate to each other.

The graphic illustrates the interaction between key high-level definitions and especially how the wider definitions incorporate narrower ones. For example, Sustainable Finance (which if achieved, should lead to greater economic and financial stability) is the widest definition incorporating all the others – Green Finance, Social Finance, Impact Finance, ESG Investing and Climate Finance. White boxes aim to identify key differentiating themes between the definitions e.g. social issues & outcomes in relation to Social Finance. Double arrowed boxes identify definitions that interact dynamically with themes or other definitions e.g. Impact Finance in relation to Green Finance and Social Finance.

Achieving a net zero carbon economy by 2050 requires transformation of business models as well as wider economic and social transformation by companies, banks, investors, insurers and also households. In this context sustainability reporting has a key role to play, meaning reporting by companies on the extent to which they have set and achieved sustainability objectives and targets. While not all sustainability issues such as rising temperatures and water droughts and water shortages may be deemed immediately existential by some companies, they can nonetheless have severe impacts on companies and their stock market value.

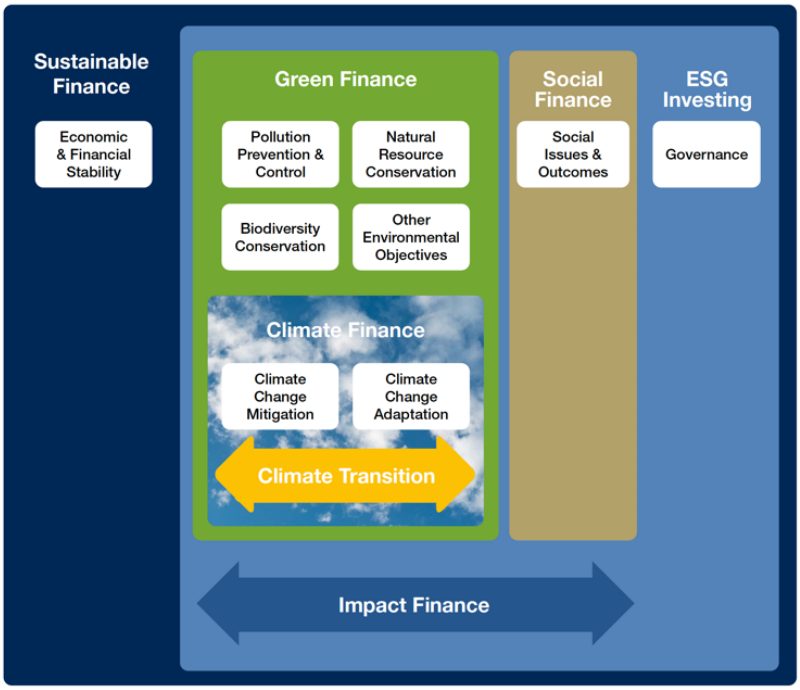


Figure 3.1 Wide and narrow definitions within sustainable finance

[13] https://www.icmagroup.org/sustainable-finance/ and https://www.icmagroup.org/assets/documents/Regulatory/Green-Bonds/Sustainable-Finance-High-Level-Definitions-May-2020-051020.pdf

[14] https://www.icmagroup.org/assets/documents/Regulatory/Green-Bonds/Sustainable-Finance-High-Level-Definitions-May-2020-051020.pdf

Sustainable finance

Sustainable Finance incorporates climate, green and social finance while also adding wider considerations concerning the longer-term economic sustainability of the organisations that are being funded, as well as the role and stability of the overall financial system in which they operate.

Climate finance

Climate finance is the financing of the changes in the real asset structure of society required to transition the world from carbon-based energy to carbon-free energy and to finance adaptation of the world (households, corporates, cities, countries) through for example protective sea walls (such as Venice requires), air conditioning to cope with increased temperature and changes in road surfaces to minimise tar melting. Climate finance is about re-wiring the financial system which we consider in the next chapter.

Climate finance seeks to support mitigation, adaptation and other actions that will address climate change. While climate-change is clearly the existential issue right on the horizon (the Tragedy of the Horizon that Mark Carney has talked about) and is the focus of this book there are many other environmental issues that may impact the sustainability of the planet and therefore individual companies and individuals and even countries.

Air pollution in the home (from natural gas or wood cooking), in city air (in cities with vehicle smog problems), water pollution in rivers (from dumping of dangerous/carcinogenic materials), plastic in the ocean (from plastic being dumped in rivers and flowing into the oceans), are all environmental issues. Today, however, maintaining biodiversity of plants and animals, keeping the air fresh and ensuring the quality of water, soils and minerals which in aggregate means preserving and enhancing natural capital, are aspects of sustainable finance which is an all-encompassing term.

Green finance

Green finance has been defined as any financial activity that has been designed to ensure a better environmental outcome. Thus, as well as climate finance, we would be concerned with financing which helped maintain biodiversity of plants and animals, keeping the air fresh and ensuring the quality of water, soils and minerals which in aggregate means preserving and enhancing natural capital.

OECD Definition of Green Finance is finance for "achieving economic growth while reducing pollution and greenhouse gas emissions, minimising waste and improving efficiency in the use of natural resources."

People's Bank of China (PBoC) definition of Green Finance Policy refers to a series of policy and institutional arrangements to attract private capital into green industries such as environmental protection, energy conservation and clean energy through financial services – including lending, private equity funds, bonds, shares and insurance."

Impact Finance

Impact finance refers to investments in corporates or funds and loans to corporates, made with the intention of generating a beneficial social or environmental impact as well as a financial return

The Transition Finance Market

This can be described as finance that supports an organisation to credibly decarbonise its activities, respond to any climate-related risks and opportunities, and to contribute to a global and economy-wide net zero transition. This includes financing of hard to abate and high emitting areas of the global economy.

Net Zero

Net zero refers to the balance between the amount of greenhouse gas produced and the amount removed from the atmosphere. We reach net zero when the amount we add is no more than the amount taken away. How to achieve this is the focus of this Guide.

Net zero in financial asset portfolios is highly contentious in practice even though the general definition above would probably be widely accepted for corporates. One area of contention is around the concept

of 'offsetting' by, for example, buying carbon credits or planting tree saplings. Also, what we need is 1.5 °C maximum rise in temperature with net zero being only one possible mechanism for achieving it.

The first observation we have to make on Green Finance is that any type of financing can be green. However, particularly for investors with a 'green mandate', a means of knowing that a particular security is 'green' is vital. It is for this reason that the Principles Community described below, set out to develop Principles and Guidelines for 'Green Bonds'. This was some years prior to the EU's Taxonomy and Green Bond Standard (EU GBS) being launched.

The second observation is that in the future, all financing whether bond, bank or equity, *should* become 'green' i.e. there should be no financing of fossil fuel assets i.e. so-called financed emissions should cease. If this is not achieved, then it is unlikely that the Paris Agreement temperature objective will be achieved.

Most 'green' financing i.e. financing which follows principles laid out by the 'Principles Community', is bond financing and in large size. Although there are also green loans, they would not generally be for financing the development of new technologies but for project or corporate finance. Thus green bonds are much more likely to be used as a financing instrument in scaling existing proven technologies.

Many investors have preferred to purchase green bonds (as defined under ICMA's Green Bond Principles or EUGBS) to help achieve one aspect of their ESG mandate and may even accept a slightly lower return on them than on other bonds although at present green bonds comprise only a very small proportion of the global debt market. As the EUGBS become established, bonds meeting this standard's requirements are likely to predominate. However, ICMA has also developed Principles and Guidelines for other types of bond such as social bonds, sustainability bonds, sustainability-linked bonds. More information on all these bond types can be found at https://www.icmagroup.org/sustainable-finance/

There are two other private sector sustainable finance initiatives. The first, known as the ICMA Sustainable Finance Committee, is part of ICMA though it is separate from the Green Bond Principles initiative. We consider its role first. Launched in September 2019, the ICMA Sustainable Finance Committee (SFC) brings together various ICMA committees and constituencies (i.e. Asset Management and Investors Council, Corporate Issuer Forum, Financial Institutions Issuer Forum, Legal & Documentation Committee, Primary Market Practices Committee and the Regulatory Policy Committee) all of which have a growing interest in sustainable finance. Included in this Committee is the Steering Group of the GBP SBP SLBP Executive Committee.

The SFC is designed to:

- Provide a forum for discussing sustainable finance developments;
- provide a wide platform for ICMA regulatory responses and dialogue on sustainability-related topics;
- liaise with the Global Sustainable Finance Council (GSFC) to further leverage cooperation among leading financial industry associations; and
- provide input into specific cross-cutting sustainability-related deliverables.

The other initiative is the GSFC which we cover below:

> The Global Sustainable Finance Council (GSFC) was created in 2017 (originally as the Global Green Finance Council) with the objective to bring together key global and regional associations and other stakeholders involved in green and sustainable financing. The industry joined forces to coordinate efforts to promote sustainable finance, facilitate cross-fertilisation between related markets and asset classes, and with the ambition to act as a representative counterparty to the official sector on sustainable finance policy matters.
>
> In 2017 GSFC published a directory to provide policymakers and global and regional market participants with a simple, easy-to-use reference guide as to which international and regional governments and industry bodies have implemented or are implementing major initiatives on green, sustainability and climate change.
>
> The GSFC was also the forum where the need for guidance for green loans was identified which led to the publication in 2018 of the Green Loan Principles (GLP) by the LMA, together with the APLMA and the LSTA with the support of ICMA.

The Principles Community

The Principles Community consists of more than 400 members and observers. Members are institutions that have issued, underwritten, placed, or invested in a Green, Social, Sustainability or Sustainability-Linked Bond. Members elect the Executive Committee, decide on changes to the Principles governance, and provide input into their updating. The International Capital Market Association (ICMA) operates the Secretariat for the Principles and provides advice on governance and other issues.

3.2 Green Loan Principles

In the international loan market, the use of similar principles to those developed for the international bond markets would clearly be beneficial for debt markets as a whole. For this reason, the European trade association for loans. The Loan Market Association (LMA), the Asian loan market equivalent (APLMA) and the US equivalent, the Loan Syndication and Trading Association, have jointly produced two sets of Principles, one covering Green Loans and one covering Sustainability Linked loans.

The Green Loan Principles (GLP) build on and refer to the Green Bond Principles (GBP) of the International Capital Market Association (ICMA), with a view to promoting consistency across financial markets. The GBP are the internationally recognised voluntary issuance guidelines that promote transparency, disclosure and reporting in the green bond market. The GLP are intended for broad use by the market, providing a framework within which the flexibility of the loan product can be maintained, and will be reviewed on a regular basis, in light of the development and growth of the global green loan market.

Chapter 4: Financing Case Studies: Innovative Structures

4.1: Castor Gas Storage Project[15]

Figure 4.1 provides a description of the Castor Gas Storage Project. The refinancing of the Castor gas storage project off the coast eastern Spain was one of the most significant deals in European energy finance in 2013. It marked the debut of the **project bond credit enhancement product from the European Investment Bank (EIB)**, a risk mitigation product designed to make private sector investment into the energy transition more attractive.

Note that the currency of the transaction below is euro.

Watercraft Capital SA/Escal UGS

STATUS
Priced 25 July 2013

SIZE
1.7 billion

DESCRIPTION
1.9 billion cubic feet gas storage facility located 15km off the coast of Spain

SPONSORS
ACS (67%); Castor UGS (33%, with Dundee Energy representing 74% of that shareholding)

EQUITY
252 million

DEBT
1.43 billion

BOOKRUNNERS
Bankia, BNP Paribas, La Caixa, Crédit Agricole, Natixis, Santander, SG

MATURITY
2034

COUPON
5.76%

PROJECT BOND ENHANCEMENT
EIB

MONITORING ADVISER
Trifinium

SPONSOR LEGAL COUNSEL:
Watson Farley & Williams

BOOKRUNNER LEGAL COUNSEL
Allen & Overy (Luxembourg, English and Spanish law)

EIB LEGAL COUNSEL
Clifford Chance

TRIFINIUM LEGAL COUNSEL
Ashurst

INDEPENDENT ENGINEER
Gaffney Cline

INSURANCE ADVISER
Willis

MODEL, ACCOUNTING AND TAX ADVISER
Deloitte

Figure 4.1 A description of the Castor Gas Storage Project

We can learn a lot from the above information. First, Watercraft Capital which is what the SPV is called, is quite a highly leveraged company with most of the financing being debt. Part of the enhancement of the credit rating of the bond is a result the new EIB project bond initiative to encourage such bond issues. There is a wide range of advisors and lawyers involved in the transaction. The information below, provided by IJ Global, is very helpful in helping to understand some of the issues in project financing:

> Special purpose vehicle Watercraft Capital issued €1.43 billion in bonds, and lent the proceeds on to project company Escal UGS, the concession company for the construction and operation of an underground gas storage facility. The concession company used the proceeds of that loan to refinance the bank loans that funded Castor's construction. The original financing closed in 2010 and stretched the bank market in the aftermath of the 2008 financial crisis. Banesto Santander, Caja Madrid, Crédit Agricole, SG, WestLB, BayernLB and Natixis headed up a 12-strong group of mandated lead arrangers.
>
> The bond issue priced at the end of July of 2013 and settled on 2 August 2013. The deal was the first fruit of the European Commission and EIBs project bond initiative. The initiative is meant to attract additional long-term lending to infrastructure from institutional investors such as insurance companies and pension funds.

[15] Case study based on material from IJ Global website
https://www.ijglobal.com/articles/127334/european-refinancing-deal-of-the-year-2013-castor-ugs#comments

Spain has a troubled economic climate and its energy sector faces severe structural weaknesses. But Castor featured minimal remaining construction, and a regulatory framework that allowed the project company to recover its costs and a return on equity from gas network users.The EIB provides a €200 million liquidity line that can be drawn in a variety of scenarios that reduce the project's ability to service its debt.

The global coordinator was Santander. The structuring banks were Bankia, Credit Agricole, Caixabank, Natixis, Santander and SG. Bookrunners were BNP, Bankia, Crédit Agricole, Caixabank, Natixis, Santander and SG.

The coupon on the amortising bonds, which mature in 2034, is 5.756%, equivalent at pricing to around 100bp over the 20-year Spanish government bond. The bonds have an average weighted life of 12 years, and a rating from Fitch of BBB+, a notch above the Spanish sovereign rating.

The order book comprised institutional investors (61%), agencies (including the EIB, with a €300 million ticket, 25%), fund managers (10.1%) and banks (3.9%). The investors' geographical split was Germany (28%), Benelux (including the EIB, 23%), Spain (18%), France (11%), the UK (10%) and Italy (10%). The EIB was pleased, in particular, at the level of response from insurance companies, which had been following the project bond initiative closely.

After the bonds closed, the sponsors detected seismic activity in the region. The Spanish government again delayed final cushion gas injection and acceptance, while additional seismic surveys took place.

The vagaries of the Mediterranean's geology, then, could test the termination provisions of Castor's concession contract sooner rather than later. But these delays should not put investors off the entire bond initiative. Since then, the EIBs bond enhancement has featured on the Greater Gabbard offshore transmission operator project, which runs under the more seismically stable North Sea. If the product can help refinance a project with Castor's technical challenges and history, it should be able to enhance a variety of other unusual credits.

In December 2017 (4 years after its original analysis above), IJ Global published the following information[16]:

The Earth Moved!

WHAT WE SAID THEN: "Construction and geological risk, particularly in the latter part when the cushion gas is being inserted, is difficult to quantify".

Possibly the most dramatic instance over the last 20 years of event risk, the Castor underground gas storage project in Spain failed because it unexpectedly caused earthquakes. But while the Castor facility was closed down, bond holders and sponsors were not left out of pocket, making it a fine example of good structuring.

The original financing for Castor closed in July 2010. It funded construction of the underground gas storage facility which included both onshore and offshore elements. Wells were located 21km off the east coast of Spain providing around 1.9 billion cubic metres of storage capacity. The gas was compressed 15km inland from the coastal town of Vinaros for injection into the gas reservoir, ready to be pumped back into the Enagas-operated pipeline network.

The project carried no market or volume risk, but as we said at the time, pretty high geographical risks which were not compensated by high returns as in oil exploration development. To provide some protection, the sponsors agreed a five-year window in which they could walk away (on reasonable grounds) and receive compensation.

Construction began in 2008 and was completed in 2012, although when the debt was refinanced in July 2013, the project was still waiting for Spanish government approval after running into trouble. The lingering impacts of the financial crisis led the government to increase the debt tenors on the project from 10 to 20 years, in order to reduce the burden of charges on Spanish gas and power users.

Then in January 2013, the Spanish authorities approved a ministerial order allowing the project to use gas from a third party. Enagas, the operator of Spain's gas grid, and Castor's largest customer, had already agreed to buy half of ACS' stake at completion. It now took over from the project the role of buying cushion gas.

The bond refinancing represented the first ever use of the European Investment Bank's project bond credit enhancement (PBCE) product. This attracted criticism due to the lack of demand risk or much completion risk on the transaction. But the EIB seem very keen to test its new toy, which saw it take a €200 million unfunded commitment that starts out as equivalent to 14% of senior debt, eventually decreasing as the bonds amortise, and staying equivalent to 20% of outstanding principal.

With the PCBE, Castor achieved a rating from Fitch one notch above Spain's sovereign rating a raised a total of €1.43 billion. Then Castor was shut down in 2014 after the detection of more than 200 minor earthquakes the previous year. Full repayment of the bonds happened in November 2014 following a Royal Decree which terminated the

[16] IJ Global *The Earth Moved: Castor gas storage* 03 October 2018 https://www.ijglobal.com/articles/135619/the-earth-moved-castor-gas-storage?home=1

Castor concession. Escal was forced to relinquish the project, but the decree ordered Enagas to arrange around €1.3 billion in bank debt from a club of lenders to repay the concessionaire.

The government got the banks to compensate Escal and in exchange for payment rights from the Spanish gas system. The banks agreed to receive payments for 30 years (around €80 million per year) from January 2016. Spain was stuck with a significant bill, but the deal participants were compensated quickly and in line with legislation. In May 2017 the Spanish government confirmed that it has no plans to reopen Castor.

4.2: Case study of the financing structure for a typical offshore wind farm

Press announcement 18 July 2017: British Wind Energy GmbH a wholly owned subsidiary of Northland Power Inc., has today successfully achieved Financial Close for the approximately EUR 1 (one) billion project financing of its offshore wind farm project Deutsche Bucht ("DeBu"). The non-recourse project financing has been entered into with a consortium of 10 commercial banks. It has a capacity of 252 MW and comprises 31 MHI-Vestas V164 wind turbines. Green Giraffe[17] was the sole financial advisor to DeBu, which was initially owned by Highland Group Holdings and is now 100% owned by Northland Power Inc.

The Green Giraffe graphic below show British Wind energy GmbH (the SPV) at the centre of a 'nexus of contracts'. The SPV organises contracts with counterparties to achieve its objectives. It has no cash or revenue of its own but raises cash from equity investors (including Northland Power) and from 10 banks.

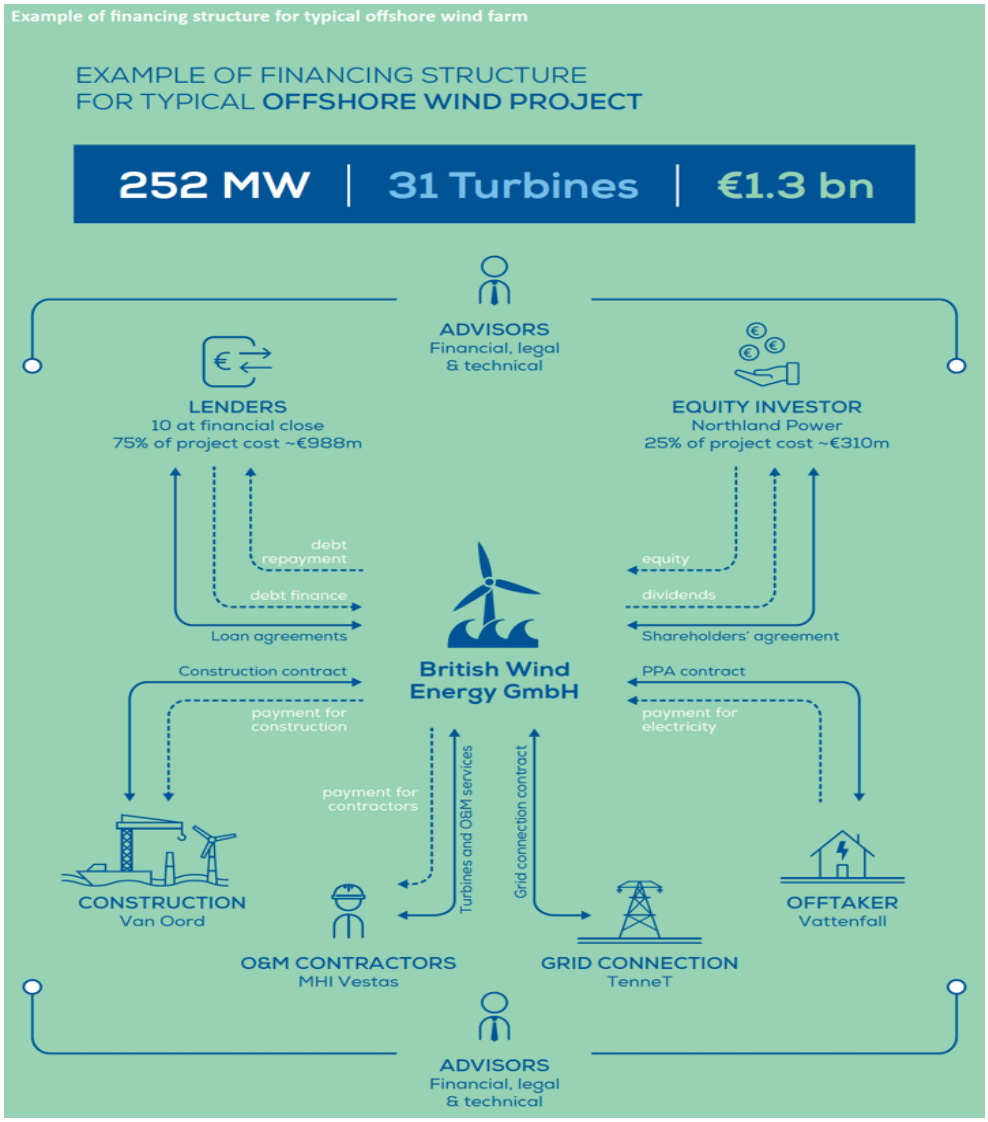


Figure 4. Financing structure for an offshore wind project

The SPV does not construct the wind farm nor operate it (it does not have the staff, or the skills to undertake either of these tasks) but writes contracts with others to undertake this work. It also writes contracts to provide

[17] Courtesy of Green Giraffe. https://green-giraffe.eu/

risk *mitigation* which is necessary to make a project 'bankable' i.e. to lower risk to a level at which banks feel comfortable lending. Thus the construction contract will have a penalty clause under which Van Oord will have to cover some of the additional costs or delays during construction. This means that the costs involved if there is an over-run or a delay in completion, don't all hit the P&L or the balance sheet of the SPV (and hence impact lender credit risk) but instead, the P&L and balance sheet of the construction company. Similarly the offtaker contract with Vattenfall compels them to take the amount of power they have contracted for even if they don't need it. If they won't take it, they still have to pay for it (a take-or-pay contract). Such risk mitigation by, in effect, paying more than otherwise to the construction company and offering power to the offtaker for less than without the take-or-pay contract, means that the lenders to the project have greater protection than otherwise (in effect the equity shareholders have paid for insurance).

July 2017 was financial close for this project which then commenced operation in 2020. In 2021 it was re-financed with Green Giraffe acting as the sole financial advisor on the non-recourse refinancing. The re-financing transaction closed in March 2021, with the sponsors successfully arranging a EUR 886 M senior debt refinancing package. This will reduce the interest rate on the project's senior debt to 2.3% all-in (from approximately 2.6% all-in) and accordingly improve cash distributions from the project over the remaining 13-year senior debt term. The refinancing also included the addition of a Debt Service Reserve Facility, which released €50 million of cash, thereby enhancing Northland's corporate liquidity. The refinancing won the 2021 Inspiratia award for Top Refinancing Deal of 2021.

In terms of the timing of different types of financing from different types of investor in the Figure 4.2[18] it can be seen that the developers are the first to put cash into the project (in the form of equity). A little later, PE funds may come in and later still some risk taking investors (perhaps hedge funds). Some independent power providers (IPPs) may come in as may contractors on the project and utilities which, once operational, may wish to buy power from the SPV. The lenders come in at the bottom of the graphic after the decision is made to go ahead (FC/FID). Finally, once commercial operations commence (COD), financially conservative investors i.e. pension funds and insurers may enter [FC/FID is financial close/ final investment decision. COD is commercial operation date.]

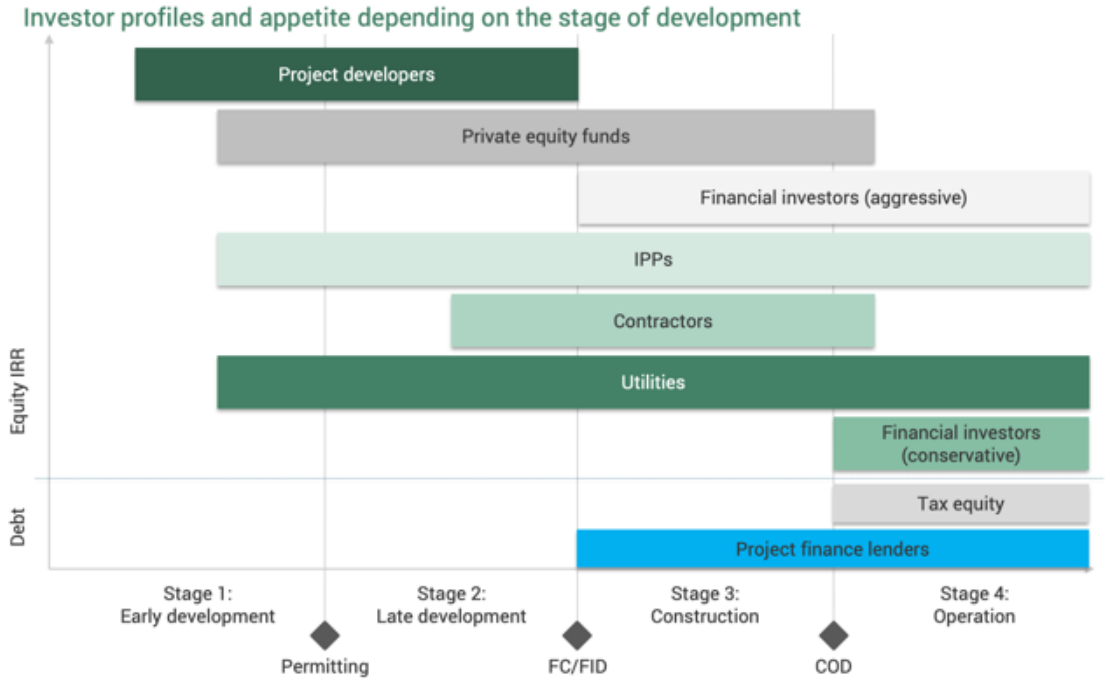


Figure 4.2 Investor Types in the financing of a project at different stages up to the operational stage

4.3 Ørsted: Case Study of Corporate Financing of Wind Power

We have not written much on corporate financing as traditional corporate finance is well covered in university finance courses and in many textbooks. Instead we look at a few particular features that were utilised by Ørsted before the wind industry as a whole faced significant changes in the environment in which they operated after 2022 due

[18] https://green-giraffe.eu/

in particular, to Covid and to energy cost rises due to Russia's invasion of Ukraine. These significant challenges included above general inflation rates of increase in their input costs. Not only were material costs much higher, but high interest rates also suddenly became an issue as well as supply chain bottlenecks. This has led many developers reducing their investment targets and postponing, and in some cases abandoning, large projects while taking substantial financial losses/impairments.

Ørsted has been a leader in the decarbonisation of the power industry mainly from developing North Sea wind farms and has made a high level of investment in *new real (green) assets*. Ørsted was originally a state-owned oil and gas company set up in 1972 to exploit the new oil and gas fields which were being discovered in the North Sea. In 2018 it changed its name from DONG (Danish Oil and Natural Gas) to Ørsted after it sold its upstream oil and gas activities to Ineos, a private UK company whose main products are now chemicals, polymers and oil and gas. The new name for Dong, honours the Danish scientist Ørsted who discovered electromagnetism (on which principles, power generation is based). By 2018 they had shifted the company to their 85/15 vision i.e. 85% of their power production being green from only 15% in 2009. Ørsted serves as a holding company for almost 300 subsidiaries (in effect projects) with all major business activities conducted through those subsidiaries which is a common practice in most corporates but with many fewer such subsidiaries.

Corporates frequently use project finance where the assets of a project are initially on the balance sheet of a separate company, an SPV which also is the issuer of equity, bonds and is the counterparty in bank lending. In Ørsted's case, borrowing was mostly on the balance sheet of the parent company (on-balance sheet corporate borrowing) and thus there *was* recourse to the parent. It is unusual in that it finances each project from its Group corporate financing but each project is structured within a separate subsidiary and thus it can sell shares in a project to its partners to reduce its total capital commitment and its risk.

> "Its financing strategy[19] is to concentrate group borrowings at the group parent level to simplify its liability profile while avoiding any structural subordination issues."

Thus for Ørsted, its financial structure protected those who lent at the group parent level and thus reduced their risk and hence the credit spread on loans as a result of this technique raising credit ratings compared with a structure with structural subordination. This is because if all financing of a subsidiary comes from the parent then lenders to the parent know that they do not need to be concerned with debt raising by other lenders to the Group through its subsidiaries which, given there are 300 subsidiaries, would involve a great deal of work by a lender to the Group.

They suggested at one point that by avoiding the use of off-balance sheet financing (project finance structure) they reduce their cost of debt by 100 basis points (1%). Financing comes from retained earnings, a range of types of debt issue, joint ventures and so-called farm-downs. This last expression means selling equity stakes in a project pre-construction to outside investors to free up their own equity capital for further projects. This is more commonly known as asset rotation or build, sell, operate which enables Ørsted to take out profits from being a developer and from operating facilities without having to have 100% ownership in them in them. Under their partnership model on wind projects they average 50% Ørsted ownership of future *offshore* wind farms. On the other hand they intend to sell down more *onshore* projects in order to invest more in offshore and hydrogen manufacture.

Their approach of not financing each project separately as they would have to if they used a project finance approach also means that the parent company of the subsidiaries holds a diversified (real asset) portfolio which enables risk reduction for lenders without some of the costs of normal project finance risk mitigation measures. This in turn results in both a lower cost of equity and lower project costs. They also chose to reduce risk further compared with project financing structures by avoiding the high levels of leverage seen in project finance which can be of the order of 75%. With lower leverage, the variability of profits is much reduced. Thus overall WACC is reduced.

They reduce risk further by having a natural hedge for much of their currency risk which means that rather than using derivatives, instead they achieve it by having borrowings in the currency of denomination of the Funds from Operations (FFO) on each project. Thus the currency of denomination of the asset side of the balance sheet is

[19] https://orsted.com/en/investors/debt/funding-strategy

approximately matched with the currency of denomination of its debt (liability side of the balance sheet). This reduces the variability in total costs relative to revenues compared with unmatched financing.

Before 2022, following an assessment of their balance sheet and financing strategies by Moody's and S&P and Ørsted having chosen a BBB+/ Baa1 as the optimal one for their company given its substantial debt requirements, they have then chosen not to try to raise their rating to the next highest level – single A. As their rating was heading up towards single A, they have therefore reduced their Funds from Operations to Debt Ratio (FFO/ adjusted net debt) from 30% to 25% thus enabling them to raise more debt for a given level of cash flow while maintaining their BBB+/ Baa1 rating. However, in February 2025 Ørsted was downgraded by Moody's to Baa2 (close to 'junk' status'). In August 2024 it had been reported that the company had abandoned an e-fuel project. But the final straw for the company's downrating may have been US President Trump's view of wind power to which he seems to have some emotional objection and which led to a large project having to be discontinued. In consequence the share price has suffered having peaked in 2021 and declined since. Nonetheless, Ørsted remains a great and successful company.

4.4. A Model for Nuclear Financing: Thames Tideway Tunnel Regulated Asset Base (RAB) Financing & Risk Mitigation through a "Contingent Financial Support Package"

The traditional proven solution to financing capital intensive infrastructure in many countries is to use the Regulated Asset Base (RAB) model which is used in electricity, water, gas and telecommunications worldwide. The UK's Department for Business, Energy and Industrial Strategy (BEIS but since 2023, split into two departments – the Department for Energy Security and Net Zero (DESNZ) and the Department for Science, Innovation and Technology (DSIT) has been having an ongoing Consultation on the use of the RAB Model for Nuclear. In a document the UK government published in 2019 there is a key paragraph which states that "there will still be a crucial role for low carbon 'firm' power i.e. always available power, in 2050". The UK Committee on Climate Change includes 38% firm low carbon in their further ambition scenario – "to meet net zero while maintaining security of supply and keeping costs low." As there are only very limited technologies for the provision of firm low carbon power apart from nuclear, this confirms that nuclear remains a key component of the UKs energy policy.

BEIS has also stated that: **"Our assessment has concluded that, by providing regulated i.e. virtually guaranteed, returns to investors, a RAB model has the potential to reduce the cost of raising private finance for new nuclear projects, thereby reducing consumer bills and maximising value for money for consumers and taxpayers." As economic regulators are required by statute to ensure that investors obtain the agreed rates of return, this financing mechanism has already been successful in enabling private sector investment in other infrastructure sectors such as the National Grid's UK investments in gas and electricity and in the US, is used for nuclear, coal and gas power stations.**

There are many types of risks in new nuclear and if these can be successfully mitigated, the overall cost of financing (including the cost of risk transfer which is in the nature of insurance) will be lower than without such mitigation. A "megaproject" in the UK which achieved a record low financing cost was the construction of a tunnel under the Thames to bring sewage and excess surface drainage water from places where it was flowing directly into to the River Thames, into the tunnel which would be constructed under the Thames to take the effluent directly to a sewage works further down the river. The project construction commenced in 2014 and was successfully completed ten years later in late 2024. Beyond the use of RAB financing and specifically for the Thames Tideway Tunnel project, the UK government developed a new type of UK Treasury risk mitigation model for certain risks which could be an archetype for an SMR/AMR project.

The so-called "Contingent Financial Support Package" known as the GSP was designed to cover certain risks on the TTT project. We cover it in detail because of its applicability in nuclear and hence to governments aiming to achieve their legally mandated GHG reduction targets for example through the use of nuclear power. The government's provision of the GSP to Bazalgette (the SPV) is aimed at mitigating some risks and transferring liability to the taxpayer if those risks materialise. The government concluded that private delivery of the project would not be financially viable without some form of government support, because of the scale of the project risks and the implications for financing costs that customers would face. Under the terms of the GSP, which expires when the three-year post-construction project-testing period is expected to be complete in February 2027, the Department has agreed to provide financial support in certain limited circumstances:

- *Cost overruns*. The project's target price is £3.2b but if Bazalgette's costs exceed this threshold, the Department agrees to either provide equity to Bazalgette or to discontinue the project and pay compensation.

- *Disruption to capital markets.* A 'Market Disruption Facility' commits the Department to providing short-term liquidity but on appropriate commercial terms.

- *Indemnity.* The Department agrees to provide an indemnity to Bazalgette to cover for liability claims exceeding the company's insurance limits.

- *Compensation for discontinuation*. In the event that the Department decides to discontinue the project, compensation will be paid..

The regulatory license from Ofwat (the UK regulator) has, according to Ofwat, "strong incentives in the form of clear rewards and penalties to complete the project within time, to quality and at minimum cost". This is a not a typical incentive on a project company to minimise risk to investors.

A further interesting aspect of the Thames Tideway project is that a means was proposed and implemented, to enable the project to pay dividends to the equity investors during construction - i.e. through the ten-year construction period, even though there would be no revenue generated by the tunnel during this period. This was achieved through an agreement that Thames Water customers would pay an additional charge on the water and sewerage bill during this period. This additional charge started at £7 on annual water bills and rose to between £20 and £25 in the period up to the mid-2020s. As Thames Water is the monopoly supplier to most of its customers, it was able to impose this additional charge. In a competitive retail and wholesale power market on the other hand, it would be more difficult. But just as interest on loans is capitalised during construction by adding it to the loan principal, so too could the equity dividend be capitalised into the total loan amount on this project or on a nuclear power project. A dividend could thus be provided to investors during the construction phase with the cash coming from the increase in the loan principal. This whole structure - RAB, GSP and dividend payments resulted in the weighted average cost of capital (WACC) for the whole project being well below expectations at only 2.497%. Given that for many such infrastructure projects including a nuclear power plant, the cost of debt service is likely to be the largest annual cost element, a low cost of capital is critical. In addition, in 2025 interest rates are well above those that pertained at the time the financing package for TTT was set up. In a nuclear project a structure such as that used for TTT would almost certainly reduce the cost of capital substantially by enabling a fall in the investor required return on equity. The package as a whole would also almost certainly also reduce the cost of debt.

Given the success of the TTT financing as shown in Figure 4.3 and that there are no other proven low carbon solutions apart from nuclear to meet the need for firm power (except for storage which still requires investment in power generation), it is clear why the UK is trying to improve the financing options for nuclear.

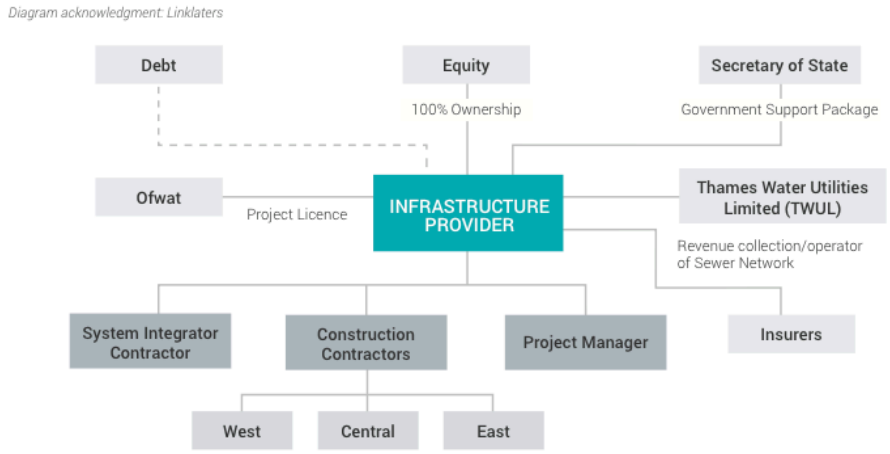


Figure 4.3 The project structure for the Thames Tideway Tunnel project

Chapter 13 covers nuclear and nuclear financing in detail.

Part II: Carbon Accounting, Disclosure and Risk Reporting Frameworks for Environmental and Social Factors

Chapter 5: Task Force on Climate-related Financial Disclosures (TCFD) Strategy, Scenarios and Risk

[For those readers who are not well acquainted with the science of climate change, we suggest that it would be very useful to read Appendix 1 on the science of climate change prior to covering the remainder of the book.]

5.1 The Carbon Questions and Financial Risk: What Does Net Zero Mean; What are Science Based Targets; Net Zero or 1.5 °C; Carbon Budgets, Externalities, Social Cost of Carbon (SCC), Carbon Taxes, Alternative Routes to Paris

Before we can consider the legal and other frameworks/directives/regulations/accounting rules and societal trends within which corporates and financial firms must operate and the risks of not complying, we must consider climate science and our understanding of what needs to be done to achieve the desired objectives of society in relation to climate change. We cover the science of climate change in Appendix 1. It is also difficult to ensure effective decision making in corporates or in financial firms without **understanding the carbon background which we call the 'carbon questions' before moving on to frameworks**.

In relation to the finance industry, climate change and in particular rising temperatures and water shortages presents risks to banks, investors and insurers not just from its direct impacts on companies but also from the potential **impact of a changing climate on mass conflict and mass migration** from areas of earth which become uninhabitable.

Efforts to mitigate rising temperatures on Earth did, in fact, begin many years ago. One of the first formal efforts was in 1994 when the United Nations Framework Convention on Climate Change (the UNFCCC) entered into force. Today, it has near-universal global membership. The 197 countries that have ratified the Convention are called Parties to the Convention.

At a Conference of the Parties (COP 21) in Paris, on 12 December 2015, the 197 Parties to the UNFCCC reached a landmark agreement to combat climate change and to accelerate and intensify the actions and investments needed for a sustainable low carbon future. The so-called Paris Agreement builds on the Convention and brings all nations into a common cause to undertake ambitious efforts to combat climate change and to adapt to its effects and to assist developing countries to do so. In 2025, for the second time, the US withdrew from the Paris Climate Agreement under an executive order from President Trump.

The Paris Agreement's central aim is to strengthen the global response to the threat of climate change by limiting global temperature rise. The Agreement has three aims of which the third is most relevant to those in the finance industry:

> (a) Holding the increase in the global average temperature to well below 2 °C above pre-industrial levels and to pursue efforts to limit the temperature increase to 1.5 °C above pre-industrial levels, recognizing that this would significantly reduce the risks and impacts of climate change;
>
> (b) Increasing the ability to adapt to the adverse impacts of climate change and foster climate resilience and low greenhouse gas emissions development, in a manner that does not threaten food production;
>
> **(c) Making finance flows consistent with a pathway towards low greenhouse gas emissions and climate-resilient development**.

To enable the world to achieve the target the Paris Agreement developed the concept of nationally determined contributions:

Nationally determined contributions (NDCs) are the core of the Paris Agreement and the achievement of these long-term goals. NDCs embody efforts by each country to reduce national emissions and adapt to the impacts of climate change.

What does 'net zero' mean? Would it achieve the '1.5°C' target? What is the finance industry's role in contributing to climate change mitigation?

What does the finance industry have to do to achieve its full potential contribution to climate change mitigation (based on the above targets), to enable it to ensure the realisation of the third of the Paris Agreement's three objectives – *Article 2.1c '*Making finance flows consistent with a pathway towards low greenhouse gas emissions and climate-resilient development.'

When we talk about net zero as a target of the world's governments such targets for the real sector of the economy are not yet embedded in the world's financial systems as an objective that shapes their decision-making. Instead, **the financial sector is much more concerned with mitigating risk in financial asset portfolios as distinct from mitigating climate change risk to the Earth**. **They are thus more concerned with management of climate risk to portfolios whether or not it actually mitigates climate change risk for the world**. A key debate is whether or not the actions and decisions being taken by the finance sector will, in practice, drive the decisions that need to be taken on reducing emissions in the real economy which requires a massive increase in "climate finance". Climate finance is new financing provided by financing institutions for the creation of new clean energy assets. Instead, financial institutions are still providing a high volume of funds to enable the creation of new carbon emitting assets by corporations. Such financing is known as "financed emissions" i.e. additional emissions enabled by the provision of financing from the issuance of new equity, bond issues and bank loans for large global emitters.

This last paragraph is emphasising that net zero is about the **real economy** – ensuring decarbonisation and hence reducing the extent to which the Earth is warming and the risk to mankind. On the other hand, management of portfolio climate risk by financial institutions is, at present, mostly about minimising the risk of loss to asset owners on existing portfolios of financial assets from physical and transition risks. More particularly it is about trying to avoid the risk of finding that a portfolio holds a high proportion of 'assets at risk' so-called, stranded assets. To the extent that demand for oil and gas may peak by 2026/7, some assets owned by oil and gas companies will ceases to have little economic value.

THE PARIS AGREEMENT TARGET IS 1.5°C which is a different objective from achieving net zero by a particular date. Net zero was simply a proposed, and agreed, means to achieve the 1.5°C target. In practice there can be **"many a slip between net zero and 1.5°C".** Net zero is a good target in some ways since achieving net zero is the only way to stop a continually rising temperature on Earth – which would eventually wipe out most life on Earth including humans. But as the net zero target is not on track, there is no reason to believe that by the time we do achieve net zero (which we are now unlikely to achieve by 2050) that we will have in fact have limited the global temperature rise to 1.5°C or even to 2°C or 2.5°C. Even achieving it by 2050 does not guarantee success with the temperature target for the world. Thus net zero is not by itself an objective but just a possible means if companies and households had kept on track with decarbonisation targets.

Zero or 'Net Zero'?

Zero emissions is different from 'net zero'. Net zero accepts (whether rightly or wrongly) that it is not possible to eliminate all emissions. Instead, it is about 'off-setting or cancelling out any carbon emissions that a company or country produces. Net zero is achieved when the amount of greenhouse gas produced in a country is no more than the amount removed. Countries, cities and companies need to have a plan as to how they intend to achieve net zero and to have set targets such as planned and actual progress each year towards their target.

Offsetting

Increasing GHG removals through *activities external to the company (or organisation) involved* to compensate for 'hard to eliminate' GHG emissions so that its net contribution to global emissions is reduced or eliminated is known as offsetting. Offsetting can be through a market for carbon credits, tree planting or other mechanisms.

Offsetting claims are only valid under what is supposed to be a rigorous set of conditions, including that the reductions/removals involved are **additional**, not over-estimated and that they are exclusively claimed and not

double-counted. Fortescue Future Industries (https://ffi.com.au/) claims to be the only major heavy industry company with a real plan to get to **'real zero'**. It aims to have invested US$6.2 billion by 2030 to eliminate fossil fuel risk and reduce annual operating costs by US$818.

So what is to be done? The answer (in brief) is that strategy in banks, issuing houses, asset owners, asset managers and insurers (the asset side of the balance sheet) and underwriting strategy in insurers (the liability side of the balance sheet), must be aligned with a decarbonisation pathway consistent with a maximum 1.5°C of warming. But what does this statement mean? Such alignment involves several assumptions including the choice of decarbonisation pathway that each financial firm chooses and the particular way in which it sets targets and achieves the required targets each year. A financial firm could choose a lesser reduction before 2030 and in the decade to 2040 and hope to catch up by 2050. But clearly such a firm's portfolio would have resulted in greater emissions and temperature rise on Earth by 2030, 2040 and 2050 than one in which a massive effort was made by the asset owner ensure that its portfolio companies took powerful steps to cut carbon emissions immediately.

Carbon budgets, Net Zero and "pathways"

In December 2020, the UK's Climate Change Committee (CCC) published its 1000-page study entitled "Sixth Carbon Budget – the UK's path to net Zero" which is a report required under the UK's Climate Change Act. Amongst other things, this study provides government ministers with advice on the volume of greenhouse gases that the UK can emit during the period 2033 – 2037.

The study suggests that a 100% cut in GHG emissions would cost 0.5% of 2050s GDP. The same Committee in 2008 estimated that to achieve an 80% cut in GHG emissions would cost 1 – 2% of GDP. The cost of action has thus reduced very substantially (in part due to falling costs of both solar and wind power). The cost in inaction is likely to have risen substantially.

To stabilise global temperature at any level vs the pre-industrial level, there is a finite number of gigatons of emissions that can be released before we achieve net zero. The 'e' after CO2 means CO2 + the impact of other greenhouse gases converted to their *equivalent (e)* damaging effect if they were in fact CO2. This is referred to as a carbon budget.

When Bill Gates when he was interviewed on BBC radio put it very simply: We have to reduce global GHG emissions today from 53 Gt per year to zero by 2050 if we were to have any chance of achieving a temperature rise of less than 2°C. We also needed to have 2030 targets and 2040 targets rather than having a *transition pathway* that involves little action until 2040. With current concentration of 427 ppm (0.0427%) in 2024, we will exceed the IPCC's estimated 2°C warming threshold of 450 ppm in the next 10 to 20 years though very possibly more quickly.

Figure 5.1[20] from Science-based Targets shows that to limit global warming, we need to limit the concentration of carbon and other GHGs in the atmosphere to achieve the temperature objective. The relationship between emissions and carbon concentrations is a well-established one so it's not difficult to work out the remaining 'budget'. To avoid breaching the 2°C target, the maximum amount of CO2 in the atmosphere (the stock or cumulative emissions) must not exceed 2,900 gigatons (1890 + 1010).

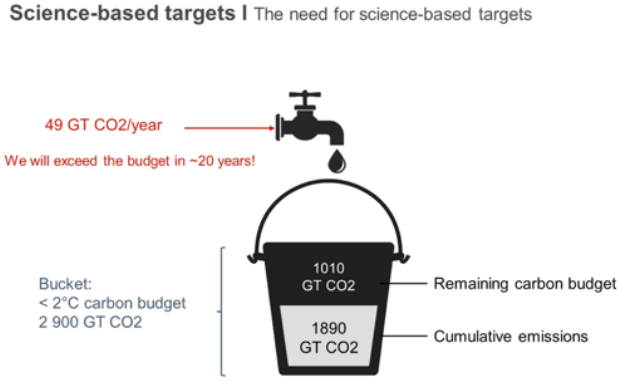


Figure 5.1 The relationship between annual emissions and carbon concentration in the atmosphere

[20] Corporate Citizenship (SBT, CDP, Diageo) https://vimeo.com/218954934

Since, in the graphic above, there is already 1,890 GT in the atmosphere, the remaining budget for 2°C is 1010 GT. Since the emission level globally was at 49 GT when this graphic was prepared (or 53 by the time Bill Gates wrote his book) this means that if we continue at this rate of emission, in 20 years' time () practice probably within ten years) we will breach the 2°C target and temperatures will continue to rise beyond 2°C. A 1.5 degree target gives a carbon budget of only around 350 GT. At the present rate of emission (and emissions have grown each year since the Paris Agreement), this gives the world less than 7 years before breeching the 1.5°C target. Today (2024) we may indeed already be past the 1.5 degree point. If there is any budget left then the question is 'over what time should it be spent' and 'who should be entitled to spend it' and how can countries agree on such budgets. This was the subject of political bargaining such as at the COP 26 Glasgow in November 2021 (with little agreement). With the US Federal government seeming to have abandoned any attempt to limit emissions, and indeed intent on increasing them, the situation is now worse than it was in 2024. Note that statistics for global carbon emissions vary depending on what is included and whether it is a CO2e (equivalent) amount which includes other GHG gases as well.

Externalities

An externality is a consequence of an industrial or commercial activity which affects other parties (more commonly in a negative way but could also be in a positive way) without this being reflected in market prices. It is a cost, or occasionally a benefit, that affects someone who did not choose to incur that cost or benefit. In the case of climate change, it is the use of the atmosphere as a 'sink' in which to dump carbon dioxide, methane and other GHGs from vehicles, power stations, home heating, agriculture etc. which imposes costs on everyone in the world which they did not choose to incur. In the case of river pollution, it is the impact on people downriver from the polluter who do not ask for this pollution to be dumped (obviously not) and the externality is that the company is not paying for the cost of the resulting damage of their pollution. This externality can be internalised (that is put into the company's own costs) either through a change in the law to make dumping of waste materials in a river or dumping CO2 into the atmosphere illegal or through a tax which makes it economic to find alternative means of disposal of toxic materials such as dyestuffs used for dyeing fabrics to supply to 'fast fashion' companies or in the case of CO2 emissions, either converting to zero carbon sources of energy or payment to have emissions "sequestered" underground for example in old coal mines or in empty gas fields in the North Sea or elsewhere.

In the case of CO2 emissions, internalising the externality would generally involve taxing such emissions thereby transferring a cost currently imposed on all individuals across the world. This then raises the question of what is the cost to society of such emissions – the Social Cost of Carbon? Most models are very conservative in their assumptions and neglect many likely costs that will be incurred in the future. But without some agreement on what these costs are, it is difficult to develop government and regulatory policy in this area. Effective climate policy must balance the cost of taking action (such as regulation and taxes) against the economic damage from inaction. Thus we need to *balance costs against benefits*.

The Need to Scale Up Climate Finance

A key question that must be asked at this point is how, in practice, the world can achieve the "real economy" transition that is required and **the maximum contribution from the financial sector in helping to achieve the scale-up of efforts to avert the worse impacts of climate change'**. Put another way, the question is how to eliminate fossil fuel assets and replace them with "clean" assets at the lowest possible cost and at the maximum possible speed. The real asset sectors – electricity supply, transport, agriculture, manufacturing (in particular the energy intensive industries such as cement, steel, aluminium, paper) need to increase the speed of their transition to minimum CO2 emissions. This is proving difficult technologically.

The Chairman and CEO of the world's largest asset manager, Lawrence Fink's has, in the past, referred to '*long-term prospects'* and "*a reallocation of capital"*. If investors and banks agree with Fink that the financial world is on the edge of a revolution, then they need to show a determination to reallocate the inflows from investors from funds which finance new fossil assets ones which are financing the necessary carbon-free assets within the needed timespan. However, Fink has greatly reduced his emphasis on the need to reduce carbon emissions rapidly in line with the new "thinking" in the White House.

Institutional equity investors will need to assess the carbon mitigation plans of companies in which they might invest and banks will need to move away from re-financing corporate bonds and loans of corporates (and sovereigns) with high exposure to (and hence risk from) climate change. Importantly, financings of all types will

increasingly move towards companies taking advantage of climate-related *opportunities* – such as the new and existing technologies which help mitigate climate change. As we noted above, insurers will do the same on the asset side of the balance sheet where they hold the assets that hedge the risks on the liability side (insurance policies). On the liability side, they will avoid covering risks where it becomes no longer possible to evaluate the climate risk they would be talking on and hence it cannot be fairly priced. An example of this can be seen in an announcement made on March 1, 2021, by Amanda Blanc, CEO of the large insurer, Aviva, when she said that the firm would no longer underwrite insurance insurance for firms which have more than five percent of their revenue from coal, tar sands, shale gas, deep water oil and oil shale – unless they have signed up to SBTI (see Chapter 6.4)

5.2 Task Force on Climate-Related Financial Disclosure (TCFD): Disclosure of Strategies and Scenarios [https://www.fsb-tcfd.org/]

TCFD, which is concerned with the disclosure of relevant Science Based climate-related information, was created by the Financial Stability Board (FSB). This is an international body that monitors and makes recommendations about the global financial system and, through its efforts, a route to achieving relevant disclosure by corporates and financial firms has been created. The FSB was established in 2009. The Board includes members from all G20 economies and the European Commission. Hosted and funded by the Bank for International Settlements, the board is based in Basel, Switzerland at the premises of the Bank for International Settlements (BIS). The creation of the FSB itself represented the G20 leaders' first major international institutional innovation. Former U.S. Treasury Secretary Tim Geithner has described it as "in effect, a fourth pillar" of the architecture of global economic governance, alongside the International Monetary Fund, World Bank, and the World Trade Organization.

As climate change increases climate change risk and hence potential financial impact on companies and financial firms, **it also has the potential to create global financial instability**. In accordance with its mandate to minimise risks to financial stability, in 2015 the FSB created the Task Force on Climate-related Financial Disclosures (TCFD) to improve and extend reporting of climate-related financial information. The reason for doing so is that **one of the essential functions of financial markets is to price risk to support informed, efficient capital-allocation decisions**. To carry out this function, financial markets need companies to provide accurate and timely disclosure of climate risks just as they do on other aspects of their performance. Without the right information, investors and others may incorrectly price financial assets such as equities, bonds and loans leading to a misallocation of capital. The FSB created the TCFD to develop recommendations on the types of information and hence disclosures that companies should provide to support investors, lenders, and insurance underwriters in appropriately assessing and pricing a specific set of risks—risks related to climate change. **These disclosures are then intended to guide decision making in financial firms.**

It was intended by TCFD that its 'Recommendations' **should drive changes in the structure of real assets and also financial asset portfolios.** It may be, however, that taking an approach which, as a result of operating principally through changes in financial asset prices, is simply too slow a mechanism to achieve the Paris (time and temperature rise) objectives. **The TCFD framework is a gradualist approach similar to that favoured by the IFRS Foundation (see below) which focuses on enterprise value**. All frameworks that rely on information disclosure leading to price changes and thus in the cost of capital, with those price changes then intended to lead to real economy changes, raise the question of whether such frameworks can on their own effect the necessary change in the real asset structure of an economy **rapidly enough** to mitigate the climate crisis. In practice, given that the transition is well behind schedule, the answer to date must be no. Nonetheless, TCFD has played a valuable role in 'kicking off' the need for disclosure. It has become the basis for other organisations involved in climate change risk and for governments and the EU. Information flows are expected to lead to appropriate shifts in capital flows. Such disclosures are expected to lead to Corporates and the Financial Sector (banks, asset owners, asset managers, insurers) making different decisions than they would have in the absence of a consideration of sustainability disclosures.

Changes in decision-making processes and outcomes on (real) capital allocation start with the corporate becoming more aware of the risks and opportunities in their company as the environment changes because of climate change. Companies also must consider **other environmental factors as well as changes in social norms and values** which are relevant to wider Environmental and Social risks. Such changes are expected to result in changes in the real sector's investment decisions on capital allocation to physical assets (away from fossil fuel dependent assets

and towards non-fossil dependent assets) and thus lead to a 'reorientation of capital flows towards a more sustainable economy'. This is what the Paris Agreement meant by "making finance flows consistent with a pathway towards low greenhouse gas (GHG) emissions and climate-resilient development".

Although we remain doubtful that disclosure of information by itself will have the desired impact and that further legislation may be necessary to speed up the transition, nonetheless, all action will start with information and understanding. Understanding is particularly germane to the good governance of climate change risks as the issues are generally not well understood or else treated cursorily. Understanding the information derived from data is relevant not only for the board of a company but to all levels in the organisation. Good governance is a central aspect of TCFD. What is likely, however, is that the increased transparency and the introduction of taxonomies (classifications of different types of asset) will lead to litigation against companies that are not conforming with the new requirements. The likely outcome will be that more such cases will succeed – though probably slowly. The European Commission's new Directive on Corporate Sustainability Due Diligence (CSDD) is likely to be a game changing Directive which will also encourage more, and more successful, litigation against corporate laggards (see Appendix 2).

The first key objective of *TCFD* is to avoid a sudden and dramatic policy response by governments and regulators such as the "Inevitable Policy Response" which could lead to financial instability. This is the almost inevitable, response of governments which will be forced to act much more aggressively to limit physical climate damage as the realities of climate change become ever more obvious to voters. The policy response will lead to a much higher level of transition risk than companies might have expected (as well as physical risk). Corporates, banks, investors, asset owners and insurers, could suffer substantial losses as assets became valueless (for example, oil and gas reserves not currently being exploited but on their balance sheets at a substantial value). Such assets would then be what are known as 'stranded assets'.

The objective of disclosure, therefore, is to reduce risks related to climate change. This is to minimise potential losses to corporates and financial firms from climate and environmental change by making the previously unrecognised risks taken on by insurers, investors and lenders, more obvious both to management of these companies and to companies in the finance sector. In the case of lenders it should discourage them from lending to or refinancing high GHG producing companies and in the case of equity investors from increasing their holdings in such companies or participating in additional equity fund raising. It is hoped that, combined with regulatory action such as limits on CO2 emissions from electric power generation (e.g. less than 100gm of CO2 per kilowatt-hour) and a rising cost of insurance on fossil assets, that companies (and lenders and investors) will reduce or cease to invest in new projects which are high GHG emitters and insurers will cease to insure such projects.

The second key objective of TCFD is to encourage corporates and financial companies to think about the **opportunities** presented to them by contributing to the reduction in global emissions by taking advantage of the transition to a low-carbon economy. Thus in parallel they hope, that by requiring financial firms to follow TCFD, they will be encouraged to expand their provision of 'green/climate finance'. This is once again in line with the statement in the Paris Agreement that one of its objectives is "*making finance flows consistent with a pathway towards low greenhouse gas emissions and climate-resilient development*".

The amount of work required by corporates and financial companies to produce carbon statistics and work out the implications for the profit and loss account and balance sheet of a number of scenarios, is seriously high. But while corporates are, of course, already required to provide detailed information on all the material risks they face until recently there was no requirement, or only a very limited requirement, to specifically consider the **financial impact of climate change as a risk factor**. But as a result of today's new appreciation of these risks, corporates, asset managers, asset owners, bank lenders, non-bank lenders, securities underwriters and insurers who are all exposed to such risks, need to face up to the new reality.

Climate change risk (and, importantly, other environmental risks including natural capital risks), and also social and human capital issues, have become differentiating factors in any investor assessment of corporate risk and hence security pricing. The good news is that appropriate responses to climate change by corporates or the lack of such responses should lead to the potential for those investors focussed on such issues to generate alpha for their clients, relative to those investors that are not focussed on these risks.

TCFD: Recommendations, Industry Sectors, and the Four 'Themes"

Financial markets live off information. It is information that enables investors to assess risk and then price shares and bonds. It also enables lenders to underwrite loans appropriately i.e. assess risk and price that risk in terms of the economic interest rate to be charged and for insurers to charge an appropriate premium for cover. Whilst companies in most countries have been obligated to disclose all risks to which they are subject and this should include climate-related risks, in practice this has generally been perfunctory or absent.

Financial firms themselves do not have a large environmental impact (bitcoin miners are an exception) but their portfolios of corporate equities, bonds and loans may include companies that do have such an impact. Only if such risks and opportunities are properly disclosed (and in a consistent way) can financial companies make an assessment of the risks and potential opportunities in their portfolios. In turn, it is only through such disclosures in the published financial reports of banks, that the bank's own equity and bond investors can calculate the risk premium that should be attached to a bank's equity and debt.

TCFD focusses in the first instance on those industry sectors that are most subject to climate change and hence to physical and transition risks. It categorises such industries in four groups

Energy;
Materials & Buildings;
Transportation; and
Agriculture, Food and Forest Products.

In addition to the corporate sectors above, TCFD covers the financial sector. A key reason for this is that companies in the four sectors above are likely to need to use the services of lenders and investment banks for financing or re-financing – either borrowing from a bank or arranging for an equity or bond issue though a bank (ECM & DCM services) and also insurance services. **Thus banks and insurers are being asked to include climate risks in their underwriting processes and thereby to be gatekeepers protecting a nation's savers (and the nation's banks and insurers) from being subject to undisclosed risks**. Investors in equities and bonds also need to be able to evaluate the climate-change related risk to which companies in these sectors are subject before making investment decisions. The reason for these types of financial institution being included in TCFD is that both the expected and the unexpected impact of climate change on corporates in which they have a financial investment can have a large effect on the value of their portfolios. Investors and lenders need to be able to consider this risk in their investment decisions which requires that they have the relevant information available to estimate the appropriate price for the asset relative to the perceived risk.

The intention of the TCFD Recommendations was to establish a global framework for disclosure within the *Mainstream Financial Filing*s (such as the Annual Report and Accounts) of corporates but based on the existing standards set out by various bodies such as the Sustainable Accounting Standards Board (SASB) now incorporated into the International Sustainability Standards Board (ISSB). Essentially the central banks are trying to unify voluntary disclosure principles which are gradually becoming mandatory rules or, in the case of the EU, Directives. Thus while its recommendations are 'voluntary' and also incorporate the principle of materiality, such 'voluntary' approaches generally become mandatory over time.

Types of Risk
The TCFD Report divides climate-related risks into two categories:

- Transition risks i.e. risks related to the transition to a lower-carbon economy, and;
- Physical impact risks of climate change

But it is also concerned that companies (and financial institutions) must take advantage of the opportunities that mitigation and adaptation will offer. In consequence, "Opportunities" is a key theme of TCFD since if the finance industry does not seek out such opportunities and work with the entrepreneurs, corporates and others trying to develop the new types of green asset, then the energy transition will not happen. The Figure 5.2[21] shows how

[21] https://assets.bbhub.io/company/sites/60/2021/10/FINAL-2017-TCFD-Report.pdf

risks and opportunities feed into the strategic planning and risk management of a business and hence have a financial impact on the balance sheet and profit and loss account. We explain these different risks below.

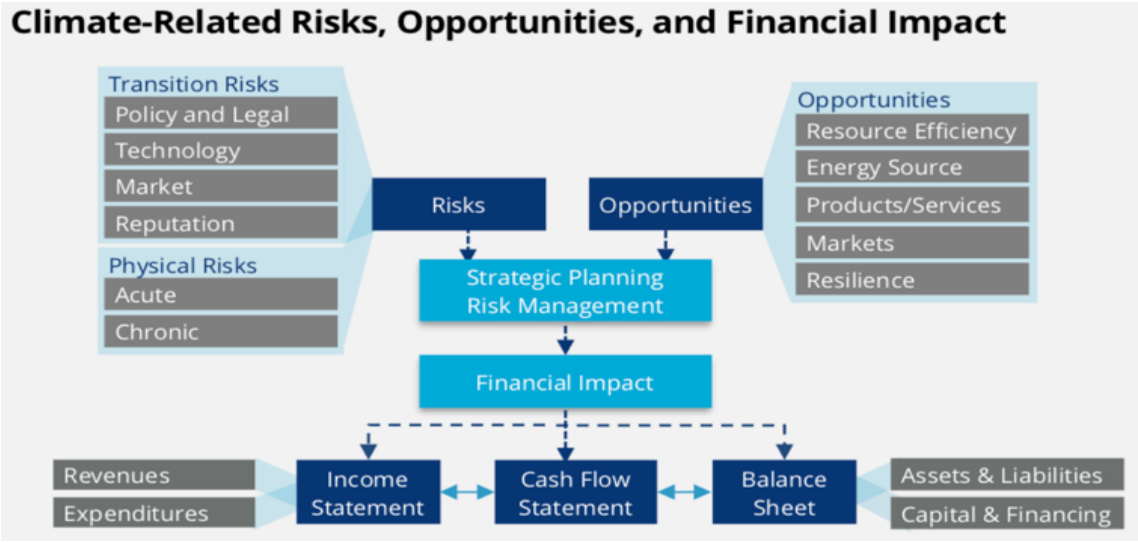


Figure 5.2 TCFD climate related risks, opportunities and financial impact

Transition Risks

Transitioning to a lower-carbon economy may entail extensive government policy and legal changes and will also result in technology, and market changes. Policy will be designed to address mitigation and adaptation requirements related to climate change.

Policy Risks

Policy actions around climate change continue to evolve. Their objectives generally fall into two categories—policy actions that attempt to constrain actions that contribute to the adverse effects of climate change or policy actions that seek to promote adaptation to climate change. Some examples include implementing carbon-pricing mechanisms to reduce GHG emissions, shifting energy use toward lower emission sources, adopting energy-efficiency solutions, encouraging greater water efficiency measures, and promoting more sustainable land-use practices. The risk associated with, and financial impact of, policy changes depends on the nature and timing of the policy change.

Legal (litigation) Risks

Another important risk is litigation or legal risk. Recent years have seen an increase in climate-related litigation claims being brought before the courts by property owners, municipalities, states, insurers, shareholders, and public interest organizations. Reasons for such litigation include the failure of organizations to mitigate impacts of climate change, failure to adapt to climate change, and the insufficiency of disclosure around material financial risks. As the value of loss and damage arising from climate change grows, litigation risk is also likely to increase. There are currently some hundreds of potential cases being considered by lawyers and courts.

Market Risk

While the ways in which markets could be affected by climate change are varied and complex, one of the major ways is through shifts in supply and demand for certain commodities, products, and services as climate-related risks and opportunities are increasingly taken into account.

Technology Risk

Some new technology changes will result in some companies thriving while others disappear. For example, many brands of automobile which are unable to match the best new electric vehicles will not survive..

Reputation Risk

Climate change has been identified as a potential source of reputational risk tied to changing customer or community perceptions of an organization's contribution to or detraction from the transition to a lower-carbon economy.

Physical Risks

Physical risks resulting from climate change can be event driven (acute) or longer-term shifts (chronic) in climate patterns. Physical risks may have financial implications for organizations, such as direct damage to assets and

indirect impacts from supply chain disruption. Organisations' financial performance may also be affected by changes in water availability, food security, and extreme temperature changes affecting organisations' premises, operations, supply chain, transport needs, and employee health and safety.

Acute Risk
Acute physical risks refer to those that are event-driven, including increased severity of extreme weather events, such as cyclones, hurricanes, droughts or floods.

Chronic Risk
Chronic physical risks refer to longer-term shifts in climate patterns (e.g., sustained higher temperatures) that may cause sea level rise or chronic heat waves.

Opportunities
Opportunities arise during the transition such as the conversion of power plants from coal, oil and natural gas to the low carbon energy produced by nuclear. ANT-HTMRs have a very low carbon impact as well as overcoming the 'intermittency' problem of wind and solar and are well able to **'re-power'** existing coal, oil and gas power stations using the same steam turbines and electricity transmission capacity. Opportunities also arise from hydrogen production, solar, wind, nuclear, storage and other technologies and the need to reduce the energy intensity of space-heating in buildings through insulation, ground source heat pumps and other means.

TCFDs Four Themes

Figure 5.3[22] shows the four themes and related disclosures which TCFD has developed. Ultimately the aim is greater transparency in reporting by the financial industry and non-financial groups to improve the market pricing of securities and loans by reflecting the risks that climate change presents.

The **governance** theme asks companies to ensure that their governance structure from board level down, enables them to assess climate change risk. The effect may be negative (e.g. reduction in the value of assets) or positive (creating opportunities).

The **strategy** theme asks companies to consider how material climate change is to them and its impact on the business, its strategy and its financial planning. This should be conducted through forward-looking scenario analysis. We cover 'scenarios' in a following section.

The **risk management** theme stresses the importance of not considering climate risk as something separate from other risks to which a company is subject but to integrate it into their existing risk framework.

The **metrics and targets** theme importantly says that financial reporting should include climate-related metrics and that targets should be set in line with the risks and opportunities identified through the themes of the recommendations.

[22] https://assets.bbhub.io/company/sites/60/2021/10/FINAL-2017-TCFD-Report.pdf

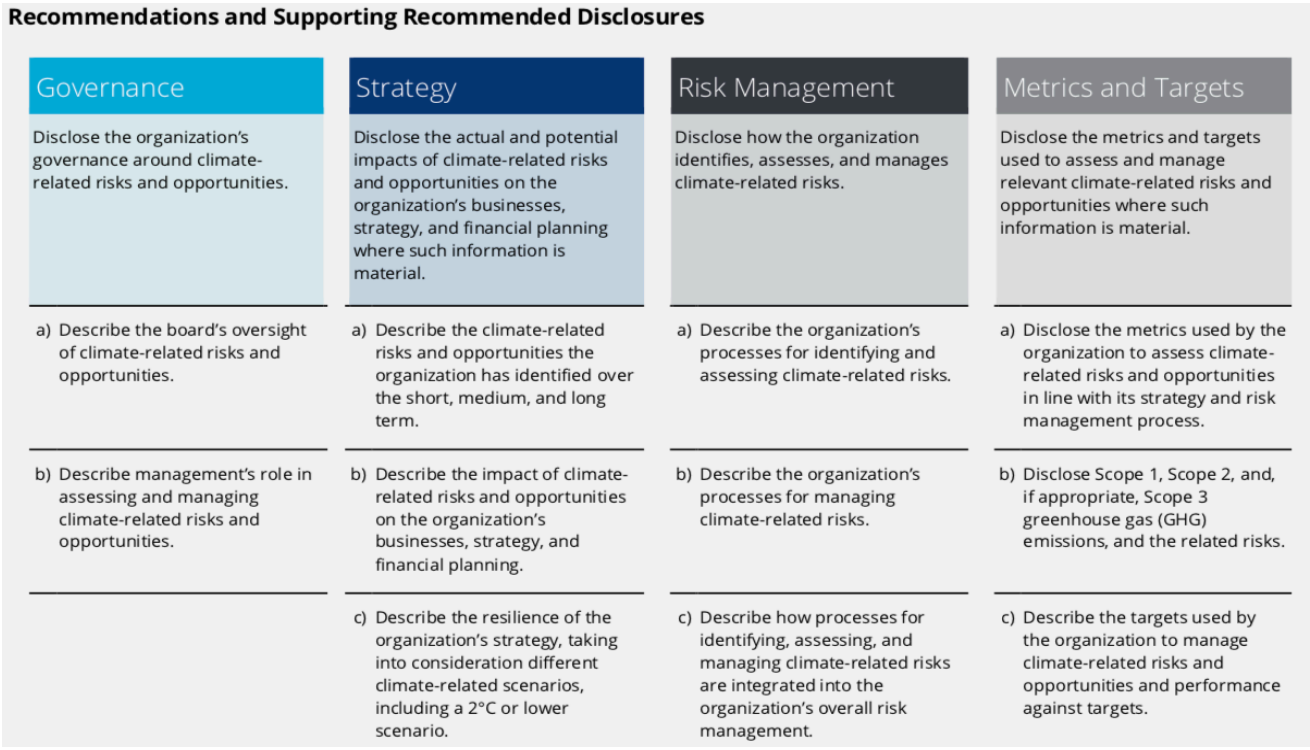


Figure 5.3 TCFD recommendations

TCFD Strategy: Strategy and Scenarios in corporates and financial firms

We focus on the middle two sets of Recommendations above – TCFD Strategy and Risk Management rather than the other two themes as these two are the key ones needed as a foundation.

Before it is possible for a corporate or a financial firm to follow the Recommendations of TCFD or any statutory framework i.e. the recommendations in the graphic above, it is necessary to understand what are the so called "scenarios" which may come to pass and which will affect different firms in different ways. While normally in finance we use probability distributions when thinking about the future, given the uncertainties of how GHG emissions will impact different industries and different firms, it is more appropriate for each firm to analyse its operations based on what might happen under different circumstances (scenario analysis).

So rather than thinking in terms of probability distributions, we should be thinking about what are the different possible outcomes that a company faces which then enables its management to consider how they would **respond to different temperature outcomes over future years.** This leads on to developing strategies based on a variety of scenarios.

What is Strategy and What are Scenarios

A strategy is a plan to achieve a long-term goal under conditions of uncertainty. It can be applied by corporates (corporate strategy) or by governments planning strategy for climate change mitigation or for future pandemics.

In the case of climate change strategy, we are not talking about strategic planning as conventionally understood in business but about how to understand the new uncertainties and in particular the changing climate environment that will shape business and society over the next 80 years. This could be likened to making decisions today in view of what might happen beyond the next quarter rather than as many managements do, focussing on the next 3 months revenue and profit figures at the possible expense of the longer term.

The core of TCFD is Strategy, based on Scenario Analysis in which Multiple Plausible Future Scenarios "build contrasting stories around key uncertainties". Following on from this, the issue is how resilient the chosen strategy will be for a particular corporate or financial firm under different climate temperature scenarios.

For investors, TCFD suggests that scenario analysis may be applied in different ways, depending on the nature of the asset(s) being considered:

- Some investors may develop energy transition pathways that they believe to be either optimal and/or likely and use those pathways to measure individual potential investments and drive engagement activities
- Other investors may consider how climate-related scenarios relate to the future performance of particular sectors, regions, or asset classes
- The results may show that some portions of a portfolio are set to benefit from a particular scenario, while others face a loss in value
- Such results, while not conclusive, can be a useful additional factor in determining where to prioritise risk management activities and where to consider making additional allocations.

Financial companies are recommended by TCFD to disclose their strategy in relation to the impact of climate change on their business (normally their financial asset portfolio), in line with the following:

> (a) the company has to identify the possible impacts over short, medium and long-term time periods;
>
> (b) it has to describe how the risks and opportunities identified will impact their strategy;
>
> (c) finally, it has to disclose the resilience of its *strategy under a range of different scenarios*, in this case different °C of global warming including one where future temperature increases by less than 2°C through the period up to either 2050 or 2100.

Disclosure (c) above includes a 'less than 2°C scenario' because that would imply a very substantial **transition risk** for many companies as a result of the 'inevitable policy response. This is regulation which forces the pace of climate change mitigation efforts much more strongly than before and would result in a substantial re-pricing of many securities. Physical risks would be correspondingly reduced if such a response were to materialise.

In summary, TCFD is concerned with climate-change risks (CCRs) and opportunities, strategic planning, risk management and the potential financial impact of these so-called, 'non-financial risks' ('sustainability risks') on the financial aspects of a company i.e. revenues, expenditures, cash flows, assets and liabilities and capital and financing. The key to this whole exercise is trying to identify uncertainties and establish the types of risk to which an entity is subject *under a range of scenarios* and then manage these risks. Equally important is a mechanism for taking advantage of opportunities. TCFD aims to minimise the risk of 'the tragedy of the horizon' (Mark Carney's expression), i.e. the use of too short a time horizon in evaluating climate change risk.

Taking such a forward-looking view (based on scenarios) is very different from just looking at today's 'carbon footprint'. The forward-looking view is critical in climate change risk as it is *a unique category of risk* where actions taken today have financial implications not just a decade away but two or even three decades away and has many other features that make it different from customary risks. Thus, a forward-looking approach, beyond the normal horizon of corporate boards, investors and bankers is necessary.

A scenario is not a 'best guess' i.e. not a forecast of the future but a tool to encourage critical thinking about the range of possibilities of some variable in the future, in this case the Earth's temperature. What we need when we are in a world of uncertainty is the possible future and their impact compared with Business As Usual (BAU). It also notes that the purpose of scenario analysis is to consider, and better understand, how a business might perform under different future states (i.e., its resiliency/robustness). Climate-related scenarios allow an organization to develop an understanding of how the physical and transition risks and opportunities of climate change might plausibly impact the business over time.

What we are saying, in summary, is that traditional methods of dealing with uncertainty based on probability distributions around a best guess of the future are unlikely to be appropriate.

5.3: TCFD: Risk Management: Black & Green Swans, Radical Uncertainty, New types of risk caused by a warming Earth

Whilst the shift from human and animal labour then to fossil fuels and electric power has brought the world to its present atmospheric GHG concentration level in August 2024 of approximately 427 parts per million (PPM), that level cannot be reduced quickly and indeed will continue to increase for many years and perhaps decades. Even if all emissions stopped today, the concentration level in the atmosphere would take many decades to decline. Thus the climate would continue to deteriorate over that time. Climate warming changes the frequencies of hot and record hot weather, the change in the shape of the bell curve resulting in an increase in the mean and the variance of temperature on Earth [with increase in the mean and variance along the bottom axis and probability of occurrence on the vertical axis].

This graphic implies much more hot weather and also much more record hot weather. Unfortunately, this climate effect (induced by humans) cannot be reversed in the way that damage to the financial system as a result of large credit losses can be overcome over time. The likelihood of a future repeat of the financial crisis of 2007/8 can also be reduced by more effective regulation, improved supervision and enhanced equity capital levels.

Climate change risk is different and requires a shift in thinking about this type of risk which at one time was thought of as being similar to credit cycle risk but is not. **Climate change risk actually increases year by year and does not cycle back**.

Climate Change risk is different from traditional risks such as credit or market price risk which vary through the business cycle (cycle up and down) and **which can be estimated by reference to past lending and investing experience**. **In the case of climate change risk, historical data is of little relevance**. This is because what we are concerned with is **secular** (ongoing), rather than **cyclical**, changes in the state of the world which are unprecedented historically. Instead of using the assumption that 'the future will be like the past' or reversion to the mean as most finance risk models assume, we have instead to rely on the work of climate scientists and the outputs of their climate models to tell us the most likely climatic impact of secular changes due to greenhouse gas emissions. In this regard climate change impact is very similar to the Coronavirus pandemic impact in that it is only scientists who can advise on likely impacts under **different scenarios**.

Climate models warn us that warming trends are not only a secular (rather than a cyclical) phenomenon, but that the impacts are also **potentially nonlinear**. They warn that as warming has been seen to accelerate and vary considerably over most of the twentieth century, then future abrupt, and possibly irreversible, change is a key risk. We thus have to be concerned not only with predicting the future of climate change based on what has happened to date on this front but be even more concerned about what happens if we have what, in finance, is known as a 'Minsky Moment'.

This is a reference used by the former Chairman of the Financial Stability Forum, Mark Carney, when talking about climate risk in relation to the financial sector. A Minsky Moment refers to Dr Hyman Minsky's financial instability hypothesis. This recognises that the nature of the capitalist financial system means that it is always prone to financial crises such as in 2007/8 when the financial economy shifted abruptly from a seemingly benign stable system to an unstable one. He was correct in respect of the Global Financial Crisis though few believed his hypothesis until financial instability hit the world unexpectedly.

Unfortunately, climate change risks include a Minsky Moment risk i.e. the risk that we move abruptly from a stable to an unstable system because of, for example, the melting of the polar ice cap or the melting of the Russian tundra which would result in huge volumes of previously stored GHGs being released into the atmosphere. Such risks require that investors and lenders make decisions based on a range of possible futures (scenario planning). This can only be achieved by **basing decisions on scientific evidence (using climate models) rather than, in the case of credit risk, the use of expected loss models based on historic loss data.** This raises the need to introduce scenario planning for possible future climate change scenarios (such as a range of 1.5 – 5°C rise in global temperature). Doing this in practice is a difficult transition for many companies.

The 5°C rise in temperature is, hopefully, not what will happen. But it is a warning to everyone in the finance industry that preventing this catastrophe is on their shoulders as well as on the shoulders of governments, regulators, corporates, the farming industry, consumers and others.

Types of Risk – White Swans, Black Swans and Green Swans

As climate change risk is different in nature from 'traditional' financial risks, we must consider what these differences are. In the field of finance, the usual assumption when modelling risk is that financial risk follows a 'bell curve' and that this curve is stationary in the long-term. Models based on the bell curve are used in finance to determine Value at Risk (VaR) in a portfolio. Such (Bayesian) models work well over the space of short periods of time such as a day and also when markets are behaving 'normally'. Unfortunately, although such models allow us to predict our maximum loss with, for example, a 99% likelihood, what we should be really worried about is what might happen in the 1% tail of the distribution of possible losses (this tail may not be part of a Gaussian bell curve). Within the 99% likelihood bound, conventional statistical techniques based on historical evidence are reasonably reliable. Beyond this, there is unlikely to be any statistical evidence since it is concerned, essentially, with extreme events that have not happened before. In this region of the distribution, probability statements are meaningless. So *within* the 99% bound, a financial company will be taking and holding risk (which is indeed a key function of financial companies) in order to generate an expected return. Beyond the 99% probability point, the aim of a financial firm is survival.

Nassim Taleb, the author of "The Black Swan"[23] which was published just before the GFC in 2007 was one of the very few people who predicted the collapse in asset prices along with John Paulson (see his Wikipedia entry) who personally made a profit of $4bn from shorting the US sub-prime mortgage market in the lead up to the GFC. It is likely that it was no surprise to Taleb that Goldman's risk models were producing 'nonsense' information. His insights are highly relevant to the 'forthcoming' climate crisis. A climate crisis is not unforeseeable (but when, in what form, where and with what consequences is not foreseeable) while action to forestall the likely damage to the Earth, human society, animals and plants is urgently necessary yet action on this front to date is well behind what is necessary.

In just over a decade everyone in the world has been impacted by two global crises – the Global Financial Crisis and the Global Covid Pandemic and many people in the world have already been badly affected by Global Climate Change. Global Climate Change is likely to cause another global crisis that may either overwhelm the world or may be mitigated by actions taken by governments, financiers, businesses, consumers and others. Central banks can help to prevent the crisis leading to financial instability (that is part of their mandate) but they cannot prevent mass migration from land which becomes uninhabitable – the type of migration we consider using the expression "tipping points".

Now to more consideration of 'types' of swan. The first type of 'swan' we consider is the 'white swan'. This is the type of event that financial firms and others can protect themselves against through adequate equity capital on their balance sheet (as in the graphic above) and by the employment of standard risk management techniques. 'Adequate' capital has to be calculated by regulators. Value at Risk (VaR) is one approach to determining how much equity capital is necessary. Value at risk (VaR) is a measure of the risk of loss on a portfolio. It estimates how much a portfolio might lose (with a given probability), **given normal market conditions**, in a set time period such as a day.

However, the real existential risks are 'black swan' or 'green swan' risks. Black swans (the actual birds) were once thought not to exist. Thus there was no need to consider them as a possibility. But just as someone, one day spotted a real black swan on a lake, subsequently many others were then noticed by other people.

Black swans are events that are initially thought to be so unlikely that they will not occur in any living person's lifetime and so can be disregarded. The financial crisis showed that such events in finance did exist and indeed were more common than had been believed. Nassim Taleb was the first person to discuss events like the GFC in terms of what he called Black Swan risk even before the GFC had hit the world.

[23] Taleb, Nassim *The Black Swan: the impact of the highly improbable* https://www.amazon.co.uk/Black-Swan-Impact-Highly-Improbable/dp/0141034599

What we call a Black Swan event is one with the following three attributes:

> *First,* it is an outlier outside the realm of regular expectations as the past is not a guide to such an event
>
> *Second,* it carries an extreme 'impact'.

He summarises these attributes as: rarity and extreme 'impact', with only retrospective and not prospective predictability. He goes on to say that a small number of Black Swans explains almost everything in our world, from the success of ideas and religions to the dynamics of historical events, to elements of our own personal lives such as choosing a marriage partner.

The term Green Swan is defined as:

> "… a climate event that is outside the normal range of expected events. Green swans are different from black swans because there is some certainty that climate change risks will one day materialise."

Decision Making under Conditions of Risk & Radical Uncertainty

For financial analysts and boards of directors, having to operate today and make decisions in a world where historical information on asset price changes may be quite irrelevant to future asset prices and where, therefore, traditional Value at Risk models have little relevance, is a totally new experience. We don't know with certainty what the mass of future GHG emissions will be (tons of carbon dioxide equivalent (tCO2e) although we can make reasonable forecasts based on the past. We don't know how any particular degree of warming will translate into physical damage (impact assessment). We also don't know what the socioeconomic consequences of policy, regulatory and social changes will be. Coronavirus impact, regional wars, displaced people and many other factors will determine actual outcomes for the planet. We also don't know what impact there will be from technological developments such as the possibility of low-cost power generated from nuclear fusion or from policy changes (such as a very high carbon tax imposed very broadly across industries in a majority of countries). But we can make informed guesses based on as to what *could* happen under various scenarios (not forecasts). That is going to be critical for success both in terms of avoiding the worst impact of climate change on corporates and financial firms *and in mitigating global warming.*

The Precautionary Principle is an approach to decision making under uncertainty and is in fact a core principle of EU environmental law. It is also part of the 1992 Rio Declaration on Environment and Development which states:

> Where there are threats of serious or irreversible damage, lack of full scientific certainty shall not be used as a reason for postponing cost-effective measures to prevent environmental degradation (UNEP 1992).

This Principle is certainly one that could be sensibly applied in the case of climate change where there are great uncertainties but also 'tail events' which could wipe out life on Earth or at best make life on Earth very difficult for people, animals, sea-life and plants. Insurance is about low probability, high impact events. By insurance we don't mean an insurance company policy but what actions 'the world' should be taking now given how very bad the worst-case scenario of global warming could be. We have to be most concerned about known unknowns. But there are also unknown unknowns, (with credit for this expression to former US Secretary of Defence, Donald Rumsfeld). And if one looks throughout history, it is the unknown unknowns that tends to be the difficult ones.

Chapter 6: TCFD Metrics and Targets – Carbon Accounting, and the Role of Financial Firms

IFRS S2 (Climate-related Disclosures) came into force in January 2024. The objective of IFRS S2 is to require an entity to disclose information about its climate-related risks and opportunities that is useful to users of general purpose financial reports in making decisions relating to providing resources to the entity. It also requires an entity to disclose information about climate-related risks and opportunities that could reasonably be expected to affect the entity's cash flows, its access to finance or cost of capital over the short, medium or long term.

The objective of TCFD is for companies to tell their investors whether and how they have addressed climate related risks and opportunities in their strategic planning and decision-making, how it is resourcing its plans, and how it is progressing against the plans that it has in place. An example of the sort of disclosure we mean here is providing information about any transition [to net-zero] plans that a company has in place, such as for its greenhouse gas emissions targets—what those targets are, how it plans to achieve them, and what progress it is making over time. To achieve this, it is necessary to be able to measure those emissions.

One of the four 'themes' of TCFD, therefore, is 'metrics and targets' as shown in an earlier graphic. However, TCFD does not specify how companies should measure GHG emissions or how they should set a target. In the following chapter we cover International Financial Reporting Standards (IFRS) in relation to climate change risk as this is likely to be the framework used by most corporates worldwide [US corporates are one of the exceptions] for meeting the requirements for disclosure laid out by TCFD.

As IFRS has stated that its 'standard', S2, fully incorporates the TCFD concepts we have the same issue with IFRS that it does not specify 'how' to achieve the objectives it sets. Like TCFD, IFRS asks for information to be provided about metrics and targets as well as about governance, strategy and risk management. IFRS also says that if a corporate follows TCFD recommendations they will be very familiar with those concepts because they took them and incorporated them into S2.

While neither TCFD nor S2 specify what metrics should be used or how targets should be set they do say that Scopes 1, 2 and 3 GHG emissions as in the **Greenhouse Gas Protocol** (see below) should all be disclosed and the risks to the company of such emissions.

TCFD does, however, in its Guidance for All Sectors in relation to Metrics and Targets, note that:

> Organizations should provide the key metrics used to measure and manage climate-related risks and opportunities.
>
> Organizations should consider including metrics on climate-related risks associated with water, energy, land use, and waste management where relevant and applicable.
>
> Where relevant, organizations should provide their internal carbon prices as well as climate-related opportunity metrics such as revenue from products and services designed for a low-carbon economy.
>
> Organizations should provide their Scope 1 and Scope 2 GHG emissions independent of a materiality assessment, and, if appropriate, Scope 3 GHG emissions and the related risks.
>
> GHG emissions should be calculated in line with the **GHG Protocol** methodology. Organizations should describe their key climate-related **targets** such as those related to GHG emissions, water usage, energy usage, etc.
>
> Other goals may include efficiency or financial goals, financial loss tolerances, avoided GHG emissions through the entire product life cycle, or net revenue goals for products and services designed for a low-carbon economy.

In describing their targets, TCFD says that organizations should consider including the following:

Whether the target is absolute or intensity based, time frames over which the target applies, base year from which progress is measured and key performance indicators used to assess progress against targets.

6.1 Greenhouse Gas Protocol (GHGP)

The starting point for all measurements of GHG emissions and their impact is work that has been undertaken by the Greenhouse Gas Protocol. This body has set the globally agreed standards to measure and manage emissions. This is a critical starting point since it has been relatively easy for suppliers to claim they offer carbon free power when it in fact such a claim may be dubious.

As an example, some energy suppliers offer what they call a clean energy Carbon Free Electricity (CFE) Power Purchase Agreement (PPA) even though it is not a **24/7** CFE PPA. The issue here is knowing what this actually means. However one looks at it, it is very different from true 24/7 Carbon Free Electricity (CFE). The reason that they can make the claim they do is that the rules that were set by the Greenhouse Gas Protocol many years ago have allowed it. The Greenhouse Gas Protocol is a framework to enable standardised accounting for, and reporting on, GHG emissions. It was, somewhat surprisingly, launched as long ago as 1998 by the World Resources Institute (WRI) and the World Business Council for Sustainable Development (WBCSD) as a voluntary system of accounting and reporting.

The view taken by the joint promotors of this key initiative was that by obtaining the agreement of companies that they would account for their emissions and report them thus making them widely available, this would lead to **impact** by incentivising more companies to reduce their GHG emissions and utilities to close gas and coal power stations and replace them with solar and wind. At that time, it was probably not particularly relevant to set the rules in such a way that they prohibited the contradiction of a 100% CFE **supply** through a Pay-as-Produced (PAP) PPA. They key to this issue is that **purchasing** 100% CFE power may not mean, and often does not mean, that the purchaser was actually **consuming** 100% CFE **24/7**. The purchaser may have been selling surplus CFE power from the PAP PPA in the summer but using those sales as a credit to enable it to claim to be using CFE power in winter when in fact it was using power produced using a high percentage of fossil fuel.

Today this issue has become critical. In July 2024, GHGP issued a study entitled "Market-based Accounting Approaches Survey Final Summary Report"[24]. Market-based means using 'offset' type instruments to reduce reported emissions. Accounting for such market-based approaches is a complex business but the GHGP hopes to issue new rules based on the results of their survey. The GHGP also offers internships worldwide for those with relevant masters degrees with a view to future employment.

While the GHG Protocol started as a voluntary programme it has become the worldwide basis for GHG reporting. But more than this, it has become the basis for mandatory reporting in the EU commencing in 2024 and at some point also in the United States through the expected requirements from the financial market regulator in the US, the Securities and Exchange Commission.

The protocol is a framework for the accounting and reporting of ***Full Lifecycle Emissions*** of a company throughout its value chain. It divides emissions into those that can be attributed to the company's direct emissions and indirect emissions which are those which from its suppliers (including its power supplier) on the one hand (upstream in the value chain) and customers (downstream in the value chain) which would include the burning of an oil and gas company's petrol in a motor car. Figure 6.1[25] shows the complete product cycle in more detail starting with upstream activities (inputs to the production process) coming into the company and then downstream activities which is everything that happens after production through to 'end of life' such as the scrapping of a motor car, the demolition of a building or the storage of nuclear waste from a reactor. Also, it should be noted at the top of this graphic all the GHGs which this standard considers such as methane and nitrous oxide.

Scope 1

Direct GHG emissions that occur from sources owned or controlled by the reporting company, i.e., emissions from combustion in owned or controlled boilers, furnaces, vehicles, etc.

[24] https://ghgprotocol.org/blog/now-available-market-based-accounting-approaches-survey-final-summary-report

[25] https://ghgprotocol.org/blog/you-too-can-master-value-chain-emissions

Scope 2: Indirect GHG emissions from the generation of purchased or acquired electricity, steam, heating, or cooling consumed by the reporting company.

Scope 3
All other indirect GHG emissions (not included in Scope 2) that occur in the **value chain** of the reporting company.

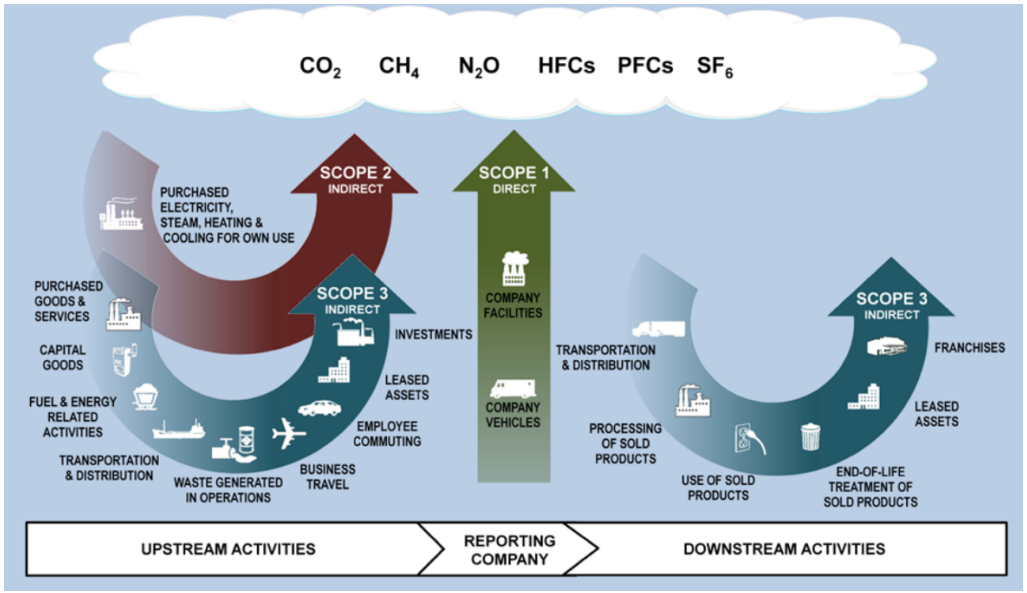


Figure 6.1 This shows the complete product cycle starting with upstream activities (inputs to the production process) coming into the company and then downstream activities and related Scopes 1, 2 and 3

Scope 3 can be broken down into upstream emissions and downstream emissions. Upstream emissions include all emissions that occur in the life cycle of a material/product/service up to the point of sale by the producer, such as from the production or extraction of purchased materials. Downstream emissions include all emissions that occur as a consequence of the distribution, storage, use, and end-of-life treatment of the organization's products or services. In particular, in the case of an oil company producing petrol (gasoline), diesel and kerosine (for planes) its reporting of Scope 3 emissions would cover emissions by the customers for its products i.e. the CO2 produced when people drive their petrol of diesel cars or travel in an aircraft using kerosine etc. Frequently, however, oil and gas companies at least in the past have reported only their own emissions which are much smaller than those of its customers burning its fossil fuels.

One consideration when trying to calculate the relevant numbers for reporting on a company's emissions is the risk of double counting. All of a company's Scope 3 emissions are someone else's Scope 1 and Scope 2 emissions. Not only is there the risk of double counting. There are also very considerable difficulties in obtaining data on many of the emissions which are required to be attributed in particular, inputs to a company's value chain which are produced overseas. However, our main concern at this point is with "purchased electricity for own use" in Scope 2.

6.2 The Role of Energy Attribute Certificates (EACs)

The difficulty with knowing the attributes (in particular the carbon content) of power coming to a customer through the transmission and local distribution systems arises because an electron is an electron is an electron and does not carry any indication of how it was generated. It is not possible to 'tag' electrons as they enter a customer's premises. As a result, without some mechanism which can link the source of supply to the point of use it is not possible to know what type of energy, fossil or carbon-free, produced the electricity being used. The solution that was arrived at some 20 years ago was the development of what are known as Energy Attribute Certificates (EACs) which are a means of attributing a supply of electricity from a particular source with a consumer purchase. However, until recently, this could only be achieved on an annual basis which meant that surplus solar energy produced during the day in summer could be claimed for use as clean power even during the darkest winter evenings.

There are three different certificates according to the region in which they were issued.

Guarantee of Origin (GO): instruments of the European Energy Certificate System used in 30 European countries

Renewable Energy Certificate (REC): in the United States and Canada

International Renewable Energy Certificate (I-REC): used in countries outside Europe and North America

On the demand side of the market for EACs, this comes mainly from corporations wishing to show that their power comes from 100% renewable sources and can then be accounted for using the GHG Protocol Scope 2 emissions methodology to enable them to provide disclosures, normally through the Carbon Disclosure Project now known simply as CDP, for climate disclosures or RE100 for renewable energy usage.

What is required in the **next transition to 24/7 CFE**, which is currently under way is a shift to what is known as **temporal granularity** meaning the scale or level of detail in a data set which. This means being able to 'timestamp accurately, and regularly (down to 15 minute intervals)

In summary, the complete value chain in the power industry – regulatory authorities, 'green' certificate issuing bodies, generators, TSOs/DSOs, wholesale and retail consumers, and others – is currently working towards standards for the future in order to enable the transition 24/7 CFE. This requires new processes and tools to manage ***Time Based Energy Attribute Certificates.***

6.3 Energy Tag[26] time-based certificates

We could not, until recently, know if an offer of power by a supplier of power to provide 24/7 clean power was a genuine offer of clean power at any time of day and in any season or just 'greenwashing'. To help overcome such problems, *EnergyTag* has been developing an hourly (granular) matching system. Their goal has been to establish the Granular Certificate (GC) as a common, tradable instrument that provides traceability across markets for power, flexibility and carbon. The GC Scheme Standard provides the first version of the criteria that those implementing Granular Certificate Schemes need to meet to prove, through a third party audit, that they are *"Compliant with the EnergyTag Standard".*

Energy Tag is an independent non-profit organisation which is defining and promoting Net-Zero aligned electricity *carbon accounting*, enabled by real-time tracking with *Granular Certificates.* Its Initiative is seeking to define a framework for adding a timestamp to energy attribute certificates (EACs) for both clean power and clean hydrogen which will make them more reflective of the physical availability of clean power or clean hydrogen. Participants believe this may improve public perceptions of clean energy claims and could incentivise energy storage and support new carbon accounting methodologies.

The problem as they see it, and the reason behind Energy Tag's creation is that the more successful the world is in deploying renewables, the harder they are to integrate into the grid. Current methods for procuring clean energy as considered in the sections above, **only match supply and demand over a year and over large geographical boundaries. This results in a disconnect between carbon accounting and grid realities and fails to address the hardest hours of decarbonisation i.e. at night or in daytime when the wind speed is low or the sun is obscured more than usual by cloud cover. This is clearly a critical issue.**

Even when claiming to offer '100% renewable an organisation may still be reliant on fossil fuels for a significant portion of the day particularly in winter, enabling substantial underreporting of real-world emissions. The lack of data about the *time of electricity production* on Energy Attribute Certificates is a key root cause of this issue. Given that corporates purchase over 50% of the world's electricity, fixing these carbon accounting issues is crucial for achieving deep grid decarbonisation. **The failure of companies to contribute towards additionality despite their REC purchases and yet seemingly achieving their science based targets** is **a key issue.**

[26] The information in this section is based on Energy Tag's publication *Carbon accounting & tracking for 24/7 clean grids* https://energytag.org/?utm_medium=email&_hsmi=80439457&_hsenc=p2ANqtz-_VHsRqYCEZ8BaTsv0DDKtTmpIY-RML56J0qryLPrWdylUKP6qIMhMWcOYX-3bTWPZd3nvqghX5MlAGWwyZA0Ycw4t_tq&utm_content=80439457&utm_source=hs_email

Energy Tag's Granular Accounting [accounting in ever smaller time steps] is designed to ensure consumption is matched with clean energy in the same hour [or half-hour or quarter-hour] from a location where electricity can be delivered and *encourages new clean energy supply*. This is next frontier of clean energy sourcing and will be enabled by time-stamped Granular Certificates also known as T-EACs. It is system is designed to enables consumers to drive deep decarbonization of the grid – every hour of every day. This is a way of:

- Enabling the production of clean products, not only power but hydrogen and its derivatives such as liquid hydrocarbon fuels and ammonia, without any fossil reliance.
- Providing more transparency by linking production to consumption in 'real-time'.
- Supporting storage and flexibility by providing a new price signal by ensuring the market price for power in times of shortage, even over short periods, reflects this scarcity and thus makes the provision of energy storage on the grid or the provision of generation capacity at such times provides an adequate reward to investors.
- Facilitating better risk management by avoiding volatile fossil-based electricity pricing.

6.4 Science-Based Targets Initiative (SBTi)

TCFD (and ISSB) on their own do not provide protocols to measure carbon emissions or achieve targets for emissions reduction. The initial work of the SBT initiative was to enable corporates (the main GHG emitters) and other emitters to have a science-based target which provided a clearly defined pathway that would be sustainable. It would achieve this by specifying how much and how quickly a corporate needed to reduce its GHG emissions to be 'Paris-compliant'.

Nearly 1,000 companies in 50 sectors from coal and gas to pharmaceuticals, with a cumulative market cap of USD $15.4 trillion, have pledged to align their de-carbonization plans with the Paris Agreement by adopting SBTs. But no such target framework was available for banks or other financial institutions – until 2020. We need to reduce emissions, well beyond even what was agreed by all the world's governments in Paris in 2015 to achieve a <2°C target. There is a limited 'carbon budget' (see Chapter 5.1) left for the world as a whole and if one company uses more of it than calculated as allowable under its 'Science Based Target', there is less available for other companies which have few or no options for carbon reduction. The consequence then would be that the 1.5°C target would not be achieved nor even the 2°C target resulting in substantial costs to everyone from physical damage.

Targets adopted by companies to reduce greenhouse gas (GHG) emissions are considered "science-based" if they are in line with what climate science says is necessary to meet the goals of the Paris Agreement – to limit global warming to well-below 2°C above pre-industrial levels and pursue efforts to limit warming to 1.5°C. SBTi calculates this target or 'mini budget'. SBTi makes clear the benefits from financial institutions being actively involved with corporate de-carbonisation through the influence they can exert on corporates to which they lend or in which they invest. They see financial institutions as being the vital link in enabling the kind of system-wide change we need and enabling the re-direction of capital to support the carbon free energy technology solutions which we cover in later chapters. The financial sector Science Based Targets framework *is designed to enable financial institutions to accelerate the transformation required* by aligning lending and investment portfolios with the level of ambition required by science. But note that the focus of this initiative is less on the risk to financial firms' portfolios but on the extent to which financial firms can drive climate change mitigation through the lending and investing decisions. A graphic from SBTi showing a typical target for a corporate is shown below:

An obvious question is whether corporates generally are in practice actually cutting their emissions by the amount required to hit Paris targets particularly given many of the necessary tools, such as SBTs, are now available. It would seem that in practice many have not set targets and others which have, have nonetheless found it impossible to meet them. This is part of the explanation why greenhouse gas emissions continue to rise almost every year.

In June 2021, Science Based Targets initiative (SBTi) published a report entitled "TAKING THE TEMPERATURE: ASSESSING AND SCALING-UP CLIMATE AMBITION IN THE G7 BUSINESS SECTOR". SBTi notes that a dramatic gap in ambition remains when it comes to 2030 targets set by corporations and countries. Recent analysis shows a significant difference — a gap of 20 to 23 gigatons of carbon dioxide (GtCO2e) — between the level of GHG emissions expected in 2030, under current policies and country commitments and where they should be to limit global warming to 1.5°C.

SBTi has now advanced one of its objectives which is to provide a **Net Zero standard for financial companies**. The framework was introduced in 2020 for lenders, investors and insurers. These will now be able to verify that their climate targets are Paris-Agreement-aligned through the Science-Based Targets Initiative (SBTi). One of these approaches is the **temperature rating approach** which financial institutions can use to determine the current temperature rating of their portfolios and take actions to align their portfolios to long-term temperature goals by engaging with portfolio companies to set targets.

Approved targets should result in financial companies committing to reduce their Scope 1 (direct) and Scope 2 (power-related) emissions by 4.2% annually under the 1.5C pathway and 2.5% annually under the 'well below 2C' pathway. Scope 3 target requirements will vary based on the kinds of assets which the organisation has holdings in. It should be noted that a revised standard was introduced in May 2024. To help speed the transition, one change in the revised framework is that validated targets will have a reduced timeframe window for Scope 1 and 2 emissions, down from 5-15 years to 5-10 years.

6.5 Partnership for Carbon Accounting Financials (PCAF): financed emissions standard and Carbon Disclosure Project (CDP)

There is a considerable difference between the issues in accounting for the GHG emissions of a corporate and accounting for the carbon intensity of a lender or investor's financial asset portfolio. Another organisation, PCAF, has developed a standard for **financed emissions**[27] (emissions resulting from financing provided by a financial institution).

The *Partnership for Carbon Accounting Financials* (PCAF) is an industry-led initiative. Created in 2015 by Dutch financial institutions (FIs), PCAF extended to North America in 2018 and scaled up globally in 2019. More than 300 banks and investors participate in PCAF. As the graphic below from the PICAF publication *Global GHG Accounting and Reporting Standard for the Financial Industry* shows, without agreed standards there are many possible "metrics" that corporates might choose to use in their company's GHG emissions reporting. In the case of financial firms they would be reporting on the carbon intensity of their portfolio (lender asset book or equity and bond funds). In the case of financial firms, the input into their own disclosure is, of course, based on the reporting of the corporates that they hold in their portfolios.

A wide array of intensity metrics is applied in the market and each has its own merits. The table below includes a list of the most common metrics.

Metric	Purpose	Description
Absolute emissions	To understand the climate impact of loans and investments, and set a baseline for climate action	The total GHG emissions of an asset class or portfolio
Economic emissions intensity	To understand how the emissions intensity of different portfolios (or parts of portfolios) compare to each other per monetary unit	Absolute emissions divided by the loan and investment volume, expressed as e.g. tCO_2e/€M invested
Physical emissions intensity	To understand the efficiency of a portfolio (or parts of a portfolio) in terms of total carbon emissions per unit of a common output	Absolute emissions divided by an output value, expressed as e.g. tCO_2e/MWh, tCO_2e/ton product produced
Weighted Average Carbon Intensity (WACI)	To understand exposure to carbon-intensive companies	Portfolio's exposure to carbon-intensive companies, expressed as tCO_2e/€M company revenue

Figure 6.2 Financed Emissions Metrics

Using PCAFs Financed Emissions Standard equips financial institutions with standardised methods to measure financed emissions and enables them to:

[27] PCAF (2022). The Global GHG Accounting and Reporting Standard Part A: Financed Emissions. Second Edition. https://carbonaccountingfinancials.com/files/downloads/PCAF-Global-GHG-Standard.pdf

- Assess climate-related risks in line with the recommendation of the Task Force on Climate-related Financial Disclosures (TCFD).
- Set science-based targets (SBTs) using methods developed by the Science Based Targets initiative and other science-based methodologies.
- Report to stakeholders like the CDP.

CDP

Originally called 'Carbon Disclosure Project' but now using as its strapline Disclosure, Insight, Action, CDP runs the global environmental disclosure system. Each year CDP supports thousands of companies, cities, states and regions to measure and manage their risks and opportunities on climate change, water security and deforestation. It does do so at the request of their investors, purchasers and city stakeholders.

CDP believes that improving corporate awareness through measurement and disclosure is essential to the effective management of carbon and climate change risk. It requests information on climate risks and low carbon opportunities from the world's largest companies and others. Over 23,000 companies representing two thirds of global market capitalisation report through CDP on climate change, forests and water security.

CDP has also produced a position paper on carbon credits – a highly contentious issue – in which they provide clear global guidance and accounting on what constitutes a 'credible' carbon credit and a standardised approach to the quality and use of such credits. Without this it is difficult to evaluate efforts towards deep decarbonization and whether companies are falling short of where they need to be. If the use and reporting of carbon credits is not credible, comparable and consistent, it risks damaging the integrity of emissions accounting more widely.

PCAF and CDP collaborate closely to enable transparent disclosures allowing stakeholders to better understand the climate impact of financial institutions' portfolios. CDP's disclosure framework integrates *The Global GHG Accounting and Reporting Standard for the Financial Industry* into its Climate Change Questionnaire for Financial services, which asks for disclosure of Scope 3 portfolio emissions.

While it might be thought that through the various target setting, disclosure and reporting organisations that we have covered above, all corporates and financial firms would have clear targets for emissions reduction, this has not so far been the case for banks.

6.6 'Big Tech' Case Studies Google & Microsoft: 24/7 Clean Baseload PPAs

Many case studies of 24/7 CFE use a 'tech' company as an example. In particular the tech companies that provide 'cloud' services, cryptocurrency companies and more recently Chat GPT type services using large language models which consume **multiple amounts of power** compared with more traditional services. This would be the case, for example, for the increase in power required for large language model (LLM) type searches compared with the traditional Google search model. In the case of Ireland, the International Energy Agency has forecast that datacentres will account for 32% of Irish electricity demand by 2026. Such data centres also need cooling using water. Thus shortages of power and shortages of water around the world remain critical for such companies' growth and for the Green and Energy transitions. As we noted in the previous chapter, however, **while the tech companies have tried hard to achieve their targets, much of their reporting seems to have been based on the use of 'offsets' rather than in reductions in real GHG emissions. [Note: we cover the sudden increase in interest in 2024 by the 'tech' companies in advanced nuclear technology for generating carbon free power in Chapter 13.5].**

As a result of competition for power supplies between households in a region and data companies, local authorities have in some cases refused to allow new data centres unless the owner of the intended datacentre can source all its power from **additional clean generating capacity and ideally 24/7CFE**. As a result, the 'big five' US tech companies - Alphabet (Google), Amazon, Apple, Meta, and Microsoft and Alibaba, Tencent, and Xiaomi in China – have all been involved in 24/7 CFE initiatives. It has been reported that in London, a Datacentre Task Force has been set up to try to limit the issues that can arise through competition for power supplies. This can be exacerbated if a potential location has good fibre-optic cable provision which is critical for any datacentre.

Google

Google is an example of a company that is supporting Energy Tag's work. As more organizations shift to hourly clean energy purchasing, Google emphasised that there needed to be a way to track and certify their progress. To contribute to this effort, Google teamed up with M-RETS, an energy certificate registry in the United States, to pilot Time-based Energy Attribute Certificates (T-EACs). These instruments track how, where, and when electricity is produced. T-EACs are designed to bring transparency to granular clean electricity production data and consumer claims. Google is also supporting efforts on a global industry scale including a €530,000 grant from Google.org to EnergyTag, which is the leading organization developing an international standard for T-EACs.

Figure 6.3 [28] shows how the energy needs of one of Google's datacentres one day in March were met over each of the day's 24 hours. The straight line shows the steady demand from their datacentre. The coloured bars show what type of energy met this demand. Their objective here, as elsewhere, is to eliminate the black bars which they are suggesting could be achieved by storing the surplus contracted wind power and using it to power their datacentre at times when the grid is using fossil fuels.

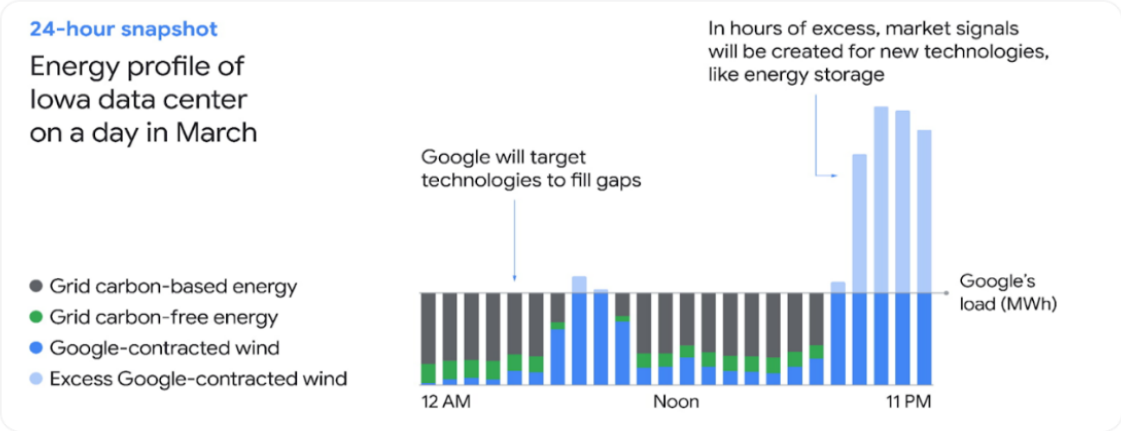


Figure 6.3 Hourly snapshot pf the energy profile on one Google datacentre on one day in March

Engie and Google Enabling 24/7 Carbon-Free Energy Solutions

The energy company, ENGIE, signed a contract in 2021 to provide Google's datacentres in Germany with a **guarantee to provide 80% carbon-free energy (CFE) supply on an hourly basis**. This requires ENGIE to develop a carbon-free energy portfolio on Google's behalf.

Unfortunately, the advent of Large Language Models (LLMs) such as Google's Deep Mind model, Gemini (or Open AI's Chat GPT) has resulted in a surge in electric power requirements. As a result, it has been reported by Google that their GHG emissions have increased by nearly 50% over the five years from 2019 - 2024. The reason is that a Generative AI system (LLM) can use 30 times as much power as computers running task-specific software. The IEA expects that by 2026, data centres could be using as much electricity as Japan.

Microsoft Cloud for Sustainability (MCS)

In 2020, Microsoft made a commitment to shift to 100 percent renewable energy supply in its buildings and datacentres by 2025 although sadly this no longer seems achievable even in the medium term. On this journey, it recognises that how it tracks its progress is just as important as how it gets there. It is now the first hyperscale cloud provider to track hourly energy consumption and renewable energy matching in a commercial product using the Vattenfall 24/7 Matching solution for its new datacentre regions in Sweden. Microsoft itself has committed that by 2030 it will have 100% of the energy consumption of its data centres, 100% of the time. globally matched by zero carbon energy purchases - its 100/100/0 by 2030 goal.

The critical issue as the world aims to move towards 90 – 100% clean power every hour of the day is 'how to fill the gaps'. Thus for Google, Microsoft and others attempting to achieve this aim, it remains an aim except in a small number of regions of the world which have dispatchable power from geothermal or run-of-the-river power or nuclear power.

[28] https://cloud.google.com/blog/topics/sustainability/t-eacs-help-drive-around-the-clock-carbon-free-energy

Location based versus market based carbon footprint

It is clear that tech companies have committed to making progress on reducing emissions related to their activities (and to reducing water consumption for cooling). But as always, 'the devil is in the detail'. Although some of the big tech companies **do have targets** to eliminate CO2 emission by a specific date, the evidence suggests that while reported numbers may have been declining, actual physical emissions have been increasing dramatically. There are two approaches to measuring carbon footprint which can give entirely different results. The GHG protocol has the following definitions of the two approaches:

> A **location-based** method reflects the average emissions intensity of grids on which energy consumption occurs (using mostly grid-average emission factor data).
>
> A **market-based** method reflects emissions from electricity that companies have purposefully chosen (or their lack of choice). It derives emission factors from contractual instruments, which include any type of contract between two parties for the sale and purchase of energy bundled with attributes about the energy generation, or for unbundled attribute claims.

While the evidence suggests that Big Tech is not doing nearly enough to decarbonise, in November 2024, a number of the big tech companies announced nuclear power initiatives which are likely over the course of the next decade to outdate the current discrepancies between the two approaches. We cover nuclear power financing, including initiatives by Big Tech in Chapter 13.5.

A useful article with an excellent set of graphics demonstrating the difference between location-based and market-based methods in relation to large tech firms can be found at:

Big Tech's bid to rewrite the rules on net zero FT August 2024 https://www.ft.com/content/2d6fc319-2165-42fb-8de1-0edf1d765be3

Chapter 7: Accounting & Disclosure Frameworks: International Accounting Standards Board, IFRS and ISSB. **European Commission Double Materiality, CSRD and ESR**[29]

In the chapters above, we covered some of the risks to which any investment may be subject with a particular focus on climate change risk. The TCFD Recommendations covered the approach that needs to be taken by both corporates and financial firms in trying to assess the impact of such risks on their business. Once the risks have been assessed then there is a need to report them in an appropriate way in order that investors can make decisions based on a full disclosure of risks including climate risk that the company foresees. Of course this is all easier said than done. We now consider the reporting side of such risks.

7.1 IASB, IFRS & The Creation of the ISSB

The International Accounting Standards Board (IASB) is an independent, international, group of experts which is responsible for the development and publication of International Financial Reporting Standards (IFRS). The Board is also responsible for approving Interpretations of IFRS Standards as developed by the IFRS Interpretations Committee. IFRS Standards are currently required in more than 140 jurisdictions and permitted in many more.

In November 2021, IFRS issue a press release in which they announced:

> "The formation of a new International Sustainability Standards Board (ISSB) to develop—in the public interest—a comprehensive global baseline of high-quality sustainability disclosure standards to meet investors' information needs;
>
> The publication of prototype climate and general disclosure requirements developed by the Technical Readiness Working Group (TRWG), a group formed by the IFRS Foundation Trustees to undertake preparatory work for the ISSB. These prototypes are the result of six months of joint work by representatives of the CDSB, the International Accounting Standards Board (IASB), the Financial Stability Board's Task Force on Climate-related Financial Disclosures (TCFD), the VRF and the World Economic Forum (Forum), supported by the International Organization of Securities Commissions (IOSCO) and its Technical Expert Group of securities regulators. The TRWG has consolidated key aspects of these organisations' content into an enhanced, unified set of recommendations for consideration by the ISSB."

These announcements mark a major step forward in achieving a unified standard for sustainability disclosure globally (though the US is not a party to IFRS). They also follow closely the TCFD approach to disclosure.
In March 2022, two Exposure Drafts were issued by IFRS.

Objective

> The objective of [draft] IFRS S2 *Climate-related Disclosures* is to require an entity to disclose information about its exposure to significant *climate- related risks and opportunities*, enabling *users* of an entity's *general purpose financial reporting*:
>
> (a) to assess the effects of significant climate-related risks and opportunities on the entity's *enterprise value*;
> (b) to understand how the entity's use of resources, and corresponding inputs, activities, outputs and outcomes support the entity's response to and strategy for managing its significant climate-related risks and opportunities; and
> (c) to evaluate the entity's ability to adapt its planning, *business model* and operations to significant climate-related risks and opportunities.

Strategy

> The objective of climate-related financial disclosures on strategy is to enable users of general purpose financial reporting to understand an entity's strategy for addressing significant climate-related risks and opportunities.
>
> To achieve this objective, an entity shall disclose information about:

[29] The material on IFRS and ISSB is all available at https://www.ifrs.org/sustainability/knowledge-hub/introduction-to-issb-and-ifrs-sustainability-disclosure-standards/

(a) the significant climate-related risks and opportunities that it reasonably expects could affect its business model, strategy and cash flows, its access to finance and its cost of capital, over the short, medium or long term;

(b) the effects of significant climate-related risks and opportunities on its business model and *value chain*;

(c) the effects of significant climate-related risks and opportunities on its strategy and decision-making, including its *transition plans;*

(d) the effects of significant climate-related risks and opportunities on its financial position, financial performance and cash flows for the reporting period, and the anticipated effects over the short, medium and long term—including how climate-related risks and opportunities are included in the entity's financial planning and,

(e) the *climate resilience* of its strategy (including its business model) to significant physical risks and significant transition risks.

It is likely that the new climate-related disclosure requirements will be implemented in many countries by around 2024. This will lead to the need for auditors, compliance staff and others to understand what is behind the introduction of these rules as well as the rules themselves.

In April 2022, ISSB held a live webinar on its proposed standards on general sustainability-related disclosures and climate-related disclosures. This included the following slides. It will be noted that the focus is INFORMATION NEEDS OF INVESTORS **not** the 'double materiality' of EU's approach but purely 'financial materiality.' In due course this approach may change due to public and investor pressure. It is also noticeable that it refers to the TCFD structure which is now the overarching framework in disclosure.

In May 2022 The Chairs of the IFRS Foundation's International Accounting Standards Board (IASB) and International Sustainability Standards Board (ISSB) announced plans for the two boards to work together to agree on how to build and integrate the Integrated Reporting Framework into their standard setting projects and requirements. The Chairs of the IASB and the ISSB actively encourage the continued adoption of the Integrated Reporting Framework, given the importance of its principles and concepts to both high-quality corporate reporting and the connectivity between financial statements and sustainability-related financial disclosures.

The IFRS Foundation has said that the standards being drawn up by the ISSB **do take account** of the impacts of a company's activities [on society as a whole] if they are material to its value. They have also said that while enterprise value is the key issue in their standards, it is also the case that this can be affected by the reverse impact of a company on society. This, however, is quite different from reporting on "Environmental & Social Materiality as required by the EU.

Graphical representation of the IFRS approach

Figure 7.1 shows an interesting graphical representation of the amalgamations into the IFRS framework of five bodies originally involved independently in developing sustainability standards. The diagram is called 'the house'.

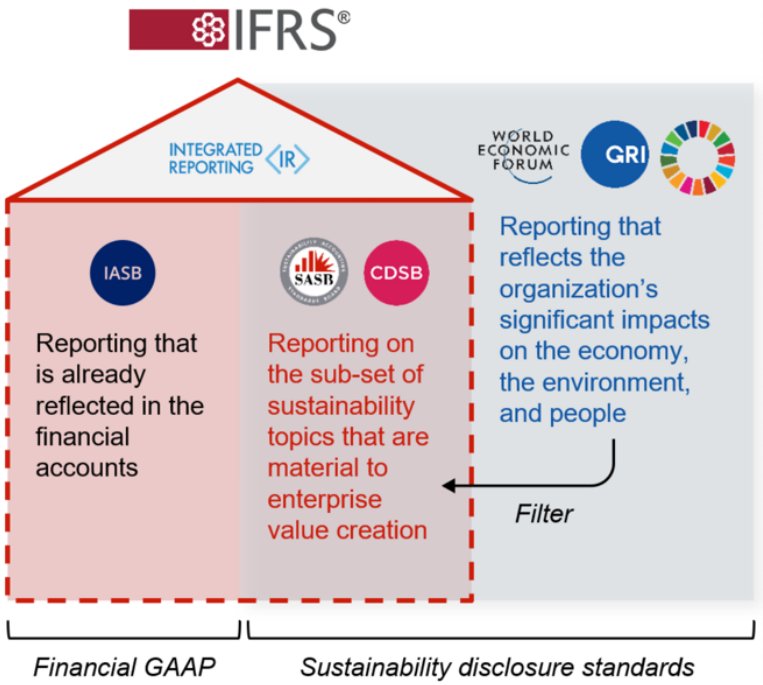


Figure 7.1 [30]A Coherent, Comprehensive Corporate Reporting System

On the roof of the house is the IIRC (Integrated Reporting) whose aim is to be an 'umbrella' organisation acting to integrate all information relevant to economic decision making within one report. Thus it is acting as an umbrella over, on the one-hand, the financial accounting standards (IASB and FASB) and on the other hand over the sustainability framework and standards organisations (SASB and CDSB). In turn, these latter two organisations build on data provided by GRI. GRI (Global Reporting Initiative) is the independent, international organization that helps businesses and other organizations take responsibility for their impacts, by providing them with the global common language to communicate those impacts. It provides the world's most widely used standards for sustainability reporting – the GRI Standards.

The five bodies referred to above which have since been integrated. They acknowledge that there are two types of user with different *materiality* needs. For a wide group of potential users, what is material to them (what matters to them) is how any particular company impacts the environment, social systems (people) and the overall long-term health of the economy (sustainable development) in a material way. Thus for this large group of people, anything which materially impacts these factors is important. This is the European approach we covered above. While acknowledging this type of materiality, this does not mean that ISSB will adopt the double materiality approach and at present it has not.

For the second group (mainly investors), what is relevant is factors which *materially* impact enterprise value creation/destruction. What is new here is that it is now widely accepted that 'non-financial factors' impact corporate value just as financial factors do. So just as financial accounts recognise financial factors which materially impact corporate value, material sustainability topics are also relevant to enterprise value creation/ destruction. One of the five bodies (IIRC) has been focussed on the integration of financial and non-financial reporting into a single report using the International <IR> Framework. Indeed, The Integrated Reporting Framework (International <IR> Framework) has now become part of the IFRS Foundation. Its purpose is to bring both sets of material factors affecting long (or short) run enterprise value into the Annual Report and Accounts of a company. While this might seem like an acknowledgement of the EU double materiality concept, in fact it is not. It is only an acceptance that environmental factors impact corporate value. While they acknowledge that corporates also impact the environment they do not agree at this point that ISSB should require that this impact should also be reported.

[30] https://sasb.ifrs.org/blog/ifrs-foundation-aims-for-coherence-not-complexity/

7.2 The EU's double materiality concept, its Corporate Sustainability Reporting 'Directive (CSRD) and the European Sustainability Reporting Standard (ESRS)

There is a key area of controversy over the approach taken by TCFD, IASB/IFRS/ISSB on the one hand and other approaches, in particular the approach taken by the European Union in the European Commission's *Corporate Sustainability Reporting 'Directive (CSRD)* and its European Sustainability Reporting Standards (ESRS) which we cover below. The controversy is on the question of whether sustainability-related disclosures should focus **only** on the possible impact of climate change on the profit and loss account or balance sheet of an entity or if the reverse impact should also be considered i.e. the impact of the entity (normally a corporate) on the environment and hence on society. In more detail:

> what is the impact of sustainability issues such as climate change risk or water supply risk on the entity and its financial position or;
>
> should it also disclose the opposite side of the equation namely the impact of the entity itself (the corporate) on civil society, employees, consumers and presumably also investors.

This divide on this issue between these two perspectives is considerable and even fundamental. The EU has been very clear in its view of the duality of disclosure required.

The concept of 'materiality' is important in this debate in order that both corporates and society should be able to avoid focussing resources on activities that may not contribute much to mitigation or, alternatively, failing to focus resources on things that are widely believed to matter. Thus the meaning of the word 'material' in the context of finance is something we need to consider. One definition would be any data or information which if not provided, might prevent investors or others making appropriate decisions.

The **financial impact of climate change on a corporate** arises firstly from physical risk (material physical damage to its real assets) and from transition risk (loss in value of its real assets and from material changes in regulations and technology). In addition risk arises from impacts on populations (including employees and consumers) from the environment where they work and live which may be a result of productivity loss from climate change due to heat, insufficient food or drinking water and the possibility of new diseases. **Thus if such potential impacts are 'material' to the financial well-being of the company (however that is defined) information on such potential impacts should be reported in standard IFRS Annual Report and Accounts.**

In the view of many, appropriate action on climate change risk means concern not just with the impact of climate-change on a company's value **but also the impact on society of a company's behaviour**. The European Commission argues that corporates need to be concerned with not just one perspective on materiality, the traditional one i.e. the impact of climate change on the company, but *two perspectives (double materiality as they call it)* meaning also covering the impact of the company on society. In fact it is being increasingly recognised that both impacts affect sustainability and hence *Enterprise Value* and are therefore material. So to the extent that both of these impacts are material (under the IFRS definition), they must be reported.

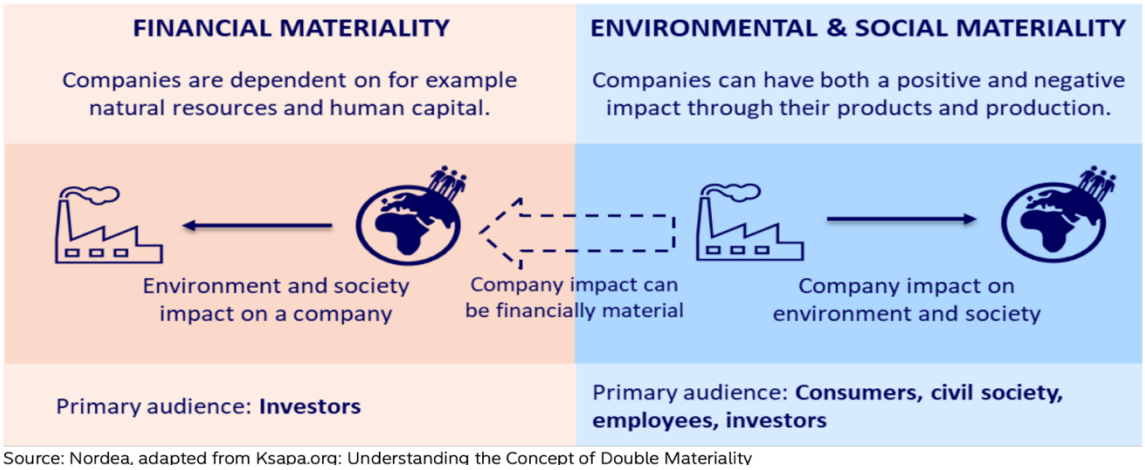
Figure 6.5 [31] is from a large investment management company, Nordea. Nordea itself must follow the reporting and other requirements of the European Union CSRD and ESRS even though Norway, its home country, is not in the European Union. The two perspectives on materiality in the graphic below are:

1. On the RHS, the "external impacts of the company's activities" (labelled as "environmental & social materiality", whose audience consists of consumers, civil society, employees, and investors too);

2. On the LHS the potential or actual impacts of climate-related risk and opportunities on the "performance, development and position" of the company indicated as "financial materiality", with a focus on the interests of investors (TCFD's main focus).

[31] Nordea, *How Sustainability Risks Affect Investments* 2023 https://www.nordea.com/en/doc/final-sustainability-risks-and-how-they-affect-investments-q2-2023.pdf

The double materiality perspective of the Non-Financial Reporting Directive (NFRD) and, since 2021, the Corporate Sustainability Reporting Directive (CSRD). When reporting environmental and social information an entity must use the European Sustainability Reporting Standards (ESRS) as covered below:



Source: Nordea, adapted from Ksapa.org: Understanding the Concept of Double Materiality

Figure 7.2 Understanding the concept of double materiality

Corporate Sustainability Reporting Directive (CSRD) and European Sustainability Reporting Standards (ESRS)

In April 2021 the European Commission (EC) published its proposals for a new directive on corporate disclosure (CSRD) which amended and greatly strengthened the previous sustainability reporting requirements in the EU under the Non-Financial Reporting Directive (NFRD). It is a major step change in corporate reporting which will affect businesses not only in Europe but worldwide. Corporates, regulators and auditors will need to be ready for implementation from 01 January 2023 (SMEs from 01 January 2026). The additional reporting requirements focus on reporting in more detail on the double materiality (inward and outward impact), alignment to the Paris Agreement and forward looking sustainability information. At implementation date, large businesses will need to disclose their strategy & business model in relation to sustainability-related risks and climate scenarios, their sustainability targets & indicators and the extent of progress towards achieving them, environmental-related matters that are affected by or affect their business, social considerations such as equal opportunities, their approach to business ethics, their governance processes and their control mechanisms and risk management approach.

The ESRS (Reporting Standards) are to be used to meet the requirements of the CSRD. The CSRD provides the reporting requirements and obligations while the ESRS provides a framework and methodology for undertaking the reporting.

Chapter 8: ***SDGs, DEI & ESG:*** *Financial Rating Agency Approach to ESG and other risks*

8.1 The United Nations (UN) *Sustainable Development Goals (SDGs)*

There are many societal issues which are relevant to achieving the best possible basis for life for all people who live on the single Earth and might, therefore, be relevant to those who run corporates and manage equity or bond portfolios. Achieving, or not achieving, these wider goals is likely to impact all countries and also corporates and financial companies as goals such as minimising hunger and poverty, reducing inequality and having strong and diverse institutions governing countries, are all likely to reduce the likelihood of wars and mass migrations all of which can improve corporate profitability. Even without wars or migrations they affect corporates and in particular the large multinational corporates based mostly in Western economies and in China.

Just as the United Nations was central in the achievement of the Paris Agreement, it has been very active in other initiatives designed to make the world a better place for humanity. Below we consider areas which are highly relevant to lenders and investors.

An asset owner, an asset manager, an insurer or a bank needs to have an underlying set of principles or goals to operate under when it comes to trying to define its 'purpose' and its strategy for long-term success. Trying to develop a set of principles or goals or defining a company's 'purpose' is difficult in the absence of some generally agreed framework to aid decision making. Having a framework has been made easier by two things:

> First the development by the United Nations of 17 global goals agreed by 193 countries at UN General Assembly in September 2015, and;
>
> Second, the Paris Agreement in December 2015 which we covered earlier.

The ideas behind the *Sustainable Development Goals (SDGs)* were developed by Paula Caballero, Director of Economic, Social and Environmental Affairs at the Ministry of Foreign Affairs in Colombia. The fact that the most participatory intergovernmental process in human history ended up with 17 agreed goals is nothing short of miraculous.

We now have, as one of the expressions of 'society's goals', the framework (the SDGs) which **was used as an input by the 23 banks which in 2019, developed *Principles for Responsible Banking*** (Principles for Responsible Investment came into existence four years before the SDGs were developed).

Figure 8.1 The UN sustainable development goals

The United Nations notes[32] that:

- **"Sustainable development has been defined as development that meets the needs of the present without compromising the ability of future generations to meet their needs.**

  - Sustainable development calls for concerted efforts towards building an inclusive, sustainable and resilient future for people and planet.
  - For sustainable development to be achieved, it is crucial to harmonize three core elements: economic growth, social inclusion and environmental protection. These are inter-connected and all are crucial for the well-being of individuals and societies.
  - Eradicating poverty in all its forms is an indispensable requirement for sustainable development. To this end, there must be sustainable, inclusive and equitable economic growth, creating greater opportunities for all, reducing inequalities, raising basic standards of living, fostering equitable social development and inclusion, and promoting integrated and sustainable management of natural resources and ecosystems."

A number of investment managers offer SDG focussed funds which employ strategies which focus on investing for positive outcomes. This is likely to be companies that are providing solutions to address the world's challenges and are helping to meet the environmental and social targets set out in the UN Sustainable Development Goals. Water and Food security are two key SDGs.

[32] https://www.un.org/sustainabledevelopment/

8.2 Diversity, Equity and Inclusion (DEI)

DEI is a concept that dates back to the mid-1980s in particular in the United States. The aim of the DEI movement is to promote the fair treatment and full participation of all people and in particular those who have historically suffered from discrimination on the basis of their skin colour, disability, sexual orientation, gender etc. particularly in corporate business but more widely as well including in universities. In the corporate environment the argument is not just about fairness but the belief, supported by research, that a diverse workforce should be seen as a competitive advantage rather than just as a legal constraint.

Today, the term is increasingly seen as a demographic and economic imperative, in particular in the US where traditionally white people (mostly of European descent) have constituted a majority of the population. It is now the case, given already existing demographic trends in the United States that by 2027 there will no longer be any majority ethnic or racial group in the under-30 age range. Over time that trend will extend across all age groups and increasingly in other countries also. What the DEI movement is, in effect, saying is that corporate America's workforce should look like America. This would mean that every level in an organisation from new hires to the C-suite (those with 'chief' in their title such as CEO) should be representative of the country as a whole and hence of the potential customers for the company's product or services. Looking at DEI as "Demographic and Economic Imperatives" essentially means that a business must be aware of the characteristics of its customers, suppliers and other stakeholders when considering hiring and training policies. If not, it may have a reduced financial performance compared following DEI considerations. Many Republican politicians disagree with the principles behind DEI while in Europe it is much more accepted as quite normal to apply such principles.

8.3 What are ESG principles and how are they applied by financial rating agencies?

The abbreviation ESG, stands for Environmental, Social and Governance factors. ESG is now a widely used term in the finance industry. Our particular interest in ESG is in the creation of new assets which meet genuine criteria for sustainability and provided ESG funds with suitable assets. Figure 8.2[33] summarises some of the factors considered in ESG assessments many of which will be relevant to any company.

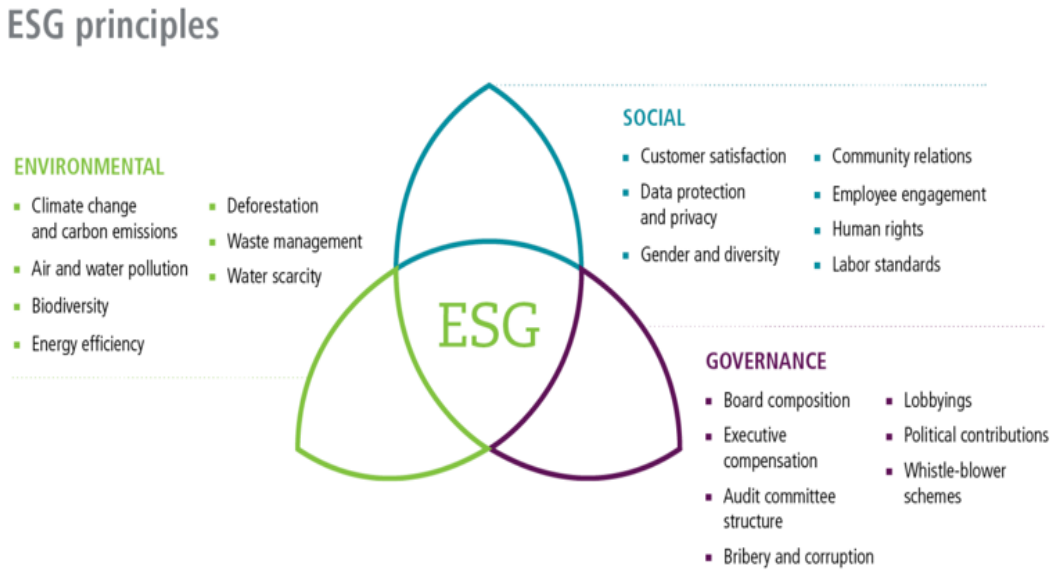


Figure 8.2 ESG Principles

The word "*Sustainable*" at the centre of the Figure 8.3[34] below, could be defined as integrating E&S criteria as well as normal economic/ financial criteria (achieving the investor required rate of return for an asset in the particular risk class), into **financial investment, lending, insuring and corporate (real) investment decisions for the long-term benefit of investors, beneficiaries such as pension plan holders and society at large**.

The implication is that those investors (and corporates) which do not consider E & S principles and the G principle (governance) of the company **including governance of climate-change risks**, may not generate as high returns **in**

[33] Trade Financial Global https://www.tradefinanceglobal.com/posts/sustainable-finance/
[34] https://www.yelp.com/biz_photos/microgreensla-hermosa-beach?select=WNSb0ZJmVFmJD9aT3zUK-g

the long-term, or even in the short-term, as those who do. Indeed, the long-term versus short-term perspective is critical in this new approach to investing/ lending and to the viewpoint of bank prudential regulators – hence one aspect of a required **transition in thinking**.

Figure 8.3 The Three P's People, Profit, Planet

Sustainability (as in the centre of the above graphic) **is most often defined as *meeting the needs of the present without compromising the ability of future generations to meet theirs***. Its three main pillars are social, environmental and economic (a company must make a profit to survive). These three pillars are informally referred to as people, planet and profits, or **3Ps** as in the graphic above.

An alternative way of looking at **interdependencies between people, planet and profit** is to consider the dependence of economic systems (manufacture, electric power generation, food production etc.) on the maintenance of life support ecosystem services such as climate regulation, land quality, water resources, biodiversity etc. The Earth needs to operate within its "sustainability limits".

The first use of ESG ratings is to enable **security selection decisions** by investors and criteria on IPOs and secondary issues of new equity and appropriate loan and insurance **risk underwriting**. The availability of an ESG rating on a company i.e. the company as a whole (unlike a credit rating which is on a particular security) enables an investor to make use of both financial and sustainability information that is relevant to the valuation of that company's securities. But just as the *availability* of data remains a serious issue so too is the issue of the particular *metrics* used, their *reliability* and *comparability* across companies. The production of indices based on this data and ESG funds which track these indices is also an area full of 'minefields'. We look at some aspects of rating agencies approach to constructing their ESG indices in Chapter 8.

Most companies, banks, insurers and investment managers have a Sustainability Department devoted to matters relating to environmental [E] and social [S] issues along with those related to how the company is governed and to what **purpose** [G] collectively known as ESG. While climate change may be the **existential issue for mankind**, in terms of analysing the potential return on an investment, social and governance issues as well as other environmental issues, are also critical. In the EU, the European Commission has made it clear in its Directives and regulations that it believes that social [S] issues are almost, or as important for *society* as environmental issues [E].

While this book is focussed on climate change since this is the existential issue for mankind it is only one of several issues that has attracted investor attention in recent years. **Other environmental [E]** issues are biodiversity, drinking water contamination, river pollution, water shortages for domestic and agricultural use, air pollution, loss of rainforest, ecosystem destruction, land degradation and damage from mining and disposal of waste, many of which issues reflect a loss of **natural capital**. All these issues are environmental [E] issues.

Social [S] issues in particular have come to the fore more recently in financial and corporate and financial circles in recent years. The 'S' in E&S includes matters such as diversity (ethnic origin, gender, sexual orientation, disability etc.) in boardrooms, in senior management and in other areas of business, equal pay (males and females doing

the same job to be paid at the same rate) human rights, health & safety, consumer protection, income and wealth distribution, the terms and conditions of 'gig' workers such as those working for Uber and minimum pay rates.

Some objectors to including social issues such as workforce diversity consider that such issues are irrelevant to financial performance and that the job of directors of a company is simply to maximise profit with no mention of the risks involved in focussing on absolute profit alone and not also on risk or timescales of impacts. Recent research has, however, demonstrated fairly conclusively that in the case of gender balance in companies (proportion of females in a company), diversity pays off handsomely.

Whether companies with more gender-balanced workforces outperform their least-balanced peers remains a contentious subject. But some evidence suggests that they have outperformd by as much as 2 percentage points. However, this could also depend on which year or years were chosen for research.

However, ESG is not yet a science and may never be and an investor should not make any assumptions about high-scoring ESG companies or ESG funds providing higher returns over any specified period than low-scoring ones. Indeed, some recent research suggests that returns may be comparable with other types of asset rather than exceeding them. But the past may be no guide to the future as the world's climate changes.

Investors are increasingly demanding E&S information from companies before deciding whether to invest or not and companies are becoming more aware of the impact of their behaviour on both of those fronts if they lack concern for, and do not take action where appropriate on, such issues. Investors are generally seeking alpha from the companies they invest in (unless they are index investors) by using 'non-financial information' as well as financial. Conversely, corporates are concerned about the impact, whether positive or negative, on their share price if they neglect such issues. As we explain later, the EU now uses the term **"sustainability-related financial disclosure"** as it is more meaningful than "non-financial disclosure".

The increasing focus on diversity in boardrooms in many countries, in management and on the 'shop-floor' is a result of a belief that companies **should** reflect the diversity of the population, not only in routine jobs but right up to CEO level. The importance of treating all people equally, often described as Equality, Diversity and Inclusion (EDI) policy arises also from a belief that different ways of thinking from diverse populations is likely to lead to better decision making and possibly also a higher level of profit. This is partly because the customers of most firms are households and households are diverse. Thus having a corresponding diversity in the workforce through all levels in an organisation may result in relevant factors that were not considered before coming to be a normal part of the process of designing, marketing and selling new products.

A Morgan Stanley research piece, as well as other analyses, suggests that "Our research suggests that using racial and ethnic board diversity factors can improve U.S. large-cap equity stock selection. There may be additional benefit in tilting toward more diverse companies across all four developed markets." As a result of such evidence of the impact of social issues on securities' prices, understanding and analysing non-financial factors (sustainability factors) as well as financial factors, is likely to be a means of achieving improved financial performance in investment portfolios and for banks, a higher return on capital from lending. For insurers, it should lead to a lower loss-ratio through better pricing of risks. High scoring companies in an industry are more likely to be protecting natural capital and considering social factors which in many cases could lead to outperformance relative to peers.

Corporate governance (G) The third letter [G] in ESG representing Corporate Governance is always considered along with E&S in the analysis of a company to create a set of ESG principles. Governance can be thought of as how the company tries to ensure that its '*purpose'* is achieved. One key purpose will be achieving the appropriate risk-adjusted return for shareholders. But it also includes ensuring that the company survives profitably **in the long-term in the interests of its shareholders and other stakeholders**. This points up the distinction between profitability over relatively short periods (quarterly reporting) and longer-term risks and returns to shareholders. As many of the large investors are pension funds with long-term liabilities, such funds need to be concerned with risks perhaps one or two decades away which implies that they are concerned to know that returns will be sustainable long into the future.

A specific governance focus on environmental and social risks is included in TCFD's Recommendations where issues of governance is one of the four areas it focusses on. It requires that companies develop a framework specifically for ensuring appropriate **climate risk governance**.

The objective of all four of TCFDs sets of disclosures is to 'nudge' directors and officers of a company to think harder about climate change risk rather than fudging the issue by issuing bland statements which might amount to no more than 'greenwashing' i.e. pretending that something was being done about these issues.

Ratings, Rating Agencies and Users of ESG Ratings

S&P describe how they undertake an ESG assessment immediately below and also a joint venture by S&P with Boston Consulting Group (BCG) called Climate Credit Analytics [35]:

> "S&P Global Ratings' environmental, social, and governance (ESG) evaluation is a cross- sector, relative analysis of an entity's capacity to operate successfully in the future and is grounded in how ESG factors could affect stakeholders, potentially leading to a material direct or indirect financial impact on the entity. ESG factors typically incorporate the **entity's impact on the natural and social environment and the quality of its governance**. **Our definition of stakeholders goes beyond shareholders to include employees, the local community, government, regulators, customers, lenders, borrowers, policyholders, voters, members, and suppliers, among others.**
>
> First, we establish an ESG Profile for a given entity, which assesses the exposure of an entity's operations to observable ESG risks and opportunities, accounting for the governance structure in mitigating risks and capitalizing on opportunities. Second, we assess the entity's long-term *Preparedness* namely its capacity to anticipate and adapt to a variety of long-term plausible disruptions."

Clearly S&P has thought about the sustainability issues that arise if a company is or is not, prepared for eventualities and they consider the degree of *preparedness* of a company as being an important contributor to the ESG rating which may raise or lower the final evaluation. It is important to note that the ESG Evaluation is **not a credit rating, a measure of credit risk, or a component of its credit rating methodology**. However, the information it gathers for an ESG Evaluation *can inform its credit analysis of rated entities.*

ESG factors may impact industry risk, competitive position, cash flow/leverage and management and governance. They identify these as the areas where ESG factors can influence the capacity and willingness of issuers to meet financial commitments (interest and principal payments on debt securities or loans).

The next step after the production of an ESG rating on a company is the aggregation of companies in an index based on their ESG performance. For example, FTSE Russell Indices (part of the London Stock Exchange Group as is Refinitiv) overcome the difficulty many investors would have in trying to select appropriate stocks for a large diverse portfolio based on ESG characteristics. An example of the use of such indices was demonstrated recently by Blackrock when one of its new funds – *Blackrock US Transition Readiness* exchange traded fund (ETF) launched in April 2021 – was soon the largest ever ETF launch as it quickly reached $1.2bn. The fund was based on the Russell 1000 index. The fund was created by taking this index and then assigning weights to its components to reflect their 'carbon transition readiness score' rather than the usual E, S and G components.

Chapter 9: Financial Portfolios and Climate, Temperature and Paris Alignment Case Study of InfraRed Climate Aligned Fund

9.1: What is the meaning of climate alignment?

The data, metrics and targets and the corresponding climate-related disclosures recommended by TCFD that we considered earlier are all designed to ensure "alignment". Alignment is a much used, but somewhat ill-defined

[35] https://www.spglobal.com/_assets/documents/ratings/ratingsdirect_environmentalsocialandgovernanceevaluationanalyticalapproach_1176513_apr-10-2019.pdf and Climate Credit Analytics https://www.spglobal.com/marketintelligence/en/solutions/climate-credit-analytics

expression. The *Center for Climate Aligned Finance*[36] (part of the Rocky Mountain Institute) based in New York provides the following useful definition: "Climate-aligned finance is the process of bringing a financial institution's portfolio of lending and investment activities into alignment with a 1.5°C consistent emissions pathway." It's a short and simple definition but achieving it requires an arduous journey and, in fact, a transition in thinking by those working in the finance industry.

The risk to lenders and investors is that many assets that they hold may not be valued correctly as lenders and investors may not be pricing assets in anticipation of this real economy transition. 'Market efficiency' may be low in this regard because of the myopic view that many investors seem to have about the longer term.

A way ahead? Portfolio temperature metrics/ portfolio climate alignment

The Investment Leaders Group (ILG), a group which includes large financial companies such as Pimco, Robeco, Zurich, State Street and HSBC has, like many others has concluded that reporting a temperature score for a fund, based on standardised metrics, is the best way for stakeholders to know if the fund manager is "aligned with the Paris Agreement" i.e. 2°C or less. But why use this particular metric? One good reason is that non-specialists can understand it in an intuitive way and it is relatively simple measure.

As an increasing number of countries do legislate for net zero in all sectors of the economy, corporates, financial companies, households and central, regional and local governments will all need to adapt. For investors and lenders, this will mean identifying risks and opportunities in the transition to a net zero economy. This will require a framework for measuring the alignment of companies and bank and investor portfolios with net zero or with a temperature target as well as targets for households and sub-national authorities.

With such a framework:

- Investors and lenders will be able to assess the position of companies and portfolios on the path to net zero – to compare them with peers and track progress over time.
- This will in turn increase the ability of financial institutions to effectively allocate capital in a way that supports climate goals, assess transition risks, and engage with companies to facilitate real economy emission reductions.
- This will also increase the ability of financial institutions to track their own contribution to limiting global temperature rise and reaching net zero.
- While a number of methods exist today, their fragmentation and methodological differences prevent the different methods from fulfilling their fundamental purpose, as outputs across companies and portfolios lack consistency and comparability.
- Methodological differences could also increase greenwashing risks as the metric could be adjusted to fit different purposes.

The text above is from "The Portfolio Alignment Team" publication "Measuring Portfolio Alignment"[37] led by David Blood and Iren Levina. Portfolio Alignment is, indeed, the objective we have focussed on as it should enable more effective action to be taken in respect of the real economy. We have, however, changed the text from 'without such a framework' to 'with such a framework' and thus we are listing the positive benefits of having such a framework rather the negatives of having no framework. It is important to understand however, that as the fourth point above notes, at the present time there is no settled way of measuring alignment. We are dealing with a relatively new field of endeavour – minimising the risk to humanity of climate change. Everyone is learning as they go but based on understanding different approaches that have been developed.

There are three possible approaches to measuring alignment with net zero as the Portfolio Alignment Team (PAT) explains:

- Measuring the percentage of the total value of the portfolio which reflects companies in the portfolio with net zero targets. This does not, however, measure the alignment of the part of the portfolio without targets nor different de-

[36] https://climatealignment.org/#:~:text=Climate%2Daligned%20finance%20is%20the,%C2%B0C%2Dconsistent%20emissions%20pathway.
[37] https://www.tcfdhub.org/wp-content/uploads/2020/10/PAT-Report-20201109-Final.pdf

carbonisation transition paths of different companies. It also reflects only a 'commitment' and does not consider the 'transition' i.e. it does not tell us where the company is on its path towards net zero.

- Measuring the deviation of the portfolio from a target or sector benchmark which then involves measuring the deviation of each company within the portfolio. For example, the EU Climate Benchmarks for portfolios from which index companies can then create indices for investors to replicate, requires them to decarbonise at an average rate of 7% per year as well as being aligned with the IPCC 1.5°C scenario.

- Using degree warming metrics to which we referred above. This can use the output from the second approach above – 'deviation from a sector benchmark' which can be converted to a temperature score to provide *degree warming metrics.* SBTi provides guidance on how to set targets and also on how to convert to a temperature score.

Degree warming metrics seem to be more 'intuitive' to most people than deviation from a benchmark. Those investors who have already started experimenting with the degree warming approach such as the insurer and asset manager Axa finds that the approach is useful to assess climate related risks but also to facilitate capital allocation in line with climate objectives and, importantly, to provide a good starting point for engagement with a company on their decarbonisation strategy, plans and actions.

When we think about the meaning of a single company's contribution to global warming, we are faced with the awkward fact that if only one company in a sector de-carbonises, it will make little difference to global emissions. As a result, when it comes to translating a carbon budget to a benchmark, what we are in fact doing in assessing a company is comparing it with its peers around the world. What we need is for all companies in the sector to decarbonise at the scientifically calculated rate that will enable the target to be met. So the benchmark sets what the industry needs to do and the company position relative to this index tells us if it is acting at the rate at which, if all companies in the industry acted, then the industry would keep within its carbon budget.

9.2 Portfolio Temperature, System Transformation & Alignment and Paris

Paris Agreement alignment is not just about asset portfolios. It is really about a system transformation and alignment. It is also an approach to ensuring that governments of developed and developing countries institute policies that should ensure 'better development'. The slogan "build back better" has been used by politicians in several countries and while the use of these three words as a slogan may be just 'politics', it is nonetheless important to ensure that what is being achieved by the finance industry does serves people and the planet in a wider sense. It is not just about a narrow concept of project-based finance flow alignment. It's about making countries and the planet less susceptible to crises, and more resilient in the face of crises which do arise such as global pandemics, tsunamis, water shortages, famine, sudden rises in the cost of energy and other events that disrupt human, animal and plant life. A good summary of this approach would be the United Nations "Sustainable Development Goals (SDGs)".

Specifically on climate finance, we emphasise once more that there are two key desired outcomes. One is for investors and bankers to reduce the 'temperature' of their portfolios. But also, to shift their portfolios in a transition over time by **increasing their allocation to new issues of shares and new bond issues or re-financings of and lending to, companies creating climate mitigation and adaptation assets for the real economy.** Without that second step while we would see falling emissions we would also see a fall in living standards which would arise if we simply shut down GHG emitting companies without 'building back better'.
The development of ESG ratings and the challenges that this has involved, is a good starting point for considering the issue of how to transition the world to achieve the Paris objectives: 1.5°C/2050 Net Zero Paris Alignment by making finance flows consistent with low GHG emissions and a climate-resilient pathway. Unfortunately there are many issues in the creation and the construction and use of indices.

Paris Aligned Funds

The EU Technical Expert Group (TEG) Final Report observes:

> Benchmarks have an indirect but important impact on investments. Many asset managers and investors rely on them as investment monitoring solutions to track their return (passive and smart beta investment strategies) or to define the investment universe and to measure the performance of an investment fund/portfolio (active investment strategy). A more recent use case for benchmarks that has emerged in the industry is their use as an incentive tool to encourage companies at scale to improve their ESG performance. Benchmarks play a significant role and can be a

key lever in aligning the investment and analyst community with long-term sustainability considerations and the transition towards a low-carbon economy. Since conventional benchmarks do not reflect low-carbon considerations in their methodologies and are not appropriate to measure the performance of sustainable investment strategies, over the past decade benchmark administrators have designed hundreds of ESG and 'low-carbon' benchmarks.

While the supply of low-carbon benchmarks is there, lack of harmonisation of the methodologies (especially lack of consensus on how comprehensive the assessment of a carbon footprint should be) and lack of clarity on the objectives pursued with regard to the impact on global warming have affected comparability, reliability and adoption of low-carbon indices. Furthermore, the varying °C of reporting hinders market players' ability to compare indices and choose the adequate benchmarks for their or climate-related investment strategy.

Finally, the benchmarks currently offered in the market do not necessarily align with the financing needs implied by the limitation of global warming to well below +2°C pursuant to the Paris Climate Agreement, as current benchmarks are likely to be more aligned with a 'business-as-usual' scenario, where temperature rises range from 4°C to 6°C, leading to catastrophic damage to the Earth.

The TEG arrived at a need for two different but related benchmarks which asset owners or asset managers could use to develop indices and/or funds, based on the requirements of the two indices. The two types of climate benchmark pursue a similar objective but have different levels of restrictiveness and ambition. The so-called Paris Aligned Benchmarks (PABs) are designed for highly ambitious climate-related investment strategies and are characterized by stricter minimum requirements, while EU Climate Transition Benchmarks (CTBs) allow for greater diversification and serve the needs of institutional investors in their core asset allocation. A summary of the minimum standards is shown below in a Standard and Poors graphic in Figure 9.1

MINIMUM STANDARDS	CTB	PAB
Minimum Scopes 1, 2, and (3) GHG Emission Intensity Reduction Compared with the Underlying Benchmark	30%	50%
Scope 3 Phase-in Period	Up to four years	
Baseline Exclusions	Controversial weapons and societal norm violators	
Activity Exclusions	None	• Coal (≥ 1% revenues) • Oil (≥ 10% revenues) • Natural gas (≥ 50% revenues) • Electricity producers with lifecycle GHG emissions higher than 100 gCO2e/kWh (≥ 50% revenues)
Year-on-Year Self-Decarbonization of the Benchmark	At least 7% on average per year: in line with or beyond the decarbonization trajectory from the IPCC's 1.5°C scenario (with no or limited overshoot)	
Minimum Green Share/Brown Share Ratio Relative Compared with Investable Universe (Voluntary)	At least equivalent	Significantly larger (factor 4)
Exposure Constraints	Minimum exposure to sectors highly exposed to climate issues is at least equal to equity market benchmark value.	
Corporate Target Setting	Weight increases considered for companies that set evidence-based targets under strict conditions to avoid greenwashing	
Immediate Disqualification	If misaligned with trajectory for two consecutive years	

Source: S&P Dow Jones Indices LLC, The EU Technical Expert Group on Sustainable Finance. Table is provided for illustrative purposes.

Figure 9.1 EU Paris-Aligned and Climate Transition Benchmarks

The main users of EU CTBs are likely to be institutional investors such as pension funds with the objective of protecting assets against various investment risks related to climate change and the transition to a low-carbon economy, i.e. transition risks. The main users of EU PABs are likely to be institutional investors at the forefront of the immediate transition towards a +1.5°C scenario. The differentiating features are:

- EU PABs allow for a higher decarbonisation of the investment relative to the investable universe;
- EU PABs have additional activity exclusions;
- EU PABs have a stronger focus on opportunities with a significantly enhanced green share / brown share ratio

A benchmark is not an index or a specified list of securities which match requirements. It is simply a 'set of rules' according to which an index provider can undertake security selection and portfolio allocation to create a benchmark compatible fund. As an example of indices based on the EU benchmarks i.e. following the requirements of the CTB and PAB benchmarks, Standard and Poors (S&P) is one of several index providers that have produced such indexes based on these benchmarks. S&P call these Paris Aligned Climate Transition (PACT) indices

"To meet the EU's proposed minimum standards for CTBs and PABs, we have launched two new index series: the S&P Paris-Aligned Climate (PA) Indices and S&P Climate Transition (CT) Indices, respectively, collectively referred to as S&P PACT: the S&P Paris-Aligned & Climate Transition Indices. Both methodologies meet the respective EU requirements and other climate objectives to become compatible with a 1.5°C scenario and recommendations from the TCFD."

9.3: The Impact Investment Spectrum

Portfolio Alignment with the Paris Agreement is critically important for the Earth and the humans, animals and plants living on it. But there are other important environmental and social issues that should be considered when making investment decisions in what we might call, in shorthand, an ESG world. These considerations apply as much to corporates making real asset investment decisions as to financial firms.

While climate change may be the existential issue for companies and for mankind, in terms of analysing the potential return on an investment and also the impact it may have on the lives of those impacted by it (impact), social and governance issues as well as other environmental issues, are also critical.

There is a wide range of approaches to creating sustainable portfolios but all of them should be designed to have some 'impact' on the wider environmental and social issues that society faces. Figure 9.2 [38] shows that there is a wide range of investor objectives. If there is no impact of investment policy on the wider world outside of a company, a project or fund management house, then it might as well to simply apply a classic approach. This can be thought of as an approach where impact creation is not considered at all and the only concern is financial return. On the other hand, the subsequent approaches are all designed to have impact in varying degrees.

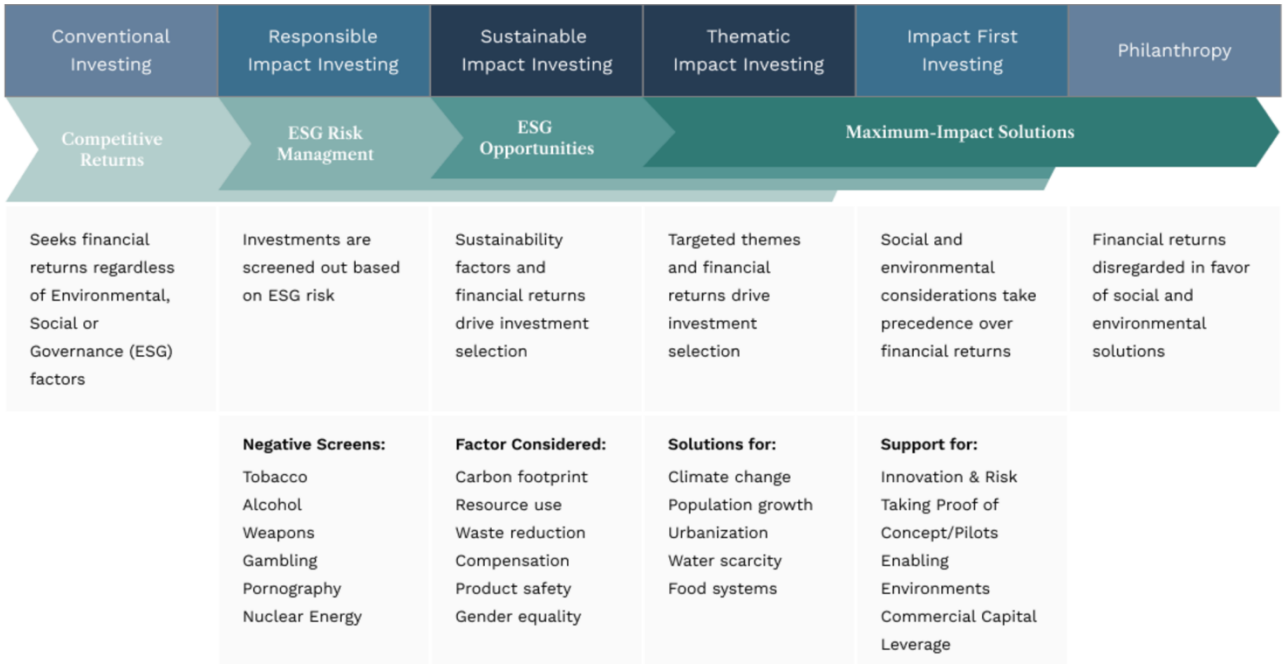


Figure 9.2 A range of investor objectives from conventional to philanthropy

The financial system's role is to allocate capital toward long-term value creation. In well-functioning markets, private financial value converges with social value. In reality, externalities, i.e. unpriced social and environmental costs or benefits, break that alignment, distort investment signals, and increase systemic risk. These misalignments can lead to capital misallocation, underinvestment in public goods, and overexposure to systemic risks that threaten financial stability. Climate risk illustrates the point: Firms that offload the social

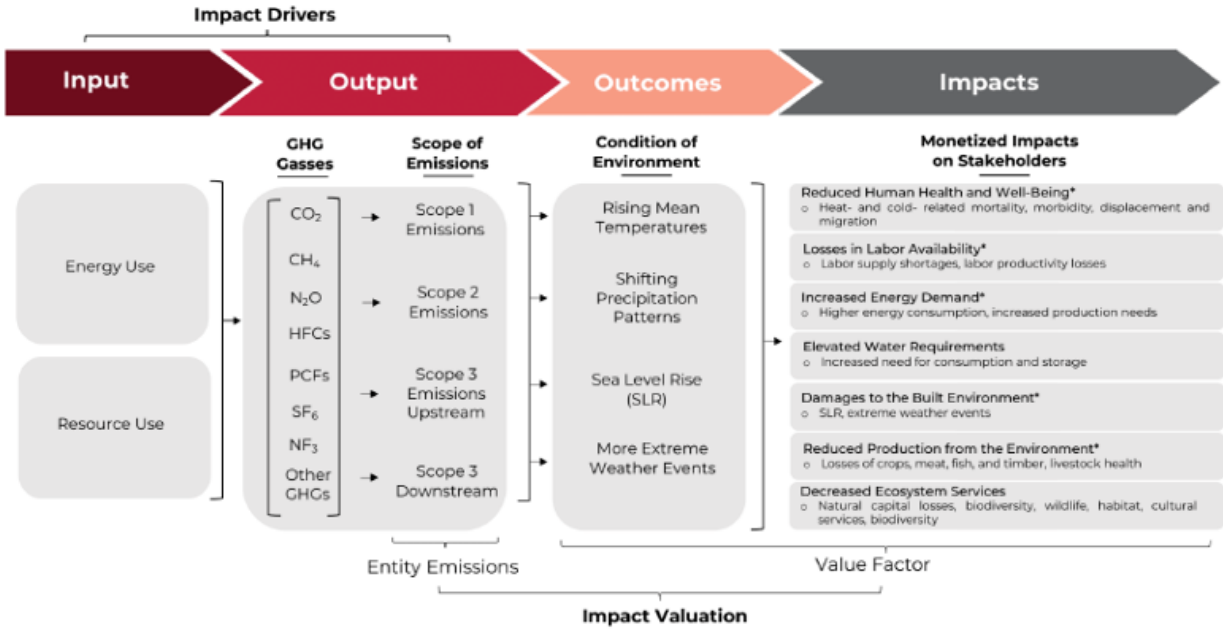
[38] https://sonencapital.com/impact/our-approach-to-impact-investing/ & info@sonencapital.com

cost of carbon may appear more profitable today but embed transition and physical risks that can impair asset values and cash flows[39].

It is necessary to be able to measure how impact affects the value of a company. Impact valuation translates externalities (such as impact on climate, water quality etc.) into monetary inputs for finance, such as cash-flow adjustments, hurdle rates, and risk premia, enabling more accurate capital allocation. This is particularly critical in transition finance, which channels investment into decarbonisation and other transformations in hard-to-abate sectors. The *OECD* emphasises that credible transition finance depends on robust corporate transition plans, which must include quantified impacts, science-based milestones, and governance mechanisms to ensure accountability and transparency.

Impact valuation monetises the positive and negative effects of business activities, such as GHG emissions, air pollution, water use, health outcomes, and knowledge spillovers, so they become commensurable with financial metrics. This enables capital budgeting to adjust hurdle rates and investment appraisals, corporate steering to inform KPIs and incentive structures, improve disclosure to enhance transparency and comparability, and to define mechanisms to value intangibles and their impact on share price ratios.

Impact valuation methods increasingly seek to correct distorted financial signals caused by unpriced social and environmental externalities. These externalities, such as pollution, resource depletion, or health impacts, are often excluded from conventional financial analysis, leading to misallocated capital and overstated profitability. By internalising externalities through taxes, shadow prices, standards, or monetised metrics, firms and investors can reflect true societal costs and benefits in financial decision-making. These approaches support impact-adjusted P&L and balance sheets, an integration into enterprise risk management and governance, an alignment with SDG-linked performance metrics and transition planning and scenario analysis.



Sonen Capital defines Impact investing as the full range of investment opportunities which address an investor's desire to realize financial return while achieving meaningful and measurable positive social and/or environmental impact. The **Impact Investing Spectrum** in the graphic above, after Conventional Investing' shows different categories of Impact Investing

Responsible Impact Investing avoids negative impacts. It was once the main policy of fund managers who to avoid investing in 'bad' things such as tobacco, oil & gas, weapons manufacture etc. This approach was often driven by the underlying asset owner (a pension fund, an endowment etc.) stating that this was a requirement of the mandate. Note that 'nuclear energy' is a 'negative screen' in this diagram. However, many managers are now reversing that exclusion and instead including nuclear. Weapons manufacture may also start to be included rather than excluded.

[39] This paragraph, the three following ones and the graphic are from a draft whitepaper by Martina Macpherson, Thomas Wulff, Andreas Gintschel and Christian Heller entitled *Impact Valuation, Externalities and Transition Finance: Demonstrating financial Relevance.* September 2025

Sustainable Impact Investing involves the integration of ESG considerations into investment decisions. In the past the data available for doing this was very limited but increasingly it is becoming available. Factors that would be considered include the impact of a company on climate change risk and the wellbeing of the employees of the company.

Thematic Impact Investing considers what main activity the company is involved in rather than just how they ensure that their investments have a sustainable impact. Here we are concerned only with themes with a [positive] social or environmental impact.

Impact First Investing considers the impact of an investment to be more important than the ROI of the investment. As with thematic investments, there will be a clear connection to a particular E or S impact theme.

Philanthropy generally involves providing a grant to a project. There is no expectation of any financial return. The only consideration is impact. But many private philanthropists are willing to give money to projects they support.

Once environmental and social issues are considered as factors in investment decisions, all approaches apart from 'classic' involve *avoiding harm*. When we move on to 'sustainable' and 'thematic' then we have an approach that *considers all stakeholders* (investors, suppliers, customers, employees, the local area and its residents) and tries to benefit them or at worst, not disadvantage them in some way. With 'impact first' investment the focus is on contributing to 'solutions' to particular issues. Two issues on which an *impact first* investor or fund might focus would include the energy transition and ensuring fresh water supplies.

*9.*4 Case Studies: InfraRed Capital Partners, Gore Street Energy Storage Fund, Gresham House, Schroders Greencoat UK Wind and NextEnergy Capital

Infrastructure Equity

Most clean energy projects start life as unlisted assets (project finance company). Institutional investors can access such unlisted projects either directly or subsequently through funds which can also be listed. We focus on unlisted investments. Private market assets may have different characteristics to both listed infrastructure assets (such as utilities) and the companies which underlie indices such as the S&P 500 or the FTSE 100. Particularly relevant is that their liquidity is lower than that of listed assets and thus there is no "market price" as there will normally be with listed equity. In theory they should offer a higher expected return for a given level of risk to compensate for their illiquidity but in practice may not.

InfraRed Capital Partners

We have covered many aspects of financing clean energy infrastructure in the previous chapters. We have also covered Project Finance (infrastructure bonds) and financing companies at a development stage prior to TRL9. We now have a case study of a large infrastructure manager which, amongst other private (unlisted) investment funds for institutional investors, manages a large London Stock Exchange Listed Clean Energy Infrastructure Fund (**TRIG**) as well as a general infrastructure fund (HICL).

The company was established in 1990 when there was little clean energy infrastructure available and still manages such assets. But today it is an international infrastructure investment manager, with more than 190 professionals operating worldwide from offices in London, New York, Sydney and Seoul. InfraRed manages US$14bn+ of **equity capital** for investors around the globe, in **listed and private funds** across both income and capital gain strategies. It has invested in over 300 infrastructure assets across 18 countries and realised more than 90 investments.

The Renewable Infrastructure Group (TRIG):
This fund, managed by InfraRed Capital Partners is FTSE-250, London-listed UK investment company. It was one of the first renewable energy investment companies to be listed on the London Stock Exchange (LSE).

It is important to understand the difference between this listed fund and 'private market funds' as both types of fund are managed by InfraRed. Private market funds are for investment on a wholesale scale and secondary market sales of participations involve bilateral agreements between buyer and seller as there is no market for pricing such share participations. On the other hand, TRIG's shares are priced daily on the LSE. Private market funds should generate an "illiquidity premium" over listed shares which is compensation for the illiquidity. However, depending on demand and supply balance, there may or may not be such a premium and indeed a quick sale might require a discounted price and hence an illiquidity discount.

Trig's portfolio is comprised of established renewable technologies. 48% of the portfolio is onshore wind, 34% is offshore wind, 13% is solar and 5% is flexible capacity. The factsheet above gives more information on TRIG. TRIG's purpose is to generate sustainable returns for its shareholders from a diversified portfolio of renewables infrastructure that contributes towards a net zero carbon future. TRIG's highly diversified portfolio consists of onshore and offshore wind farms, solar parks and flexible capacity in the UK and across Europe. These assets generate revenues from the sale of electricity and from government-backed green subsidies and benefits. TRIG has a differentiated approach to portfolio construction, assembling a mix of assets that contribute to the decarbonisation and security of energy in Europe whilst aiming to provide investors with long-term, income-based returns with a positive correlation to inflation.

We noted above that TRIG had 5% of its assets in 'flexible generation'. This type of generation is an absolute requirement in systems transitioning to renewables as it can 'fill in the gaps' in renewable generation power output. It can be provided by energy storage or by natural gas turbines or, in the case below, storage as an instant source of electricity and natural gas generation afterwards. An example of this type of company that was in its portfolio (but has now been sold to realise a capital gain) is Statera Energy. For more information on flexible production, see Chapter 10.2.)

Gore Street Capital & its Energy Storage Fund

Just as Infrared Partners is the investment manager for The Renewable Infrastructure Group (TRIG), so Gore Street Capital (GSC) is the manager for Gore Street Energy Storage Fund. Gore Street Capital was one of the first asset managers in energy storage and is the only UK-listed energy storage fund with an internationally diversified portfolio. Since the inception of Gore Street in 2015, it has increased its assets under management at the time of writing to 1.7 GW spread across five grid systems, including in Ireland where the Company manages assets for the Ireland Strategic Investment Fund (ISIF).

The company's internationalisation seems to have helped it weather falling UK revenues in 2023 when storage capacity on the UK grid almost tripled and prices in a number of market sectors fell. In a similar way to diversification policy in equity or debt securities as a means of improving the risk/return profile of a portfolio, Gore Street has taken a very active approach to diversifying sources of revenue over different revenue streams within a country's market structure and through revenue streams from a range of countries.

In terms of geographical diversification, it has storage assets in the UK, Ireland, Germany, Texas and California. It is this approach to asset management (physical assets) that has enabled it to differentiate its performance from many competitors. Its target portfolio includes Decarbonisation, Digital inclusion, Regeneration, Health & Education, Resource efficiency, Waste solutions.

Gresham House – specialist investment

Gresham House is a specialist alternative asset manager providing funds, direct investments and tailored investment solutions, including co-investment across a range of differentiated alternative investment strategies. They note that significant investment is needed in new Sustainable Infrastructure that can support and accelerate the changes and innovative solutions required to address key environmental and societal challenges. They identified six subsectors where real asset based solutions, and their 'modern' approach to infrastructure, have the best opportunity to create the biggest impact - and therefore the biggest financial reward.

Investing in Sustainable Infrastructure with Gresham House offers appealing characteristics to buy-and-hold investors and provides the potential for stable, long-term returns. They note that Sustainable Infrastructure investments can (meaning may or may not) benefit from:

- Downside protection through substantial asset-backing
- The potential for capital growth
- Attractive long-term cash flow yields
- Material positive social and environmental impact
- Returns with a low correlation to the broader economy or financial markets

Schroders Greencoat UK Wind

Schroders Greencoat LLP is a specialist manager dedicated to the renewable energy infrastructure sector. It has offices in London, Dublin, Germany, Netherlands and Chicago and approximately GBP 10 billion under management, making it one of the largest dedicated managers in Europe. It was founded in 2009 and currently has fund mandates with strategies investing into bioenergy, renewable heat, solar and wind energy infrastructure in the UK, Europe and the United States.

Greencoat UK Wind invests in operating UK wind farms. Given the nature of the Company's income streams, the board has increased and intends to continue to increase the dividend in line with retail price index (RPI) inflation. The Company also aims to preserve capital on a real basis by reinvesting excess cashflow in additional operating UK wind farms, and through prudent use of portfolio leverage.

The Company offers exposure to UK wind generation through a 'premium' listed vehicle. Greencoat UK Wind is invested solely in operating onshore and offshore UK wind farms which are currently producing income. Greencoat UK Wind was the first renewable infrastructure fund to list on the LSE main market and is the only infrastructure fund or renewable infrastructure fund domiciled in the UK. The Company is a constituent of the FTSE-250 index. Greencoat UK Wind is incorporated in England and Wales and is a UK **Investment Trust.** The Company has a broad investor base that includes numerous blue-chip institutional investors and also retail investors.

NextEnergy Capital

This fund manager in total has $4bn assets under management (AUM). It has one listed fund and five institutional funds (unlisted). In total it holds 400 assets with a total capacity of over 3GW and distributed over 9 countries.

NextEnergy Solar Fund Limited (NESF) **Listed on the London Stock Exchange**

This is a specialist solar and energy storage climate impact fund, which is listed on the premium segment of the London Stock Exchange. It currently has an installed capacity of over 983MW spread among 102 individual operating assets in the UK and Italy, with an asset value of over £1bn. NESF is one of the largest listed solar and energy storage investment companies in the world.

NextPower **Private ESG Funds**

NextPower V ESG is Next Energy Capital's fifth investment vehicle focused on the solar sector. It is targeting mid to double-digit returns. But beyond the financial returns it generates, it expects to contribute to the decarbonisation of the power generation sector, reducing electricity prices and increasing energy security. It restricts itself to OECD country solar assets and energy storage expertise. It is a 10-year closed-ended vehicle that qualifies as an Article 9 Fund under the EU SFDR.

NextPower UK ESG is a private unlevered fund investing in greenfield subsidy-free solar projects, with PPA's, in the UK. It is a 10-year closed-ended private fund launched in December 2021 and has raised total commitments of £733m. Interestingly, the UK Infrastructure Bank provided cornerstone match-funding of £250m. The fund is classified as Art 9. Under EU SFDR.

Part III: The Energy Transition Challenge: Converting Variable Renewable Energy into 24/7 Dispatchable Power, Grid versus Decentralised Systems

Chapter 10: What is the Energy Transition and what are its Technologies?

In this, and subsequent, chapters we move on to consider the actual issues in the energy system and in particular in the electricity system that require the very large volume of funding we referred to in an

earlier chapter. Without some understanding of the existing system and its characteristics and the structure of the New Energy Economy and its quite different characteristics, it will be difficult for debt and equity financiers to undertake due diligence on projects which are offered to them. Equally, for entrepreneurs, intrapreneurs, corporates, utilities and others, without some understanding of the financing side and how due diligence is undertaken which we covered earlier, it is less likely that they will convince financiers to provide the necessary capital for their project at the lowest possible cost of capital. Following this chapter, we cover the individual technologies and their techno-economic features.

10.1 A Key Issue – the loss of 'Firm Power': integrating variable renewable energy (VRE) into the electricity **system,** 'nameplate power', capacity factors, LCoE, and the energy trilemma

The current energy transition (a key component of the 6th wave of innovation in the graphic in Chapter 2.4) has been ongoing for some decades, in fact since 1990 and will hopefully have achieved society's objectives for mitigating further climate change by 2050. It must be remembered that in 1990 (some 35 years ago), the world did not have solar power or wind turbines that could be used at scale to generate power. The technologies had to be developed. The speed of development has been impressive relative to earlier transitions. Nonetheless, the current transition is going so much more slowly than is required to even come 'within the last mile' of the Paris Agreement objectives. It is for this reason that a massive step-up in investment in electricity systems and other clean energy technologies requires a massive increase in the funds allocated by investors to this task But as we have noted before, this will only be achieved either by a massive increase in regulation which forces companies to replace their fossil fuel assets or through much higher and broader carbon taxes or by governments being willing to take on many of the risks to which investors would be subject if singlehandedly they speeded-up their investment in carbon free assets.

In the absence of fossil fuels to provide energy for heating, cooking and lighting, transport and industry (as fossil fuels are planned to be phased out by 2050, there is a need to find alternative means of generating electric power and producing heat, enabling transportation and continuing to have an industrial base but without using coal, natural gas or oil. Electrification will have a major role, for example electric transport (cars, trucks, rail) , electric heat using heat pumps for space heating and electric industrial processes. In addition it is likely that hydrogen will be used across all sectors though currently its price is very high relative to the price of fossil fuels. Hydrogen will not only be used directly but will, to a limited extent, also be used to manufacture synthetic liquid hydrocarbon fuels for use in transport.

Hydro, solar, wind, nuclear, geothermal and biomass as sources of primary energy have to be converted to energy carriers as shown in the graphic in Chapter 1. The first, and main, such carrier is electric power. Clean hydrogen can then be manufactured from a proportion of that power along with water and a proportion of that hydrogen can then converted to hydrogen derivatives including synthetic methane, methanol, ammonia and synthetic liquid hydrocarbon fuels.

In the past, the whole power system was based on using heat energy generated from fossil fuels to produce steam to drive a steam turbine which then drove an electricity generator. Such systems provided continuous 24/7 generation of power (dispatchable power) but solar and wind which will eventually make up a high proportion of power generation, are not continuous but intermittent i.e. variable renewable energy (VRE). Today the only large sources of non-fossil continuous power generation are geothermal, hydro and nuclear.

In more detail, generating companies provide electric power with one of two characteristics:

> **firm power** (dispatchable, round-the-clock, 24/7 supply) from: fossil fuel heat sources (coal, oil, natural gas,) connected to steam turbine generator sets, flow of the river hydropower, nuclear, concentrating solar power (CSP) and geothermal heat sources, or;
>
> **variable renewable energy** (VRE): Wind turbine, solar photovoltaic (PV), concentrating solar power (CSP) which are non-dispatchable without energy storage and/or Demand Side Response (DSR). These sources of power are not constant and indeed in the case of solar, clearly produce no power after sundown.

The key question in today's power system design is how to convert VRE into firm power. This is necessary if power systems are to be able to "keep the lights on" at all times. As all countries and regions need to take steps to mitigate climate change risk which must involve changes in their energy mix and sources of power, they must, at the same time, **ensure security of the energy supply**.

The term "ensure energy security" is a key one but can refer to two different causes of energy 'insecurity':

> *First,* it means **minimising the probability of 'the lights going out' (brownouts or blackouts) by ensuring that countries have adequate 'firm power sources' i.e. 'dispatchable' power/ with 24/7 availability**.
>
> The energy transition requires the elimination of the use of petrol, natural gas, diesel internal combustion Engines (ICEs) and of course coal, all of which have provided dispatchable power for many decades (the old energy system now being phased out). In addition to providing power, the turbo-generator sets associated with generation using steam to drive a steam turbine which then drives a power generator (turbogenerator) have also provided a key service to the grid by having 'inertia' i.e. the weight of the turbogenerator enables it to continue to turn and provide what are known as 'frequency response services' to the grid. This is required to avoid damage to the power network when power generation on the network fluctuates over short time periods. In the UK, in one period when too much solar and wind (which do not provide inertia) were generating too much power, the National Grid still had to request gas turbine generators to start up to provide inertia. National Grid then had to pay a high price to encourage European countries to absorb our surplus power.
>
> *Second,* in Europe for many years it has meant avoiding the risk of Russia (or other countries) turning off gas supplies to Europe or the Middle East cutting oil supplies (although many politicians chose to neglect these risks). One of these risks was realised after Russia invaded Ukraine and in March 2022. The European Commission then announced that countries in Europe would cease buying oil and gas from Russia as soon as possible. We are not covering the issues involved in coping with the second type of energy security but only with the first.

VRE sources can, however, provide dispatchable power if linked to an energy-storage facility co-located with generation or located separately on the grid. At present storage makes economic sense only for relatively short periods – up to 4 or sometimes 8 hours. But long-duration energy storage (LDES) often described as making power available for 8 hours or more but also defined as being available for at least 100 hours, is likely also to be required.

Even in sunny locations, the power generated by solar may also be much lower than the stated maximum power output of the solar panels for quite long periods. As a result, without longer duration storage, power users may have 'brown outs' or 'blackouts'. Indeed, solar produces only around 25% of its stated maximum power over a 12 month period. The fact that solar panels do not generate any power when it is dark, is well known. But they also generate less than their stated power for much of the time. Their stated (nameplate) power is also called peak power (MWp). This is the theoretical maximum production of the panel on a clear sunny day with the panel at a right-angle to the sun. In addition, the stated nameplate power in the northern hemisphere is the power that will be generated at latitude 35° (35 degrees) North (N) in summer. No part of Europe falls into this category with Morocco being the closest to Europe that falls within this definition. In addition, the temperature of the panels which heat up under sunshine has a considerable effect on output. The peak power will only be generated at a temperature of 25 °C or below. Thus even global warming will affect output. It is is for these reasons that the actual output of solar panels known as their capacity factor is much lower than MWp and is around 25% as shown in the graphic below.

System Costs – the true cost of different power sources: the energy trilemma

What the above makes clear is that there are substantial issues with intermittent power as a result of costs which grow disproportionately in a non-linear way as the proportion of VRE sources of power increases. This could be partly ameliorated by reducing the historic standards of electricity reliability and on-demand availability by accepting the economic, socio-industrial and environmental consequences of more frequent optional interruptions and non-optional brownouts or blackouts. The high capital cost for solar and wind arises from the fact that their low capacity factors means that the average fraction of time for which they are generating at their rated full output stated on their 'nameplate' can be as low as ~25%. This is known as the **capacity factor** and the low capacity factor of VRE (wind and solar) means that for a stated maximum MW power output, much more VRE plant investment is required to achieve the required megawatt hours during a year. Thus if the capacity factor of

solar is ~25% as shown in Figure 10.1[40], then four times the amount of plant must be built to achieve that stated amount of output *and even then, only when there is adequate sunshine.* In addition, that output is often not produced at a time when demand for power is at its highest.

A low-capacity factor also means more land must be used to achieve a given true capacity which in Europe is an issue. In the case of offshore wind, the capacity factor is likely to be higher than shown below and may, indeed, reach 50%. However, on those days or weeks when there is little wind even offshore right

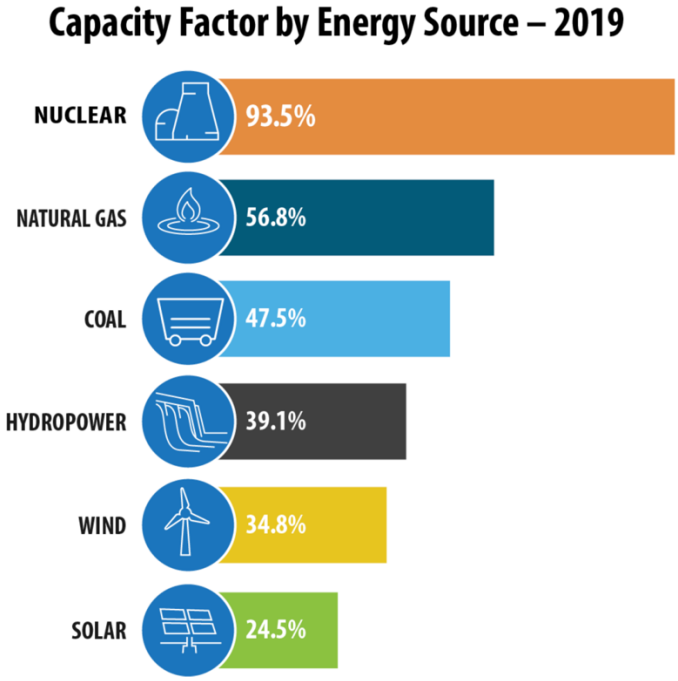


Figure 10.1 Capacity factor by energy source

across the European mainland, any electricity system has a severe problem if it is too reliant on wind. Indeed, Mads Nipper, a previous CEO of Ørsted which is one of the largest North Sea wind generators, said that April, May and June 2021 "had been the 3rd worst quarter in the last 88 quarters. It was an extraordinarily poor wind quarter. It's very serious. It is like being a farmer and it doesn't rain." Lost earnings were said to have amounted to $220m.

An example of the difference that location can make is shown by statistics on the location of an e-fuel plant in Chile. It has been reported that in Chile, a wind turbine runs at full load for an average of 270 days a year whereas in Germany it runs at full load on 66 days a year due to the geographical and meteorological conditions. At 74 percent, the utilization rate of the wind turbine in Chile is therefore four times higher than in Germany at 18%.

The Energy Trilemma[41] shown in Figure 10.2 is a graphical representation of the need for decision makers to balance different features of a power generation system. As with most things in life, there is a need to optimise in determining an energy and electric power policy. The interconnectedness of the three goals shown in the graphic below, underscores the relationship between economic growth, social equity, and environmental stewardship, including tackling energy poverty.

[40] https://www.energy.gov/ne/articles/infographic-capacity-factor-energy-source-2019
[41] Source: World Energy Council Trilemma https://www.worldenergy.org/assets/downloads/Trilemma-what-is-the-energy-trilemma.jpg

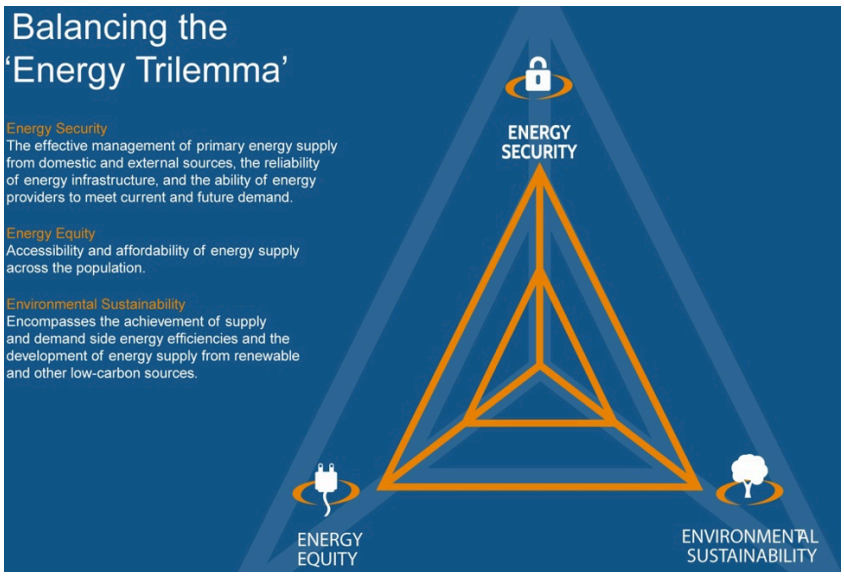


Figure 10.2 Balancing the 'Energy Trilemma'

System cost is the additional amount that must be added to the 'farm gate price' of any form of power generation to find the full cost of that source. Such additional costs include long-distance transmission costs, local distribution costs and the cost of 'balancing the grid' as covered below. In the case of a high proportion of wind and solar on a grid system, balancing costs can be high. Transmission costs may be high since the source of generation is likely to be distant from industrial demand locations. Large amounts of battery energy storage will be required, the costs of which are substantial. There is also a need for the grid operator to pay considerable sums of money to wind and solar sources, known as 'constraint payments' when there is potentially too much power flowing into the grid and wind and solar sources are paid to switch off (when it is sunny and windy). The national grid was only ever intended for fossil fuel power stations that are situated near urban centres, not remote offshore wind farms.

As an example in the UK, on a windy day in June 2025, one wind farm (Ocean) in Scotland received £72,000 to switch off for 30 minutes yet because of the lack of transmission capacity (a common problem in many countries now) a natural gas plant in the south of England was paid £43,000 to produce more power. £65 million was paid out by the grid operator last year to Seagreen which is Scotland's largest wind farm for being asked to restrict output for more than half of the time. Total payments to renewable generators to stop generating when there is too much wind and solar and to natural gas generators to 'switch on' when there is too little wind and solar amounted to over £1bn in 2024 and this additional cost is expected to be considerably higher in 2025.

While the cost of electricity from each generating source is frequently evaluated using a measure known as 'Levelised Cost of Energy' or Levelised Cost of Electricity' (LCOE) which we call the farm gate price, a much more appropriate measure is Levelised Full System Costs of Electricity (LFSCOE). This would show that, for example, nuclear as a source of carbon-free power which can be located near customers (in particular so-called Generation IV small modular reactors covered in Chapter 14) and therefore requires less transmission capacity and little or no battery energy storage, may in fact have a full system cost slightly lower or slightly higher than natural gas. In addition, Generation IV nuclear can be 'embedded in industrial facilities or within cities thereby cutting down both transmission and distribution costs substantially and eliminating government taxes on electricity

10.2 The 'Holy Grail' of 24/7 Fossil-Free Electric Power

In the previous chapter we covered the convertibility of one type of energy to another – mechanical, electrical, thermal. In addition we have potential energy sources including chemical, nuclear and gravitational which are also relevant to power systems. There are two main problems with achieving the 'holy grail' – i.e. the need to overcome insufficient power generation at some times of the day, month or season and excess power generation at other times of day. This is the **'balancing problem**'– how to balance supply and demand of a commodity (power) which has traditionally been thought of as not being capable of being 'stored' on a power grid system except in small amounts using traditional lead-acid batteries as used in fossil-fuelled road vehicles. The balancing problem

has always been with us it's just that if we have a high proportion of renewable energy, the problem becomes much more severe.

Solar energy is the most abundant energy resource on earth – nearly 200,000 terawatts of solar energy strikes the Earth continuously. That's more than 10,000 times the world's total energy use. Offshore wind has a resource potential of some thousands of Gigawatts. Thus there is a huge amount of energy available on the Earth without the need to use fossil fuels. However, neither the generation profile of wind nor of solar matches the profile of demand. One partial solution to this problem can be achieved by combining wind energy (with often has an evening peak) and solar energy (which usually has a daytime peak), to create a 'hybrid' wind-solar complex which can combine to better match daily demand. It might also require battery storage and indeed many projects today are now combining solar, wind and storage as the cost of storage has declined.

Many people would argue that even solar, wind and storage does not provide the **optimal trade-off** between energy security (a guaranteed continuous supply regardless of the weather), cost (the cost of a purely renewable solution has rising marginal costs) and environmental sustainability. They would argue that an electricity **system** needs to have a proportion of dispatchable power from nuclear and geothermal to achieve the best trade-offs.

Balancing the Grid: Dispatchable Power and Flexibility

Managing the power transmission system in a country or region is necessary because electric power cannot be stored in the same way as a water or gas supply. Both water and gas can be easily stored in 'containers' such as reservoirs for water or caverns for gas. Without some means of 'storing' power, if demand rises unexpectedly at short notice a country or region will suffer a 'brown-out' or even a 'blackout' i.e. we will not achieve the energy security objective in the trilemma. Sudden absence of power cannot be prevented unless an additional supply of power from dispatchable generation or so-called balancing technologies such as energy storage is made available immediately or if some users agree to have their power switched off (demand response).

Equally, if too much power is generated relative to demand, this can overload and damage the grid and consequently the Transmission System Operator (TSO) will need to request that some producers curtail their production in return for a pre-agreed **curtailment** payment. As a result of this need to balance supply and demand, a country or region must manage the system on a second-by-second minute-by-minute, hour-by-hour, day-by-day, month by month basis. This management process involves forecasting both demand and supply at each future time period starting with "day ahead". **With increased wind and solar power feeding into the network, there is also an ever-greater need for accurate forecasts of expected available supply of energy from sunlight and wind.** Equally, the TSO will be trying to estimate demand at each future moment in time based on weather conditions (need for home heating or air conditioning), whether an important sports fixture is going to be on television at a certain time, the likely average time that people will go to bed which may be later at weekends. The need to continually **balance the grid** existed long before renewable energy became an important source of power – it's just, as we noted above, a much larger problem today.

The issue with ever increasing proportions of VRE is how to provide the same ability as fossil to balance the grid when wind and solar are the main source of power but are not dispatchable sources. This is becoming an ever more critical factor as the proportion of VRE (solar and wind) rises.

An FT report on 13th September 2021 noted that the UK had seen the worst (low) wind conditions in the North Sea for more than 20 years. Most of Europe suffered from the same phenomenon. Unfortunately the stated capacity factor for wind (which is much higher than for solar) is an average over a whole year and thus tells only part of the story and not the most important part. More important is the issue of wind droughts which can last for a week or a month even if the average capacity factor of a windfarm over a period of 12 months is high. Wind speeds can also fall dramatically for longish periods. For example, while wind accounted for 18% of the UKs power in 2020, by September 2021, it was only 2%.

To date, gas generators have been 'fired-up' during wind droughts to enable the grid to be balanced but in future, the other solutions which we cover below will hopefully ensure that fossil fuels do not have to be used at such times. In addition to the need to balance supply and demand there has always been a need for so-called ancillary services that are frequently needed by the grid operator to enable the smooth transmission of power. These services include frequency control and voltage stabilisation.

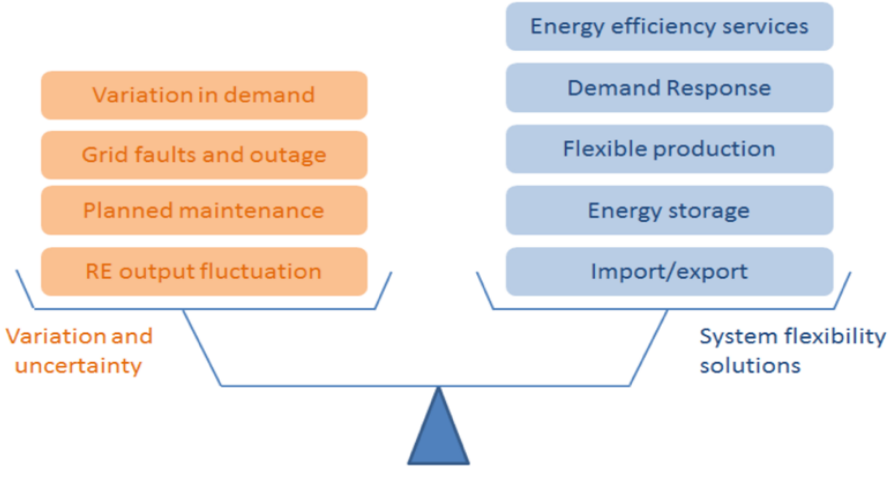
The first solution to the variability of renewables is to have an adequate supply of clean **dispatchable power** perhaps of the order of magnitude of 20% or more of total demand. In the graphic below, it can be seen that storage which is one suggested means of converting variable power to 24/7 power is only providing 1% of power demand in the UK which is similar to most countries. In many countries most storage is **pumped hydro** which is stored potential energy in a reservoir at a height above sea level. For reasons of diversification of technologies, having a substantial proportion of nuclear in a country may be an economic means of ensuring that 'the lights remain on'. However, even nuclear can go off-line as happened in France on a large scale in 2022 due to the need for maintenance of, and repairs to, a high proportion of the nuclear fleet.

There are also limits on both wind and solar quite apart from the high capital cost. The first reason for this in the case of wind is a shortage of suitable land and sea-bed areas for wind and also planning issues. In the case of solar, not only are planning issues a cause for concern but also low solar irradiance in Europe measured through both summer and winter. But a further reason for the limits to solar and wind a sources of electricity supply is that as the proportion of variable renewable energy exceeds about 70% of total power generation, the cost of achieving 100% carbon free energy using renewables rises rapidly. It is widely believed that the cost of using nuclear which is a dispatchable power source with a **high capacity factor** is likely to be on a par with, or lower, than the alternative VRE route as the proportion of VRE on a grid increases towards 100%.

The high energy density of nuclear means that it can take up less land per unit of energy output than existing fossil plants or wind or solar. For example a 1000 MWe (one Gigawatt) nuclear plant could be built on a factory-sized plot whereas the equivalent solar plant would require 50-100 sq. miles and a wind plant 200-300 sq. miles. Such large acreages of land (or seabed) in Western Europe as would be required for high proportions of solar and wind would be unlikely to be available. In addition, due to its exceptional energy density, nuclear has an extremely low ecological footprint to build, to run, to fuel and to decommission. This applies in particular to Generation IV reactors (which we cover shortly) and to the plant itself, its fuel/material mining and final waste streams. In contrast the manufacture, transport and combustion of methane gas used by a gas turbine all create considerable emissions of methane from accidental leaks and other releases in addition to CO2. In addition, methane is 20 – 30 times more potent as a greenhouse gas than CO2 so 'fugitive' emissions (leaks) do matter. At present gas turbines powered by methane gas (natural gas) are the main alternative to nuclear for providing the necessary dispatchable (firm) power with Li-ion and other storage mechanisms being an alternative for relatively short periods.

While nuclear clearly raises issues of disposal of waste fuel, even windfarms raise environmental issues when the turbine blades start to fail, perhaps within 20 years, and have to be buried and left to decay while still ensuring that the plasticisers and other chemicals used in their construction do not contaminate the water table.

Imports of power can change the percentage of power generated by natural gas. This is enabled by inter-connectors which are power cables connecting countries or regions. However, if the whole of Europe is suffering a wind drought, it is unlikely that any one country can supply any other with power. The conclusion we can draw from the above is that much more so than a power system using mostly firm power sources (as was the case when most sources were based on fossil fuels), power systems based on VRE require 'flexibility' which is the capability of a power system to cope with the variability and uncertainty of solar and wind and thereby avoiding curtailment of power and reliably supplying all customer energy demand.



Challenges to system balancing and options to increase flexibility in the electricity system

Figure 10.3 Mechanisms to balance supply and demand in an electricity system

Figure 10.3 [42] (above) shows some of the '**system flexibility solutions'** are applicable to fossil sources of power but all are relevant to VRE. On the LHS it shows some of the factors that can result in the need to balance the grid. The first, **variation in demand**, is an obvious one. On the left hand side we have a list of causes of imbalance in the system. On the right hand side, some ways of ameliorating such shortages namely system flexibility solutions.

System Flexibility Solutions

Energy Efficiency Services.

There is no doubt that if total demand can be reduced, for example through replacing incandescent filament lighting in offices and homes with LED's, a lot of power will be saved. But all that does is to reduce the total need for generation capacity but does little for balancing a grid which has a high proportion of VRE.

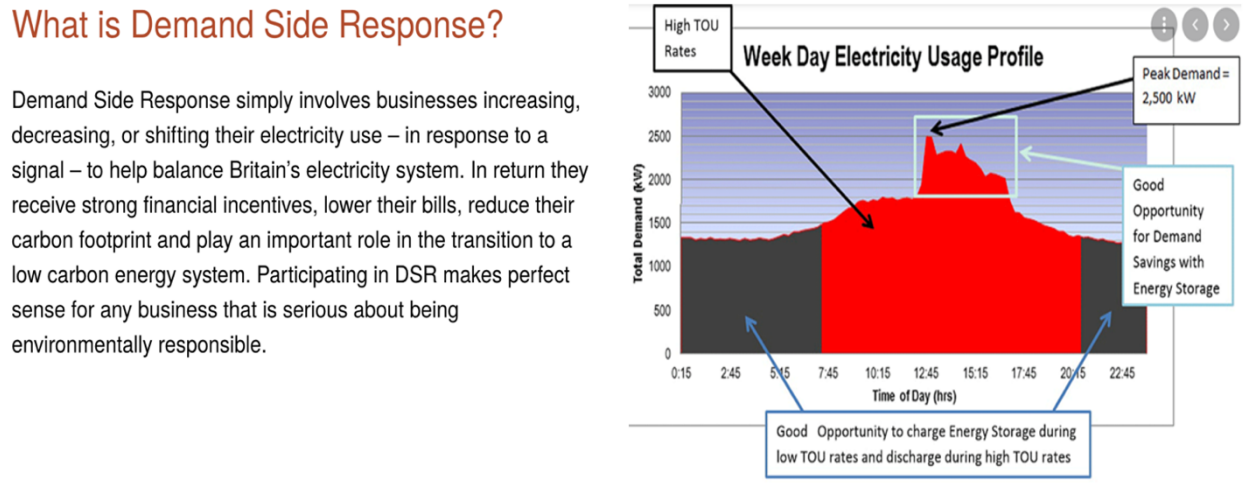


Figure 10.4 Demand-side response and weekday demand profile over 24 hours. Also known as Time of Day/Time of use (TOU) pricing.

On the left hand side of Figure 10.4 is a generic graph [43]of power demand against time of day. On the right hand side is an explanation[44] of DSR from the UK National Grid. Traditionally the peak load is the most expensive to provide since 'peakers' which use open cycle gas turbine generators are only operating for a short time each day and are paid a higher price per MW than baseload generators. In this (US) graphic, peak demand arises in summer from air conditioning. To enable the power distribution system operator (DSO) to supply air conditioning demand *without additional generating capacity*, it is necessary either to reduce other demand during peak hours or to use energy storage or ideally both. The objective is to 'shave the peak'. Increasingly peak demand can be met by battery storage rather than from natural gas.

Time-of-use (TOU) pricing can make the profile of demand flatter. However, some dispatchable peaking power may still be required. The benefit to the network of such time shifting of demand is to reduce the total power generation assets required on the system and thus to cut the capital costs of providing round the clock (RTC) power including the high demand for the peak times of day. It can be seen in the graphic that there is an arbitrage available for owners of storage systems. They can buy power before 7.45am cheaply, store it (see storage below) and sell it at peak times at a higher price as if they were a dispatchable generating source In this example, power is being stored during low demand periods (power prices are low) and supplied at peak times (prices are high).

Demand response is designed to encourage customers to shift demand from periods of potential excess demand on the grid to times when there is potentially a surplus of supply. This can be achieved either through TOU pricing or through monetary incentives to allow the DSO or TSO to disconnect certain power demands automatically. For

[42] European Parliament Energy Storage: *Which Market Designs and Regulatory Incentives Are Needed*? https://www.europarl.europa.eu/RegData/etudes/STUD/2015/563469/IPOL_STU(2015)563469_EN.pdf

[43] https://www.nrel.gov/state-local-tribal/blog/posts/when-does-energy-storage-make-sense-it-depends.html

[44] https://www.nationalgrideso.com/industry-information/balancing-services/power-responsive/demand-side-response-dsr#:~:text=Demand%20side%20response%20(DSR)%20is,shift%20demand%20in%20real%2Dtime.

example, some heating might be cut or car charging delayed. Some large technology companies with high demand for power may avoid undertaking non-urgent tasks during peak demand periods. However, TOU pricing does not increase the supply of power and may cause inconvenience and even raise production costs in industry.

When coupled into a power grid, the new technologies which use power or store power can also become components of the power system. For example, electric vehicles (EVs), electric boilers, heat pumps and electrolysers for hydrogen production can provide flexibility to the power system either by adjusting their demand profile based on price signals (using market forces) or by making any integrated storage (in particular electric vehicles) a source of energy storage for the power system. By using these resources on the grid, it might be possible to decouple the timing of demand for a proportion of final energy from the total electricity demand.
The approach of the International Renewable Energy Agency (IRENA) is to incorporate demand-side flexibility, grid reinforcements, storage and sector coupling as additional flexibility sources and potential game changers in the action needed to increase the proportion of non-fossil energy in the power system. They note that making a system flexible is likely to require two sources of flexibility – one is coming from the generation side (supply) and the other is coming from the usage side (demand). In addition, the two sides need to be digitally connected (sector coupling) to enable control of power users' demand by the supply side.

Demand side response (DSR) also known as load shifting is essentially the process of moving electricity consumption **from one time period to another**. For example, postponing an industrial process to another time. The idea is that by shifting the load to another time, the returns generated through energy cost savings or DSR participation are greater than the loss of production. Unlike many energy cost saving strategies, load shifting tackles the "when" rather than the "how much" conundrum.

is shown in relation to heating, hydrogen production and electric vehicles. The first two are demand side users who can be switched off for short periods at peak times while the third (EVs), can be requested (through an automatic AI system) to either switch on charging (when excess power is available) or to switch off at peak times.

Flexible Production (supply-side response)

This means the ability to call on a resource to increase or reduce its supply at short-notice – milliseconds, seconds or a few minutes and normally for a relatively short period of time. Currently that often means calling on open cycle gas turbine generator sets to start providing power. They have an advantage over the more efficient combined cycle gas turbine (CCGT) of being simple and flexible with quick start-up times and lower capital cost. But in times of excess supply, flexibility can mean requiring wind or solar generators to cease injecting power into the grid to avoid it being overloaded.

As the proportion of variable renewable electricity on the grid increase, the issue becomes much greater. Thus the need for **flexible energy generation and storage technologies** both of which can help balance the grid, is likely to continue to increase. Storage includes pumped storage in a reservoir, electrochemical battery and hydrogen. Flexible energy at the present time cannot be easily achieved without resorting to natural gas. As an example of this, Statera Energy, as part of its portfolio of assets, includes pumped hydro storage but also flexible natural gas generation.

We note here that the UK transmission system operator (NESO) has reported that while the UK grid will have reduced the share of unabated gas generation to below 5% on average over a year by 2030 (under one of its pathways) the country will need to retain the same capacity in natural gas plants as a strategic reserve for windless days.

Energy Storage

Storage in a form which can almost instantaneously be converted into electricity is a key supply side flexibility technology. **Storage does not generate energy** as the grid has to supply the power in the first instance unless the storage facility is located contiguously to the generating source rather than charging from the grid. A storage system can re-charge at times when there is excess power coming onto the grid which is available at a much lower price than when the imbalance is in the opposite direction. Most storage mediums and in particular battery storage have the great advantage of being able to provide power almost instantaneously much like a torch which lights up immediately it is switched-on. What it does is to **'bridge time gaps'** by separating the production of power from its use. It may soon be the case that large-scale battery storage rather than natural gas use is the first choice for peaking services, based on flexibility, cost, services to the network and emissions reduction."

Import/Export

This is about extending grids to connect with grid systems in other states (as in the United States) or areas such as neighbouring countries in the EU or, more recently, distant countries. *Interconnector power lines* between regions of a country ideally in different time zones, or with different wind conditions reduce variability in power available relative to power demand. **The portfolio effect of combining sources of generation which are not correlated can enable reductions of investment in power generation required to ensure a continuous supply of power.**

An example of a new interconnector is ElecLink which was opened in 2022 and goes through the Channel Tunnel. Unlike other interconnector projects, which are fully underwritten by electricity consumers, ElecLink bears 100% of the investment risk. The project has been financed entirely through private funds. ElecLink is a wholly owned subsidiary of Getlink, the company that owns the Channel Tunnel concessionaire, Eurotunnel.

It can happen that in Europe, for example, all regions are suffering from a lack of wind or sunshine or both and sometimes for many days. All those connected to interconnectors may therefore be seeking power at the same time. Thus traditional interconnectors, **not being storage facilities**, may not be available just at a time when they are needed. However, interconnectors attached to **additional** generation capacity in areas with high levels of incident solar radiation, hydropower or geothermal power including, also, countries outside Europe, provide the possibility of what is effectively 24/7 dispatchable power from VRE. In the case of, for example, solar and wind projects in Morocco, it is likely that if they were also manufacturing hydrogen, they could cease such production for periods when power prices in Europe were high and they could sell the power very profitably compared with manufacturing hydrogen. They would then resume the manufacture of hydrogen when power prices in Europe fell and thus such manufacture would be more profitable than selling power. The other means of providing power when wind speed drops for long periods would be long-duration energy storage (LDES). This could be either from electrochemical storage or from hydrogen storage linked to electrolysers.

An example of an interconnector with a country distant from Europe and which was at an advanced planning stage at the beginning of 2025 (X-links) but has now been told by the UK government that they will not support it.

Xlinks[45] – long-distance high voltage DC transmission of VRE power from Morocco to England

Europe is not the best location for solar or wind power generation due to relatively low levels of sunlight and wind strength compared with other locations. Morocco, as well as generating power for local use is also seen as a source of power generation for use in Europe with up to three times the intensity of sunlight and the benefit of the strong trade-winds from the Atlantic. The Xlinks proposal is one such attempt to bring North African power directly to the UK rather than using the power to produce hydrogen locally, converting the hydrogen to ammonia, shipping it to the UK in tankers then converting it back to hydrogen, either for direct use or for converting back to power. The project's estimated will cost is $22 billion. It would comprise 7 GW of solar and 3.5 GW of wind, along with on-site 5GW/ 20GWh battery storage, in Morocco. The cable would start in Morocco's Guelmim-Oued Noun region with a UK landing point in Devon. The cable would be around 2.400 miles long (3,800 km). This would make it the longest power cable in the world (currently the longest is a 2,539km overhead cable in Brazil). It should be noted, however, that during the night in Morocco, no power will be generated and thus the power supply at such times, if required, would come from energy storage.

10.3 How Clean are National Power Systems and is 100% Clean Actually 100% Clean? Time Matched/ Location Matched 24/7 Carbon-Free Energy/electricity (CFE)

National Grids across Europe and elsewhere in the world are certainly nowhere near being clean. Not only that but power sold under a 100% renewable PPA will often not in practice be 100% CFE. The need for much speedier action to decarbonise power systems will be clear after looking at the graphic below. Figure 10.5 which reflects the average CO2 emissions over the 12 months of 2023 is provided courtesy of Electricity Maps[46]. This company provides users with actionable data quantifying the carbon intensity and origin of electricity. This data is available on an hourly basis across 50+ countries and more than 150 regions.

The map below shows that only two geographical regions – the Nordic region and France are anywhere close to having a 100% 'clean grid supply while Poland, which relies mainly on coal, is at the 'dirtiest' end of the graphic with a carbon intensity of 800 grammes of CO2 per kilowatt- hour (kWh).

[45] https://xlinks.co/

[46] https://app.electricitymaps.com/map/24h

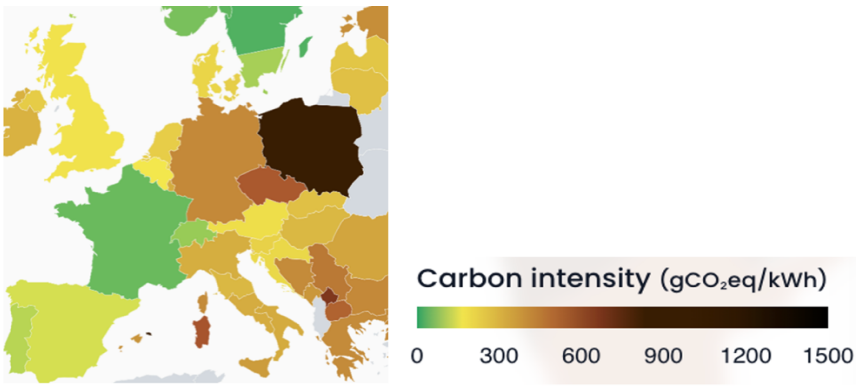


Figure 10.5 Average emissions intensity across European countries

If we look at the overall carbon emissions intensity of power in different parts of the transmission grid across Europe we find that they differ substantially by country and by region. In some cases regions are connected to each other to give an average carbon intensity which is the same across the country but in others, these regions are not connected. In addition, as we noted earlier these intensities depend not just on the local generation in that country but also on the carbon intensities of the countries from which they import power through interconnectors. It can be seen that the UK is very far from being de-carbonised (sometimes defined as less than 100g CO_2/kWh) despite the fact that it imports from France (mainly nuclear) and from Norway (mainly hydro) which are two of cleanest countries. France has a very low-carbon power system with around 75% of total French electricity generation being from nuclear. As a result, 93% of electricity generated is carbon-free. In France in 2019, the emission of CO2 was only 35 g CO_2/kWh compared to an average of around 350 – 500 across Europe. Poland is around the 800g intensity. From having intended to reduce the proportion of nuclear its fleet, they now plan to maintain or increase it.

Some retail suppliers in a country claim that their power is 100% clean i.e. Carbon Free Energy. In practice this is not likely to be the case. Customers must take power from the grid unless they have a direct connection to a renewable source combined with storage or their supplier is using storage to cover gaps in the availability of solar or wind. Otherwise only a few national or regional grids (in France and Scandinavia) are close to being 100% CFE.

Power is 'fungible' i.e. any one electron coming down a wire looks like any other electron. When charging an electric vehicle, in many countries the power has been produced by generators using on average around 40% fossil fuel or in some cases a much higher fossil percentage. Thus at present, electric cars are not "100% clean" though many purchasers think they are. They are clean 'at the tailpipe' and do not emit health-endangering substances such as particles less than 2.5 micrometres in diameter, also known as fine particles or $PM_{2.5}$, which are emitted by diesel engines. The only ways of making electric cars 100% clean is if the **grid** is 100% clean at every hour of the day or if the companies providing commercial charging facilities have what we examine below, namely a granular 24/7 CFE contract with a power company.

Time Matched/ Location Matched 24/7 Carbon-Free Energy/electricity (CFE)

Increasingly customers want 100% or near 100% clean power either because they are willing to make an effort to limit climate change or because carbon taxes are high or because regulations require them to report the carbon intensity of the power they consume. If the latter is the case, while it may not be illegal for a corporate to consume high carbon power, it may be that investors and customers of the company will take action against it for doing so.

We now consider what this means in practice. We need to know how PPAs in particular can be adapted to make them equally relevant to the new world that is within reach if we can develop appropriate frameworks and contractual arrangements which will also encourage ***additionality*** i.e. creation of new assets.

According to LDES[47]

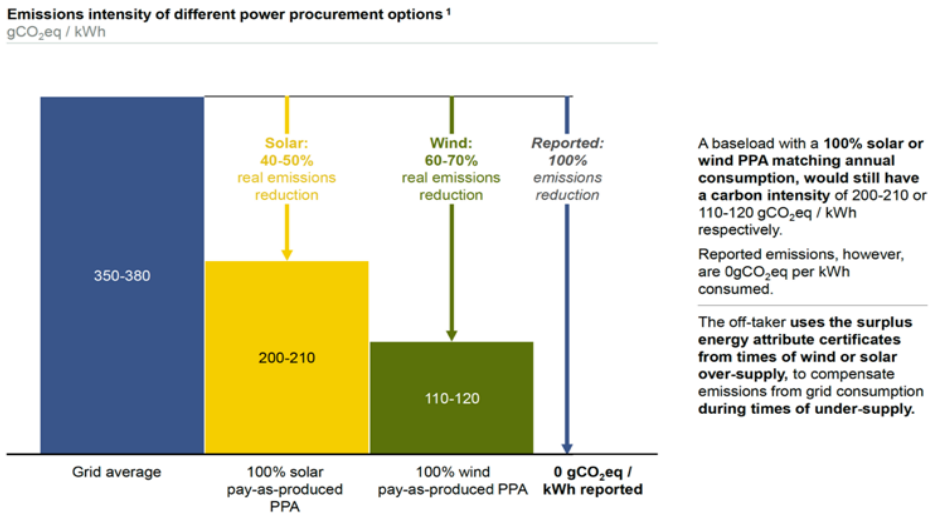
[47] LDES *A path towards full grid decarbonisation with 24/7 clean Power Purchase Agreements* https://www.ldescouncil.com/assets/2205_ldes-report_247-ppas.pdf

A new form of Power Purchase Agreement (PPA) — the **24/7** CFE PPA — promises dramatic increases in the effectiveness of emission reduction efforts in the power generation sector and for corporate off-takers. 24/7 clean PPAs have the potential to address key problems in balancing supply and demand for power as renewables contribute an increasing proportion of generation capacity. ***In the process they can help providers and off-takers accelerate the overall decarbonization of the power sector.***

24/7 clean power PPAs measure electricity consumption and greenhouse gas emissions on a **granular** basis — *e.g.,* by the hour (or half or quarter hour) — and provide *time-matched* clean power. They represent an improvement from today's Scope 2 decarbonization with pay-as-produced renewable PPAs which balance supply and demand on an annual basis but only achieve 40 — 70% decarbonization of the off-taker's *actual* electricity consumption.

Time-matched clean power provided through 24/7 clean PPAs addresses both problems but requires a combination of renewables, dispatchable generation and flexible capacity such as storage and DSR to create a more granular match between renewables supply and demand. **Current barriers to wider adoption of 24/7 clean PPAs include cost and a lack of agreed standards,** but these can be overcome.

Traditional clean baseload contracts oblige the seller to enter the power market (including the fossil-generated power market) to purchase power to meet the contract's requirement of a continuous power supply. This may still be called a 100% clean energy contract since all the CFE power purchased by the buyer over 12 months is credited to the PPA contract counterparty. Even when some of the power was produced in excess of the buyer's need in summer its gives the buyer a **credit** in winter when it is actually using 'high carbon' power much of the time.



The **credit** is the Energy Attribution Certificates (EACs) from this generation (covered below) which can be used in winter when as an 'offset' to the use of fossil fuels. This enables it to claim the summer certificates against its total power consumption for the year. In consequence, much of the power may be generated by fossil sources as shown in Figure 10.6 from LDES

10.6 Emissions intensity of different electricity generation sources

In contrast, under a **24/7 clean baseload contract (24/7 CFE) contract, clean power must be supplied almost every hour of every day rather than just on average over a one-year period**. However, at present almost no energy provider will guarantee 100% CFE at every hour. Thus the PPA will state perhaps 70% or 80%. Energy Attribution Certificates EACs generated in the summer from CFE under this type of hourly contract **cannot be used in the winter "to make up the numbers to 100% CFE".** A PPA from a CFE generator to supply CFE on a 24/7 basis will almost certainly require energy storage including long duration energy storage to cover wind 'droughts' and other interruptions to supply of electricity.

The United Nations Energy Compact defines 24/7 CFE as every kilowatt hour being provided by CFE every hour of every day. At the present time, the world does not have 24/7 granular time matched 100% CFE PPAs except on an experimental basis. What can be provided by a supplier is a 24/7 granular time matched PPA but not necessarily, or likely yet to be providing carbon free electricity during every hour of the contract i.e. not CFE 24/7. Google was one of the first to set a target of 100% CFE but in 2020 it was still only able from a techno/economic point of view to achieve 67% matching on an hourly basis.

Google first said it was carbon-neutral in 2007 through carbon offsets. In 2017 it started purchasing carbon free electricity through PPAs in 2017 and is now trying to transition from 100% **annual** renewable energy matching to **24/7 matching** by 2030. This involves achieving **hourly** load matching instead of annual, volume-based matching.

By 2020, it had achieved only 67% matching. Since then, the demands of AI large language models (LLMs) has dramatically increased its need for power making the achievement of improving matching much harder to achieve. The question today is how to transition towards the point where buyers of power can be guaranteed that they are buying power with an agreed high percentage of carbon free power (CFE). *Eurelectric* has shown how this organisation thinks we could move towards "Co-establishing commitment to a minimum % threshold of hourly CFE matching." Eurelectric describes itself as "the federation of the European electricity industry." Its membership comprises more than 3,500 European utilities active in power generation, distribution and supply. It recently produced a publication entitled "Contracting Principles for Moving Towards Location and Hourly Clean Energy Matching" which can be found on their website.

Achieving a system for 100% CFE on an hourly basis is part of a process in which electricity systems move from purely fossil energy to close to 100% carbon free energy. Such a system would only allow for clean energy consumption claims if proof of **generation** origin was matched with electricity **consumption** on an hourly basis and sourced from the local grid. This would be achieving both carbon-free electricity for consumers of power but also additionality of CFE assets on the grid.

*10.4: The United Nations **24/7 Carbon-free Energy Compact**[48]*

According to the Carbon-free energy compact, 24/7 Carbon-free Energy (CFE) means that every kilowatt-hour of electricity consumption is met with carbon-free electricity sources, every hour of every day, everywhere. It is based on the following principles:

1. **Time-matched procurement:** matching hourly electricity consumption with carbon-free electricity generation. Hourly matching helps connect clean energy purchasing to underlying electricity consumption.
2. **Local procurement:** purchasing clean energy on the local/regional electricity grids where electricity consumption occurs. This is the only way to drive the electricity-related emissions that a consumer is directly responsible for to zero.
3. **Technology-inclusive:** recognizing the need to create zero-carbon electricity systems as quickly as possible. All Carbon-Free Energy technologies can play a role in creating this future.
4. **Enabling new generation:** focusing on delivering additional Carbon-Free Energy to drive the rapid decarbonization of electricity systems.
5. **Maximizing system impact:** addressing the dirtiest hours of electricity consumption where the most fossil fuel is used in generation.

Note in the list above, No. 3 is an approach that includes any carbon free energy source including nuclear thus not just renewables. No. 4 refers to the fact that this system would provide better price signals to the system as the price of hourly certificates would vary in time, depending on the availability of clean electricity in the system. It should therefore help generate confidence in those considering 24/7 CFE power projects as they move towards the financial investment decision (FID) on the assumption that more such power supplies will be required. But granularity can, with benefit, go further. The European Network of Transmission System Operators for Electricity **(ENTSO-E) has recommended that the granularity target should be set to 15 minutes**, in line with the target imbalance settlement period in Europe. This would enable the certificate market to be more closely aligned with the physical grid reality.

For the following chapters, an important matter to keep in mind is that with a few exceptions, it is not yet possible in fact to have 24-hour-a-day 100% carbon free electricity across a country or region. We have the technology to achieve it but scaling-up that technology **at a cost acceptable to voters**, is what is difficult. Thus 24/7 CFE across the grid, and hence also for the power consumed by electrolysers to ensure that it is 100% CFE hydrogen, is the objective but remains some years away yet. Like so much of the technology we cover, it is an ongoing project requiring the collaboration of many parties involved in the power industry value chain. However, confusion can be caused by the different meanings attributed to the expression '24/7' by different parties. This also makes it more difficult for investors to commit to projects that might move the world closer to being carbon-free.

[48] This material is sourced from: https://www.un.org/en/energy-compacts/page/compact-247-carbon-free-energy#:~:text=24%2F7%20Carbon%2Dfree%20Energy%20(CFE)%20means%20that,hour%20of%20every%20day%2C%20everywhere.

10.5 Decentralised Electricity and Heat Systems and their Financing

Back in 2020, Bill Gates was agonising over the inability of countries or districts with lower than average incomes to take advantage of low cost renewable energy and the lack of improved technology for such countries being developed in the higher income countries to facilitate falling costs. In one of his early "Gates Notes"[49], he explained what he called the "Green Premium" as:

> the difference in cost between a product that involves emitting carbon and an alternative that doesn't. This difference in cost is what he calls the Green Premium, and understanding it is key to making progress on climate change.

He continues by noting that:

> they help us measure our progress toward eliminating carbon emissions. The bigger a Green Premium is—especially for lower-income countries like India and Nigeria whose energy needs are growing—the further we are from a zero-carbon future. They also serve as a guide to action. In cases where the Green Premiums are big, we know we need innovations that will close the price gap. In cases where they're small—or where clean products are actually cheaper than the polluting version—it suggests that something other than the cost is keeping zero-carbon products from being deployed, and we need to understand why.

Up to this point in the text, we have been considering the production, distribution and financing of power in the context of high income countries and regions which have traditionally employed large, centralised coal or gas power stations with electricity transmitted through a grid system and then through local distribution networks. This chapter covers some of the technology possibilities offered by decentralised electricity provision which can be used in countries or regions that do not have connectivity to a national network or where such connectivity would be economically unattractive. Such electricity generating systems have fallen dramatically in cost since Gates wrote the above words, but today the barriers are still as much in the financing arena as in the required technologies.

Decentralised energy systems, which describes the generation of electricity and heat off-grid closer to where it is consumed, are crucial in progressing the energy transition to enable a zero-carbon global economy and, potentially, to achieving energy equity. Such systems can be deployed more rapidly and with less upfront cost than a conventional centralised energy model. They enable renewable generation, distribution, and consumption at a community level and, if using low carbon generating systems in particular solar, wind and battery storage, reduce reliance on fossil fuels, enable shorter distribution connections and no high voltage transmission network thereby minimising energy losses and cost. As well as introducing electricity for the first time to new neighbourhoods, such projects may also involve replacing generating systems currently using diesel generators. They can also enable the elimination of home cooking devices using gas or wood burning stoves which are both harmful to health and also emit CO2. Replacement with electric stoves would then also require a capital investment in such home hardware which would need financing.

Despite the advantages it could bring, both to households and to global decarbonisation, the adoption of decentralised energy remains underfinanced. International climate finance still favours investments in large-scale fossil fuel utility companies or large-scale centralised wind and solar projects. Such capital flows tend to concentrate on high-income countries and high-ticket centralised systems thereby excluding decentralised renewables which could speed up energy access and the global energy transition. Innovative financing could facilitate the acceleration of the energy transition by extending electrification to parts of the world where energy utilities would find it difficult to justify investment. But progress in this regard is slow and is only likely to proceed with more investment by national development funds and international organisations such as the World Bank.

[49] https://www.gatesnotes.com/meet-bill/accelerate-climate-innovation/reader/introducing-the-green-premiums?ref=carbonrisk.world

While aspects of green bonds, yieldcos, and securitisation in renewable energy financing have been studied (though mostly in the context of developed markets), most existing research focuses on centralised renewable energy projects and mechanisms for the integration of large-scale renewables into centralised transmission systems, and the operational and the planning challenges for such integration. On the finance side this has mostly focussed on the potential of green bonds as an instrument to promote renewable energy.

Decentralised Energy Systems

Decentralised energy systems include any system where energy production occurs at or near the point of use including micro and mini grids, stand-alone projects (e.g., PV rooftop solar panels), battery storage, mini-hydro, biogas plants, and micro nuclear reactors. They imply not only distributed and dispersed generation and spatial proximity to its consumers but also new ownership, financial, and governance models. Decentralised systems can expedite energy access, particularly in remote and not densely populated areas, enable climate resilience, contribute to energy equity, affordability, and environmental protection.

Decentralised energy systems have the potential to enable energy equity (as they can provide clean energy to underserved rural and urban communities), energy resilience (as decentralised grids can be more resilient to climate change than large power plants and transmission lines) and empowerment (enabling communities to become energy producers as well as being consumers i.e. prosumers.

Traditionally, centralised, utilities are often known for being cost inefficient, conservative, carbon intensive, lacking transparency, and operating as natural monopolies. They are vulnerable not only to climate change but also to geopolitical risks. Decentralised energy systems, on the contrary, imply a shift off-grid generation facilities, from large utility companies to medium size and individual enterprises and to private residents as both producers and consumers of energy.

Such structures are very relevant in developed markets. But in addition, they should be increasinly relevant in emerging markets in countries or districts where GDP per head is low. In such areas at present they may require official financial support to be viable. But in such cases, the increase in small-scale manufacturing and other industry that may become possible in consequence of electrification may enable such local systems to become economically self-supporting in the longer run and also to contribute to the economic growth of a region or country.

Decentralised models require new financial, ownership, governance and business models. Rather than being state owned companies or large utilities quoted on a stock exchange they would have widely dispersed financial ownership structure. These types of ownership include:

- Private Ownership (plug and play model), where the microgeneration unit is owned and financed by the end user, who may also export energy into the grid (e.g., BeON, the system available from the French utility EDF).

- An energy service company (ESCO) where energy providers use microgenerators to substitute for central power generation. Consumers provide the site for the microgenerator, which is owned and operated by the energy service company. ESCOs can reduce upfront capital investment as they finance the project and thus customers don't need to pay upfront. Payments to the ESCO's may also be contingent on achieving energy savings if the ESCO is replacing an existing (possibly diesel fuelled) generator system. By taking on the financial risk, ESCOs can make energy efficiency projects more accessible for customers, both domestic and larger organizations such schools or hospitals.

- Community/corporate ownership (micro grid model), where all microgeneration units are connected within the microgrid and can be partially owned by the consumers. This ownership model combines the benefit of corporate control with consumer involvement and partial ownership, which incentivises the system to operate smoothly.

The current features of international climate funds discourage investment in small-scale decentralised projects. These funds incentivise large scale investments in high and middle-income countries, particularly grid-based and fossil fuel energy, over decentralised renewable energy projects. International climate finance does not prioritize

decentralised energy projects. Its institutions, such as Multilateral Development Banks (MDBs), are not designed to, nor do they have the instruments to invest is small scale decentralised renewable energy projects.

The biggest international climate funds, Clean Technology Funds (CTFs), employ concessional loans to finance utility scale projects in middle income countries, which offer better returns and lower transaction costs than funding many individual decentralised energy projects.

Barriers to the Financing of Decentralised Energy

Decentralised energy projects are perceived as less bankable and require a different range of financing and risk mitigating instruments; they are perceived as having higher rates of failure and frequently operate in a context of an uncertain regulatory framework. Because of their small size, they may have high transaction costs and, most difficult to resolve, high credit risk customers. Even established companies such as Mobisol, a Berlin-based company that provides solar home systems became insolvent in April 2019 despite achieving 500.000 users. It struggled with low consumer repayment rates and an inability to scale a pay-as-you-go (PAYG) off-grid model without ongoing subsidies.
Barriers for decentralised energy also include policy frameworks that favour investment in centralised energy systems, financial instruments that enable only larger investments, lack of reach of Special Purpose Agencies which can channel funds to small-scale projects and lack of capacity building. These risk characteristics are barriers to increasing the appetite of private capital for such projects. Despite all the barriers referred to above, decentralised projects do involve less bureaucracy and their smaller fund size can be appealing to some types of impact investor.

Financing Mechanisms for Decentralised Energy

Green Bonds

We covered green bonds earlier in the context of large scale projects. They were created to fund projects which have positive environmental or climate benefits. Proceeds are earmarked for green projects but can be backed by the issuer's entire balance sheet when the issuer is a corporate as most have been to date or they may be project financings through a special purpose vehicle (SPV). Green bond terms are based on ICMA Principles and must offer investors transparency, impact assurance, and alignment with global ESG standards. They can provide a powerful bridge between the demand for sustainable investments by particular categories of investor and the need to scale up community-level carbon-free infrastructure which is a necessity if the world is to come within reach of the Paris Accord's objectives.

However, for decentralised energy projects their utility is limited given their high issuance and transaction costs, including legal, verification, and certification requirements. These fixed costs are burdensome for small scale projects. Costs are increased further with the post-issuance reporting and monitoring requirements for green bonds.

Green bonds are most suited to large scale infrastructure projects as institutional investors seek a minimum ticket size investments. Decentralised energy projects, driven by local enterprises, community-based organizations and local developers, thus have limited access to the bond markets, given their lack of a credit rating and limited market visibility and thus capacity to issue bonds. Aggregation challenges limit their use for small projects as they do not achieve economies of scale and require complex aggregation structures like securitisation and green asset-backed securities (ABS).

As decentralised energy projects are diverse, fragmented and not standardized it reduces their attractiveness in the bond markets. Finally, there is a lack of uniform standards on the legal terms and some issuers label green bonds without offering credible or verifiable environmental benefits. This clearly undermines investors' trust i.e., leads to greenwashing risks. The limitations of green bonds as financing instruments for decentralised energy systems always comes back to the small scale of projects, lack of creditworthiness and lack of liquidity or even just a secondary market. Without a credit risk guarantee they would be unlikely to attract investors. In future, however, it might be possible to

develop a ‘green type’ bond which was suitable for smaller projects and attractive to impact investors wishing to improve lives in a community.

Yieldcos

Yieldcos, short for yield companies, are publicly traded investment vehicles that own and operate renewable energy assets (e.g., solar farms, wind, and hydro plants) that have entered long-term energy delivery contracts with customers and are therefore, in theory, able to generate predictable cash flows. Yieldcos are structured to deliver regular dividends to investors in a similar way to REITS and Masters Limited Partnerships (MLPs). Yieldcos are structured with the specific purpose of acquiring operating energy projects from the sponsor and from third parties. They generate stable revenues through power purchase agreements (PPAs), mitigation of development risks, and price certainty of renewable investment assets arising from the long-term contracts to deliver energy to consumers at fixed prices.

To be attractive, decentralised energy projects must deliver large, diversified, stable cash flows via long term PPAs in order to provide dividend-like returns to attract investors. Yieldcos can aggregate portfolios of small-scale distributed assets, providing scale, standardization, liquidity and bundling across geographies, type of consumer, and technologies. They allow capital recycling, as developers can sell operating assets to Yieldcos and use the proceeds to fund new projects, providing an exit route for early-stage investors and acting as platforms for refinancing early-stage projects.

Yieldcos in decentralised energy have, however, high aggregation and transaction costs, as collecting customer payments, achieving standardization, and undertaking due diligence are all administratively more intensive and costly than for one large centralised plant. In practice, many decentralised energy projects have revenue volatility as they often lack long terms PPAs and depend on small customers’ payment reliability.

Securitisation

Securitisation bundles renewable energy loans into tradeable securities. It aggregates them to achieve scale, diversification of risk and lower capital costs. In theory, securitisation should provide a creditworthy structure attractive to long term investors. They are a mechanism to aggregate cash flows from multiple energy assets to create a large-scale investable asset making these projects accessible to institutional investors. Securitisation allows early-stage financing and capital recycling, as it converts illiquid, long-term project returns into tradeable securities, attracting private capital into emerging market and off-grid projects.

However, securitisation still involves considerable risks for investors, as observed during the 2008 global financial crisis (GFC). First, many potential customers of decentralised energy systems are low income with no formal credit histories which makes it difficult to make cash flow predictions. To overcome the lack of creditworthiness of many end users, guarantees are needed (e.g., power purchase agreements, PPAs with corporates or some official backing). On the other hand, if supplying retail customers, guarantees are difficult if not impossible to obtain. Second, securitisation involves high transaction and structuring costs, represented in legal, rating, and structuring fees. Further, there is still a limited track record of securitisation in the decentralised energy space, which are still perceived as high-risk investments.

In summary, these financial instruments should be aiming at increasing the ability of decentralised clean energy projects to raise capital, facilitating interest cost reduction and speeding the process of raising capital. At the present time, however, particularly when interest rates in most countries are higher than they were some years ago, even the most innovative instruments may not be able to overcome the essential problem in those countries where consumers have very limited income and hence creditworthiness. On the other hand, these instruments, or modified versions of them, can still work for decentralised projects in high income countries and over time as incomes rise in other countries, it should be increasingly possible to bring energy equity to many more people.

This chapter has covered decentralised renewable electricity systems. There is another type of decentralised electricity provision which is currently being developed and is based on advanced nuclear technology (ANT) small and micro reactors. Their inherent safety even compared with legacy reactors means that they can be located close to demand in an SMR campus or an industrial zone or even be embedded within a datacentre itself. This might also require a connection to the grid though normally its energy storage system would take it through most interruptions to supply. Such a system is also known as the “private wire” approach and overcomes many issues and in particular the shortage of capacity on transmission lines in developed markets which are required to take distributed solar and wind from where it is generated to where it is needed. The private wire approach should also enable power to be provided at well below the price that users would need to pay if they relied on grid power. We cover the private wire approach in much more detail in Chapter 15.2.

Chapter 11: Energy Storage as a Contributor in the Conversion of VRE into Carbon-Free Dispatchable Electric Power:

> *To be of greatest value to consumers, CFE power needs to be available continuously even when not being produced at night or when the wind force is low. The "conversion process" from VRE to 24/7 CFE power becomes ever harder as the proportion of CFE on the grid increases. Storage of energy in, for example, chemical form in Li-ion batteries which can be converted to electric power virtually instantaneously has been a key focus for technologists, developers and financiers. In addition, as the proportion of VRE rises, there is an ever greater need for long duration storage for the times when wind force is low for periods of weeks. In the UK, for example, winter electricity demand is twice that of summer (as the UK does not yet have much domestic air conditioning. However, without a proportion of dispatchable power on the network, the cost of power is likely to be much higher than it would otherwise be if dispatchable power were a significant part of the power mix. In practice transitioning from 95% clean power to 100% is likely to involve the use of backup natural gas storage and gas turbine generation (hopefully with carbon capture and storage) and would almost certainly be much less costly than attempting to achieve 100% carbon free electricity using mainly VRE.*

11.1 Storage Technologies for 24/7 Dispatchable Electricity from Variable Renewable Energy (VRE)

There is still much debate about how best to "fill in the gaps" as we have already observed earlier. The debate is between those who believe that the lowest cost means of achieving 100% clean energy is to provide a certain percentage of firm (dispatchable) clean, or at least low carbon, sources of power such as nuclear, flow of the river hydro, geothermal, and fossil plants capturing and storing CO2 as we have noted above. The reason for this is that trying to achieve 100% renewable by using storage only is not likely to optimise the energy trilemma. The outcome may be a much more costly power system yet one which is not as reliable as should be expected by consumers. **Batteries and demand flexibility do not substitute for firm resources.** Without a certain percentage of dispatchable power, electricity costs rise rapidly as a system approaches 100% CFE. Trying to achieve 100% CFE without firm power is not a least cost strategy.

Below, we cover a wide range of technologies for energy storage. Some use the force of *gravity,* for example hydro, to release the energy in water which has been raised to a height by the energy in the heat of the sun through evaporation of water which then returns to Earth as rain. Some use **energy stored as *heat*, for example Rondo,** which can then be used when required to generate power or provide industrial heat. Indeed the **Terrapower Natrium** reactor, as well as being able to generate dispatchable power all day also has a 'boost' function through which heat is stored for release at times when power prices are low. The heat is then used to generate and sell power additional to its normal output when power prices are high. Some storage mechanisms use the energy stored in *hydrogen* which can then be released into the grid by passing it through an electrolyser or even an internal combustion engine (ICE). Others use *compressed gas* such as air which when released can then power a turbine.

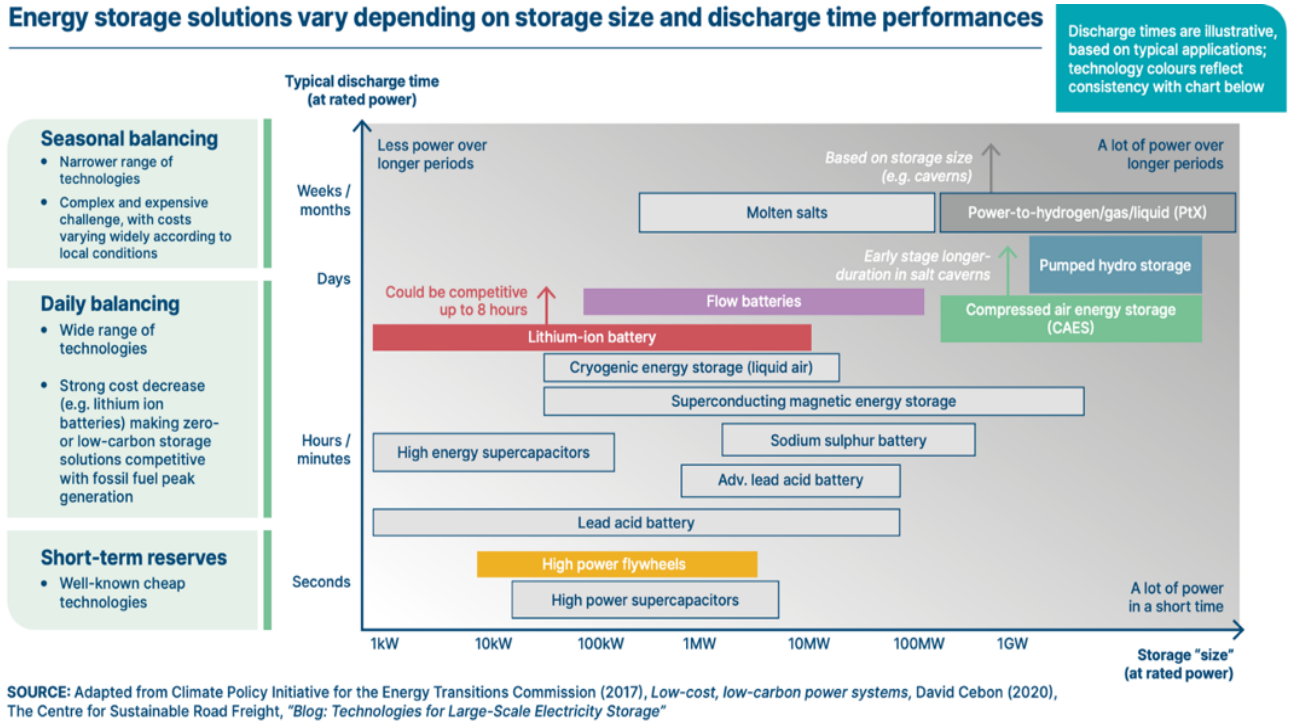
Some of the storage mechanisms noted above are different from the type of storage that is growing fastest at the present time and that is electrochemical storage in some form of 'battery' or battery energy storage (BES). Much of the electrochemical battery storage capacity 'earns a living' by means of 'financial arbitrage' in power markets by purchasing power at a lower price than it is subsequently sold at (time of day arbitrage).

It is important to be clear that electrochemical storage does not generate any power in contrast to, for example, 'flow-of-the-river' hydro generation which is a source of electrical power. In contrast, 'pumped' hydro, like battery storage, also involves a type of financial arbitrage and also does not generate power. Instead, it buys power when it is cheap in order to pump water uphill and then releases it to generate power when the price of power is high. It operates essentially like an electrochemical battery.

In addition, there is a 'balancing market' which is power supplied to help balance the grid at certain times. Traditionally this has been provided by gas turbine generation. The ETC graphic[50] shown in Figure 11.1 summarises

[50] Energy Transitions Commission (ETC) *Making Clean Electrification Possible: 30 years to electrify the global economy*, April 2021, https://www.energy-transitions.org/wp-content/uploads/2022/07/ETC-Global-Power-Report-Final.pdf

some of the main technologies in existence at present, many of which are still being developed and are not yet completely proven either in terms of their technology or their cost effectiveness.



SOURCE: Adapted from Climate Policy Initiative for the Energy Transitions Commission (2017), *Low-cost, low-carbon power systems*, David Cebon (2020), The Centre for Sustainable Road Freight, *"Blog: Technologies for Large-Scale Electricity Storage"*

Figure 11.1 Duration of energy storage mechanisms against storage capacity

On the vertical axis it shows the maximum *period* of storage at present which is from seconds to months. On the horizontal axis is the maximum *amount* of storage that can be provided by the technology at present at an economic cost. It can be seen that hydrogen has the potential to be at the multi-gigawatt scale because large amounts of hydrogen can be stored and also, therefore, used to cover seasonal needs and not just intra and inter-day gaps in supply and demand as is currently the case for battery storage. The only problem with hydrogen at the present time is that green hydrogen is expensive and not easily available in large quantities. What this graphic does not show, however, is the cost of such storage and hence which type of most cost-effective for each type of use. Although some technologies are presently costly, in many cases the unit cost of storage will fall, perhaps dramatically, as technologies develop and as economies of scale and learning come into effect. It also does not show the improvement in flow-battery technology which now potentially enables 100+ hours of storage at low cost nor some new 'heat battery' solutions.

At the bottom of the graphic below, very short-term reserves of energy are shown. They are useful for the provision of auxiliary services to the grid such as voltage or frequency stabilisation. We will not cover short-term storage mechanisms but focus on those which potentially provide storage for hours rather than minutes or longer duration economically. Although we have categorised technologies this way, with technology moving so fast, a year from now storage durations, cost effectiveness and technologies could all have changed. For this reason, we are not going to classify any particular technology as either daily balancing or longer duration or seasonal balancing. Some can already provide all types of service while others may be able to do so within a few years.

Revenue Stacking

The key to the massive deployment of batteries is to make the investment profitable. To achieve that, so-called revenue stacking is necessary. Batteries projects need diversified sources of income (different layers of revenue from providing diverse functions) and be able to participate in any service that can generate income.

Price arbitrage in the day-ahead market and also in intraday markets is the basic function of storage i.e. buy cheap and sell dear as in financial markets which is 'time arbitrage'. But in addition there are so-called Ancillary Services which need to be provided to the TSO which makes payments for such services as we covered above. But critically important in today's market is payment for providing 'capacity' to the grid. This so-called Capacity Market ensures security of electricity supply by providing a payment for reliable sources of capacity to provide power when called upon by the TSO. This source of revenue is provided by the TSO simply for making power available.

The actual provision of power, if required by the TSO, is paid for separately. The profitability of such a service may, however, be quite variable.

11.2 Electrochemical energy, grid-scale.

By far the best known and mostly widely used at present, power storage technology is the Lithium-ion (Li-Ion) battery which can provide power which can be dispatched immediately. However, such storage is currently relatively expensive and therefore the economic time for which it can provide power is generally limited to four hours. However, development of Lithium Iron Phosphate (LiFePO4) batteries known as LFP may provide lower cost and safer, large-scale, longer duration batteries for grid storage.

The UK's largest battery storage project at the time of writing is at DP World London Gateway on the Thames Estuary. This project which will provide 320MW/640MWh of capacity [i.e. it can provide 320 megawatts of power for two hours or other combinations of power versus time].

StorTera

StorTera is a provider of energy storage systems [we cover its AI system (Traicon) designed to take in data, analyse it and make 'optimal' decisions in Chapter 12.3]

StorTera produces a non-flow battery using lithium iron phosphate (LFP) technology. They also offer a flow battery which uses their lithium/ sulphur technology. They predict capital costs of approximately £75/kWh which would make it potentially an economic technology. The StorTera battery has received grants which have enabled it to develop the technology. They were also the prize winner in an international competition sponsored by the UK and the Canadian governments. The project took place in a town in Canada and the statement below from a Canadian government minister was made at the end of August 2022 when the prize winner was announced:

> "Our government challenged innovators to find breakthrough cleantech solutions to help solve some of Canada's biggest problems — and they delivered. I'm proud to award the grand prize for the Power Forward Challenge to Equilibrium Engineering and their UK partner StorTera."
>
> *The Honourable Jonathan Wilkinson Minister of Natural Resources Canada*

Fluence

Fluence is a global provider of energy storage products, services and digital applications for renewables and storage originally launched and owned by AES and Siemens Energy and now publicly listed on the NASDAQ (ticker: FLNC). It is a large provider of these services having deployed, contracted or managed 6.2GW of storage, 225 energy storage projects in 44 markets and provided more than 18 GW of AI optimised renewables and storage. It provides an ecosystem including modular, scalable, energy storage products such as Gridstack™. It does not manufacture battery cells but outsources this requirement to the most appropriate provider.

Electrochemical energy, Vehicle to Grid (V2G)

This potential source of storage is not yet developed fully but could become important. The logic of this type of storage is that one day millions of vehicles around the world will be battery-powered and will mostly be parked somewhere and therefore not using their battery power. That provides the potential to sell that power to the grid as 'peaking' power or as ancillary grid services (voltage and frequency regulation). At the present time we already have a halfway step to this in the concept of smart charging which means the charging system will decide when there is most surplus power available to the grid and try to charge a vehicle then and will avoid charging at the peak. This would reduce the need for peaking generation. The next step is a two-way flow of power between a vehicle and the grid (V2G) whereby the charging unit would sell power to the grid at peak times (if the car was not being driven and had sufficient battery capacity for the next journey home). This would result in an even larger reduction in the need for peaking capacity which remains the key problem of de-carbonised grids.

Electrochemical energy Storage – Flow batteries for long duration storage

These use electrochemical storage as do Li-ion batteries but use different electrolytes. The interconversion of energy between electrical and stored chemical energy takes place in an electrochemical cell. A great advantage of flow batteries over more traditional batteries such as Li-ion is that while conventional electrode structures couple the energy storage capacity and the power rating of the battery i.e. it is not possible to increase the duration of energy storage without also increasing its power rating which requires more anodes and cathodes, this is not the

case with flow batteries. By increasing the size of the separate electrolyte tanks in a flow battery, it is possible to achieve a long duration battery without also having to increase its power output at the same time. Li-ion batteries currently have a limited ability to provide long-duration flexibility economically as their storage capex increases in a direct proportion to required duration.

ESS Inc.

There are many flow battery technologies (including one of Stortera's technologies but first we focus on ESS as one of the companies that has developed such systems over the last decade with financial backing from the Advanced Research Projects Agency – Energy (ARPA-E), a US government agency that advances high-potential high-impact energy technologies that are too early for private-sector investment. ARPA-E gave grants to ESS Inc. in 2010 to enable it to develop its long-duration storage system. In May 2021, it announced that it had entered into a definitive agreement for a business combination (a SPAC) and thus became a publicly listed company on the New York Stock Exchange.

A flow battery is a system that stores the electrolyte as liquid in external tanks. Currently, flow batteries account for less than 1% of the grid-scale energy storage market because of their high system costs to date. ESS uses iron in its innovative design which increases power density and enables a smaller and, importantly is less costly than other materials. ESS sells flow battery systems based on iron, salt, and water without requiring the use of hazardous chemicals or rare-earth metals (it uses two chemical components dissolved in liquids that are pumped through the system on separate sides of a membrane).

Hugh McDermott, ESS senior vice president of business development and sales has observed that ESS calculates that Patagonian utility Edelaysen to which they have sold a system will be able to save 12 years' worth of diesel emissions and $3 million in fuel and maintenance costs over 25 years by using a 300-kilowatt, 2-megawatt-hour *Energy Warehouse™* flow battery. Flow batteries could last for 20 years with no major servicing, unlike lithium-ion units, which degrade over time. This could allow ESS to lease its batteries using *asset-backed financing* to cover the costs of battery manufacturing. ESS Inc. has also partnered with Munich RE to launch an industry-first insurance coverage of their flow batteries. The battery modules in their storage solutions come with up to a 10-year extended warranty backed by this global investment-grade insurer.

Form Energy

Its first commercial product is an iron-air battery capable of storing electricity for 100 hours at system costs competitive with legacy power plants. Made from iron, one of the most abundant minerals on Earth.

In its most recent equity fundraising in August 2021, Form raised $240m. Form has now raised more than US$360 million, including its earlier Series C round which raised US$76 million. The investors are:

TPG Rise Climate and NGP Energy Technology Partners III,
Perry Creek Capital (a PE firm)
Coatue (an Investment manager)
Temasek (Singapore state investment group)
Energy Impact Partners, Breakthrough Energy Ventures, Prelude Ventures (impact VC funds)
MIT's The Engine,
Arcellor Mittal (iron and steel manufacturing group)
Capricorn Investment Group (mission-aligned investor)
Eni Next (the VC arm of Italian energy company Eni)
Macquarie Capital (part of Macquarie Group which has advisory, capital raising and financing functions)

11.3 Kinetic/gravity, compressed air, cryogenic and heat energy storage

Pumped Hydro Storage

This uses pumped hydro storage aka closed-loop pumped storage in which water pumped into a high level reservoir converts the potential energy (energy storage) in a water reservoir into mechanical/ kinetic energy through the power of gravity which is then used to turn a turbine generator. Thus unlike 'flow-of-the river' hydroelectric systems which are used for the generation of firm i.e. continuous power, pumped storage can be used where a river has inadequate flow or where there is no river at all but water can be made available. An upper and a lower reservoir can be created using traditional mining processes enabling pumped hydro storage using low-

cost off-peak nuclear, surplus solar or wind power, to pump water from a lower reservoir to a higher reservoir. Such 'off-river storage' has a much lower environmental impact compared to a dam on a river as it does not flood a valley and it does not require a continuous supply of water as it uses the concept of recycling of water which has already run through a turbine many times before.

Rye Development

This company is a good example of pumped hydro storage. It is best suited to operating as a 'peaker' but is less appropriate for intermittent, unscheduled and unpredictable matching needs. The economics of this type of system depends on an arbitrage between the purchase price of low-cost power at night from nuclear or around mid-day from solar, and the selling price of high-cost power to the Transition System Operator (TSO) at peak times. Note that pumped storage currently accounts for 95% of all utility scale energy storage in the US. While the UK's pumped storage projects were originally for peaking and balancing to support fossil fuel power stations and nuclear, with wind, solar replacing fossil fuels, these storage facilities can no longer work on the traditional daily cycle i.e. pumping at night when power prices are low and generating when required for peaking and balancing. Instead they now have to ramp and cycle frequently, sometimes as often as 50 times a day. This is in order to 'fill in the gaps' caused by the loss of so much dispatchable power as a result of it being replaced by VRE the supply of which will rise and fall many times during the day.

Kinetic (gravity) energy using weights.

Pumped hydro storage uses low-cost power to raise water to a height when power is inexpensive and then releasing it to fall by gravity to pass through turbines and generate power which can be sold at a high price.

Gravitricity

GraviStore – which is Gravitricity's energy storage system uses existing mineshafts to support 1,000s of tonnes of mass (e.g. concrete) to store energy. This can be converted from potential energy to electric power through a generator. Underground gravity energy storage technology uses the force of gravity to offer some of the best characteristics of both lithium batteries and pumped hydro storage. In practice it would use abandoned deep mine shafts to allow the concrete block weights to fall a considerable distance thus storing a substantial amount of energy. Gravitricity's energy storage system functions by suspending a large weight, up to 2000 tonnes, in a vertical shaft and, using low cost, off-peak, electricity to winch the weight to the top of the shaft. The kinetic energy can then be released when required.

Advanced Compressed Air Energy Storage (A-CAES)

This uses some aspects of other compressed gas systems (which we cover below) and is also similar in some respects to pumped water storage as it uses water pressure from gravity, but not to directly provide the power for turbines but to provide 'back pressure' to the expanding air flowing through the turbine.

Hydrostor

This is a private company founded in 2010 and based in Toronto has developed a product which improves on existing mature Compressed Air Energy Storage (CAES) technology by eliminating the use of a gas turbine. It does this through storing the heat generated during the air compression process and then reusing it to heat the air on its way to the turbine.

Hydrostor is said to provide long duration (up to 100 hour) energy storage enabling direct replacement of fossil generation. Hydrostor is not a capital intensive business but one which sells its technology skills to others. In January 2021, the Private Equity and Sustainable Investing business within Goldman Sachs Asset Management (GSAM) provided a preference share financing commitment of US$250 million to Hydrostore.

Highview Power Cryogenic Storage

Air as a storage medium can be compressed which enables it to store energy. When compressed and cooled to a low temperature it turns to liquid which can subsequently be heated (with waste heat if available) to vaporise and release the phase change energy (liquid to gas). This results in its volume increasing by 700 times. The gaseous air then powers a gas turbine connected to a generator. Highview provides an example of a company that has developed a liquid air system and has already sold its system to a number of customers. The graphic below shows the process.

Air turns to liquid when cooled to -196°C (-320˚F), and can then be stored very efficiently in insulated, low pressure vessels. Exposure to ambient temperatures causes rapid re-gasification, which is then used to drive a turbine and create electricity without combustion. This technology is said to allow storage from 4hrs to 4 weeks unlike batteries which are for short-term storage only. It has also been described as having the same giga-scale, long duration capabilities as pumped-hydro but without geographical constraints sometimes referred to as "pumped hydro in a box". In this respect it is similar to Hydrostore.

Compressed air energy storage (CAES) is also based around the gas turbine cycle. Power is used to compress air using a rotary compressor and then store it, often in an underground chamber. When power is required, air is released from the chamber and passed through a turbine that generates electricity from the flow of the high-pressure air. Output from the plant can be boosted by burning natural gas in the high-pressure air before it enters the turbine, as would happen in a conventional gas turbine. However this has the penalty of producing carbon dioxide emissions. More advanced plants can store heat during air compression and release it during the expansion phase (as in the Hydrostore system described above).

Compressed CO2 energy storage – Energy Dome

One issue with cryogenic energy storage is the difficulty of liquifying air. An alternative medium for liquefaction and subsequent use by re-gasifying and passing it through a turbine, is the gas, carbon dioxide. According to Energy Dome which has developed such a storage system, CO2 is the perfect fluid to store energy cost effectively in a closed thermodynamic process as it is one of the few gasses that can be condensed and stored as a liquid under pressure at ambient temperature allowing for high density energy storage without the need to go to extreme cryogenic temperatures.

On 25 July 2025, Energy Dome, announced a global commercial partnership with Google using Energy Dome's CO2 battery technology to enable carbon-free energy for the grids that power Google's operations. Alongside the commercial agreement, Google has made a strategic investment in Energy Dome. In contrast to Li-ion battery storage which provides backup power for 4 -6 or sometimes 8 hours, Energy Dome is capable of continuously dispatching energy for periods of 8 to 24 hours, providing enough firm electricity to meet both the baseload and flexibility requirements of large energy users. An additional advantage of this technology is that it uses a turbine to produce its electricity output. These turbines have 'inertia' just as those in power stations using coal or gas with a turbine and as inertia has always been necessary to stabilise the grid (some alternative methods are also being developed), it reduces one of the issues arising from large amounts of non-inertia based power on a grid.

Hydrogen as an Energy Storage Medium

Hydrogen Storage – electrical energy converted to chemical energy

Hydrogen can be stored in pressurized vessels, or solid metal hydrides or nanotubes can store hydrogen with a very high energy density but only up to a few MWh. Very large amounts of hydrogen can be stored in underground salt caverns at high pressure which might be able to store 100 GWh of energy. Balancing seasonal variations might be possible with this system.

Power Generation – chemical energy converted to power

Hydrogen gas can be converted back into power using fuel cells but there is an energy loss in the form of heat. This gives the possibility of creating a combined heat and power solution using a fuel using the waste heat from power generation to provide space heating in homes and offices. The efficiency of this conversion process from gas to heat and power is higher than from a system which produces only power.

Hydrogen energy storage is shown in the top right-hand corner of the ETC graphic at the start of this chapter. This is the corner where a technology provides long discharge times (LDES) and large storage size (a large amount of power). In terms of cost per kWh of storage, it is likely that the cost of hydrogen will have to continue to fall to make it economic but in terms of the global climate crisis, cost is only one factor to be considered.

Hydrogen is likely to have the greatest potential among technologies for **seasonal** energy storage. In the 2050-2070 timeframe, hydrogen with as much as two weeks of stored energy is forecast to be a cost-effective storage method. Total global hydrogen use would grow substantially if it were to be used for energy storage in the way that natural gas has been used for this purpose.

Rondo Heat Battery: Heat as an Energy Storage Medium

The Rondo Heat Battery uses electric heating elements, like those in a toaster or oven, to turn power when it's available at low cost into high-temperature heat. Electrical heaters (so-called Joule heaters) convert electrical energy into heat at 100% efficiency, and interact smoothly with grid and off-grid generation. Thousands of tons of brick are heated directly by this thermal radiation, and store energy for hours or days with very low loss (less than 1% per day). When heat is wanted, air flows up through the brick stack and is heated to over 1000°C. The heat delivery rate is adjusted easily by changing air flow. Heat is delivered as either superheated steam or as hot air.

We have already noted that one problem companies developing new technologies have is persuading either venture capitalists or infrastructure investors to back their projects. In Rondo's case, they have persuaded the Breakthrough Fund (originated by Bill Gates) to provide the necessary funding to enable them to offer not just "first of a kind" but 2nd and 3rd projects. If these are successful, Rondo will be well on its way to full commercialisation of its technology.

Heat energy can also be stored in molten salts. This is the mechanism by which the heat from TerraPower's nuclear reactor (see Chapter 13.5) can be stored in its Natrium storage plant when the reactor output exceeds demand for electric power and this heat can boost the power output of the generating system when the grid is in most need of it.

Part IV: Innovation in Systems/ Networks: Digitalisation, Business Models, Market Design and AI in Electricity & Heat Systems

Chapter 12: The Other Transition: Digitalisation – The Vital Requirement for the Green Transition

12.1 System Re-design and Innovation Necessary for the Energy Transition

There are three key factors in the energy transition that are leading to the need for new business models for producers of power, transmission and distribution network providers and electricity retailers. These factors are the **decentralisation** of generation, the **decarbonisation** of generation, and the **digitalisation** of networks and of retail distribution models, often known as the "3Ds". In addition, there are constantly changing frameworks within which companies in these industries must operate a result of changes in 'market design' resulting in part from regulatory change and partly from innovation by 'disruptive' companies operating in the electricity and other energy markets competing against those who are less innovative.

Much of the innovation in energy systems has been in energy storage, transmission and distribution of electric power and 'demand side management' systems which enable demand to be reduced to better match supply. In addition energy efficiency such as a reduction in the power required by individual appliances such as vacuum cleaners, washing machines and room lighting (LEDs) in the domestic context and power consumption in manufacturing processes can reduce the need for constructing new generation capacity. However, as power systems becomes more complex, it will increasingly be impossible for human beings to make all the necessary decisions to optimise 'the system' i.e. to achieve the best trade-off between energy security, affordability, and sustainability or the Energy Trilemma as it is known (we covered this in Chapter 10.1). Instead, it requires the abilities of an AI system to take in thousands of pieces of data every second and process it to make instant decisions on system optimisation. In addition, digitalisation and AI systems provide the opportunity to create business models which generate more value both for the company which has developed a new model and for their customers.

The international Renewable Energy Agency (IRENA) sees the need for innovation not just in the 'hard asset' and 'soft asset' technologies but in business models (private sector) and in market design (state regulation) as shown in the upper half of Figure 12.1[51],. Business models and regulation that will enable these new technologies to be used in an economic, i.e. least cost, fashion, are critical. There are numerous definitions of the term 'business model' but a useful one is '*organizational structures designed to enact a commercial i.e. profitable, opportunity*'. The business model can also be viewed as the way in which a company creates value which is demonstrated through the rate of return on the investment that it is able to achieve.

A new and innovative Market (Re) Design to reflect the different nature of the new power market to ensure that the new technologies can be attractive to investors is of key importance. The European Commission, for example, issued its proposals for the Revision of the EU's 'Electricity Market Design (EMD)' rules in March 2023. These are intended to accelerate the energy transition which is currently well off-track. Reform of the current EMD which is structured around pricing based on 'merit order', which is the sequence in which power plants are designated to deliver power to optimise the electricity supply in terms of the price of electricity, may not be achieving the lowest cost for consumers if they are unable to reap the benefits of low-cost renewables. The merit order is based on the lowest marginal cost producers being given priority in supplying power to the grid. Another issue is whether a region (such as Scotland in the UK) which produces low-cost wind power, should be allowed to have lower tariffs than a region which has to import power from such regions at a considerable cost in transmission fees.

[51] https://www.irena.org/Energy-Transition/Innovation/Systemic-Innovation and https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2019/Feb/IRENA_Increasing_time_granularity_2019.pdf

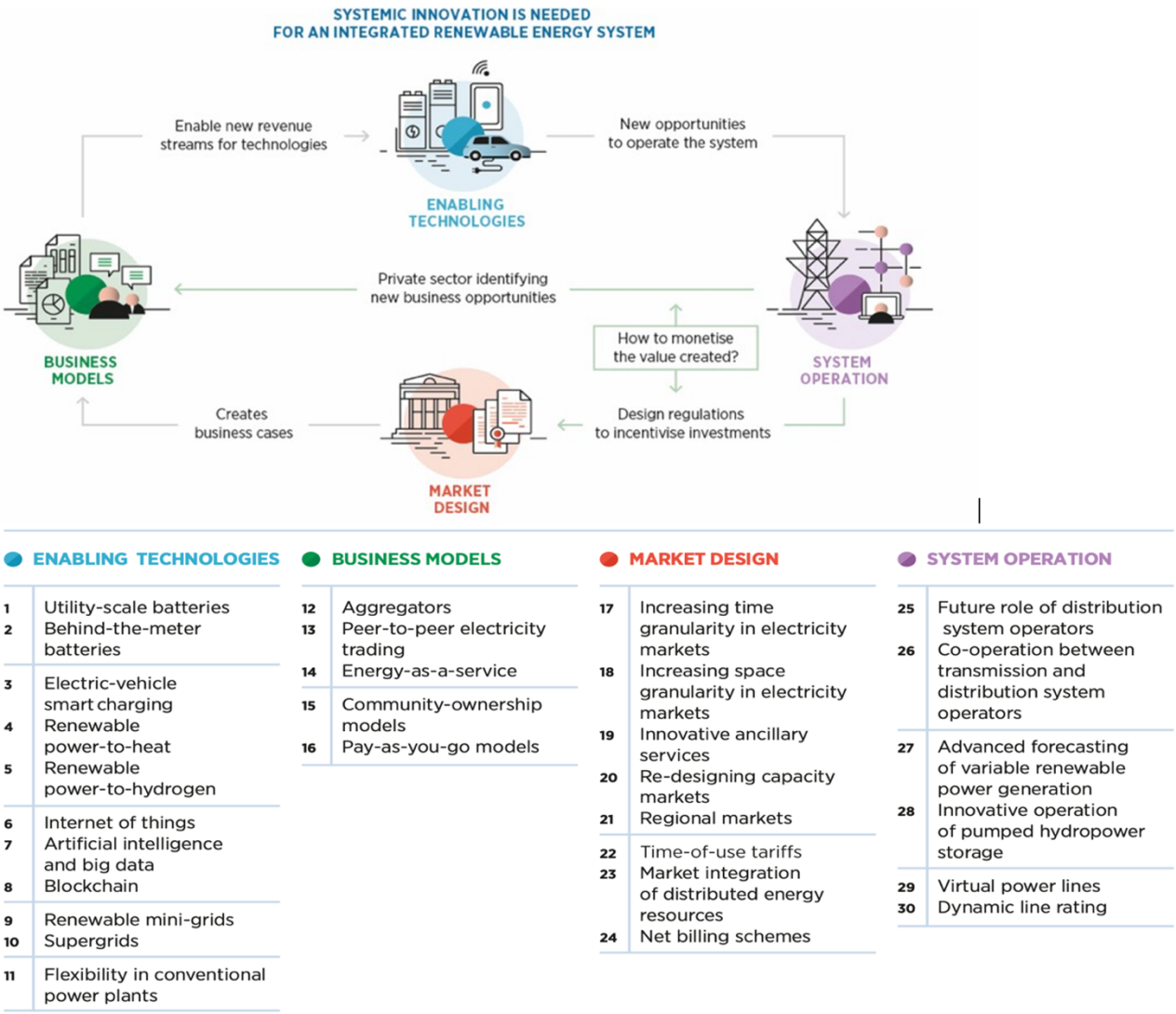


Figure 12.1 Systemic innovation in an energy system and what is necessary to achieve it

In the bottom half of Figure 12.1 under the heading 'Market Design', is the expression 'net billing'. Net billing is a way to charge but also compensate retail customers who supply power to the grid based on the actual market value of electricity (rather than just a notional figure), balancing what they consume against what they inject into the grid. It then raises a key issue for existing electricity utilities, what is known as 'Grid Defection'. This is the phenomenon of retail customers leaving the grid (going off-grid) at the point in time when technology has reduced the cost of photovoltaic (PV) panels and battery storage to a level at which 'home-made' power is actually lower cost than the supply from a utility. In some countries electricity prices vary substantially. For example, in the US, San Diego the rate is $0.69/kWh, while in Nebraska its only $0.06/kWh. Thus Nebraska utilities are much less likely to suffer defections from the grid than those in San Diego. For many countries this is likely to be a key issue for regulators and the design of their electricity market.

In the graphic the technologies and new regulatory frameworks are predicated on an increasing proportion of renewable energy in the electricity system and a much more 'distributed' system. At the top of the graphic are the new technologies and expanded and improved enabling technologies which we covered in subsequent chapters. On the right of the graphic is 'System Operation' which is having to change to reflect the new technologies. In turn this requires innovation by Transmission System Operators (TSOs) and innovation in business models by power companies if the investors who finance innovations and expansion in the existing solar and wind technologies are to be satisfied with the expected return from such new assets at the level of risk associated with those expected returns – the traditional, risk/required return investment framework. At this point, we are focussing on 'Enabling Technologies' 6, 7 and 8 which are principally digitalisation and IT. But as can be seen in the graphic every aspect of the energy system requires innovation. Note that under Market Design 21 and 22 refer to

a consumer pricing system where location and time of day (or imbalance on the grid) can determine the price paid for electricity or paid by the distributor.

Electricity market re-design in any country will need to be one which can best integrate variable renewable energy (VRE) projects into the national power system of a country without risking brownouts or blackouts. It also needs to be compatible with, for example, an ability by companies to develop a business model for pricing longer duration storage at a rate which provides an acceptable return to investors and can be used within the revised market structure. AI will be the key to optimising systems. To supply the AI system with the necessary data, all components of the physical system must transmit information to the 'central brain' and this requires the Internet of Things (IoT) i.e. sensors, which can pick up the required information from the physical assets including solar and wind generators and energy storage assets and transmit it to the centre. In addition there needs to be a system to request, receive and transmit information from satellite observation, prices from electricity markets, from the TSOs and from providers of weather data such as the **Met Office** in the UK and the **European Centre for Medium-Range Weather Forecasts (ECMWF)** which is based in Reading in the UK and will shortly move on to the University of Reading campus close to the university's Meteorology Department. This data must then be transmitted to the AI computer system.

*12.2 **EU Green and Digital Transitions**: Green Power and Digitalisation for 24/7 CFE*

Digitalisation is thought to be so important by the European Commission that it devotes a key section of its 2022 *Strategic Foresight Report* to energy markets and digitalisation.

Going back a little in history, at one time, power generation was centred in cities with one large power station to meet the needs of that city (often owned by a local authority rather than a private company). Gradually as power stations grew in size and number, they were located outside cities and were later connected to a national grid. Today **the major shift is to de-centralised power** with lots of small generating facilities (solar, wind and storge). By 2035 there are also likely to be many small modular reactors (SMRs) and advanced nuclear technology (ANT) providing power directly to users (rather than to the grid) or on the grid.

The problem this raises is the need to integrate renewables, particularly from small scale generation such as from rooftops and also and from SMRs/AMRs, into the power system. The issue becomes more challenging to resolve as the number of assets on the grid (solar farms, rooftop systems, wind farms, SMRs, AMRs, energy storage) grows. As a result, the need for systems comprised of power generators, storage systems and transmission lines (the physical assets), and cyber components (including sensors, IoT, and connections to external data sources) that control the operation of the grid has grown dramatically. This, of course, also provides investment opportunities. Controlling the interplay between such many supply sources (generation), Demand Management Systems (DMS) and storage would be virtually impossible for humans to undertake without these intelligent control systems.

What is happening in the energy field is one example (and a particularly important one) of what has become known as the Fourth Industrial Revolution. In Figure 12.2[52] we are focussed on the New Technologies of Today. The best way of looking at what is happening in the power industry (but also in many other industries) today is the implementation of what has become known as **Cyber-Physical Systems (CPS)**.

[52] World Economic |Forum (WEF) *3 ways to be a good leader in the Fourth Industrial Revolution* https://www.weforum.org/agenda/2019/01/the-fourth-industrial-revolution-needs-new-forms-of-leadership/

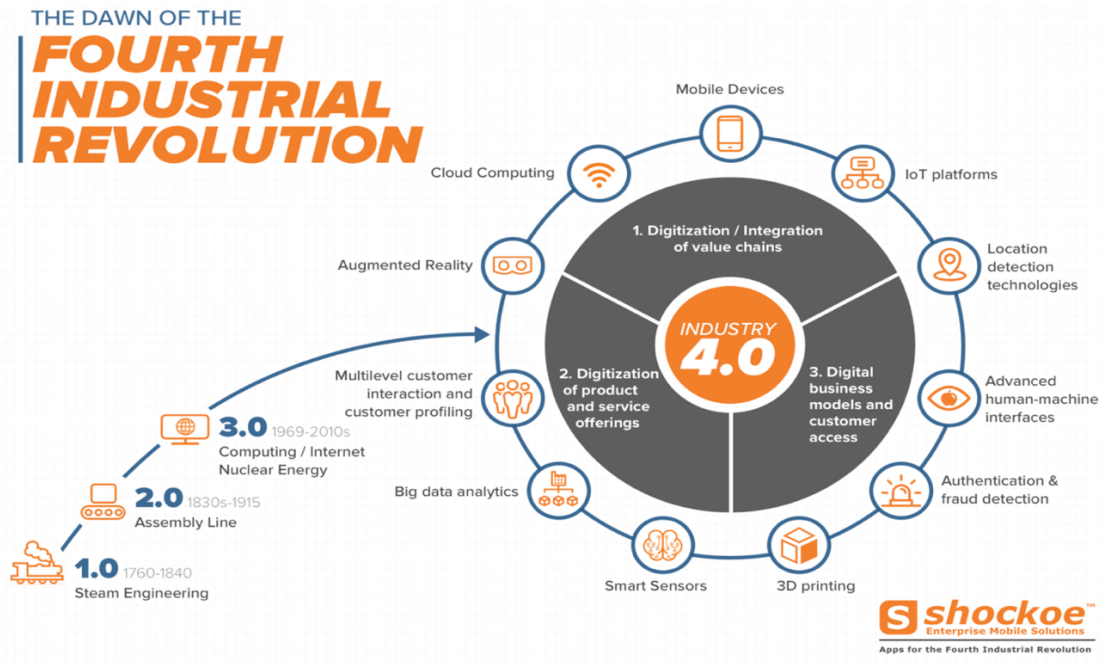


Image: Shockoe.

Figure 12.2 Where we are today – the 4th industrial revolution

These are systems that integrate sensing (for example, temperature, wind strength, power flows) and transmission of that sensor data through the internet (IoT). This then enables computation, control and networking of physical objects and infrastructure via their internet connections to each other. Such systems can also take in other information through the internet i.e. from sources other than sensors and can thus also integrate weather data (including forecasts), power market price data, data from transmission system operators (TSOs), cloud cover pictures from satellites etc. to enable decision making. Humans could in theory make these decisions making but in practice they have to be made by artificial intelligence (AI). The new technologies involved are listed under Industry 4.0, below.

Machine learning is often used to identify patterns in data and provide specific results. Machine learning (ML) has been improving by leaps and bounds, not only because of the skills of those designing ML systems but also because of the falling cost and increasing power of computer systems. This has led to the cost of 'compute' (to use the industry jargon for computing power) falling substantially and thus enabling much greater compute to be applied to decision making in, for example, the power industry.

Energy management systems have traditionally been rule or logic-based and specialised for the function of managing a collection of assets and their links to other systems. Recently some energy companies and project managers have been experimenting with the use of more advanced ML systems – Generative Pre-trained Transformers (GPT) such as Chat GPT which we covered above. However, it is important to understand that the use of GPT in the energy field (apart from its use in call centres) is at the research end of the spectrum and not at this time being let loose except experimentally on decision making for operating energy networks.

12.3 Case Studies of Software Companies Contributing to the Energy Transition

Below is a selection of relevant companies. Many others are also making a substantial contribution.

StorTera

StorTera, is an energy storage innovator in Edinburgh, which was awarded the $1 million prize along with their partner Equilibrium Engineering in the highly competitive Canada-UK Power Forward Challenge. One of the key issues in the competition was how to bring multiple stakeholders together, such as grid network operators and consumers, to share both the costs and benefits of upgrading an energy system to handle the integration of renewable energy, electrification of heating and charging of electric vehicles.

StorTera's innovative solution utilised an artificial intelligence platform to optimise the performance of the smart grid in real time by having access to **accurate forecasting of weather and energy demand for the day ahead**. The installation included intelligent energy storage, energy efficiency upgrades to buildings, the ability to control electrical loads such as hot water heating (demand side response).

Figure 12.3 shows a graphic of StorTera's TRAICON[53] (tri-layer artificial intelligence controller) platform which enables AI control and optimisation. The TRAICON system utilises neural network-based machine learning to analyse large datasets, monitor, optimise and proactively control all the connected assets. The controller learns local energy usage and storage patterns and uses cloud based machine learning to integrate weather forecasting and other available APIs allowing networked synchronisation of multiple systems and continually optimised energy performance. The key innovation is the use of an artificial intelligence (AI) platform that manages data and control ranging from battery cell information to sophisticated algorithms that predict the solar energy and energy demand of customers for the day ahead. The graphic below shows the data inputs and associated APIs required for optimum AI decision making and the links to power users.

Figure 12.3 AI control and optimisation of an energy system

Fluence IQ & Nispera

Fluence IQ, delivers AI-enabled digital applications for managing and optimizing renewables and storage *from any provider i.e.* it is technology and storage cell provider agnostic. In 2022 it acquired Nispera AG advanced technology which enables customers to monitor, analyse, forecast, and optimize the performance and value of renewable energy assets. Its flagship offering is an AI-driven utility-scale asset performance management platform.

In combination with Fluence IQ's digital application offerings, Nispera's suite of five applications enable the combined firm to offer its customers a user-centric "manage app". Nispera's technology portfolio includes a predictive maintenance app, an operations and maintenance app, and a portfolio management app. The company also offers a fifth application that provides enhanced power generation forecasting services for energy assets that is particularly attractive to energy traders, large asset managers, grid operators, and financial institutions.

[53] www.stortera.com

Virtual Power Plants (VPP)

ABB

AI systems can control a wide variety of de-centralised assets (demand side and supply side) of different types over large areas to aggregate them into 'a virtual power plant'. The collection of assets owned by different people and companies can then function as if it is a 'power station' providing continuous power.

In more detail, VPPs are cloud-based data control centres that aggregate data from distributed energy resources (DERs). Examples are solar PV plants, battery storage facilities, residential storage systems, electric battery-powered road vehicles and other sources of power demand and supply which are connected to the grid. The data centre uses the internet and 5G mobile services along with the information supplied by sensors connected to internet of things (IoT) sensors to gather data. An operator can then monitor and control the production of each plant to optimise security of supply and cost. Alternatively, this optimising will be undertaken by AI algorithms. A number of different power sources such as solar, wind, small hydro supported by battery storage and also by demand side management systems can create a 'virtual power plant' or VPP. ABB is one of the companies which provides this type of software. A VPP can overcome pressure on the grid at times when demand would otherwise exceed supply by aggregating various energy sources (and enabling a reduction in demand from demand management) into a virtual power plant coming on to the grid almost instantly when required.

Next Kraftwerke

Another provider of the services needed to provide VPP services is Next Kraftwerke. The strapline under its name on the logo explains succinctly what it is and hence the service provided by such a company: THE POWER OF MANY as it enables many types of asset to be connected to the network. That network can respond either to the grid operator when it demands flexibility by supplying it or (and) second, it can supply that flexibility to the power system by trading its ability to provide flexibility on an energy exchange.

Piclo

This company is a "flexibility marketplace provider". It offers two platforms. The first is Piclo Max which is for flexibility service providers to help enable them to maximise revenue stacking opportunities for flexibility. It offers buyers and sellers a 'one stop shop' for all electricity markets globally. For transmission system operators, it offers Piclo Flex which is a marketplace for local flexibility currently in six countries. Piclo Flex provides an end-to-end solution, for flex buyers (system operators) world-wide. Piclo is headquartered in London.

GridBeyond

GridBeyond defines its role as intelligently dispatching flexibility into the right market at the right time. By achieving this, it supports asset owners and energy consumers to unlock new revenues & savings, achieve resilience and manage price volatility helping support the transition to a Net Zero future. Grid Beyond connects up distributed energy resources and electricity markets. It bridges the gap between distributed energy resources around the grid and the requirements of the grid at any point in time.

The need for such Optimisation Solutions arises from the ever increasing proportion of renewables on the grid whose power outputs vary continuously creating an ever increasing need for grid balancing services. This, in turn, requires real-time solutions to manage this new power market structure. A technology platform is needed to connect distributed energy resources (DER) and grid-connected technologies that consume, generate or store electricity. When optimised across traded power markets and flexibility programmes, DERs can lead to significant cost savings for grid operators and utilities.

GridBeyond provides a link between renewables generators and energy buyers, enabling them to hedge against risk and gain benefits through corporate PPAs. By accurately forecasting forward power prices and a company's business demand its AI-powered optimisation solution runs Monte Carlo simulations to match site demand with managed renewables generation and battery storage assets, helping manage price risks and make savings per MW. The linkages are shown in its graphic below:

The company works with existing and soon to be operating renewable energy projects to structure long-term offtake agreements that match a customer's demand and renewable energy goals. While there is no direct, physical connection between the power generator and the customer the PPA guarantees that the energy the

customer offtakes from the grid can be traced back to a specific wind or solar farm that feeds the equivalent power into the grid. This is evidenced with certificates, which Grid Beyond issues for every MWh generated. The certificates prove that the electricity purchased is renewable and comes from a specific renewable energy asset, for example a wind farm or a solar farm, to support a company's Scope 2 GHG reporting.

GraphCast – a new way of forecasting solar power generation

Traditional forecasts such as those from the Met Office in the UK have relied on physics based models. They solve equations based on air flow, moisture and atmospheric thermodynamics. This new approach to short-term (one or two hours ahead) is quite different. It makes predictions based on gigabytes of previous data on similar weather conditions. It is said to take only seconds or minutes to come up with a forecast rather than the hours taken by the traditional computing approach. The new approach uses "prediction engines" which feed historical data into Google's AI Deep Mind weather simulator known as GraphCast. In weather forecasting, Machine Learning (ML) systems such as this are beginning to improve on the traditional Numerical Weather Prediction (NWP)-based forecasting models while also being lower cost to run and very much faster and therefore more useful.

A 'hoped for' use of this new technology is to provide more accurate and much faster and up-to-date forecasts of wind strength and of cloud cover which reduce solar output. To cope with the often quite dramatic changes in output of wind and solar power into the grid, grid operators (TSOs) schedule, and request power-up from natural gas turbines well in advance in case power production from solar falls short of power demand. Thus if the approach proves effective it will make a significant impact on GHG emissions from power generation by reducing 'false' gas switch-ons. The alternative to natural gas for load balancing flexibility is, of course, on-grid battery storage which is immediately dispatchable and which would also benefit from more accurate weather forecasts.

Iberdrola as an Exemplar of Web 3 and Blockchain in Genuine 100% Clean Energy

Web3 is a term used to describe the next iteration of the internet built on blockchain technology. A blockchain is a digitally distributed, decentralized ledger across a computer network to record transactions. As new data is added to a network, a new block is created and appended permanently to the chain. All nodes on the blockchain are then updated to reflect the change. In consequence the system is not subject to a single point of control or failure. Web3 refers to decentralized databases and systems architecture.

Iberdrola Group has a pilot project based on using blockchain to guarantee, in real time, that the energy supplied and consumed is 100 % renewable i.e. 24/7 CFE. Using this technology, they have managed to link plants where electricity is produced to specific points of consumption, allowing the source of the energy to be traced. This increases transparency and ultimately encourages the use of renewable energy. These systems are used by energy companies to enable them to achieve the best rate of return on capital from their portfolio of energy generation assets (wind, solar and storage), given time of day power pricing and customer demand (and demand management) patterns.

FlexiDAO

This is a European climate tech startup which provides end-to-end carbon–free energy solutions for energy buyers and sellers. Its software as a service (SaaS) platform enables organisations to monitor, quantify and demonstrate to society, their contribution decarbonisation. Their vision is that we should be living in a 24/7 carbon free energy world. They envisage this happening by using their Energy and Emissions Management Platform which enables corporates to accurately measure, report and reduce their energy-related emissions. Iberdrola and Red Electrica de Espana are two of the energy companies that use FlexiDAOs SAAS.

ACCIONA Energía

In Continental Europe, ACCIONA Energía, said to be the world's largest fossil-free utility operating exclusively in renewable technologies, has become the first utility to use blockchain technology to certify the origin of its energy to consumers. It has created a commercial demonstration to trace renewable generation from five wind and hydro facilities in Spain through to four corporate clients in Portugal. The company used a platform and software developed by the Energy Web Foundation (EWF). EWF believes that blockchain technology has the potential to reduce transaction costs, enhance data security and transparency, open markets to a larger number of active participants (consumers and devices), and accelerate the transition towards a digitalized, decarbonized, decentralized, democratized and resilient energy system.

Energy Trading – the shift from manual trading to algorithmic trading

We noted that controlling energy systems is only practically possible using AI assistance. Figure 12.3 provided an example in a graphic of the Traicon system which uses AI as a control and optimisation mechanism. While such systems are used for controlling **physical** assets in an energy (electricity) network, there are also energy traders who may, or many not, work for an electricity utility who just want to trade electricity in both **spot and forward markets**.

Many of the large investment banks have energy trading desks just as they have interest rate trading desks. Such traders will be arbitraging markets to take advantage of large price swings over short and longer periods of time. As with arbitrage more generally, this should reduce price volatility as well as generate an acceptable rate of return on their investment in trading.

As an example of available opportunities, electricity spot prices in Germany can switch from negative on days when there is too much power to more than €600 per megawatt hour on days of "Dunkelflaute" (windless and sunless periods). As renewables provide an ever higher proportion of the electricity in a system, the need for real-time balancing of energy markets is increasing. In addition, energy markets and thus the price of electricity responds quickly and sharply to geopolitical tensions, macroeconomic news, the regulatory frameworks, new technologies and their expected impact, the extent to which demand flexibility is available in a system and, of course, the weather.

12.4 Developing Data Standards to Enable Inter-Operability

BRIDGE

This is an EU initiative to gather all the varied Horizon 2020 smart grid innovation projects. It is designed to foster collaboration and consolidate results between the research & innovation projects in smart grids, energy storage, islands and digitalisation of the energy system funded under the Horizon 2020 program. Since its creation, it has gathered 80+ projects involving 700+ organisations from 38 countries for a total EU funding of €750m. BRIDGE is structured around 4 working groups: Business Models, Regulation, Customer & citizen engagement and **Data Management**. The Working Group on Data Management is focussed on:

- **Communication Infrastructure** – the technical and non-technical aspects of the infrastructure needed to exchange data and the related requirements, such as issues faced by TSO and DSO
- **Cybersecurity and Data Privacy**, entailing data integrity, customer privacy and protection
- **Data Handling,** including the framework for data exchange and related roles and responsibilities, together with the technical issues supporting the exchange of data in a secure and interoperable manner, and the data analytics techniques for data processing.

Trusted Energy Interoperability Alliance, (TEIA)

In the US, several organisations have come together to try to tackle the issue of standards and interoperability. The organisations involved include E.ON SE, Intertrust, JERA, and Origin Energy which in 2023 were the founders of the Trusted Energy Interoperability Alliance, (TEIA), set-up o develop and promote open technology standards for energy devices, data interoperability and security for the global energy industry ecosystem.

UK National Grid ESO (now called NESO) and the Energy Data Taskforce

The UK national energy regulator (Ofgem) along with the government established the Energy Data Taskforce to provide a set of recommendations on how data can assist with unlocking the opportunities provided by a modern, decarbonised and decentralised Energy System at the best value to consumers. This work is part of the Smart System strategy being developed by Government.

The Energy Data Taskforce's aim is to provide a set of recommendations that will support the delivery of a Modern, Digitalised Energy System unlocking the benefits of decarbonisation and decentralisation through better use of data. The Taskforce believes that digitalisation releases value, opportunity and resilience, enabling decarbonisation and decentralisation to be delivered at optimal cost for the benefit of consumers. As the system becomes more disparate, diverse and decentralised, data sharing will be crucial to coordinate the wide range of actors undertaking new roles across the sector and ensure system stability.

Part V: NUCLEAR – A New Approach to Clean Energy: Innovation in Technology, Co-Location of Supply & Demand and Nuclear Financing

Chapter 13: Nuclear and renewables have quite different characteristics

It is likely that wind and solar power along with energy storage and demand management will eventually provide more than 50% of the generation capacity in most countries. What percentage of renewables will in practice be possible or optimal, however, in the sense of ***minimising the cost of electricity and heat****, ensuring* ***supply security to customers*** *but also enabling* ***supply security from 'bad actor' actions from Russia, China or Iran*** *(the current actual and potential bad actors) is not yet clear. A clear risk in the case of offshore wind is 'cable cutting' using unmanned small submarines. Not only could such attacks give rise to power brownouts and blackouts but re-joining cables in deep water offshore is not a simple task.*

Some proportion of the balance of generation capacity will almost certainly be made up by using 'dispatchable' 24/7 power from nuclear, hydro and geothermal with a proportion of natural gas generation which may be required to cover long periods of 'dunkelflaute'. This is a German word meaning dark doldrums referring to a period of low or no wind and minimal sunlight which impacts renewable energy generation. It is also known as 'anticyclonic gloom' and can continue for many weeks over large geographical areas such as most of the European Union.

Another phenomenon, this time one which affects hydro, is a lack of rainfall which lowers electricity output as the water level in reservoirs falls. This also may require the installation of genuinely 'dispatchable' generation capacity of which nuclear is likely to be the key technology if a country is taking de-carbonisation seriously and minimising the need for back-up natural gas generation.

A good example of the type of problem we cover above is demonstrated by developments in Italy. In January 2025 renewable energy production fell compared with the previous year even though installed renewable capacity grew. This decline was mainly driven by a drop in wind (-8.7%), hydroelectric (-5.1%), and photovoltaic (-0.5%) production. In contrast, generation from coal and gas increased by 11.5%.[54] *The problem of a massive blackout in Iberia which we cover later, also suggests that electricity systems in Europe more generally need modernising as ever more power from renewables needs 'integrating' into the grid system while avoiding the risk of electricity shortages and blackouts.*

We have not given extensive coverage in this book to solar PV generation technology or to wind turbine technology as these technologies are well known and well developed while 'new nuclear' employs new technologies which are not yet widely understood. For this reason, Part V on nuclear is the most extensive part of this book.

In this chapter we focus on Europe as a region that has been trying to decarbonise the economy rapidly but is falling behind targets despite the high degree of effort that has been expended. But the same logic applies worldwide. What we expect to happen, which will make a contribution to meeting decarbonisation targets in many countries, is the use of the new nuclear technology that is about to come on-stream to overcome some of the problems that are becoming apparent in trying to de-carbonise an economy **at the pace required to keep the global temperature rise below 2°C while focussing only on renewables. It will also be necessary to** meet the electricity 'demands' of the hyperscalers' datacentres.

13.1 Embedded Nuclear and Five Key Differences Between Renewables and Small and Micro Advanced nuclear technology (ANT) reactors

Nuclear **can facilitate the integration of renewable electricity into a grid system** (filling in the gaps from dark nights and light winds), can provide baseload power (24/7/365 dispatchable electricity) can provide industrial heat and space heating (which critically and urgently require de-carbonisation) and can be used as a heat generator for energy storage in a heat battery for use as industrial heat or for subsequent conversion to

[54] Statistics from Terna, the Italian grid operator.

power. It can also provide portable power from micro-reactors which can be easily transported to where heat and/or power is needed, embedded within an industrial complex either as a single unit or in multiple units or used to generate heat or heat and power. The micro reactors we cover can also be very attractive for military use to generate electricity and heat as an alternative to diesel generated power which is noisy and emits fumes and is therefore easily detected by an enemy. Micro and small reactors can also have a role to play in maritime transport in particular for container ships. In addition, some companies are working on 'nuclear barges' which can be towed to where power is required and moored just off the shore. There are already many submarines using nuclear propulsion and also icebreakers which have not caused any dangerous incidents over many decades. In future, nuclear power is likely to be used much more widely at sea as its safety comes to be understood better.

As we noted above, the European Commission is one of the most active organisations trying to ensure the maximum contribution to global decarbonisation by European countries. On 26th February 2025, the European Commission issued a document: the **Clean Industrial Deal** (CID).

> "The EU aims to become climate-neutral by 2050, with the Clean Industrial Deal and the Green Deal Industrial Plan as key strategies to support this, focusing on **decarbonizing industries** and boosting clean-tech sectors through increased investment and simplified regulations".

While the objective of the Clean Industrial Deal is to decarbonise industry, a well-considered plan to decarbonise space heating in domestic residences and commercial premises and in industry is also required.

Much of this decarbonisation will involve **energy in the form of heat rather than electricity.** It will also require the production of much more green hydrogen (some produced using low cost nuclear **heat** and not 100% electricity) rather than hydrogen made from natural gas. In fact, electricity is only around 20% of the total energy market while heat is 40% and transport is also 40%. Depending on the temperature of carbon-free heat required, it can be provided by electricity (power to heat) using resistance heating, induction heating, electric furnace, electric boiler, industrial heat pump or from nuclear heat.

The key problem in Europe's conversion from fossil to carbon-free-energy (CFE) is the difficulty of integrating CFE **in ever higher proportions into an existing transmission and distribution system designed for very different energy sources.** The traditional energy system built many decades ago was very different from the one it is hoped to have later this century and certainly by 2050 but might benefit from a reversion to some aspects of that system (community energy). As the International Energy Agency notes in a recent Report[55]:

> *"This report emphasises that while systems at early phases of VRE integration can accelerate deployment with relatively low system impact, those in high phases face more complex challenges related to* ***stability and flexibility, which call for a transformation of how power systems are operated, planned and financed****."*

Traditionally, electricity was generated in power stations relatively close to demand which is very different from today when an ever increasing amount of electricity for European customers is generated in the North Sea or from distant solar farms. On the other hand, in Europe users of electricity are mostly many hundreds of kilometres south of the North Sea and distant from good solar or wind locations. When the existing legacy transmission systems in many countries were constructed many decades ago using pylons with overhead cables, it was relatively straightforward to connect electricity producers with electricity users and generally these were not very far apart. In those days it was also possible to overcome resistance from people living near intended transmission routes. Today the distances required for transmission are so much greater and opposition by local communities so much stronger that it will take many years to connect all generating facilities to users if it is possible at all. The cost may increase sharply if opposition to pylons and overhead cables is such that energy networks must be installed underground.

Although expanding and strengthening the network is necessary and subject to difficulties and delays, ENTSO-E cost-benefit analysis suggests that implementing the whole portfolio of planned transmission projects would create up to 33 billion euros increase in socio-economic welfare for EU citizens in 2040 (i. e.,

[55] IEA *Integrating Solar and Wind* September 2024 https://www.iea.org/reports/integrating-solar-and-wind

socio-economic welfare generated exclusively in that single year). However, what is planned across the EU is, according to ENTSO-E **unlikely to be sufficient to achieve the hoped for outcomes.**

One solution to the transmission problem, though a somewhat limited one, is greater use of rooftop solar though this would have to be connected to adequate energy storage which could be in a community energy storage facility rather than individual homes which is a less efficient solution. Such a system can provide grid benefits beyond just the power that it produces. It generates electricity exactly where it will be used and therefore does not require additional transmission capacity (unless also connected to the grid.) As well as reducing transmission costs, it can also help reduce line losses and congestion without requiring new land for solar farms.

What we are suggesting for nuclear would be on a much larger scale and would make a much larger impact on overcoming the problem of lack of transmission capacity. It would also bring benefits in terms of de-carbonisation particularly through its ability to decarbonise heat. To achieve these benefits, there is a need to revert to some aspects of the **model of yesteryear** when electricity generating capacity was sited near domestic and industrial users in cities or regions. Today it can also be **embedded within an industrial facility** or, as we cover elsewhere, next to a **hyperscale datacentre**. We propose that the best solution would be to further facilitate the use of small modular reactors (SMRs) but more particularly **advanced nuclear technology (ANT)** high temperature **small and micro** reactors **embedded in communities** [somewhat parallel to the EU idea of 'Community Energy'] and in industrial zones. A key advantage of nuclear (and also geothermal) is that unlike renewables, it provides continuous power 24/7/365 (**dispatchable power/ baseload**) and other necessary grid services which it is difficult, and can be expensive, for renewables alone to provide as the proportion of renewables on the grid increases.

In Chapter 10, we noted the issues involved in trying to 'convert' renewable intermittent power into dispatchable 24/7/365 electricity. This is a 'system' issue. In designing an electricity system with a high proportion of renewables, one key is ensuring that a sufficient proportion of the assets on the system are 'system-beneficial' assets. Battery storage clearly is but such systems do not generate power though they are of critical importance in providing instant availability of electricity when there is a short-term system imbalance but are less useful for longer periods when actual **generation capacity** falls short of demand.

Interconnectors to other countries are also invaluable day-to-day to match supply and demand over a much wider geographical area as renewables generation will vary by geography and the time of peak demand may also vary. There thus is a 'portfolio' diversification benefit of reducing risk (in this case the risk short-term fluctuations in supply of electricity, brownouts or blackouts). However, if there is a long 'dunkelflaute' (using the German term) this means it is both dark (which increases electricity demand and reduces solar supply) and a fall in wind force (also resulting in less power is being generated.) In this situation, if wind strength is down across Europe for days at a time, interconnectors are less useful.

We summarise below why today's and tomorrow's nuclear 'fits the bill' for datacentres while it can also enable the necessary integration of higher proportions of renewable power into the grid system. Nuclear, as well as supporting renewables grid integration is also ideal for providing baseload power to the system. Finally we must also mention the need for heat, both in industry and in space heating where nuclear can supply relatively low cost heat for district heating systems from its 'waste' heat.

There are five key differences between renewables and small and micro advanced nuclear technology (ANT) reactors:

1. nuclear power is **dispatchable** 24/7/365 unlike variable output renewables which require ever more storage, demand management and backup gas generation;
2. advanced nuclear technology (ANT) nuclear as well as providing electricity can also provide **high temperature heat for industrial processes** (as well as low temperature heat for district heating networks and thus can replace coal and gas which renewables cannot achieve so easily;

3. it can help **reduce the problem of how to transmit ever larger amounts of power** requiring ever more transmission capacity for example between the North Sea wind farms and industrial centres in Europe as it enables the source of electricity to be located close to demand in built-up areas as it was in the early days of coal power stations and also town gas manufacture from coal;
4. micro [high temperature or low temperature] reactors, much smaller than SMRs, can be embedded within an industrial complex which reduces some of the transmission and distribution problems that countries face today and thus should also enable the power to be provided to industry below the "zonal price" laid down by the energy regulator. This could substantially reduce the price of power for industrial use.
5. Nuclear can provide 'inertia' services and voltage stabilisation. Generators using steam turbines have spinning rotors rotate at the required frequency to help balance supply and demand and can spin faster or slower if needed. Also, the kinetic energy 'stored' in these spinning turbines provides system inertia. Turbines continue spinning even if the generator itself has lost power and thus slow the rate of change of frequency (normally 50 cycles per second) allowing the central control room more time to restore balance. This provides a dampening effect on the system. While new ways of trying to achieve 'synthetic' inertia are being trialled, having baseload nuclear mostly overcomes such problems.

13.2 The Convertibility of Heat to Power & Power to Heat

Much of the urgently required decarbonisation across the world will involve **energy in the form of heat as well as the use of electricity.** The use of heat can also enable the production of lower cost green hydrogen using some proportion of low cost nuclear heat in electrolysers rather than 100% electricity. In fact, as we have already noted, electricity is only around 20% of the total energy market while heat is 40% and transport is also 40%. Renewables can be a source of heat but they can only produce heat by passing electric current through a resistance. Nuclear is likely to be ever more necessary in its role in providing industry with high and low temperature heat, electricity and hydrogen.

Heat has three principal functions in the energy system:

1. Low temperature heat is used for space heating in domestic and commercial premises using either heat pumps or district heating systems
2. High temperature heat is required for industrial processes mainly directly as 'dry heat' or as steam.

Heat is also used to produce steam for **dispatchable** generation and this heat can be from nuclear, geothermal or concentrating solar, all of which can provide Carbon Free Electricity (CFE) using a turbogenerator set. Heat also enables relatively low cost energy storage with that stored heat later be converted to electricity.

Almost half of the final energy consumed globally is for the provision of heat itself (1 and 2 above). More than half of this heat goes towards meeting the space heating and water heating requirements of our homes, offices and commercial activities. The balance is mainly for food processing, general manufacture as well as the 'hard to decarbonise' industries – iron & steel, cement, glass, aluminium, chemicals and paper. In future new processes may be created for these using carbon free electricity or hydrogen as a fuel and as a reagent and from dry nuclear or nuclear steam or concentrated solar heat using mirrors in desert sites (such solar is discussed in Chapter 14). In addition, heat can be used for the thermochemical or electrolytic splitting of water into hydrogen and oxygen with the hydrogen being stored for subsequent use. It can then, if required, be used as an energy storage medium (to be converted back to electricity through a fuel cell on demands) though the efficiency of this process is low.

Globally, most heat is produced by coal and natural gas (and wood). But while natural gas may emit only half as much CO2 as coal there are also frequently high levels of 'fugitive' emissions of methane (itself a potent greenhouse gas) from the extraction and transport of natural gas. It can thus only be a very short-term expedient to help de-carbonisation. Today, the role of natural gas in Europe has come to centre stage after the Russian invasion of Ukraine as Russia was providing up to 40% of Europe's natural gas and a higher proportion in the case of Germany. Today, much of Europe's natural gas is imported from the United States as Liquified Natural Gas (LNG) though unfortunately its liquefaction, transport and regasification are energy intensive. Three products **– carbon free heat, carbon free green electric power and carbon free green hydrogen for industrial heat and chemicals** can all be produced using nuclear as can desalinated water. This latter point may become more important in Europe as sources of clean water become scarce due to droughts.

Heat itself can be stored unlike electricity which can only be stored as other forms of energy such as electrochemical (battery) storage. Heat stores in district heating systems typically have insulated hot water tanks holding many thousands of litres as a simple and cheap way to enable low temperature heat supply to be matched to demand. High temperature heat storage such as is used in, for example the TerraPower Natrium reactor, usually takes the form of molten salts where the latent heat of phase change (from solid to liquid) is used to keep down the size of the expensive thermally insulated storage vessels. Some Generation IV high temperature nuclear reactors use molten salt both as a heat transfer medium and as the carrier of fissile material (nuclear fuel).

Heat can be classified into high temperature >600 °C and lower temperature <250 °C. In the absence of fossil fuels, only nuclear and electricity can produce high grade heat from which industrial value can be gained. Low temperature heat is useful for space heating and for horticulture. Very high temperature heat above 750 °C can also energise direct thermochemical decomposition of water (H2O) into hydrogen and oxygen more efficiently, i.e. at a lower cost, than by electrolysis. This thermochemical process has driven the ongoing development work on advanced Generation IV nuclear reactors which may be able to operate at 800 °C and above which also opens other essential clean energy needs such as green steel.

Heat, unlike electricity, is expensive to transport generally needing large diameter pipework for outbound and returning fluid usually superheated steam or water. This limits heat distribution to a few kilometres from the source which **favours small local energy plants typically owned by the community, municipality or the city**. Hydrogen as a gaseous fuel is easier to transport than high temperature heat although some parts of the existing gas pipe network in a post-fossil world could become so-called 'stranded assets' i.e. of no economic value. Some of the gas network may be suitable for hydrogen transmission but other parts may not be suitable for metallurgical and safety reasons.

13.3 Nuclear Desalination using Heat and Power, Nuclear Hydrogen and Regional Economic Growth

In Europe, Spain is by far the country with the largest desalination capacity totalling nearly three quarters of total capacity. It is followed by Italy, Cyprus, Malta and Greece. But the need for fresh water supplies is rapidly increasing. Cyprus would seem to be particularly exposed to water shortages not only due to drought but also to under-investment in water systems. It has been reported that in June 2025, the country's reservoirs were only around 22% full compared whereas in 2020 they were at 97% capacity in the same month of June.

Another region which may require desalination of sea water is the Oxford/Milton Keynes/ Cambridge Arc where the government aims to massively increase housebuilding and science research facilities and other infrastructure such as electric rail lines. Oxford and Cambridge have both suffered from shortages of water and electricity that have led to environmental regulators temporarily blocking housing and science developments. Generation IV nuclear might be the optimal solution to enabling further economic growth in this region by increasing the supply of electricity, heat, fresh water and, if required, hydrogen (as covered in the paragraph below).

According to the International Atomic Energy Agency (IAEA)[56] nuclear hydrogen production technologies have great potential and advantages over other sources that might be considered for growing the hydrogen share of energy in a future world energy economy. The selection of hydrogen technologies (to be coupled to nuclear power reactors) greatly depends on the type of the nuclear power plant itself. Some hydrogen production technologies, such as conventional electrolysis, require only electric power. Others, such as thermochemical production may require only process heat. High temperature steam electrolysis (HTSE) and hybrid thermochemical cycles can also be used and these require both heat and electricity. While at present the only nuclear power available is generally low temperature (around 300°C), once Generation IV reactors are available at scale and commissioned, they will be able to provide heat at 600°C and above which may eliminate the need to use electricity as well as heat in the production of hydrogen.

[56] https://www.iaea.org/topics/non-electric-applications/nuclear-hydrogen-production

13.4 But is Nuclear 'Safe'? Lifetime & End of Life Risk Management of Nuclear

Nuclear is one of the few carbon-free sources of heat. It can also, of course, produce electricity. *Decarbonising both heat and electricity (and hence domestic and commercial heating, transport and industry) is the basic requirement for mitigating climate change*.

> Do you believe humanity can invent something to stabilise the climate? *I always advocated nuclear as a good and cheap and sensible way of getting energy…. But too many people hate it.*

The above quotation, published by The Observer newspaper in July 2020, came in an interview with the creator of the Gaia concept, James Lovelock when he was age 101 (someone who was certainly 'green' and very aware of the global risk from climate change). Like him we believe that nuclear is likely to be essential to stabilising Earth's temperature – a task which requires a rapid increase in all forms of clean energy production. New reactor technologies, providing both direct heat and electricity include Gen III+ (both Gigawatt scale and SMRs) and small and micro sized Gen IV Advanced Modular Reactors. AMRs are very different from previous generations of nuclear and utilise completely new, inherently safe designs based on physical principles and not engineering add-ons which was the case with legacy reactors.

The UKs National Energy System Operator in its December 2024 Report entitled Clean Power 2030, states (in relation to electricity alone) that:

> "Nuclear power will play an important role in achieving a clean power system by 2030 and beyond into the 2030s". On SMRs and AMRs specifically, the Report states "Government could consider the value of bringing this rollout forward".

Nuclear fission as a source of heat and power remains controversial but decreasingly so in Europe. Factually, given the number of reactors in the world (over 400 in 32 countries) there have been few accidents and very few deaths compared with most human activities of comparable scale and purpose. Three – Fukishima (2011), Chernobyl (1986) and Three Mile Island (1979) have had 'incidents'. Those reactors which had incidents used what are now old and long since abandoned technologies developed at a time when knowledge levels were very much less then today and all reactors designed since have taken note of the root causes of these accidents and have achieved design and operational standards to control risks to much higher standards.

Charles Perrow's *Normal Accident Theory* describes "unanticipated interactions of multiple failures in a complex system". Perrow believes that the failure at Three Mile Island was a consequence of the system's immense complexity. Such systems are at high risk particularly if not well managed. He then suggested the need for a radical redesign of reactors. In fact, designers of Generation 3+ and *a fortiori* Generation IV, have taken note of his observations. Generation IV are generally designed **without using water for cooling and heat transfer**. Generation IV reactors are specifically smaller-sized and their resultant inherent physics enables them to function safely through their inherent design without human intervention or external power in all physically possible internal and external conditions. *This was not possible for the larger reactors of the past due to their being sized far above a critical geometrical limit for self-dissipation of their own decay heat.*

Italy had Gen II and III nuclear power plants in the 1960's and 1970's but after the Fukushima tsunami, in a referendum, citizens voted against further expansion of the nuclear fleet. Subsequent under pressure from activists, the government decided to shut down its existing plants. In practice, however, the exposure doses from radioactive iodine-131 of Fukushima releases were less than one thousandth of that from natural radionuclides. The incident did, however, create a high level fear amongst the public across Europe.

More recently, the energy security minister in the Italian government disclosed that Italy is now planning to have Generation IV small modular reactors operating within 10 years. He believes it is important to reduce dependence on fossil fuels (which includes Russian gas and oil). Italy has now set a target of at least 11% of electricity generation being from nuclear by 2050. The Italian government has also recently imposed restrictions on new solar power projects in the belief that they threaten food security by using cultivatable land. **So for the Italian government, there is a trade-off between energy security and food security.** Also in Italy, as we note elsewhere, the Italian government has been concerned about the growth of solar farms reducing agricultural output.

Some people, not surprisingly, are concerned about the risk of having a nuclear reactor in their near vicinity. Indeed many people see Chernobyl as the worst energy disaster in history. But deaths as a result of nuclear are

lower than for other types of energy generation. For example, while the UN has registered only 43 deaths as a result of Chernobyl, in comparison 240,000 people lost their lives when the Banqiao Dam in China (which provided hydropower) collapsed. Another statistic would compare the 43 deaths associated with Chernobyl with 42,514 deaths from vehicle crashes in 2022 in the United States alone. Erlend Hagen[57] suggests that we are moving quite rapidly towards technologies that will further reduce any radiation or environmental impacts of nuclear power generation that may or may not exist at present.

Understanding the full scale of potential impacts from all current and future electricity generation systems involves life cycle assessment across a wide range of environmental indicators including risks from radiation, CO2 emissions, recycling costs of the technology, toxic materials in landfill leading to water table pollution issues and many other considerations. Nuclear has always been thought by some to be the "most risky" technology. But given the current very high recycling costs of renewables waste of which there will be many thousands of tons, it seems likely that most used panels and wind turbine blades will still go straight to landfill. This may then cause leaching of dangerous chemicals into the water table. The International Energy Agency (IEA) has noted that only 3% of solar panels manufactured prior to 2009 have been recycled. Battery manufacture is highly CO2 emissive due to the electricity used in manufacture frequently produced from coal. In addition, every tonne of lithium extracted through mining is said to lead to the release of approximately 15 tonnes of carbon dioxide.

In the case of solar panels, the Harvard Business Review estimates[58] that by 2035, discarded panels would outweigh new units sold by 2.56 times. In turn, **this would catapult the LCOE (levelized cost of energy, a measure of the overall cost of an energy-producing asset over its lifetime) to four times** the current projection due to the high recycling costs that one-day someone must bear. The economics of solar would darken quickly as the industry sinks under the weight of its own trash.

The view that the European Commission has taken, on the basis of its Joint Research Centre (JRC) is that nuclear is no more dangerous than other forms of energy generation and may indeed be safer, as explained below:

Technical assessment of nuclear energy with respect to the '**do no significant harm' criteria** of Regulation (EU) 2020/852 ('Taxonomy Regulation') by the EU JRC[59]

Inclusion or exclusion of nuclear energy in the EU taxonomy was a debated subject throughout the negotiations on the EU Taxonomy Regulation. While there are indirect references in regulations to the issue of nuclear energy (including on radioactive waste), co-legislators ultimately left the assessment of nuclear energy to the Commission as part of its work on the delegated acts establishing the technical screening criteria.

As the in-house science and knowledge service of the Commission the JRC with extensive technical expertise on nuclear energy and technology was invited to carry out such analysis and to draft a technical assessment report on the 'do no significant harm' (DNSH) aspects of nuclear energy including aspects related to the long-term management of high-level radioactive waste and spent nuclear fuel, consistent with the specifications of Articles 17 and 19 of the Taxonomy Regulation. This report is the result of that JRC analysis.

- The analyses did not reveal any science-based evidence that nuclear energy does more harm to human health or to the environment than other electricity production technologies already included in the Taxonomy as activities supporting climate change mitigation.
- The comparison of impacts of various electricity generation technologies (e.g. oil, gas, renewables and nuclear energy) on human health and the environment, based on recent Life Cycle Analyses (LCA) shows that the impacts of nuclear energy are mostly comparable with hydropower and renewables, with regard to non-radiological effects.
- Management of radioactive waste and its safe and secure disposal is a necessary step in the lifecycle of all applications of nuclear science and technology (nuclear energy, research, industry, education, medical,

[57] https://www.linkedin.com/pulse/from-waste-watts-europes-fast-reactor-roadmap-net-zero-erlend-hagen-v3g9f/?trackingId=BUdtf1wnSh%2B%2FKzqkCiBGGA%3D%3D

[58] Harvard Business Review *The Dark Side of Solar Power* https://hbr.org/2021/06/the-dark-side-of-solar-power June 18, 2021

[59] The Joint Research Centre (JRC) is the European Commission's science and knowledge service. It aims to provide evidence-based scientific support to the European policymaking process. The text in this section is from a 387 page *JRC Science for Policy Report* published by the European Commission in 2021 under the title of the Chapter heading above. Available at https://publications.jrc.ec.europa.eu/repository/handle/JRC125953

and other). Radioactive waste is therefore generated in practically every country, the largest contribution coming from the nuclear energy lifecycle in countries operating nuclear power plants. Presently, there is broad scientific and technical consensus that disposal of high-level, long-lived radioactive waste in deep geologic formations is, at the state of today's knowledge, considered to be an appropriate and safe means of isolating it from the biosphere for very long time scales

- Measures to ensure that radioactive waste does not harm the public and the environment include a combination of technical solutions and an appropriate administrative, legal and regulatory framework. Although there remain contrasting views, it is generally acknowledged, that the necessary technologies for geological disposal are now available and can be deployed when public and political conditions are favourable. No long-term operational experience is presently available as technologies and solutions are still in demonstration and testing phase moving towards the first stage of operational implementation. Finland, Sweden and France are in an advanced stage of implementation of their national deep geological disposal facilities, which are expected to start operation within the present decade.

  The comparison of environmental impacts of various electricity generation technologies on human health and the environment, leads to the following main findings:

  Average lifecycle GHG emissions determined for electricity production from nuclear energy are comparable to the values characteristic to hydropower and wind;

  Nuclear energy has very low NOx (nitrous oxides), SO2 (sulphur dioxide), PM (particulate matter) and NMVOC (non-methane volatile organic compounds) emissions. The values are comparable to or better than the corresponding emissions from the solar PV and wind energy chains.

  Land occupation of nuclear energy generation is about the same as for an equivalent capacity gas-fired power plant, but significantly smaller than wind or solar PV.

In terms of cost, some forecasters report that once the new generation of Advanced Nuclear Technology (ANT) reactors has production lines operating in a mass production system like that for car manufacture, they should be able to generate power 'at the farm gate' at a lower cost per unit of output than gigawatt power stations such as Hinkley Point C in Somerset in England. Furthermore, small nuclear once standardised and manufactured at a mass-production scale is likely to be able to supply electricity and heat at a lower cost to the end customer without the high system costs of transmission involved when power is generated by distant wind and solar farms or legacy nuclear. It should be possible, once a mass production scale of manufacture is achieved, to provide electricity or heat without subsidy **while commanding a commercial rate of return for the investor**.

Two new factors now favour nuclear instead of natural gas. First, the passive safety systems of new nuclear technologies make them more widely deployable, responsibly, across the developed and developing world. **Second, the realisation that the world is not in practice decarbonising at the required rate of progress because heat and transport, which are the dominant carbon emitters will require a massive increase in carbon-free heat and electricity production, transmission and distribution to become carbon free. Unfortunately, these needs are unlikely to be addressable by the methods adopted based on intermittent renewable power alone.**

Solar and wind are producing no more than a relatively small fraction (just under 14% in 2023) of electricity worldwide or even in large economies despite sizable capital flows into these technologies. It is now appreciated that **a massive expansion of solar and wind would require a huge investment in energy storage to overcome intermittency and a huge investment in transmission capacity to transport power from where it is produced to where it is required. It also requires the use of large land areas which might alternatively have been used for food production, as the Italian government has already realised.** Additionally, the quantities of material needing to be mined to build a global low carbon electricity network with just wind, solar and energy storage required for electricity alone (without considering the absolute need for carbon-free heat) would be likely to wreak intolerable environmental damage. According to EU studies such material resources may not even be geologically available. **Accordingly, the EU has now positioned nuclear as a 'transition fuel' making it effectively a 'green' fuel in the EU Taxonomy**.

Chapter 14: NEW Nuclear and 'LEGACY' Nuclear Have Quite Different Characteristics

There have been several trends in nuclear design in recent years. The main ones have been:

- A great simplifying of designs to reduce the complexity of all types of reactor
- A gradual shift currently taking place to smaller designs which should overcome the complexity of attempting to construct 'Giga' scale reactors and the time taken for their construction
- The use of passive safety systems which use the principles of physics to minimise risk
- The ongoing design work not just on 'small' reactors (usually 300 MWe or less) but on much smaller (micro) designs and on low temperature reactors to produce heat alone at 100°C to 150°C
- The expected eventual shift from PWRs operating at around 300°C to AMRs operating at 600°C or above as AMRs which results in a higher conversion rate of heat to power and also provides the high temperatures required for some industrial processes. However, for immediate availability for construction, currently mostly developers must use Generation III+ pressurised water reactors (PWRs) Giga reactors and small modular reactors (SMRs).
- The development of 'heat only' reactors for district heating, industry and de-salination

Generation I
Now obsolete and mostly de-commissioned: legacy technology.

Generation II
Now obsolete and being de-commissioned or currently having life extension: legacy technology

Generation III
The mainstay of the current generation of nuclear operating worldwide

Generation III+ (Gen III+) using water for cooling and heat transfer

1. Gigawatt Generation III+
2. Small modular reactors (SMRs) using Gen III+
3. Micro Modular Reactors (MMRs)

Generation IV (Gen IV) using liquid metals, liquid salts or gases such as helium for cooling and heat transfer

1. Small Modular Reactors using Advanced Nuclear Technology (ANT)
2. Micro Modular Reactors using Advanced Nuclear Technology

We do not cover Generations I, II or III as these are no longer being manufactured. We start with Generation 3+

14.1 Generation III+ Pressurised Light Water Reactors (PWRs) – some examples

1. **Gigascale:** Westinghouse AP1000, EPR (PWR) (page 128 onwards)
2. **Small Modular Reactors** (SMRs): Westinghouse AP300 (PWR), General Electric/ Hitachi BWRX300 (PWR), Holtec SMR300 (PWR), Next Energy (PWR), Rolls Royce SMR (PWR), NewScale (PWR), Neward (PWR) Rosatom (PWR) (page 129 onwards)
3. **Micro Modular:** Deep Atomic (PWR), Steady Energy (PWR), Last Energy (PWR), Calogena (very low pressure LWR)

There are many nuclear technologies in use or in development at the present time. To date, 75% of all operating reactors have used a technology known as 'light water'. This is essentially treated tap water, in which the water is not only the coolant and heat transfer fluid but also the 'moderator' to slow down neutrons and sustain the fission process which produces the required heat. The other 25% includes, for example, gas cooled (CO_2 or helium)

reactors which can operate at a higher, and therefore more thermodynamically efficient, temperature and which predominate in the UK's legacy fleet. Also, heavy water technologies (heavy water contains a proportion of deuterium atoms which are an isotope of H2) such as CANDU and in most of India's 22 current production plants. More recent water-cooled reactors are the Westinghouse AP1000, Korean APR1400, Russian VVER, Chinese Jualong 1000 and the French EPR originally standing for European Pressurised Reactor or Evolutionary Power Reactor but now just EPR. Most of these are known as Generation III+ technology reactors. They mostly have an electrical power output of one Gigawatt or more and can be built as two or more units combined to generate a multiple of the unit size.

Some GW LWR reactors have been scaled down to become small modular reactors, such as the Westinghouse AP300 scaled down from the AP1000. NuScale manufactures small LWRs with the individual reactors generating only 77 megawatts of power but which can be assembled in groups of 12 to provide almost one Gigawatt. NuScale also uses passive safety systems which can function without power to enable a reactor to cool down after an incident. When we refer in the text to SMRs, by this abbreviation we mean small LWR/PWRs.

1. Gigawatt Reactors: Some examples of Gen 3+ LWR/PWR

Westinghouse AP1000™

This design is an 'advanced' (Gen 3+) pressurised water reactor with improved use of advanced passive (AP) nuclear safety and many design features intended to lower its capital cost and improve its economics. It can provide 1.1GWe. Twelve AP1000s are currently in operation or under construction – four at two sites in China. Six more are under construction. Nineteen more AP1000s are planned – 6 in India, 9 in Ukraine, 3 in Poland, and 1 in Bulgaria.

Simplification was a major design objective of the **AP1000™.** Simplifications in overall safety systems, normal operating systems, the control room, construction techniques, and instrumentation and control systems provide a technology that is easier and less expensive to build, operate, and maintain. These simplifications yield fewer components, cable, and building volume, all of which contribute to considerable savings in capital investment, and lower operation and maintenance costs. At the same time, the safety margins for the reactor have been increased dramatically over currently operating reactors. In December 2005, the US Nuclear Regulatory Commission (NRC) approved the final design certification for the AP1000. Since then, many have been constructed or are under construction or planned. Information based on the Westinghouse website: https://westinghousenuclear.com/energy-systems/ap1000-pwr/

EPR (also known as European Pressurised Reactor & Evolutionary Power Reactor)

This reactor was designed and developed jointly by Framatome and EDF (the French nationalised power generation company). Two such reactors are operating in China and both had commenced operation by 2019. EDF has admitted that there have been considerable difficulties outside of China in constructing these reactors. One has been under construction in France (Flamanville 3) since 2007. In 2024, fuel loading commenced and in September that year the first very low level of power was generated in test circumstances. It is hope that the reactor can be ramped-up to full power in 2025.

In the UK, the Hinkley Point C power station in Somerset currently being constructed is an EPR with a rated output of 3.2 Gigawatts generated from two reactors. Although planning for this project commenced in 2013, it is not expected to be generating its full power output until 2030.

South Korean APR 1400

Korea is an important country for research, design and manufacture of nuclear reactors and components. The Korea Atomic Energy Research Institute (KAERI) has been at the forefront of innovative reactor design. The APR 1400 is a light water reactor based on earlier successful designs but incorporating several innovative safety and control features and digital instrumentation and control. The design was licensed in Korea in 2002 and in the US in 2019. To achieve its goal of expanding clean power generation in the country South Korea will build two new APR 1400 nuclear power plants with a combined generation capacity of 2.8 GW and start operations before the end of 2038. The new plants will be constructed by the state-run Korea Hydro & Nuclear Power Co. Korea is also developing an SMR and an AMR which we cover in subsequent sections.

CANDU

Canadian nuclear power reactors, CANDU are heavy-water reactors developed by Canadian scientists and engineers. CANDU stands for **Canada Deuterium Uranium**, because it uses deuterium oxide (heavy water) as a moderator and coolant and utilizes natural (not enriched) uranium as a fuel. There are currently 30 CANDU reactors operating worldwide. Following publication of four executive orders by the Trump administration in May 2025 which set targets to quadruple nuclear energy capacity in the country by 2050, Atkins-Realis plans to start work on 10 large reactors by 2030 and accelerate regulatory approvals. Atkins Réalis engineering, is a services and nuclear organization which operates existing CANDU reactors in Canada. These new reactors are a modified version of existing reactors using a lesser amount of heavy water (thus saving cost) and are larger than the earlier generation. The new reactor is a Gen III+ with an output of 1GWe and is known as the CANDU MONARCH.

2. Small Modular Reactors (SMRs): Examples of Gen 3+ LWR/PWR

Westinghouse Nuclear AP300™

In contrast to Generation IV reactors which are mostly still in development, this Gen III+ design is based on the existing, operational reactor the AP1000™ (approximately 1 Gigawatt). The same technology that is in use for the AP1000 will be used in the AP300.

The AP300 small modular reactor is the only SMR based on an advanced, large Generation III+ reactor already in operation globally – Westinghouse's proven in use AP1000 technology. Unlike AMRs under development with first-of-a-kind technologies and risks, Westinghouse's AP300 SMR utilizes the AP1000 engineering, components and supply chain, enabling quicker licencing and the ability to leverage already available technical skills. Together, these suggest that the first operating unit may be available in the early 2030s.

In August 2024, the company received formal approval from the U.K. Department of Energy Security and Net Zero (DESNZ) to enter the Generic Design Assessment (GDA) for the AP300™ Small Modular Reactor (SMR). This is the first step in the two stage GDA. In September 2024, the AP300 SMR was one of four reactor designs selected for the current phase of Great British Nuclear competition in October 2024. Earlier that year, in February 2024, Westinghouse signed an agreement with Community Nuclear Power, Ltd. (CNP) to build four AP300 SMRs in Northeast England (see Chapter 13.9).

General Electric Hitachi BWRX – 300

This reactor is of a type known as 'boiling water reactor' (BWR). It uses light water (ordinary water) as the coolant (heat transfer medium and moderator) but rather than employing a separate steam generator, it produces steam within the reactor. The technology is simpler than that of traditional light water reactors (LWRs) which reduces its capital cost. It also has a smaller footprint. The estimated construction time is 24-36 months while achieving approximately a 90 percent volume reduction in plant layout. Reducing the building volume by about 50 percent per MW should also enable them to become operational with the use of only 50 percent of the concrete per MW compared to earlier reactor designs — a significant improvement in both affordability and size (and carbon emissions from the production of cement). It has a design life of 60 years. GE Hitachi hope to have the first commercial installation operating by 2029.

Holtec SMR-300

Like all small water cooled reactors, Holtec's SMR-300 is designed to deliver high performance but with a smaller footprint. Its design requires fewer components thus reducing the need for substantial on-site construction and lowering operational and maintenance costs. It is expected to have an 80-year service life which is considerably longer than traditional nuclear power plants. Maintenance costs are also expected to be lower. The heat from fission is used to boil water and uses the resulting steam to make power in a conventional Rankine power cycle via a turbine. This next generation reactor design incorporates passive features to ensure safety of the reactor without any human intervention and to simplify its operation. It can use air cooling (with cooling fans driven by some of the power being generated) thus avoiding the problem of obtaining water use permits where that is an issue. The plant is quite compact requiring ~30 acres for two co-located units though it uses more land than Generation IV reactors are expected to require. In the UK, it has been supported by £60m of investment of which half came from the UK government's Future Nuclear Enabling Fund award last year.

It works with Hyundai Engineering and Construction Co. of Korea and Mitsubishi Electric Co. of Japan. Holtec's plans in Europe include the establishment of a manufacturing plant in the UK to build SMR-300 components. It has already secured regulatory Step 1 endorsement in record time.

Rolls-Royce (RR) 470 MWe

This reactor does not easily fit into the usual categorisation of an SMR being 300MWe we or less. The reactors considered above are all 300 MWe while this one has an output more than 50% larger. Thus two such reactors in tandem generate almost a gigawatt of power.

This company has manufactured light water reactors for UK atomic powered submarines for some decades though they differed in many ways from those being designed today. For example, they had a higher level of Uranium enrichment. They have used their experience of developing and maintaining such reactors to design a 'small' light water reactor for civilian use. Because of its size and high output, this reactor addresses a different market to the embedded SMR or AMRs which other companies are developing. Furthermore the thermal output of the reactor exceeds that of most industrial heat loads and is at a temperature below that classified as high temperature heat. Like all the LWRs, it is less suitable for industrial heat provision. Of course, a light water reactor brings some advantages such as regulatory familiarity from legacy technologies and fuel types but it is very different in design, manufacture and construction from existing light water reactors. RR themselves claim that:

> "The Rolls-Royce SMR offers a radically different approach to delivering nuclear power, we have drastically reduced the amount of construction activities and transformed the delivery environment from a large complex infrastructure programme into a factory-built commoditised product."

We have already noted the need for mass production of SMRs/AMRs in order to make them competitive in the electricity and heat generation market. In 2022, Rolls-Royce CEO SMR is reported as saying that the investment case for setting up an SMR business was based on selling many hundreds of SMRs by 2050. In terms of when it might be expected that reactors would be 'rolling off the production line', one constraint is the time taken for the regulatory Generic Design Assessment (GDA) process. The process for the RR SMR commenced in 2022 but is expected to take four years.

The development cost of this reactor has been considerable and to help finance it, RR has brought in minority investors which have reduced its ownership of the company – Rolls-Royce SMR Limited – to 76%. Other investors now include Qatar and Constellation Energy, a US energy utility.

NuScale 77 MWe SMR VOYGR

This company was the first to obtain US certification for its small light water reactor (LWR) module. Rather than providing clients with a standard 300MWe reactor as many Gen 3+ manufacturers do, this company has designed smaller modules intended to be grouped to provide the required power output. An expected configuration for initial plants is 12 modules providing almost 1 Gigawatt of power. The first overseas installation of this configuration is expected to be in Romania. While the company has had financial difficulties at various stages in its development, it now seems well supported by Fluor Corporation and other investors. It is now a public company whose shares performed extremely well in 2024.

Nuward

Nuward is a project commenced 2019 by the French Alternative Energies and Atomic Energy Commission, EDF, Naval Group and TechnicAtome. The small modular SMR will generate 340 MWe using two pressurised water reactors (PWRs) of 170 MWe each. Its design is based on EDFs experience in PWRs. In mid-July 2024, the company changed its intended design and abandoned some of the technologies it was developing. After considering customer needs including the earliest possible availability, EDF decided to change the approach to development by shifting to proven technology only. They believe that this will provide a higher level of success by facilitating technical feasibility.

Korean SMART 100 SMR

This is a water cooled reactor with many new safety features which eliminate, for example, the possibility of a coolant pipe fracture.

Rosatom SMR RITM-200N

Russia's first small reactor is based on one which is employed in icebreakers. It is a light water-cooled reactor. Its output is 55MWe. It has both active and passive safety systems. Work has started on a power plant in Uzbekistan (see Chapter 13.7 below) which will employ six units giving a total power output of 330MW. Rosatom is also reportedly aiming to build much larger PWRs. It is also reported that Rosatom is in discussion with Myanmar for a PWR and in India for a lead-cooled fast neutron BREST-300.OD. The role of the proposed **International Bank for Nuclear Infrastructure** in relation to creating strong competition to Rosatom in international markets is covered in Chapter 13.7.

Micro Modular:

Deep Atomic MK60 LWR – **Purpose-Designed for Datacentres**

Deep Atomic is a relatively new company based in Switzerland which has as its tagline **Compact Nuclear Reactors, Purpose-Built for Compute.** It is being designed with datacentre requirements at the forefront – both for power and cooling. The design is for a LWR with an output of 60MWe + 60MW cooling. It is described as 'plug and play' in the sense that generation can start small and be increased as necessary by adding more units and thus able to provide the generating capacity for a Megawatt power station as is the case with many designs of small reactor. The three services that this design of small reactor can provide – cooling, electricity, and heat make it stand out from other designs with cooling being the additional service which others have not yet focussed on but which is becoming increasingly necessary.

Deep Atomic says it is "compact, scalable, and built on a foundation of proven technology". This last point is also a critical one at this point. While LWRs are not "the latest thing" their basic underlying technology has been in use for decades. Given the small size of individual units (giving a high surface area to volume ratio), the ability to cool down naturally in the event of an incident should be assured which is critical for a project intended to embed reactors within industrial facilities.

Steady Energy LDR 50 – **Purpose-Designed for District Heating**

This is a Finnish design which will generate 50MWth. It is not designed to generate electricity but if it were it would be around 20MWe. It will provide energy at a temperature of up to 150 °C. Its main use initially is likely to be for district heating as this is the main form of heating in Finnish cities. The first operational plant is expected to start construction by 2028 and to be operational by 2030. In June 2024, Steady Energy and Swedish project developer Kärnfull Next (see below) announced a strategic partnership to use the LDR 50 SMR in projects. According to Steady Energy:

> "With heating using 40% of all energy and the single biggest source of CO2 and in Europe, fossil fuels being the main source of district heat, the LDR-50 offers the possibility of replacement with heat to be delivered at an estimated cost below €40/MWh."

Last Energy PWR 20 – **Designed Mainly for Power Generation**

This company is set to produce an air-cooled light water 20 MWe reactor with the 'nuclear island' being installed underground. The reactor operates at a pressure comparable to that of a household espresso machine. As expresso machines are safe for use in the home, this reactor should be much safer even than the Gen III+ reactors which operate at very much higher pressure. As the technology uses a standardized balance of plant and modular construction, the company claims its reactors could be fully factory fabricated and installed on-site in as little as three months. Factory fabrication simplifies the regulatory process and the company believes it can keep project costs below $100 million. This should also make it privately financeable. Already the company has secured power purchase agreements (PPAs) for 34 PWR-20 small modular reactor units with four industrial partners in the UK and Poland. Securing PPAs with customers is a key factor in enabling nuclear to be privately financed.

One of its first projects will be in the UK where it is planning four 20-megawatt (MW) micro reactors at a site in Bridgend in Wales which they say could be up and running by 2027. It is said to have a provisional offer of over $100m from the US Export-Import bank to enable this project. The location in Wales was previously occupied by a coal-fired power station. But it sees its main market as owners of server farms and hopes to build around 10,000 units of its reactor.

Calogena **– Designed for District Heating**

According to Calogena, heat accounts for 40% of energy consumption in Europe. Nearly 70% of this is produced by fossil fuels. And 80% of this heat is for hot water and heating at less than 100°C. The two major low-carbon solutions for the transition are heat pumps and heat networks. The Nuward project is aiming at enabling existing and new heat networks to be powered by low temperature nuclear instead of natural gas or coal. Given the extent of domestic and commercial use of natural gas boilers for space heating, there is considerable scope for decarbonising this part of the economy – difficult though this is.

Gas-fired boilers emit just over 220 grams of CO2 per kWh, while coal-fired power stations emit around twice as much. Each Calogena module could decarbonize district heating by 30,000 tonnes of CO2 a year if it replaces a gas-fired installation, and twice as much if it uses coal. Gas boilers also emit other pollutants in addition to CO2 and CH4 (methane) from natural gas production. A particular source of pollutants are nitrogen oxides (NOx) which have harmful effects on human health and the environment. By comparison, Life Cycle Studies carried out by Ademe for large reactors (SMRs and AMRs would be similar) show that these have a carbon footprint of around 6g of CO2 per kWh of electricity, and 2g per kWh of heat.

The Calogena reactor operates at a pressure of only 5 bar which is less than the pressure of a domestic water heater and is therefore effectively explosion proof. It also runs at a low temperature of just under 100 °C. It will use 3.4% low-enriched uranium pellets. Because of its simplicity of its design, it requires far fewer auxiliary systems and is even safer than any reactor currently in operation or planned. Its heat output is only 30MWth (it does not produce any electricity). It is thus a design that can safely be installed close to residential or commercial premises.

In Chapter 13.10, we cover a potentially key initiative by the European Commission to encourage the creation of a European Industrial Alliance on Small Modular Reactors (SMRs and AMRs). In June 2024, the Alliance launched a call for SMR projects wishing to be considered for the Alliance's Project Working Groups (PWGs). Subsequently, the governing board, with the assistance of the Alliance secretariat, conducted a review and assessment of the 22 applications received. The CityHeat project (Calogena, Neward/Steady Energy) was one of those accepted in the initial batch which should be helpful in ensuring that Calogena can obtain further funding for the project.

14.2: Generation IV (Gen IV) Reactors have quite different characteristics to Generation 3+ (Gen 3+)

Generation IV (Gen IV) AMRs: two sizes

1. Small Modular Reactors using Advanced Nuclear Technology (ANT) known as Advanced Modular Reactors (AMRs) usually around 300MWe
2. Micro Modular Reactors using Advanced Nuclear Technology (ANT MMRs) usually 1 - 45 MWt

The new Advanced Nuclear Technology (ANT) reactors generally described as Generation IV advanced modular reactors or AMRs are different from water cooled reactors in many ways. None has yet become commercially operational except in China and Russia but test AMRs have been operated for many decades in many countries. Some are now close to being licensed and expected to be operational by the early 2030's. However, many will never achieve the large scale production which is necessary to bring their costs down to enable unsubsidised operation under truly commercial financing, but in our view some will and for shareholders in such companies we expect the rewards to be substantial.

Generation IV, Advanced Nuclear Technology (ANT) High Temperature Modular Reactors (HTMRs) are of quite a different design to Gen 3+ and use quite different physical features and technologies. Almost the only thing they have in common is that they both use nuclear fission to create heat. They are also physically much smaller, do not

require a large high pressure vessel, which is technically difficult and also very costly, have fail safe characteristics and can be located close to a customer. The most important distinction between Gen III+ (SMRs) and Gen IV (AMRs) is that the former generally use 'light water' whereas the latter use other coolant/heat transfer agents such as **helium gas, molten sodium, molten lead or molten salts**. As a result, they do not operate under high presume and cannot explode. Like all nuclear reactors, and indeed coal or gas furnaces, they produce high grade heat which is quantified as MWt (thermal) and when converted to electricity, the output is quoted as MWe. They have a lower power output (normally 100MWe – 300MWe) but some micro-reactors, are much smaller.

While AMRs are small or micro sized, by coupling several small AMRs together, as with small LWRs, it is possible to increase a plant output to the Gigawatt scale if appropriate. However, a key virtue of the physical smallness of each reactor which together create a Gigascale plant is that it enables an important aspect of SMR/AMR inherent safety. **This important safety feature of smaller reactors and from multiple smaller reactors linked in one location arises from the fact that their smaller core volume relative to their surface area means that unattended natural dissipation of decay heat to its surroundings during an upset condition is attainable. Thus off-site emergency planning requirements can be scaled down to be proportionate to those reduced risks.** For example, the Emergency Planning Zone (EPZ) can be much smaller than with conventional nuclear and can be within the plant site boundary fence rather than being 10 – 50 miles downwind of the reactor. In addition, AMRs can be sited underground with a blast resistant concrete cover and many do not require manpower to be in attendance when operating. They are autonomous, including for load-following. While human operating error has contributed to the small number of Gen II and III incidents Gen IV has both inherent safety as well as self-regulating stability. Being located contiguously to the point of use avoids the cost and environmental disturbance of transmission inherent in larger plants or indeed large wind and solar farms. This, in turn, enables them to be located close their customers' points of use.

The higher heat output temperature of AMRs compared to previous generations of reactor, also makes them suitable for many industrial processes currently using fossil fuels while conventional water-cooled high pressure reactors are not suitable for the provision of high temperature heat. Generation IV reactors such as those using molten coolants are also low pressure and thus do not require the expensive and difficult to manufacture pressure vessel of the previous generations of water cooled reactors (Gen I, II, III & 3+ PWR plants).

When we compare the capital economics of building large and small reactors we are comparing the **economies of scale** of Gen III+ with the **potential economies of mass production** of Gen IV**.** However, to enable Gen IV reactors to produce heat or power at an economic price, the manufacture of small reactors needs to employ a similar mass-production model to that used in modern automobile manufacture to counter the perceived economies of scale (large size) advantage of large reactors. Until now, it had always been assumed that the economies of scale as reactors increased in power output and physical size would reduce the cost per unit of power output. However, as reactors became larger, it was apparent that there are also substantial dis-economies of scale which include the difficulty of constructing such complex behemoths and the time value of money lost during their construction. Recently, some Gigawatt reactors have witnessed their cost doubling and the construction time more than doubling. The 'scaling effect' which was expected to hold as reactors grew ever larger and thereby reduce the unit cost of power, seems not to hold after a certain point. In addition, with the notable exception of France and Korea the fleets have been constructed only sporadically so that each new one has had a somewhat different design which has stopped costs falling by learning.

These additional capital costs of constructing Gigasized reactors and the longer construction period have greatly increased the total capital investment requirement and the time to start generating revenue. Both issues raise the final cost of power produced particularly in an era of higher interest rates. Interest rates rose after 2021. In the case of the UK for example, the Bank of England base rate rose from almost zero in 2021 to over 5% in 2023. As debt service (capital repayments plus interest) are the largest cost of nuclear (and of wind and solar), rising interest rates are a disincentive to investment. Large GW nuclear also has the problem that the time scale for the repayment of debt capital and a break-even on equity investment is much longer than for wind and solar. Large nuclear has a long build time and an operating life of up to 60 or even 80 years compared with 20 or 25 for renewables or for gas power plants.

Keeping individual units small also brings the advantage of being able to construct them in a factory and transport them to site by road or barge. It brings an additional advantage according to the US Nuclear Regulatory Commission (NRC) in that the smaller amount of fuel in the core along with the inherent safety of Gen IV eliminates

several potential emergency scenarios resulting in much simpler systems. This reduces capital cost, commissioning time, maintenance and operating complexity including in some cases the elimination of all on-site staff. In contrast, regulatory proving and testing of active safety systems and their maintenance are an escalating cost of the legacy pressurised water reactor (PWR) family of conventional gigawatt reactors which is one of several reasons why Gen IV reactors are currently being developed in most nuclear nations.

What is not yet known about any AMR is whether, in practice, their loss of the economies of scale that large reactors should in theory have can be more than compensated by the cost-reducing features of the new designs including mass production. Thus it is currently difficult to know which manufacturers' technologies will in practice enable their reactor to generate electricity and heat at a price which customers will be willing to pay while ensuring an appropriate return on capital for investors. AMRs cost-reducing features include less complex design, greatly simplified and passive safety systems, greater speed of construction (hence less interest cost accruing during construction and faster revenue generation), higher operating efficiency resulting from the greater temperature difference between the coolant used for heat transfer and the ambient (outside air) temperature and, importantly, economies of mass production.

The foregoing has predominantly dwelt on the capital and operating economics of AMRs for electricity. However, this is to miss three larger considerations for the investor:

Firstly, as stated in the previous section, the market for heat in a decarbonising world is equal or larger than that for electricity. The only tool currently known to address decarbonised heat is nuclear. More particularly, because heat is expensive to transmit, it must be generated near the point of use which is only achievable from small reactors which match the quantity and dynamics of industrial, commercial and domestic heat demand. This lack of competition except from the tail-end of fossil fuels gives small nuclear, especially AMRs with their high temperature heat output, huge market power as well as the possibility of being able to manufacture at scale.

Secondly, much discussion in decision-making circles so far has compared the output cost of electricity (or heat) *at the point of generation* (at the factory gate). However, this is not where the customer buys the product. The cost of electricity from large scale wind or large scale solar must also incur the system cost of storage systems to match the production variability of their output to the daily and annual use pattern of the customer. This is already proving problematic in Europe where in addition, the scale of transmission reinforcement is proving to be considerable and transmission capacity only likely to be available many years hence. **This additional cost of renewables increases with the increasing renewable fraction in a region's total generation of power**.

Transmission, distribution and backup are already over 20% of the cost to the end customer while there should be no system surcharge for energy generated at the point of use and distributed without the use of a transmission system. Indeed, In the US, transmission and distribution costs are already 44% of the total cost of delivered electricity according to the World Economic Forum (WEF). In the UK, it was announced in December 2024 that the country's electricity transmission network owners plan to invest up to £77.4bn per annum between 2026 and 2031. This is necessary to enable power to be transmitted from where it is produced to where it is in demand e.g. from the North Sea to users 800km (500 miles) or more distant in the south of England. The same issues arise in Germany where industry is located in the south of the country but clean power comes mainly from the North Sea. Such costs have to be recovered from customers and can be so recovered when the power generated is operating under a regulated asset base model (RAB). However, in most countries it is currently also charged equally to all customers including those consuming power close to the point of production. At some point **location based pricing** may be necessary to alter the seeming unfairness of this pricing mechanism. This would be in addition to "time of use" tarrifs to reflect imbalances in supply and demand when renewables have lower output.

Thirdly, large scale nuclear, solar and wind along with their transmission requirements are subject to political uncertainty and public perceptions whereas a small nuclear plant inside a factory fence supplying all heat and power is, except in safety terms, entirely under the control of the industrial owner just like the rest of the factory. Indeed, the outcome from universal standardisation of inherently safe small reactors for heat and power may logically be that their operation, except for transfers of fresh radioactive fuel once every 7 or 8 years, falls entirely outside of nuclear regulation if the containment can be designed to be unbreachable under any circumstances.

Cost competitiveness, whole life energy cost stability and low political risk have been key drivers of all small nuclear propositions to industrial and Artificial Intelligence (AI) customers. Thus just as technology has moved on

from using a small number of large, centralised, fossil power stations in favour of distributed solar and wind power, so we are now seeing the possibility of distributed nuclear displacing single gigasized reactors which are also more exposed to enemy action in the event of war in Europe. Small reactors can also operate as gigasized electricity and heat generators by linking them together on the same site and sharing steam turbogenerators.

14.3 Generation IV Small Advanced Modular Reactors (AMRs) using **Advanced Nuclear Technology** (ANT)

Generation IV International Forum

The Generation IV International Forum (GIF) was initiated by the US Department of Energy in the year 2000 to help coordinate international work on advanced small modular reactors and to help realise their potential. It is an international collective representing governments of 13 countries where nuclear energy is significant now and also seen as vital for the future. The original charter members of GIF are Argentina, Brazil, Canada, France, Japan, South Korea, South Africa, the UK and the USA. They have been joined by Switzerland, China, Russia, Australia and, through the Euratom research and training programme, the European Union. After some two years' deliberation and review of about one hundred concepts, in 2002 GIF announced the selection of six reactor technologies (not companies) which they believe represent the future shape of nuclear energy. These were selected based on being clean, safe and a cost-effective means of meeting increased energy demands on a sustainable basis, while being resistant to diversion of materials for weapons proliferation and secure from terrorist attacks. They are the gas-cooled fast reactor (**GFR**), the lead-cooled fast reactor (**LFR**), the molten salt reactor (**MSR**), the sodium-cooled fast reactor (**SFR**), the supercritical-water-cooled reactor (**SCWR**) and the very high-temperature reactor (**VHTR**).

Examples of the Technologies of Some Small ANT AMRs

TerraPower: Natrium (Natrium is the Latin name for Sodium)

The beginning of construction in June 2024 on the Natrium demonstration project site in Wyoming marks the first advanced nuclear technology (ANT) reactor project under construction in the Western Hemisphere!

Bill Gates, the founder of Microsoft, provided the initial venture capital (Angel Financing) for this reactor company and is still a majority shareholder in the company as well as being its chairman. Since then it has had institutional investor or corporate rounds from Series A to Series C. One such corporate investor is Arcelor Mittal the steel manufacturing company which is a large consumer of electricity and will require much more power if its use of (electric) arc furnaces increases as is expected. It is thus a **'strategic'** investor rather than just being a financial investor. Also, SK a Korean company and CRV, a US fund have contributed to financing. The US Department of Energy has also provided a grant through prize money. Pillsbury, a specialised legal practice has also facilitated the raising of $750m from institutional investors.

TerraPower has already 'broken ground' on a site in the small town of Kemmerer which will also be the demonstration project to showcase the abilities of TerraPower's Natrium energy system. When operational, the output of the plant will be provided to PacifiCorp, a state utility company, to supply Rocky Mountain Power customers in Wyoming and throughout the region. It is estimated that the project will cost $4 billion. Bill Gates will provide $1 billion with the Department of Energy's (DOE) Office of Clean Energy supplying half of the cost. In addition, TerraPower has a deal with Emirates Nuclear Energy Corporation to explore the export of TerraPower's Natrium reactors to the United Arab Emirates.

In March 2024, TerraPower was the first company to submit its application for a commercial advanced reactor construction permit to the U.S. Nuclear Regulatory Commission (NRC) for the Natrium reactor. Through the U.S. Department of Energy's Advanced Reactor Demonstration Program, which is a public private partnership, this Natrium reactor demonstration project is intended to validate the design, construction and operational features of the Natrium technology. This first AMR commercial project outside of China is being constructed near a retiring coal-fired power plant.

Natrium features a 345 MW ***sodium-cooled fast neutron reactor (no moderator required) with a molten salt-based energy storage system*.** The storage technology can boost the system's output to 500 MW of power for more than five and a half hours when needed, which is equivalent to the energy required to power around 400,000 homes at times of peak demand. Unlike the statistics quoted about the ability of wind or solar to provide the power required for x number of homes nuclear, with its constancy of output

(very high 'Capacity Factor') and low stoppages for maintenance (High 'Availability'), Natrium can provide continuous (24/7/365) power and also flex its output up and down in accordance with the customer's varying needs which many other generation systems cannot. The energy storage system in Natrium is simply to enable it to more economically balance supply and demand for power throughout the day (flex) without having to employ a larger, and hence, more expensive, reactor. Such a reactor might be less able to benefit from the higher surface to volume ratio (for incident cooling) that a smaller reactor provides. Other reactors may 'flex' output but constantly ramping a reactor up and down may reduce its life.

The heat energy storage system enables the reactor to **operate with an even load all day** (24/7) which leads to greater efficiency and less stress on the whole system. In the combined system there are two sources of heat for the steam generator and the steam turbine. There is the heat energy provided by the evenly loaded nuclear reactor plus the heat energy from storage. Effectively the steam turbine but not the reactor itself is being 'ramped up' to cover peak demand but also to cover the times of day when renewable intermittency on a grid needs to be corrected. In the first graphic below[60] (Figure 13.2 [a]), from midnight until 6am when demand from customers is low, any excess heat from the reactor can be stored (this reactor produces heat 24 hours per day). Heat is conserved until 1pm when customer demand rises and the combined system can then produce up to 500 MWe. After 7pm, power demand starts falling and by 9pm, surplus heat can once again be stored. A visual of the Natrium facility is shown in the second graphic and Figure 13.2 [b] shows the hot and cold molten salt tanks which are a key feature of Natrium.

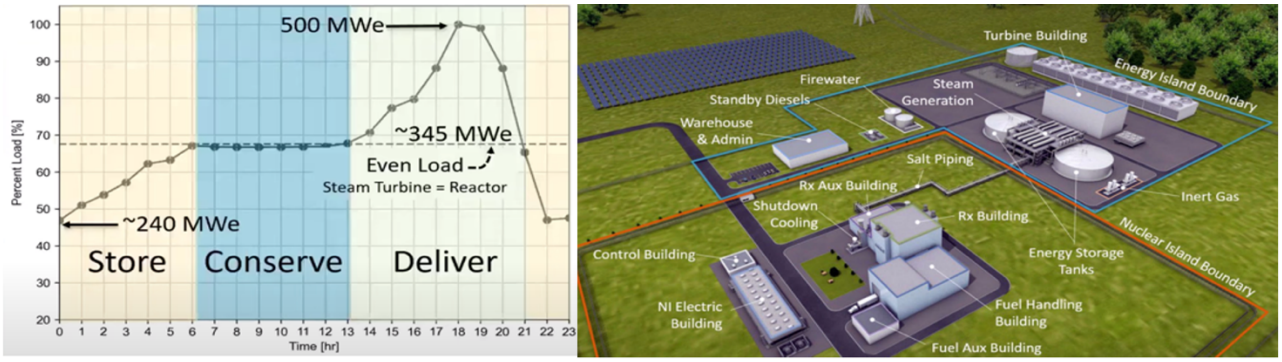
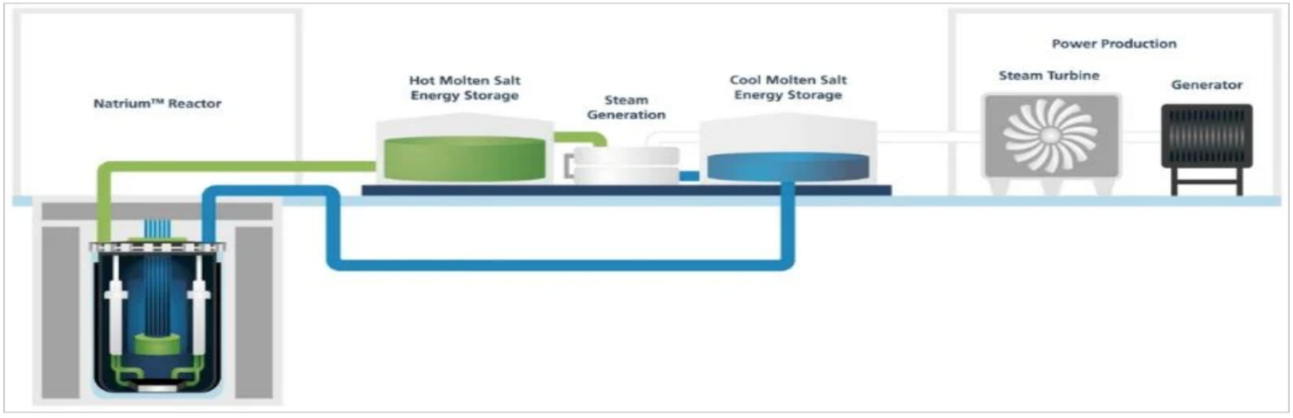


Figure 14.1 [a] How the TerraPower Natrium system functions



The Natrium heat exchange process. Image used courtesy of TerraPower

Figure 14.1 [b] The Energy Storage System

Terrestrial Energy Integral Molten Salt Reactor (IMSR)® and its Techno-Economics

This is a Canadian/US design but for European, Middle East and African (EMEA) markets it could be manufactured in Europe. It would also almost certainly be available to generate power in the EMEA region within this decade as it is at the licensing stage in Canada and the US. Speed is ever more of the essence since the European decision to phase out Russian gas and the warnings of the growing dangers of global warming in the third instalment of the IPCC's sixth assessment report published in early April 2022 has increased concern. Thus, if only for reasons of speed, it is now ever more critical for Europe to be willing to use non-European technology if this becomes available sooner than European developed ANT reactors.

[60] https://www.revistanuclear.es/wp-content/uploads/2021/03/ART-Natrium.pdf

Terrestrial's Integral Molten Salt Reactor (IMSR) reactor uses molten salts (with the fuel dissolved in the molten salts) with the liquid salts also acting as the heat transfer fluid. The moderator is graphite. The reactor is designed to provide flexible dispatchable power, grid services to balance wind and solar, voltage and frequency stabilisation etc. as well as industrial heat. It can also be used to produce hydrogen through high temperature electrolysis and could be the ideal heat source for solid oxide electrolysers (for which heat is beneficial) which we cover below. Unusually, it integrates the primary reactor components into a sealed, replaceable vessel. This unit, which contains the graphite moderator and pumps, is removed after seven years and replaced with a new core-unit thus simplifying the on-site infrastructure. During the 7-year operating life of each core-unit, it can be topped up with uranium fuel. As the liquid fuel salt does not need to be kept under pressure, there is no need for the expensive and difficult to cast pressure vessel that traditional light-water reactors require.

The greater safety of this design is based more on physics than on engineered systems which is a factor in its (and all AMRs) safe design. The IMSR approach is to rely on inherently safe design rather than complex active systems seen historically. For example, the cooling mechanism does not require backup electric power. Because the fuel is distributed through the coolant, the system avoids complex fuel fabrication processes which are required for pebble bed reactors or rigid fuel elements required for some other Gen IV designs.

It is an axiom of public safety that if a safer way of reasonably achieving the societal result is available then statute law should require it to be used. This is the political version of the investor's correct and proper aversion to avoidable risk. This type of reactor has been designed with inherent safety. In October 2018, Terrestrial Energy started Phase 2 and completed the CNSC's Vendor Design Review process in mid-2023. It now has the 'green light' for commercial operation by large industrial users.

In terms of its **techno-economics**, the IMSR's higher operating temperature improves thermodynamic efficiency from around 33% on Gen III reactors to around 46% (as it does on all high temperature reactors). According to the manufacturer, the IMSR should produce 40% more electricity than a comparably sized water-cooled SMR for the same or less fuel or waste quantity. The result is around 40% more revenue from the same reactor size. This can have a large impact on its economics. The intrinsic low temperature waste heat from such a reactor could also be used for district heating thus further improving its economics. Figure 13.3[61] shows the Integral Molten Salt Reactor (IMSR) and its range of services.

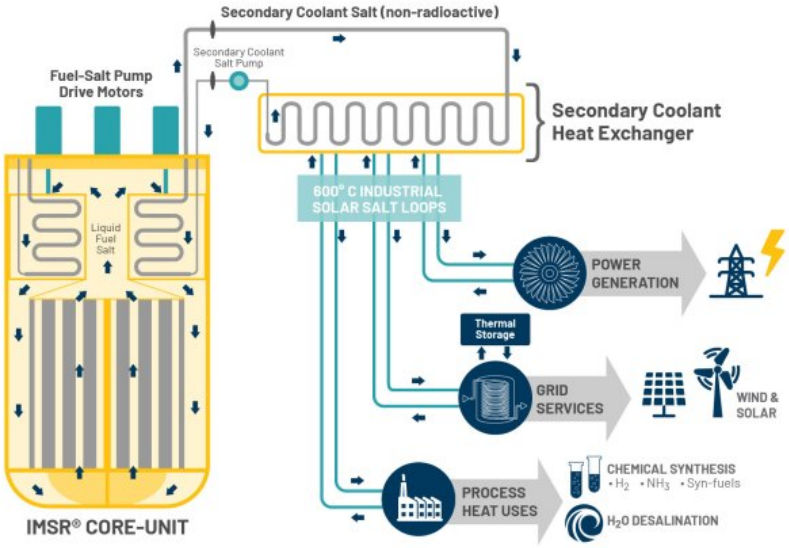


Figure 14.2 The Terrestrial reactor and its varied uses

[61] https://neutronbytes.com/2019/12/29/terrestrial-energys-imsr-to-get-joint-review-by-cnsc-and-nrc/
Courtesy of Terrestrial Energy

Attractive features of both of the above reactors and those which follow below:

Both of the above AMR reactors have a small footprint and, due to their high level of safety, can be located (subject to properly developed regulatory permissions) in or beside existing boiler houses which are usually near to built-up areas and communities. Such locations may also have suitable transmission and distribution capabilities already in place. Because they are small and manufactured in a factory and not on-site, the total capital cost is within the realms of conventional industrial and municipal projects (£200m to £1bn depending on size) and construction is much simpler resulting in a construction period expected to be only 3-4 years.

These types of reactor could be financed using, for example, the Finnish Mankala model (which we cover in Chapter 13.5 on Financing the Transition) where users of heat and power join up to own generating capacity and finance such a plant themselves. An alternative for utilities wishing to supply power to a national grid, is to use the RAB model considered in Chapters 4.4 and 4.5. possibly also with a government provided 'Contingent Financial Support Package'. Once the simplicity and performance of AMRs becomes known to investors, directly funded **merchant** plants (owners selling their power in the open market without a power purchase agreement or using the power for their own industrial use) should emerge and may become a standard part of a chemical, thermoplastic, iron and steel and other industries, factory site hard infrastructure bought off the shelf in the same way as today they buy standardised package boilers. With reference to efficiency, if the different grades of steam (different temperatures) are utilised by means of an energy 'cascade' (including providing low temperature heat to a district heating system) it is possible to utilise 80% of the heat making it very highly efficient and a much more economic system.

Newcleo

Newcleo has large teams of researchers and engineers working on a 'fast' neutron 200 MWe AMR technology that uses molten lead as the cooling and heat transfer medium (an LFR). Amongst other potential benefits, it may eliminate the need for a separate heat exchanger between the reactor and the steam generating plant. It has already completed the preparatory phase of the French regulators' licensing process. Its head office was in London but has now moved to Paris, with other offices in several countries in Europe. A key advantage of small reactors is that they can be installed next to the industrial sites or communities they supply. This is a great advantage in comparison to other clean energy technologies.

Investors include the Agnelli family's holding company Exor NV, Italian asset managers Azimut Holding SpA, as well as several wealthy Italian business people such as ex-Goldman Sachs Group Inc. banker, Claudio Costamagna. It is said that the reactors would cost around £5bn per gigawatt of generation capacity, compared to £15bn per gigawatt cost of Hinkley Point C. The small reactors could also be built and installed within four years or less whereas Hinkley will require two decades given all the problems it has encountered (though subsequent reactors of this type should not take nearly as long).

This $5bn/MWe capex estimate is similar to that of the Offshore Wind Market Report: 2024 Edition
actual capex for offshore wind which of course operates for 40 to 60% of the year whereas nuclear operates almost continuously for the same capex but does incur some fuel costs which are comparable to the additional costs of backup and transmission for intermittent wind and its much shorter operating life.

Thorizon One – can recycle waste fuel as well as Thorium

Thorizon has evolved from a spin-off of NRG (Dutch Nuclear Research Institute) into a startup with offices in Amsterdam and Lyon. The company views small modular nuclear reactors as an ideal, stable, and clean contributor to an energy system where wind and solar are the other important sources. The ambition is to develop a reactor based on molten salt as the coolant and heat transmission fluid that can be realized rapidly and smartly, is 'walk-away' safe, and *takes a first step towards circularity by using long-lived nuclear waste as a fuel source*. Thorium, uranium, and plutonium can all form stable fluoride salts.

Thorizon is working to further strengthen its financial position to be able to build a non-nuclear molten salt demonstrator in the short term and finalize the detailed design for a first-of-a-kind reactor by 2030. The output is intended to be 250GWth and 100MWe. Interestingly, like TerraPower, the 100MWe can be converted to 50-300MWe of flexible capacity by storing heat energy in liquid salt when demand is low and releasing it during peak hours.

In March 2025, the company reported that its latest funding included EUR16 million as the first tranche of its Series A round, led by the Dutch National Promotional Institution, Invest-NL, backed by an InvestEU guarantee from the European Commission for the research part, with strong backing from Positron Ventures, PDENH, and Impuls Zeeland. It said all Thorizon's existing shareholders have reinforced their commitment in this investment round.

Thorizon has also secured an additional EUR4 million grant from the Dutch Province of Noord-Brabant in consortium with VDL Groep and Demcon. The investments follow an earlier EUR10 million grant from the France 2030 Innovative Reactor Programme of the French government in 2024. In total, including its first equity round, Thorizon has now raised EUR42.5 million to drive the commercialisation of its reactor technology.

With is latest funding, Thorizon is advancing the development of the Thorizon One, a next-generation reactor **that overcomes all traditional reservations against nuclear energy**. The Thorizon One is engineered to deliver carbon-free energy while repurposing long-lived nuclear waste as fuel. Its modular design and innovative cartridge-based fuel system provide a scalable pathway to a circular nuclear economy. By harnessing molten salt technology, Thorizon is developing reactors that are inherently safe, cost-efficient, and faster to deploy than conventional nuclear plants—offering a practical solution to Europe’s clean energy transition.

Thorizon is also working closely with a range of partnerships which increase the probability of success in the shortest timescale. It has forged key partnerships with Orano for fuel development, Tractebel for engineering, and VDL Groep for prototyping, while collaborating with EPZ for early operator input, and with EDF on R&D. Dutch and French nuclear regulators have initiated a joint preparatory review of the Thorizon One design, and the company is conducting pre-feasibility studies at three nuclear-designated sites in France and the Benelux, targeting construction by 2030.

Westinghouse Lead-cooled Fast Reactor (LFR): The Next Generation of Nuclear Technology

Lead-cooled Fast Reactor (LFR) technology was selected by Westinghouse for their next generation reactor after comparing all known nuclear technologies, based on criteria supporting global commercialization and spanning safety, economics, market versatility, fuel cycle sustainability, technology readiness and other features. It is designed to output 450MWe when connected to a steam generator and turbine set. It requires only minimal pressurisation and due to the use of liquid lead as coolant, has a smaller physical size than other types of Gen IV reactors. When used to generate steam to produce electric power, it is possible to use air cooling (driven by electric fans powered by its own electrical generation) to condense the steam emanating from the turbine. It uses ‘fast neutron’ technology and can use reprocessed fuel from light water reactors. When reprocessed fuel is used, much more of the radioactive content in the fuel is used up and when it is removed from the reactor, the amount of spent fuel is smaller than in its first round of use.

Ocean-going Ship Propulsion & Floating Nuclear Power Plants (FNPPs)

FNPPs which can either be used for propulsion in ocean going ships or can be towed to a coastal location to provide power to the land, represent a game-changing approach to nuclear energy deployment. These plants can be centrally manufactured and easily transported where needed, combining advanced nuclear technology with shipyard efficiency. FNPPs can be deployed in ports and terminals, as well as in nearshore and offshore environments. The floating environment enhances safety by significantly reducing risks from earthquakes, tsunamis, and volcanoes compared to land-based systems. FNPPs can be readily moved on water, enabling fast implementation to meet urgent energy needs or decarbonisation goals. A fleet of FNPPs can be assembled and towed to customer locations.

CORE POWER

This company has focussed on the role of small advanced nuclear in marine locations. Core Power aims to extend marine use to floating nuclear generation to supply land-based needs and to civil nuclear ship propulsion and in particular, container ships. CORE POWER's mission is to improve energy efficiency and security of ocean transport and heavy industry through innovative, sustainable advanced nuclear energy

solutions that drive global progress towards a cleaner, more efficient future. Their reactor design is a molten salt fast reactor with the fuel mixed in with the salt. The company has offices in Washington and Tokyo with its head office in London.

Key Benefits of FNPPs

- Serial construction of entire FNPPs in shipyards (which are experienced in this type of construction work) reduces costs, accelerates innovation/deployment speed, and ensures consistent quality, leveraging the efficiencies of modular construction and learning from previous iterations without site.
- Simplified design can reduce materials and construction costs as FNPPs eliminate the need for extensive, site-specific civil works required by traditional land-based reactors, significantly lowering both costs and deployment timelines.
- Decoupling of the plant construction and siting means that the construction of FNPPs can commence independently from the site licensing, reducing deployment time, particularly compared to terrestrial counterparts.
- Enhanced resilience to natural disasters, including tsunamis and earthquakes, if sited offshore.
- Flexibility in deployment and location: while terrestrial plants are immovable, FNPPs can be redeployed to new locations in response to medium/long-term demand.

Key Benefits of Civil Maritime Nuclear Propulsion

The International Maritime Organization has mandated that the shipping industry eliminate 70% of its carbon emissions by 2040. The need for clean power is enormous in an industry that uses almost 300 million tons of emissions-intensive heavy fuel oil annually. Large nuclear-powered ships will be faster, have more cargo capacity, be more economical, and emit no greenhouse gases

SALTFOSS (previously known as Seaborg Technologies)

Another company developing floating nuclear power is a Danish Company set-up in 2014 with the intention of producing an innovative AMR. The company is focussed on the need to be able to 'fill the gaps' in the generation of wind and solar power and thus sees one of the uses of its technology as being to **complement renewable power**. An innovative idea that they have developed is to locate the reactor(s) on a barge which can be towed to wherever power is required whether for grid backup, industries with a high energy requirement, desalination of seawater, manufacture of ammonia or e-methanol or power to X.

The reactor modules each produce 100 MWe and are normally configured in groups of two modules providing up to 800MWe output. The output temperature of the reactor is high enough to manufacture hydrogen and its derivatives efficiently. A number of AMR designs used liquid fuels in which the nuclear fuel is mixed with the fluoride salt which is being used for cooling and heat transfer. Seaborg's design is one such design. The design also has the possibility of creating a 'closed fuel cycle' in which the waste from the reactor is recycled into fresh fuel salt with the addition of fresh uranium.

Chinese ANT

In China the question as to whether nuclear should supply a substantial proportion of power in the country has already been answered with a long-term process of building small modular reactors as well as continuing to construct Gigasized reactors to speed up the power industry transition from coal. These High-Temperature Gas-cooled Reactors (HTGRs) use graphite as a moderator and helium as a coolant, with uranium fuel in the form of 6 cm-diameter 'pebbles' which can be sorted and switched in and out for on-load refuelling. Each pebble has an outer layer of graphite and contains some 12,000 four-layer ceramic-coated TRISO fuel particles dispersed in a matrix of graphite powder. The fuel has high inherent safety characteristics and has been shown to remain intact and to continue to contain radioactivity at temperatures up to 1620°C - far higher than the temperatures that would be encountered even in extreme accident situations, according to the China Nuclear Energy Association.

A pair of these pebble bed 100 MWe reactors have been built side-by-side to share a single steam turbine and are now running commercially. The original concept was to use three of these twins to repower China's very large fleet of 600 MWe coal power stations without disrupting their infrastructure. In addition China has research programmes on small LWRs and in all of the other technologies summarised in this chapter.

Copenhagen Atomics: Thorium molten salt reactors

One commonly raised objection to uranium fuelled nuclear reactors is that they produce a large quantity of waste material after the fuel has been used. Much of the potential value in the fuel is in fact not used at all as only a relatively small proportion contributes to generating heat. In contrast, Thorium produces much less waste material and its half-life is much shorter. Thorium is also much more available in the Earth's crust than Uranium although Uranium reserves are nevertheless adequate for at least two generations.

The first Thorium molten salt reactor was developed in the US in the 1960's and worked well. However, the US Navy opposed its commercial development because it produced only a small amount of plutonium compared with Uranium 235 fuelled LWR. The military wanted large amounts of plutonium for the manufacture of atomic weapons and as a result, further work on Thorium reactors ceased.

Since then China has been first in issuing (in 2023) a permit for an experimental molten salt thorium reactor. It will undergo testing over the next few years. China also has large sources of Thorium as does India where high quality development has been continuing for decades. In Europe Copenhagen Atomics is the most advanced company in this field and believes that taking advantage of the features of such reactors it will be possible to scale-up global nuclear output to double digit Terawatt (TW) levels in the coming decades. Japan, the United Kingdom, the United States of America and other countries have also demonstrated enthusiasm for research but do not yet have an operational Thorium reactor.

Copenhagen Atomics is developing a thorium based molten salt reactor with the same footprint as a 40 foot shipping container which will deliver 100 MW thermal energy per module and is expected to reach an electricity price (LCoE) below $20/MWh in a mass manufacturing scenario. It expects to have an operational 1 MWth demonstration reactor ready by 2025. The project is being executed with Alfa Laval, Aalborg CSP and Weel & Sandvig with funding from the EU European Regional Development Fund.

Korean AMR

As well as developing the APR 1400 and Smart 100 reactors, KAERI is also developing a sodium cooled fast reactor (SFR) known as SALUS. This will have an output of 100 MWe. It is expected to have flexible applications across various purposes, i.e., on and off-grid electricity generation, district heating, industrial heat supply, replacement of coal power plants, etc.

14.4 Generation IV (Gen IV) **Micro** Modular Reactor (MMR) using Advanced Nuclear Technology (ANT)

Micro reactors are designed to be capable of providing a low-carbon, cost-effective, locally embedded, reliable source of power ***and*** heat for energy-intensive industries, hydrogen and synthetic liquid hydrocarbon fuels and also to be used in remote locations including space while also being able to provide grid support. The smallest ones are sized mainly to provide electricity 'behind the meter' i.e. not necessarily connected to the grid but electricity to be supplied at an unregulated price and directly to customers whose individual demands are rarely more than 30 MWe. The can also be mobile to enable them to be moved to locations where they are suddenly necessary and linked together, if required, to provide the required total output of heat or electricity. They can be embedded in industry or in communities and are designed to be inherently safe and do not require external utility connections for safety or cooling.

NANO Nuclear Energy

NANO Nuclear Energy Inc. is an unusually diverse nuclear technology company which is vertically integrated across five business lines:

(i) portable and other microreactor technologies;
(ii) nuclear fuel fabrication - High-Assay, Low-Enriched Uranium (HALEU) fuel;
(iii) nuclear fuel transportation;
(iv) nuclear applications for space and
(v) nuclear industry consulting services.

NANO Nuclear is also the first portable nuclear microreactor company to be listed publicly in the U.S (on NASDAQ).

Its diverse set of microreactor technologies includes:

"ZEUS", a solid core battery reactor, "ODIN", a low-pressure coolant reactor, and the space focused, portable LOKI MMR™. The Zeus nuclear microreactor features a unique design with a completely sealed core that relies on a highly conductive moderator matrix to dissipate the fission energy. Heat is then removed from outside the vessel using a novel proprietary technology. The absence of an in core fluid eliminates typical accident scenarios such as loss of coolant and loss of flow. Overall, Zeus is engineered for safe operation in remote locations. Its 'walk-away safe' feature ensures stability and safety, minimizing risks in varied environments. This aspect is critical for ensuring consistent operation without the need for active intervention.

Acquisition of Ultra Safe Nuclear Corporation

In December 2024, NANO signed an agreement to acquire select nuclear energy technology assets from Ultra Safe Nuclear Corporation. NANO Nuclear plans to highlight the high Technology Readiness Level (TRL) of its newly acquired patented Kronos MMR® and also Pylon (now renamed LOKI) technologies.

Whereas 'ZEUS' and 'ODIN' (original NANO reactors) are being designed to be portable and produce 1 to 1.5 megawatts thermal ("MWth") of energy, KRONOS MMR™ (which was an Ultra Safe Nuclear Corporation design) is a stationary installation capable of producing power up to 45 MWth. The individual MMR units can be linked to provide the level of power required and be stopped and started to address the summer/winter heat demand ratio. The MMR can be located almost anywhere. It uses no external cooling water and does not require a grid connection or other infrastructure to operate. It requires no refuelling for its 20 year expected lifetime which ensures security at the site as the reactor is a sealed unit. It has a small footprint of less than 5 acres and is designed to operate flexibly at power levels ranging up to 45 MWt (15 MWe) and can utilize either low-enriched uranium (LEU) or high-assay low-enriched uranium (HALEU).

Heat from the reactor is stored in molten salt tanks similar to concentrated solar powerplants (CSP). When electricity demand rises, hot molten salt is used to generate steam and power an electrical turbine. KRONOS MMR™ will typically store up to 10 hours of power plant thermal output and can be supplemented with hydrogen burners. The use of molten salt thermal storage allows for significant flexibility in the supply of both electricity and process heat. While the KRONOS MMR™ unit operates at constant power, electricity and heat are delivered on-demand from the power plant.

While it uses a steam generator to drive the turbogenerator set, after passing through the turbine the steam is condensed using air cooling and not water cooling. It can replace natural gas turbine generation and can match changing supply and demand if intermittent power sources (PV and wind) are also supplying power or if it operates as a heat and power island with no grid. Demonstration units are scheduled for 2026. The design is already in construction permit pre-application engagement with the U.S. Nuclear Regulatory Commission (NRC) in collaboration with University of Illinois Urbana-Champaign (UIUC). Pending Canadian governmental approvals of the acquisition, further demonstrations of KRONOS MMR™ are expected to take place at the Canadian Nuclear Laboratories. KRONOS MMR™ is the first small modular reactor to enter the Canadian Nuclear Safety Commission's formal licensing review.

Westinghouse e-Vinci Microreactor

eVinci can produce 5MWe with a 15MWth core design. The reactor core is designed to run for eight or more full-power years before refuelling. It will use TRISO fuel 19.75% enriched. It will not use any fluid for cooling or for transferring the heat externally from the reactor. Instead it uses the technology of 'heat pipes' avoiding the need for pumps. The heat output is at a temperature of 150°C as a heat source for a Brayton cycle for electricity generation. This reactor is thus using features that are new to reactors unlike its AP300.

TRISO stands for TRi-structural ISOtropic 'particle' fuel where each TRISO particle combines uranium in the centre encapsulated by a triple layer of carbon/ceramic materials This construction can avoid any release of radioactive fission products and has been tested at up to 1400 Centigrade, far above AMR decay heat temperatures. These particles are extremely small – said to be the size of a poppy seed – and can also be combined into much larger 'pebbles' around the size of a billiard ball which are then used in so-called 'pebble

bed reactors'. The heat transfer agent can be either helium or molten salt. The US Department of Energy has said that "Simply put, TRISO particles cannot melt in a commercial high-temperature reactor and can withstand extreme temperatures that are well beyond the threshold of current nuclear fuels."

Rolls-Royce (RR) Micro-reactor

RR has not revealed much about the technology of its micro reactor (in contrast to its LWR SMR). What we do know is that it is likely to generate from 1-10 megawatts of power and is likely to use TRISO fuel. Its more compact size makes it a transportable source of power. It is likely to be of value in applications including defence, remote civil power, remote industrial power and power in space such as on the surface of the Moon.

Blykalla LFR

Blykalla AB was founded in 2013 by Janne Wallenius, Peter Szakalos, and Jesper Ejenstam as a spin-off from the Royal Institute of Technology in Stockholm, Sweden, where Janne Wallenius has carried out research on design and safety analysis on lead-cooled reactor systems since 1996. The goal of this company is to construct and deploy 1,000 SMRs. It is a lead cooled 'fast' reactor (LFR). The innovative fuel is uranium nitride (U2N3) which has 40% more uranium per volume unit than standard UO2. This results in it being able to provide heat energy for 40% longer. It should also enable even better safety that some other SMR/AMR designs.

In the above sections, we have included only some of the many companies attempting to reach the stage of manufacturing small reactors (currently around 80). Which will succeed is not known at the present time. However, the industry (and society) will want to focus on those companies believed most likely to succeed to ensure economies of scale in mass production can be realised. At the present time we do not know the exact nature of their technologies as companies are reluctant to disclose some details.

Chapter 15: Financing the Global Deployment of Nuclear for Heat and Power

Financing growth in the provision of addition traditional (light water) nuclear generation of heat and power has been difficult in many countries. Problems have been common in obtaining financing at an acceptable price for large light water high pressure reactors. An example of this is the 3.2 gigawatt reactor currently being constructed by EDF in the UK known as Hinkley Point C. The problem in the case of this type of reactor is less the technology (though this is also a factor) than the evidence from build programmes in the last decade that predicted construction times have proven to be only a fraction of actual construction times and actual cost has exceeded forecast cost by a large multiple. In practice state financing has almost always been the only way of financing.

The absolute size of the financing need, a long and frequently delayed construction time and possible liability if something goes wrong has traditionally meant the need for heavy involvement of governments in providing risk mitigation instruments and frequently financing through a state-owned enterprise (SOE). SOE's can often provide finance at very favourable rates as they 'piggyback' on the high credit rating of governments. In practice it may be that a commercial lender would not achieve breakeven for up to 20 or more years and the cost of financing can be up to 67 per cent of the levelised cost of electricity of a nuclear power plant, according to the OECD Nuclear Energy Agency. Minimising that cost is critical in keeping down the cost of power from nuclear not least by using lower cost financing such as that derived from government borrowing. We will not cover the financing of gigawatt LWR generation facilities in any more detail as the more relevant issue is how to finance small and micro reactors.

Today's new types of reactor comprise a wide range of Generation IV technologies currently being developed, tested and progressing through what is known as the Generic Design Assessment (GDA) process and subsequently the site specific permitting process and then deployment. This is a different type of asset than the Gigascale reactor. While this type of asset is more likely to be able to be financed privately once developed and after many have been constructed, at the present time there is no private financing instrument that is well matched to the needs of nuclear. Project finance is not a solution at the present time (though with a state provided RAB structure it can be) but as with solar and wind, after Generation IV technologies have proved themselves and have reduced the cost of electricity per MWh, project finance is likely to become available. But a constraint that remains is that the lifetime of reactors is normally around 60 years. Investors, as noted above are not comfortable with projects that may not achieve breakeven for 20 years and therefore are unlikely to pay any dividends. If the cost of construction is amortised over only 20 years, the unit cost of generation will be much higher than if amortised over 60 years making the asset less competitive in the power market. One solution, which is not a private sector solution, is the Regulated Asset Base (RAB) model which we cover at the end of Chapter 15.3. Financing such long-lived assets through private finance, is a major innovation challenge we face.

In most countries in the past (and still in some countries), power generation (and the production of town gas prior to the use of natural gas) was provided through municipal authorities or state-owned companies. An example is the Central Electricity Board (CEB) in the UK created in 1926 and subsequently the Central Electricity Generating Board (CEGB) created in 1990 following the privatisation of electricity production and transmission in the UK. In the 1990's, after the UK electricity system was privatised, many other countries brought forward plans for privatisation. While the high absolute cost and risks of financing conventional nuclear generation have always been an issue, in the days before privatisation, financing was obtained using public funds rather than through private investors.

Since privatisation of electricity markets the subsequent discovery of natural gas such as that available from the North Sea able to be produced at a low cost, has resulted in there being very little nuclear construction during the private ownership period. For investors or utilities the regulatory and political risks, the stricter operational regime and history of cost overrun on top of a larger unit capital cost meant that natural gas exploration and production was a far more lucrative proposition than nuclear even though it was known even in the 1990s that CO2 emissions lead to global warming.

One reason for privatisation had been that some governments were no longer willing to be asset owners or to have new power plants requesting funding from the government with a consequent impact on the national accounts. While in most countries, nuclear fleet expansion took a back seat to conversion to natural gas, France took a different view. It specifically decided to go nuclear for up to 80% of its electricity supply because it had little

access to indigenous oil, gas or coal (unlike, for example, the UK which had extensive natural gas fields in the North Sea). Its intensive and standardised construction of a fleet of Gen II and III reactors is an instructive model.
It is probably true that the world will continue to need gigascale reactors. Nonetheless it now seems likely that distributed small and micro reactors and multiple small reactors linked to provide a gigawatt or more of output could be the way forward for providing much of the dispatchable 24/7 power supply (baseload) that a country requires. Along with an appropriate percentage of wind, solar, and geothermal teamed with storage and demand interruption we consider that nuclear will be a vital and large component part of the electricity system design of the future. Furthermore, non-fossil high temperature industrial and other heat can be better supplied by small nuclear (both SMR and AMR depending on required temperature) so that the number of such plants is likely to be very large indeed. The advent of much smaller and safer AMRs based on the physics of their design means they can be sited near to users and indeed embedded in industrial facilities. Adopting these new technologies has also opened the possibility of private financing coming to dominate a new model for nuclear heat and power.

15.1: Two Types of Financing Required for Advanced Nuclear Technology (ANT)

1. for AMR Development and Commercialisation:
2. for Widespread Global Deployment of AMRs/SMRs to Provision of Heat and Power

Financing AMR Development and Commercialisation

We are not covering the TRL stages up to TRL 8 in any detail as venture and other types of early stage financing are covered in Chapter 2.3. In addition, the companies on which we are focussing on are mostly already at, or close to, TRL 8. If they have managed to reach TRL 8 (and many AMR project companies will not), then they must have avoided what, in Chapter 2.3, is described as the Technological Valley of Death. After this point, however, there is an equally risky stage – "Commercialisation Valley of Death" also known as "The Missing Middle" of financing as we also covered at various points in an earlier chapter. This is the problem of transitioning from a viable product to mass production and global sales. This has been a big issue for nuclear projects in the recent past and for hydrogen projects. We also do not cover project and corporate financing in this section as these were covered for all types of infrastructure in Chapters 2.1 and 2.2.

The question we focus on in this section is how to finance companies that are **at the point when they can** enter the mass production stage (TRL 9) if only they could obtain the necessary financing. This is the point where, as Henry Ford demonstrated when he designed a production line system for making Model T Fords, there are huge economies of mass production to be obtained by using production lines. Today in the case of modular reactors, this will require a large investment in automated production systems which will involve the use of robots as in today's automobile factories.

Such mass production techniques for electric vehicles were clearly mastered by the upstart company, Tesla. Tesla's market value (MV) at the time of writing was just under $1 trillion At its peak value, it was worth more than the next fifteen global car makers combined including Ford, GM Stellantis, VW and BYD. We need the same approach to massively 'scaling-up' AMR production once companies have achieved TRL8. Elon Musk managed to achieve a large market share in electric vehicles despite Chinese competition while many other development companies have fallen by the wayside. Another such company is BYD which has mastered the art and science of low cost manufacture at scale.

Nuclear needs the same type of approach as, at the time of writing, with so many (at least 80) SMR/AMR companies said to be moving through the TRL stages in their attempt to reach TRL 8 yet most will never even reach that level, never mind TRL 9 which is when the scale of production becomes truly commercial. In the true spirit of capitalism, global competition and survival of the fittest is the most likely mechanism to enable the lowest cost designs to survive and scale in order to be able to produce competitive heat and electric power on a global scale. It has been a belief by investors in Musk's skills as an entrepreneur (at least prior to his involvement in his own and other countries' politics) that has so far enabled him to raise huge amounts of capital and build out the company's manufacturing capability. For investors it has been the prospect of a huge capital gain.

AMR's require the same type of approach to mass production. This will need both technical competence and capital. Energy utilities, conventional power generation equipment manufacturers and energy customers are often highly geared so only a few will have the balance sheet capacity or risk appetite to embark on such a development.

15.2 'Big Tech/Hyperscalers' Facilitation of Nuclear Finance through Strategic Investment and Power Offtake Contracts (Power Purchase Agreements)

S&P Global Commodity Insights held a webinar in late March 2025 which, amongst other things, provided a graphic (Figure 13.1) of the misalignment of datacentre future requirements in many countries. What this Figure notes is that **by bringing demand and supply for power closer together geographically, current and future grid constraints will be eased**. It then goes on to note that offshore wind and existing nuclear power stations are often located far from demand. On the other hand, the new designs of reactor which we cover below and which are coming to fruition at the present time need not suffer from this problem.

The first point in the 'wish list' below is that datacentres do not want to locate near renewable energy facilities if these are far distant from their customers. They need to be in a location where there is also **adequate data infrastructure i.e. capacity for their huge bandwidth demand**. Hyperscalers cannot move to a distant location from users unless they also have the necessary data capacity to meet their customer needs. The second point is that they generally are expected to have low carbon electricity for regulatory reasons and power which is additional ('**additionality'** in EU parlance) to existing supplies as many countries do not want datacentres monopolising existing and planned future low carbon power.

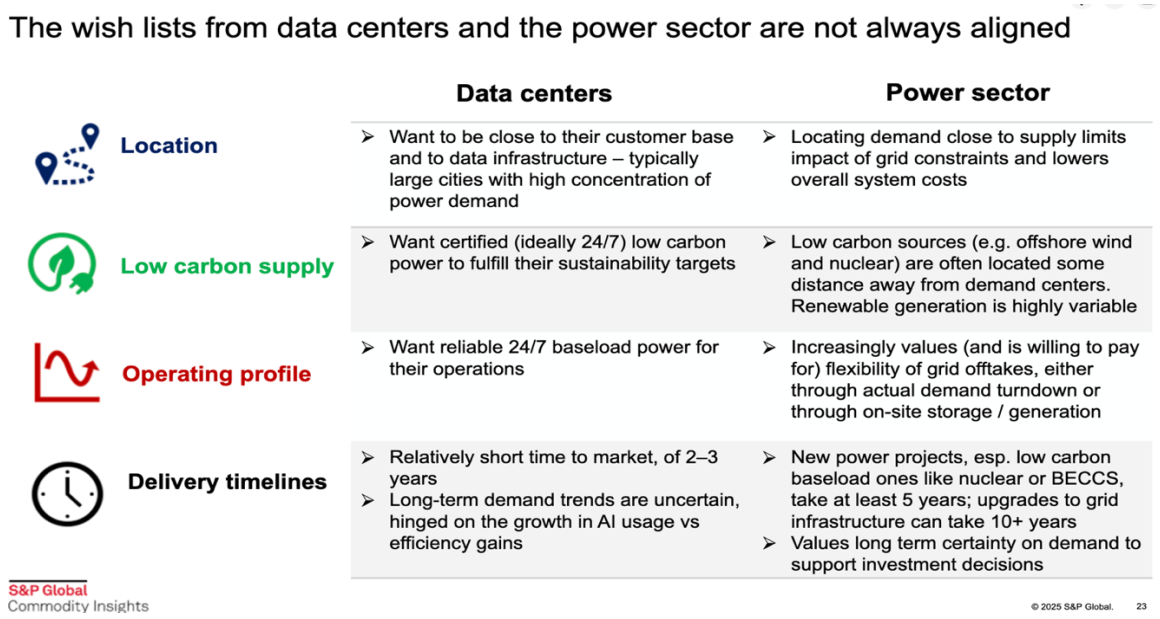


Figure 13.1 Datacentre needs and how the power sector might adapt to meet them.

The third point is that their 'operating profile' requires 24/7/365 clean power **on an hourly (or 15 minute) basis**. The final point – "Delivery Timelines" – notes that new generation assets are not yet ready to be deployed and will be at least 5 years away. The graphic also notes that updates to grid infrastructure can take 10+ years (and this is said to be the case in many countries) so the fact that advanced nuclear is probably 5 years away is much less of an issue than might be thought as it may not require the same use of the transmission network as renewables to enable it to supply power in the required locations. It can be located close to demand in an SMR campus or an industrial zone or even be embedded within a datacentre itself though it might also require a connection to the grid. This would involve the "private wire" approach which overcomes so many issues and may also enable power to be provided at well below the price that datacentres would need to pay if they relied on grid power.

Datacentres require 24/7/365 power which is difficult and expensive for renewable power companies to supply even at a high price beyond normal market prices since the investment cost of providing energy storage to cover the likelihood of a 'dunkelflaute' would be extremely high. This is the German term for a period in which little or

no energy can be generated from wind or solar because there is neither wind nor much sunlight. Periods without adequate wind can last for weeks and the gaps in supply are currently met by switching on natural gas generators. Big tech/hyperscalers/datacentre providers are therefore very receptive to proposals to link up with companies intending to offer electricity that is both carbon free and dispatchable 24/7/365 such as that provided by nuclear power.

It remains to be seen whether these SMR and AMR development companies, working with big tech companies, can also bring or buy-in the skills required for creating mass production lines which enable, for example, Chinese EV manufacturers to offer EVs at lower prices than those made elsewhere. The AMR product is in some ways simpler than an electric car build-process but it requires having to work under rigorous quality and safety regulations with which most developers and tech companies are unfamiliar (they are more familiar with the "move fast and break things" approach to development). But if they can, this will give rise to the possibility of greatly speeding up mass production and mass deployment of SMRs and AMRs worldwide and thus make a substantial contribution to mitigating climate change. At least as far as the UK is concerned, the so-called "Atlantic Partnership for Advanced Nuclear Energy" signed between the UK and US announced on 15 September 2025 may enable much more rapid certification of new reactors planned for construction in the UK. In announcing the partnership, the UK government noted that:

- Under the deal, both countries will fast-track reactor design checks, meaning if a reactor has already passed rigorous safety checks in one country, this work can be used by the other to support its assessment, avoiding duplicating work.
- This builds on collaboration between regulators for the ongoing UK assessments of GE-Vernova and Holtec reactor designs.
- This smarter streamlined approach will help speed up approvals for nuclear, to around 24 months, for advanced designs like the Rolls-Royce SMR, boosting British exports and getting Britain building. It comes as Rolls-Royce announces it has entered the US regulatory process for its SMR.
- UK and US will also work hand-in-hand to share the workload of new projects entering site licensing, speeding up delivery, to get nuclear project sites in the UK approved more quickly.

Big Tech companies that need to be certain of having 24/7/365 power available on long-term contracts (power purchase agreements) at a stable and predictable price are now investing in AMR development. Some are also forming consortia of large power users to benefit from the diversification benefit that arises from drawing power from a range of carbon free energy (CFE) electricity providers. For example, the US steel manufacturer Nucor, Google and Microsoft are working together *to develop new business models* by, amongst other things, aggregating their demand for advanced nuclear clean electricity technologies. This would provide a benefit from risk reduction through diversification of sources of their key input namely electric power and much greater security of supply if one generator goes off-line for whatever reason.

To achieve the benefits of advanced nuclear technologies, as we noted in Chapter 13.1, there is a need to revert to some aspects of the **model of yesteryear** when electricity generating capacity was sited near domestic and industrial users in cities or regions. Today it can also be **embedded within an industrial facility** or, as we cover elsewhere, next to a **hyperscale datacentre**.

Big Tech which comprises Alphabet (Google), Amazon, Apple, Meta, and Microsoft are large users of electricity to power their datacentres or chip production. The advent of Generative AI Large Language Models has dramatically increased their energy demand. As an example of this, according to the International Energy Agency, on average, a ChatGPT query needs nearly 10 times as much electricity to process as a Google search i.e. 2.9 watt-hours of electricity, compared with 0.3 watt-hours respectively. Nvidia, although it does not operate a datacentre, is at the centre of the AI boom. The demand for its chips from datacentre providers could be greater, if more clean power was available. For that reason Nvidia may also decide to invest in a nuclear AMR company to encourage greater demand for its chips. In addition, such investment could prove to be a profitable one for them regardless of its impact on demand for their chips. The only uncertainty regarding power demand by these US companies is if they are able to copy the approach taken in China by DeepSeek which is much less energy intensive.

Given the huge increase in expected demand for power by their datacentres, one solution for Big Tech would be to purchase electricity utilities around the world, outright. However, such a move would require permission from regulators and they are unlikely to agree to scarce supplies electricity being monopolised by Big Tech and diverted

from existing users. This would be at the expense of other consumers or passed into the hands of enterprises with larger financial muscle and political influence than all but the largest nations. The other answer is for these companies themselves to facilitate an expansion in the supply of carbon free electricity ('additionality in European Commission language).

Renewable supplies almost always come from a considerable distance from datacentres and in many places the installation of transmission/ distribution lines to customers is an issue. These transmission lines themselves are also a source of unreliability due to storms which can bring down power lines or ice on cables causing them to fall to the ground (c.f. the film Ice Storm). The year 2019 also saw the **First Climate Change Bankruptcy** when transmission lines owned by Pacific Gas and Electric (PG&E) fell on to parched land and caused an extensive wildfire. This resulted in claims by those whose property or other interests were affected, claiming damages or around $28bn which led to the company becoming insolvent.

The cost of transmission is clearly also an issue but the time delays in obtaining planning permission to put electricity pylons across a country is a larger issue. In Scotland, the 'unsightliness' of pylons and transmission lines required for moving renewable wind power from the North Sea to England, have apparently become an issue driving some voters to switch to voting for the UK Reform party. On the other hand, *ANT reactors can be sited near many datacentres thus eliminating the need for the transmission system operator to provide new capacity*.

In 2024 an alternative source of capital for AMR developers came into being at a very rapid pace. Several 'Big Tech' entrepreneurs who have financing capacity have decided to bring their big tech business skills and capital (and also a likelihood of signing PPAs) to AMR developers. The reason, quite simply, is that the new AI models they are developing, specifically large language models (LLMs) such as OpenAI's ChatGPT and the attempt to produce artificial general intelligence (AGI), require a very large increase in the availability of power for their datacentres. These companies are often known as "hyperscalers". In the current regulatory environment in most countries, they are unlikely to be able to obtain permission to construct a datacentre unless they can provide 'additionality' to a country's power supply. Also this additional power must be carbon-free (EirGrid, went one step further and issued a moratorium on the development of new data centres specifically in Dublin until 2028). An example of such hyper scale datacentres is Mark Zuckerberg's Meta's proposal to build a datacentre somewhere in the US which would cover an area of land almost as large as Manhattan. This facility will consume 'multiple' gigawatts while their next datacentre is expected to require 5GWe (i.e. almost twice the power produced by the latest large 3.2 GWe nuclear power station in the UK, Hinkley Point C.

Below we look at some examples of Big Tech companies, vendors of AMRs and utilities trying to overcome the problem of a shortage of additional dispatchable carbon free electricity.

Examples of Strategic Users of Carbon Free Energy in Datacentres as Investors in AMR Companies and AMR Construction Projects

Meta (Facebook, Instagram and WhatsApp)

On 03 December 2024, Meta announced[62] it will release a request for proposals (RFP) to identify nuclear energy developers to help it meet its AI innovation and sustainability objectives — targeting 1-4 gigawatts (GW) of new nuclear generation capacity in the U.S. Meta believes nuclear energy will play a pivotal role in the transition to a cleaner, more reliable, and diversified electric grid. That is why they announced that they would be issuing an RFP to identify the most appropriate nuclear energy developers to help it meet its AI and sustainability objectives. Meta hopes that the chosen suppliers can help them accelerate the availability of new nuclear generators and create sufficient scale to achieve material cost reductions by deploying multiple units, both to provide for Meta's future energy needs and to advance broader industry decarbonization. In their press release they go on to cover a number of critical financing matters that we have considered earlier:

> "Compared to renewable energy projects that we continue to invest in, such as solar and wind, nuclear energy projects are **more capital intensive, take longer to develop, are subject to more regulatory requirements, and have a longer expected operational life.** These differences mean

[62] https://sustainability.atmeta.com/blog/2024/12/03/accelerating-the-next-wave-of-nuclear-to-power-ai-innovation/

we need to engage nuclear energy projects earlier in their development lifecycle and consider their operational requirements when designing a contract."

Amazon & X-energy

X-energy's 'flagship' reactor design, the Xe-100 small modular reactor, aims to serve either as a baseload or as load-following support for intermittent renewable sources like solar and wind. Importantly, the company develops not just Generation IV high-temperature gas cooled nuclear reactors but also the TRISO-X fuel to power them. TRISO fuel is a key component of many helium-cooled reactors. Xe-100 is an 80 MWe reactor that can be scaled into a 'four-pack' 320 MWe power plant or can be even larger if required. Each unit contains 220,000 Graphite Pebbles with TRISO Particle fuel and can be refuelled on-line without interruption giving an expected 95% plant availability. The plant has been designed for a 60 year operational life..

In October 2023, Amazon, along with a utility Energy Northwest (a public utility) said it would **finance the construction of ANT reactors in Washington State and invest in X-Energy.** Like other AMRs the Xe-100 is expected to require only a 400 metre safety perimeter compared with 10 miles for conventional LWR reactors which probably means that it can be sited within the data centre's normal security fence.

Also, in March 2024, Talen Energy sold a 960-megawatt datacentre campus to Amazon Web Services a subsidiary of Amazon, for $650 million. The datacentre is directly powered by the adjacent Susquehanna nuclear power station which will provide a fixed-price nuclear power supply to Amazon.

Google & Kairos Power

Google has now made a commitment to Kairos Power to support the construction of multiple AMRs. The power output of the Kairos plant will be 150MWe generated by two 75 MWe reactors which function as baseload as it will be available 24/7/365. It will operate at near atmospheric pressure and like all ANT reactor designs, will have inherent design for autonomous safe shutdown and decay heat dissipation on loss of power. It will also use TRISO fuel in the form of ceramic pebbles similarly to X-Energy's Xe-100.

We noted in Chapter 11.3 on energy storage that on 25 July,2025, Energy Dome, announced a global commercial partnership with Google using Energy Dome's CO2 battery technology to enable carbon-free energy for the grids that power Google's operations. Alongside the commercial agreement, Google has made a strategic investment in Energy Dome. In contrast to Li-ion battery storage which provides backup power for 4 -6 or sometimes 8 hours, Energy Dome is capable of continuously dispatching energy for periods of 8 to 24 hours, providing enough firm electricity to meet both the baseload and flexibility requirements of large energy users. This is likely to be critical for the use of nuclear or any other technology for supplying electricity to hyperscalers' datacentres.

Microsoft/Constellation Energy/Helion

This company has made a 20 year power purchase agreement (PPA) with Constellation Energy, the company which owns the Three Mile Island nuclear facility in Pennsylvania. This involves re-commissioning one of the reactors to supply dedicated power to Microsoft.

Microsoft has also signed an agreement with fusion energy developer Helion Energy for the provision of electricity from its first fusion power plant. This is instructive because it shows that Microsoft's horizon is far longer than any existing utility, manufacturer or even developer who cannot underwrite such an expensive and perhaps speculative venture but a tech entrepreneur can.

Open AI/OKLO

Oklo is developing scalable nuclear power plants using liquid metal reactor technology. Utilities or others wishing to take power from Oklo power stations **will not have to invest directly in the construction process as is the case in a number of other ANT projects.** Instead OKLO will use long-term power purchase agreements (PPAs). As the company has said, this "takes the burden away from our customers" of having to invest in their own plant to use the technology. It is hoping to have its first reactor generating electricity at the Idaho National Laboratory by 2030. It calls its reactor the Aurora Powerhouse. It is

currently designed to produce 15MWe but can be scaled up to 50MWe and is expected to be able to operate for ten years or more before requiring refuelling.

The CEO of Open AI, Sam Altman, is also an investor in, and chairman of OKLO. It should be expected that he will be signing a Power Purchase Agreement with OKLO if the venture succeeds in getting to the point of generating electric power. In mid-December 2024, OKLO signed a 20 year power supply agreement with one of the largest datacentre providers – Switch. OKLA will supply reactors and generating capacity of up to 12 gigawatts (this could power the equivalent of over 7million homes 24/7/365 which wind or solar alone cannot offer. The deal is non-binding and the technology is not yet available thus the deal is an indication of intent rather than a true power purchase agreement (PPA). But OKLA is already hoping to have its first (test) reactor operating by 2027 though this test reactor will only have an output of 15MW.

15.3 Financing Global Deployment of AMRs/SMRs for the Provision of Heat and Power & case study of Nuclear RAB

How to finance the **purchase of standardised small (including multiple reactors on one site) off-the-shelf reactors by potential customers** once the production lines are working satisfactorily is the second financing issue. The facility might be for generating electricity or for the provision of heat or both. In the OpenAI/ OKLA example above, it seems that OKLA intends to finance the construction of the generating capacity itself and will contract with customers to sell them heat and power. We presume that financing would be available from banks and bond investors on the back of long-term power purchase agreements (PPAs) that would guarantee the necessary revenue flow to satisfy lenders i.e. the project would have a very good chance of being able to meet its full debt service costs and thus be bankable. Such contracts are designed to ensure that having financed the project, lenders have a high degree of certainty that the debt can be funded i.e. there is adequate 'guaranteed' revenue to service the debt (principal and interest payments). In addition, later institutional equity investors in the project such as insurance companies and pension funds will want to ensure that it will have sufficient revenue to provide an appropriate level of dividends to them.

More broadly, a nuclear generating facility (power station) or nuclear heat facility can, in theory, be owned and financed by a range of parties depending amongst other things on the final use(s) of the heat generated by the nuclear fission process i.e. power generation, industrial heat, combined heat and power for space heating, desalination, hydrogen production, powering a datacentre etc. A summary of some different approaches to financing is given below:

The Developer of the AMR
The company which has developed the AMR and is manufacturing them may also finance their deployment. This would result in such a company also acting almost like an electricity utility. The developer of the AMR might raise equity and debt and build and operate the plant to provide heat or power to a customer at an agreed price for an agreed number of years which avoids the customer having to source up-front funds for the project (see OKLO above). This is analogous to the BOO – build own operate – model extensively used for conventional power stations. This approach would involve the developer signing power purchase agreements with future customers to provide sufficient certainty of revenues to satisfy lenders.

An Existing Power Utility which could use its own balance sheet to raise the necessary financing (corporate financing, not project finance) and then sell the output to its existing or new customers the revenue from which will contribute to servicing its **general corporate debt and equity capital base**. The smaller value of each small nuclear plant (which can have additional reactors added over time) compared to the Gigascale reactors, makes this model more digestible to the utility, its shareholders and its lenders than a single large GW scale of plant.

Large Customers providing advance payments for future power to the developer and owner of the plant which provides the financing required or a proportion of it and thus minimises the need for the developer to find 100% external debt finance. Customers may be industrial firms, utilities or a new utility set up by a local authority to supply power. Customers may do this when they are not happy to take on construction risk from the developer but are willing to help finance the project developer in return for an attractive PPA. Once the plant is operating, these risks reduce so the customer may then wish to scale its investment up to 100% of the constructed capex while either leaving the developer as the operator or taking that on as well. This is similar the BOOT model – build own operate *and transfer* - which is again common in the power industry.

Retail customers of an electric power utility. It may be possible, depending on the regulatory framework, for retail customers to have an additional charge imposed on them which would be used to make dividend payments to equity investors during the whole construction period (as in Thames Tideway, discussed in Chapters 4.4 and 4.5). The additional charge could also be imposed at a level sufficient to cover interest payments during the construction period when no revenue is being generated. Thus interest would not be accruing at the time before the plant starts to earn revenue and therefore increasing the capital sum outstanding to the bank. This approach would considerably reduce the cost of power when the facility started operation. It would be a particularly good approach when interest rates are 'high' as the cost of servicing debt is the largest operating expense of a nuclear power company. This model has been the basis for GW scale nuclear in the USA for several decades.

User Ownership – joint ownership by a group of electricity users

Mankala The Mankala-model is a not an uncommon ownership arrangement in the electricity production companies of the Nordic countries. In Finland, for example, Teollisuuden Voima Oyj, owned by Fortum and Nordic Power, and in Sweden nuclear power plants in Forsmark and Oskarshamn, cross-owned by Vattenfall, Sydkraft and Fortum are organized in this way.

The purpose of the Mankala-companies is to produce electricity for the joint owners at the lowest possible cost by producing energy themselves. The owners obtain electricity in proportion to their ownership at cost price and hence the purpose of the model is to be able to sell power to the owners but without making a profit for the captive operating entity.

The model has shown itself to be a way to execute new investments so that the costs and risks of a new plant investment are shared between participants who have an actual interest in the outcome beyond the bare financial return. In Finland, construction of unit 3 of the Olkiluoto Nuclear Power Plant began in 2005. Commercial operation, originally scheduled for May 2009, began only on 1 May 2023. This points up quite dramatically, the risks in Gigascale plant construction using a first of a kind (FOAK) design, even for a very experienced nuclear owner/operator and plant contractor. In this case the same design lies behind the problems at Flammanville, Taishan and Hinkley so that the rate of learning for such a behemoth plant, the largest ever built, is slower than could be expected on simpler AMRs requiring a much reduced scope of the safety case.

***Examples of customer financing and user ownership are given below**:*

Customers - power

Both retail and wholesale customers could contribute to the financing of an AMR project. For example, long-term retail bonds offered by an SPV but underwritten by the government and intended for pension savings. Holders of these bonds could also be offered price discounts for life from the local power retailer once power started to be generated. This could be attractive to pensioners who could find the concept of lower future bills or even a whole life 'energy annuity' attractive while still obtaining a higher yield from such bonds than on their existing savings instruments.

Wholesale purchasers of power with spare balance sheet capacity could be offered attractive long-term energy offtake contracts in return for an upfront payment. An example of this approach is the Exeltium transaction under which EDF and Exeltium (a consortium of electro-intensive users) signed a memorandum of understanding on a take-or-pay contract as covered below:

Exeltium – A Consortium of Large Electricity Users Contributing to Nuclear Financing
France has the world's highest proportion of its power generated by nuclear. The fleet of reactors is in state ownership through the state's ownership of Électricité de France (EDF). Exeltium is a consortium of heavy power users through which they have contributed to the up-front capital cost of a new nuclear power station by contracting with EDF to assist with its financing (in this case Gen 3+). In exchange, EDF has provided them with a long duration fixed price supply contract which minimises price risk for the off-taking companies.

The logic behind this transaction with customers is that for electro-intensive industries (aluminium, chemicals, industrial gas, paper, steel), electricity supply and its cost is a major competitiveness issue as it represents between 15% and 50% of production costs. Industrial investments are long term in nature (10 years horizon at least) and therefore uncertainties regarding electricity price evolution threaten the sustainability of these industries particularly in the face of competition from countries with much lower power costs. Since the

liberalisation of the electricity market in 2000 in France, market prices have increased and became unpredictable so that companies' ability to invest in France in their areas of core competence, which is not power, was at risk.

In July 2005, the French government organized a roundtable with electricity producers and electro-intensive consumers to address the growing uncertainties over long term electricity prices. Following this, electro-intensive industrial companies founded a consortium called Exeltium, whose purpose was to negotiate an optimal nuclear electricity supply contract on behalf of its shareholders. Note that nuclear is uniquely able to offer this kind of contract because its main costs are initial capex and the variable operating costs are only about 10% of the electricity cost. Arguably renewables have a similar high capex but low operating variable cost structure but large capital intensive process industries cannot tolerate a potentially highly variable power supply and the cost of ensuring power supply through energy storage is expensive. The shareholders of Exeltium included Air Liquide, Arcelor Mittal, Arkema, Rhodia, Rio Tinto Alcan and Solvay. Despite difficulties in markets following the 2008 financial crisis, in 2011, Exeltium completed its raising of €2 billion.

Customers – heat

The economics of power stations improves very substantially if used for combined heat and power. ANTs can be situated safely in built-up areas to supply industrial process heat or for new or existing towns for industrial processes, power generation and finally for low temperature space heating with each energy product (electricity and heat) having its own income stream.

Individual space heating (gas boilers) for homes and offices particularly in cooler latitudes can be replaced through the deployment of a district hot water heating network. The addition of heat customers can alter the economics of small nuclear power and make it much more bankable. The technology of district heating is mature but is usually fuelled by fossil or biomass sources. Nuclear combined heat and power or pure heat offers a carbon-free heat alternative. At a conference on the proposed Oxford/Milton Keynes/Cambridge Arc project which was intended to provide housing for more than one million people and was described as "A long-term vision to secure sustainable growth and prosperity for all", the principal author of this book delivered a presentation on how decentralised power and heat for this growth could be provided by AMRs.

In July 2025, the European Commission proposed opening part of its next long-term budget - worth €2 trillion- to include funding for nuclear energy. The draft budget opens up national EU funding streams to support "new or additional fission energy capacity installed in GW." Around 865 billion euros of EU funding would be accessible under these national spending plans. In contrast this, the current EU budget explicitly excludes conventional nuclear power. While final approval will require consensus among all the 27 member states, the draft suggests institutional acceptance of nuclear as a key part of Europe's energy mix.

Banks, Institutional Investors and Sponsors finance most infrastructure projects including public private partnerships. Generally, bank finance (project finance) is used for the construction phase and the loan may be re-financed in the bond market at a later date when the early stage risks are reduced. Banks are, however, unlikely to be willing to finance the first of a kind (FOAK) of a new technology as it is not their function to take what would normally be equity risk unless, as in renewables, much of that risk is transferred to a government. Once a new reactor design has been successfully completed on time and without a cost overrun, it is more likely that banks would be willing to participate in funding.

In December 2023, the United Nations Climate Change Conference (COP28) in Dubai saw the 198 signatory countries to the UN Framework Convention on Climate Change call for an acceleration in the deployment of low-emission energy technologies including nuclear power to enable deep and rapid decarbonisation particularly in hard-to-abate sectors such as energy intensive industries requiring heat. In addition, **25 countries at COP28 pledged to work towards tripling global nuclear power capacity to reach net-zero by 2050.**

At New York Climate Week in September 2024, a group of **14 financial institutions expressed support for expanding nuclear power generation and the nuclear industry supply chain**. The institutions included: Abu Dhabi Commercial Bank, Ares Management, Bank of America, Barclays, BNP Paribas, Brookfield, Citi, Credit Agricole CIB, Goldman Sachs, Guggenheim Securities LLC, Morgan Stanley, Rothschild & Co, Segra Capital Management, and Societe Generale.

An example of one of the banks above with a strong focus on sustainability and on nuclear financing is Crédit Agricole Group. At Euromoney's July 2025 Awards for Excellence it was named World's Best Bank for

Sustainable Finance, Europe's Best Bank for Sustainable Finance and France's Best Bank for Sustainable Finance. The selection process followed a joint application by Crédit Agricole S.A., Crédit Agricole CIB and Amundi. It also received the award for "Asia's Best Bank for Sustainable Finance", recognizing its leading role in sustainable finance. The bank is also known as "la banque verte". Crédit Agricole CIB. supported Korea Hydro and Nuclear Power in issuing Asia's first green nuclear bond.

Supranational Agencies such as the European Investment Bank (EIB) will often provide some of the financing required by an infrastructure/ power project. In principle, their financing is little different from private bank financing but may be at a lower interest rate or have some different terms. Such agencies may also offer guarantees or other risk instruments. They also may be guided by the policy of their supranational region, such as the EU's recently more pro-nuclear energy taxonomy.

A Public Private Partnership (PPP) is a structure frequently used to provide a service to a government or local authority, for example, a hospital or school. In a PPP, private financing will almost always provide most of the financing to keep the debt off the public sector borrowing requirement. It also enables some or all of the risk to be kept within a special purpose vehicle or with a construction contractor or a plant manufacturer. These latter may be more willing to come forward when the absolute sum of money per AMR plant is compatible with their balance sheets. This is common for process plant and gas power stations and may well be realistic for small nuclear but less so for GW scale.

Oil and Gas Majors are under pressure from investors firstly to reduce their carbon footprint and second to use their free cash flow to diversify away from fossil fuel. But equally they are under pressure to maximise returns in the short run particularly when oil and gas prices are high. They have the potential to increase their investments wind, solar and nuclear but in general do not seem to be doing so at the rate that could be achieved. Shell and British Gas were both early movers into thermal electricity and nuclear but withdrew when power generation shifted at very large scale to gas, which was their core business. Regulators also 'punished' electricity technologies except renewables which were subsidised. Gas and oil majors have plentiful international investment opportunities in their core businesses and at the time of writing have also cut even back their relatively small funding of wind and solar assets because they perceive that their customers and shareholders are insufficiently green-sensitive.

The Asset Recycling (Australian) Model. This takes advantage of a government's low cost of borrowing which is used to finance a project up to the point when its operational revenue stream is relatively certain. The government therefore takes construction risks for which it is rewarded. It will aim to gain a return over a portfolio of projects. It does so via a high debt coupon paid by the developer and by selling-on projects to institutional investors after completion. By seeking bids to buy an operational asset with known cash flows, the price received from the sale of such an asset should exceed the cost of constructing it.

If properly managed, the internalised risk for cost or time overrun or for underperformance against expectation. should, over a range of projects, be adequately covered by the final capex mark-up on sale. The buyer benefits from the project being financed at the *sovereign borrowing rate plus a risk premium to cover the risk in its total portfolio rather than just a single project*. When the sale has completed, the government's proceeds can then be used/ recycled to finance other projects.

The key advantage should be a reduction in financing costs due to the reduction in the cost of risk capital from the start of the financing. However, the risks that are being taken on by government may or may not be compensated by the interest on the government loan depending upon how well the government is able to evaluate risk. This model is like a build operate and transfer contract (BOT).

Revenue support Mechanisms: CfDs and Regulated Asset Base (RAB)

A CfD is a contract between a power producer and a government. However, it is not an "offtaker" contract for the sale of power (unlike a PPA) and thus does not guarantee the actual sale of power. It is only a mechanism to guarantee a minimum price for the sale of power by a project over a period of some years. It is an agreement to pay or be paid the difference between the reference and strike prices while the generator is responsible for **physical power sales**. In the case of renewable power contracts, the generic CfD offers a 15 year contract term to provide **equity investors** with confidence that they will achieve their expected return and for lenders, it should make the project **bankable** meaning able to obtain debt financing. However, in the case of nuclear where the design life of the project is likely to be 60 years, it

would be desirable to have a longer contract period though some projects may choose a 'merchant' basis for generating revenue in which they sell into the market without any revenue protection in the hope of generating higher revenue. In practice, few generators would consider 100% merchant sales (and indeed this would almost certainly make the project unbankable) but many would consider holding back a proportion of sales during the life of a CfD as well as sales subsequent to the expiry of the CfD or power purchase agreement.

The Regulated Asset Base (RAB) model is the traditional proven solution to funding capital intensive infrastructure and is used in electricity, water, gas and telecommunications worldwide. The RAB model, by providing regulated returns to investors, has the potential to reduce the cost of raising private finance for new nuclear projects. As economic regulators are required by statute to ensure that investors obtain the agreed rates of return, this financing mechanism has already been successful in enabling private sector investment in other infrastructure sectors such as the National Grid's UK investments in gas and electricity. In the US, is used for nuclear, coal and gas power stations.

Like the nuclear CfD model, the nuclear RAB model has the potential to reduce the cost of private finance for new nuclear projects by sharing risk between investors and consumers (as in the **Retail customers of an electric power utility model** in a section above). Unlike the CfD model, the UK RAB model also **allows the generating company to receive payments from customers during construction**. These can then be used to provide a return (dividend) to investors or/and interest payments to lenders (instead of interest being 'rolled-up') thereby reducing the overall cost of the project.

Case Study
RAB Financing Model: Sizewell C in the UK

This is the first use of the RAB model for nuclear in the UK. Total cost estimate £38bn (at 2024 prices) for a power output of 3.2GW which represents a saving of around 20% compared with Hinkley Point C of which Sizewell is a replica second-of-a-kind project (SOAK). Output 3.2GWe which can provide the equivalent of sufficient electricity for 8m homes. It uses Generation III+ light water reactor (LWR) technology. The new Sizewell C owners include La Caisse with 20%, Centrica with 15%, and Amber Infrastructure with 7.6%. This comes alongside EDF taking a 12.5% stake in the project. as well as a proposed £5 billion debt guarantee from France's export credit agency, Bpifrance Assurance Export, to back the company's commercial bank loans.

The UK government is providing £3.8bn in equity and £36bn in debt. As the debt will be issued by the UK Treasury (gilts), the interest rate will be much lower than on bank or bond debt. £400m in equity is being provided by the French Nuclear Liabilities Fund (owned by EDF) and others to pay for de-commissioning costs which should not arise for 60 years after the plant is commissioned.

Bringing in private investors requires offering them a package which transfers of many of the risks to other parties such as overspend and delayed completion. Centrica's expected rate of return is 12% which is lower than it would be without such risk transfers. However, it is considerable higher than many other infrastructure projects which might return only 7% per year. The risk transfer involves removing the obligation from investors to provide funds beyond a project cost of £47bn but up to that point they are responsible for cost overrun.

The above description of the project's financing needs to be followed by a consideration of the different risk management mechanism used in previous nuclear projects which was a contract for difference (CfD) The key difference with RAB financing is in the funding mechanism. The RAB starts paying dividends to investors and interest to lenders from the point when investors transfer funds to the project company. A surcharge on the electricity bills of all electricity consumers until the project is revenue earning which might be ten years away should not result in more than £1 per month being added to consumer bills. But this will service debt

15.4 Project Developers, Independent Power & Heat Producers (IPHPs) and Community Projects: Kärnfull Energy, Kärnfull Next & Community Nuclear Power Ltd

Energy Communities

Recognising the potential contribution of energy communities in achieving a more secure, affordable and cleaner energy system for Europe, the REPowerEU Plan put forward the shared political objective of achieving one (1) energy community per municipality with a population of more than 10 000 by 2025. The Commission has noted that:

> *"They can contribute to increasing public acceptance of renewable energy projects and make it easier to attract private investments in the clean energy transition. Energy communities can be an effective means of re-structuring our energy systems, by empowering citizens to drive the energy transition locally and directly benefit from better energy efficiency, lower bills, reduced energy poverty and more local green job opportunities. Energy communities allow local communities to join forces and invest in clean energy. Energy communities allow local communities to join forces and invest in clean energy."*

What is not clear from the above is whether community **nuclear** would be included under the term 'clean energy' as used above.

Kärnfull Energy and Kärnfull Next

Kärnfull Energy was started by two entrepreneurs, Christian Sjölander and co-founder John Ahlberg in 2021 as a *supplier of electricity* in Denmark which has been generated 100% from nuclear. It subsequently set up another separate but related business, Kärnfull Next, as the Nordic region's first *project development company for small-scale nuclear energy.* They have built up what is already a profitable and rapidly growing nuclear energy company group as well as promoting the need for nuclear to accelerate the Nordics' (and the world's) journey to reach the Paris goals. They consider the company to be a community and industry developer. Two investors which have backed the project are Climentum Capital and Granitor Growth Management. Granitor has noted that they see great potential in Kärnfull Next's strategy to decarbonise energy intensive "difficult to decarbonise" sectors. The investment reaffirms their conviction that *nuclear energy is a key part of a sustainable future.*

In its SMR project developer role, Kärnfull Next is "tech and vendor agnostic". In practice, to have a generating project operating as soon as possible, they have chosen for now to use LWR reactors as these are currently available. It is likely in future, when the technology is commercially available and economically attractive, that they will also use AMRs in their projects and if successful, would drop the use of SMRs. The company has chosen the GE Hitachi SMR BWRX-300 (300 MWe output) for its first project. It covers the full value-chain for new-build projects company, with everything from permits to licensing, siting to land rights, design decisions and local dialogue. It takes on early stage financial risk in projects and its team creates value through replicable processes across multiple projects (which should generate economies of scale, both by lowering the price of the reactors they purchase and the cost of their construction. This is in sharp contrast to the issues that have been witnessed in the construction of Gigascale reactors where new builds generally have been relatively different to previous builds. There also seems to have been little in the way of 'learnings' from previous builds being applied to new builds.

Their objective is to set up **"SMR Campuses"** as mentioned above on which there would be the nuclear source of heat and power, electrolysers to produce hydrogen and synthetic fuel plants which would combine captured CO2 using CCUS and DAC with H2 to manufacture synthetic liquid and gaseous fuels. Such a campus could also manufacture ammonia and fertilisers without the need to use natural gas as the main input. **The campus idea is in line with the idea of "clusters' and a sector coupled economy**. which we cover in Chapter 19.

Interestingly, Kärnfull Next is not the only new nuclear company to have been set up by entrepreneurs in the Nordic region. Denmark also has Seaborg Technologies and Copenhagen Atomics, both of which have developed highly innovative AMR designs which are moving closer to achieving certification. These new reactors are even safer than the current already very safe Generation 3+ nuclear technology. In addition,

they can operate as 'burners' and thus greatly reduce conventional nuclear waste by using it as fuel – i.e. circular nuclear power.

Community Nuclear Power Ltd (CNP)

The proposed business model of CNP has similarities to that of Kärnfull Next. They are not developing a new SMR or AMR themselves but have chosen, like Kärnfull, to use a LWR that has already been developed and is ready to build. This enables them to get to market much more quickly than waiting for the right AMR to be on the market.

They see their project as a "regional development' project as well as an economic one which involves deploying a low-risk nuclear heat, power and hydrogen (and its derivatives) to enable a transformation programme in a region – initially in the North East of England industrial cluster on North Teeside. While clearly such a project must provide investors and lenders with an appropriate rate of return on their investment, they are also focussed on what it can do socially for a region as well as in economic return terms. The community perspective arises in considerable part from the background of the founding entrepreneur, Paul Foster, the CEO of the company. He worked for 20 years at the Sellafield nuclear site, in senior roles including as Chief Executive Officer. Sellafield is one of the world's largest and most complex nuclear projects which has delivered some millions of pounds in community benefit projects.

As in the above graphic from Kärnfull Next, the component parts and agreements needed to make this ground-breaking proposition happen, land, capability, technology, private capital funding and community demand are already in place and are aligned with the Great British Nuclear (GBN) roadmap. GBN is a UK government body set up to help enable the delivery of the government's long-term nuclear programme and support its ambition to deliver up to 24GW of nuclear power in the UK by 2050.

The company is working with Westinghouse to deploy four AP300s the design of which is based on well-proven technology (AP 1000) already in use around the world. The four reactors will provide around a gigawatt of energy, which is the demand expected from the green energy and chemical hub planned on the North Tees Group Estate.

15.5 Changing Mindsets – How to Speed the Scaling-up of AMR & SMR Mass Production for the Provision of Carbon Free Electricity and Heat

Although AMRs and SMRs are expected, once approved by regulators, to be safe to install in built-up areas, populations close to intended nuclear sites will still need to be convinced that a reactor near them will not be a danger. It may also be necessary to prove to people that it brings them benefits such as a lower price for power within a local distribution network.

Legislators may also need to gain a better understanding of why nuclear is so necessary to achieve climate objectives and this requires both an increase in understanding of why these new technologies both Gen III+ and, more particularly Gen IV are so much safter than even the existing global fleet of reactors but also to understand how a nation's energy system as a whole will work with increasing proportions of variable renewable electricity (VRE) being generated. There is a need for a "systems' approach to the cost of generation and a regional approach to how to achieve a locally integrated energy system.

Increasingly we think that there will be regions where wind and solar are used along with unmitigated (or eventually mitigated) natural gas to "fill in the gaps" while other regions will use principally nuclear i.e. for baseload. Baseload AMRs/SMRs will provide domestic heating and electricity while micro reactors may be embedded in factories operating in industrial zones where surplus electricity and heat will be used to manufacture hydrogen and its derivatives including synthetic liquid hydrocarbon fuels such as sustainable Aircraft Fuel (SAF), ammonia, fertilisers etc. with waste heat will provide district heating. A good example in Europe where such a configuration could work would be the proposed Oxford/Milton Keynes/ Cambridge "Arc" where it is planned to build over one million homes over a number of years and improve transport systems between these towns.
We outline below some of the 'changes in mindset' that will be necessary:

Don’t think about maximising the proportion of renewables in the generation mix. Think about maximising carbon free energy (CFE).

In 2023 in Sweden the Tidö Agreement included a change in the energy policy goal. It was "changed from 100% renewable to 100% fossil-free". Other countries need such a change in mindset to achieve climate goals.

Think both heat and electricity

Heat comprises half of the world’s total energy consumption – which is significantly more than electricity (20%) and transportation (30%) though proportions vary by country. Unsurprisingly, heat contributes more than 40% of global energy-related carbon dioxide (CO2) emissions according to the World Economic Forum. Industrial heat accounts for half of all heating but requires innovation in ‘hard to decarbonise’ sectors of industry. Space heating of homes and offices comprises most of the remainder. CFE space heating can be achieved through heat pumps, district heating systems (DHS) using heat pumps or district heating and by using waste heat from micro nuclear installed in new build, or conversion of natural gas DHS, or by installing DHS to provide heat to existing homes.

Thus the objective should not be simply attempting to maximise carbon free *electricity* production. The objective should be trying to decarbonise the electricity system AND the heat system (and of course the transport system) while minimising the unit cost of power and heat.

Don’t think of some hundreds of nuclear heat and power generation stations worldwide – think of many thousands and the size of the manufacturing market as a potential boost to economic growth

Nuclear can function either as baseload or as an aid to integrating renewable into the grid. If, however, the cost of nuclear power for a whole grid system falls substantially, it will make sense in some locations for small and micro nuclear to be used to provide a major share of electricity with renewable providing the balance. This is an alternative model for a power system network to one in renewables provide the major share of electricity and nuclear provides the balance.

Think about whole **system** cost and not price of power when the sun shines and the wind blows nor the price at the ‘farm gate’

When it is claimed that solar power is the lowest cost power supply available, while this may be true in some geographical locations **“at the farm gate”** there are other substantial costs to be considered. The first of these arises from the need to transmit electricity from where it is generated on a solar or wind farm to the customers’ point of use. Solar is generated, typically, at a rural solar farm while consumers tend to be concentrated in cities far from the source of generation. In the case of wind farms, in Europe most wind power is generated offshore and mostly in the North Sea. The costs of transmission to the coast and then from coast down to the demand location can be very high. Often objections to new routes for pylons and cables are sufficient to stop such transmission or to delay it by many years or raise its cost by forcing the cables to be laid underground instead of overhead. This ‘system cost’ can substantially raise the true cost of solar and wind and the time to generating revenue from a project due to the current long duration ‘pipeline’ of applications for connections to the grid. As we noted earlier, in the US, transmission and distribution costs are already 44% of the total cost of delivered electricity according to the WEF. Small and micro nuclear can eliminate the cost of transmission and thus on a closed network, can greatly reduce the cost of power at the customer’s premises.

An **equally important system cost** is that of converting the variable output wind & solar (which seldom provide their full so-called ‘nameplate’ power) to power that is dispatchable. Solar has a low annualised ‘capacity factor’ from as low 10% of the peak nameplate rating of its panels to perhaps 25%. The cost of short term energy storage systems for wind or solar (a few hours at most) remains considerable. Long duration energy storage is much more expensive still as the assets involved are not called on frequently and must thus charge a very high price when called into service. During “wind droughts” which can be for periods of weeks there remains a need for power and this is currently being met by ‘turning up the gas’.

A further consideration is the annual cost of **constraint payments to solar and wind farm owners** who are asked to stop production on very sunny or very windy days when supply exceeds demand. This can also be very high and in the UK, for example, is said to be of the order of magnitude of £1bn per annum. Such payments, though obviously popular with plant owners, simply increase the price charged to consumers for every kW hour of power they purchase.

As the percentage of renewables on the grid increases, the **system** costs of adding more, increases non-linearly due to the need to compensate more for temporary overproduction, intermittency, transmission costs and in some cases the rising price for suitable land or seabed.

Finally, for the investor, rising proportions of renewables are well known to '**self-cannibalise' the economics of the renewable technologies** by reducing revenue as a result of too much wind or solar wishing to offer power to the grid. So far this economic cannibalisation of renewable owners' revenue has been funded through both state taxation and higher electricity pricing. Germany and UK have Europe's highest electricity prices which is also correlated with the highest renewable penetration in the system (though there are other reasons as well).

AMRs and SMRs are a quite different reality from our feelings about legacy reactors

The differences in technology, possible locations, time to build, safety and required land area between legacy reactors on the one hand and SMR's/ AMRs on the other hand is not widely understood. It's a shift from reactors occupying a huge ground area well away from cities to a decentralised generation model using relatively small plots of land, close to or, in the case of micro reactors, embedded in user facilities or residential and business districts.

It's a reversion to the old model of a utility – local electricity generation (and town gas production from coal) within a town or city or very close by. Bringing generation closer to users is important because we cannot quickly, or affordably, put in the extra transmission infrastructure to bring wind power from, for example, the North Sea or Baltic to the demand centres in south Germany or even afford to do so.

Following on from this, the change of mindset is the need to think about **community** power generation from nuclear i.e. reverting to something like the municipal electricity (and gas and water) model of the past as covered in the paragraph above. Kärnfull Next in Denmark (see Chapter 13.10) is a good example of this approach. In the UK there is Community Nuclear Power Ltd. (also covered in Chapter 13.10). The latter has been set up by nuclear experts to try to enable people to understand that we can bring nuclear home with substantial advantage to local communities both as offtakers and potentially as investors of a small part of their savings and pensions in a real asset rather than outside their own community or even outside their own country.

Remember that de-commissioned coal power stations still have useful locations with grid connection

SMRs and AMRs are well suited to **repowering fossil fuel-fired plants,** allowing some employment in an area to be retained with well-paid high-skilled job opportunities. Unless this is widely known by members of parliament, legislators and local government, this substitution is unlikely to happen. An example of a company attempting to achieve this objective is Terra Praxis, a UK company which has well developed plans but is mainly active in the USA funded partially by a founder of Microsoft.

Small distributed nuclear gives much greater defence against foreign threats including missiles and 'cable cutting'

Today, defence security in times of foreign military threat gas has come to the fore. The Nord Stream 2 explosions from sabotage as well as cutting potential supplies of gas to Europe, also resulted in the worst release of methane gas in human history, with estimates ranging from 100,000 to 400,000 tonnes of methane released into the atmosphere.

Distributed small nuclear for heat and power in hundreds of locations is far less strategically vulnerable to concerted cyber or physical attack (such as Ukraine has experienced from Russian missiles) than a few large, concentrated installations and network hubs. More tactically, mobile microreactors which we covered earlier, have the potential to provide critical defence and societal installations as well as public services with a supply of continuous power and heat for several years without the need to refuel. Because small and micro nuclear reactors can be sited close to the point of energy consumption they are much less vulnerable to the damage which is very easily inflicted (and is already being inflicted in Europe by 'hostile' powers) on submarine cables, international interconnectors or overland transmission systems.

Regulation of nuclear needs to be regional and international

Regulation is an issue both within Europe and internationally. While EU countries can make their own decisions on the generation mix, we need to have a Europe-wide (and we include the UK in this) regulatory framework for the safety regulation requirements such as the **Generic Design Assessment (GDA)** process for nuclear. Radiation hazard itself is of course transboundary not national. A more prosaic reason is also that it is wasteful and slows down the energy transition if each country must undertake regulation of these new technology assets separately.

It should be noted, however, that the **site specific permitting** stage that takes place after the Generic Design Assessment (GDA) stage is less applicable internationally as country risks may be due to seismic considerations, weather or geotechnical issues which apply to particular locations only. A reconsideration of the types of risk that this entirely new design of reactor poses relative to our previous experience of nuclear is needed. Indeed without such a reassessment, mass production with its prize of cost certainty, lower selling prices and faster rate of decarbonisation are at risk.

Within Europe, but ideally worldwide, there is a need for common regulatory framework applying not just to reactors but, very importantly, also to the new fuels and methods of fuelling that have been developed. There are several new types of fuel and more particularly so-called high-assay low enriched uranium (HALEU), TRi-structural ISOtropic particle fuel (TRISO) and pebble bed fuelling which have not been used before in commercial reactors. It would be unfortunate if every country in Europe or world-wide were to do their own testing. It would certainly slow down, and raise the cost of, the energy transition.

Scarce land use planning – agriculture versus other uses

This issue is particularly relevant to countries with a high population density and with many people dependent on agricultural production. But it remains important for all countries in Europe. The issue is the amount of land taken up by a solar or wind farm (or even legacy reactors) and therefore taken out of agricultural use. While "agrivoltaics", which is the joint use of land by solar farms and agricultural has been suggested as a solution, to date it is not well established and may never be the hoped for "answer" to the land issue. We referred in Chapter 13.2 to the Italian government disclosing that Italy is now planning to have Generation IV small modular reactors operating within 10 years and that they have also imposed restrictions on new solar power projects in the belief that they threaten food security by using-up cultivatable land. India is another country which is seeing "a revolt against solar" because it can destroy farmers' ability to earn a living as well as reducing agricultural output.

Issues Specific to Europe

The French Nuclear Alliance

Brussels to put nuclear and renewable energies on an equal footing when it comes to funding, research, and project development The French-led nuclear alliance, which now has 12 EU member states has noted that nuclear technologies do not benefit from all types of EU funding, such as the European Investment Bank (EIB) and innovation funds. They have pointed out the need for such funding to be applied equally to nuclear**.** We hope that those who read this section of the book will take note of this and other points we have raised.

EU Net Zero Act

In the EU, the Net Zero Industry Act does not currently include nuclear as a Key (strategic) technology. As it is clearly likely to be such, the Act needs to be modified to include nuclear.

EU Global Influence: Providing an Alternative to Russian and Chinese Nuclear for Developing Countries

In our analysis. In Chapter 13.3, we covered **Rosatom reactors and fuel supply** and referred to the contract that Rosatom has just signed with Uzbekistan and planning also to sign agreements in Myanmar and India. It is going to be important to ensure that European companies have 100% assistance from Member state governments and the Commission as they try to compete in overseas markets with the technologies being developed in Europe (or developed in the US but made in Europe).

nucleareurope

This is the Brussels-based trade association for the nuclear energy industry in Europe with a membership of 15 national nuclear associations and through these associations, nucleareurope represents **nearly 3,000 European companies working in the industry.** In November 2024, it issued a press release calling on the following actions by the Commission to speed up the decarbonising of the European economy. nucleareurope calls on the next EU Commission (now the current Commission) to:

- Ensure a level playing field between net zero energy technologies (such as nuclear and renewables)
- Provide equal support for nuclear under all EU funds, EIB financing, and the upcoming Clean Energy Investment Strategy and allocate more funding for fission activities under the Euratom budget
- Support the creation of an IPCEI for nuclear (including relevant State Aid)
- Encourage further development of the European nuclear supply chain (eg fuel cycle) to reduce dependence and strengthen Europe's autonomy
- Produce an updated Nuclear Illustrative Programme (PINC) which includes the most up-to-date forecasts and concrete recommendations to support European nuclear projects
- Enable coordinated actions with other sectors to tackle both basic skill needs and STEM

World Nuclear Association

In January 2007 the World Nuclear Association established the Cooperation in Reactor Design Evaluation and Licensing (CORDEL) Working Group with the aim of stimulating a dialogue between the nuclear industry and nuclear regulators on the benefits and means of achieving a worldwide convergence of reactor safety standards for reactor designs. We hope this body will start to become much more active in furthering its objectives.

While this organisation is not specific to Europe, we hope that European companies and the Commission will help ensure that this Association can contribute to the objectives that the European Industrial Alliance on Small Modular Reactors has set out.

15.6 Economies of Manufacturing Scale to Overcome Diseconomies of Small Size

As we covered earlier, there are currently some 80 designs for modern nuclear at various points on the TRL scale. With so many companies trying to develop new designs, nuclear safety regulators are unlikely to have the resources to evaluate all the designs that they are asked to certify.

The uncertainty of which technology/technologies is/are likely to win out in the end could impact the time to market for every new design. Large-scale and global markets are needed to justify the investment in large scale factory fabrication facilities and fragmented regulation and excess competition will not assist in ensuring that a small number of manufacturers and technologies have the necessary production scale to enable a selling price that will accelerate the energy transition. The scale of the required energy transition is such that finally an oligopoly of just 4 or 5 or perhaps somewhat more companies may emerge (and perhaps should be encouraged) as has happened with similar multi-trillion dollar endeavours. Therefore a degree of well-informed regulatory steering would be helpful in reducing the time to reach the oligopoly stage from the present '**let a thousand flowers bloom'** stage which is both wasteful of talent and slow.

European Industrial Alliance on Small Modular Reactors

One potentially key initiative by the **European Commission** in this area has been its encouragement of the creation of the European Industrial Alliance on Small Modular Reactors (SMRs and AMRs). Under the heading "Why Small Modular Reactors Matter", the European Commission issued a statement in early 2024 on its view as to the importance of SMRs/AMRs:

> "Accelerating SMR development in the EU demands a robust and efficient nuclear supply chain. Collaboration among stakeholders, including project promoters, financial institutions, regulators, researchers, training centres, civil society organisations, and policymakers is essential. An alliance creates a European platform to enable and streamline such a collaboration."

The Commission document (https://ec.europa.eu/docsroom/documents/57934) on the Terms of Reference for the Alliance confirms that its work will focus on:

- reinforcing the European nuclear supply chain by identifying and addressing gaps and promoting a close cooperation among stakeholders
- supporting SMR project promoters to develop, demonstrate and deploy their projects in the EU market and beyond, by identifying and removing barriers and working together with financial investors, industry and local authorities
- establishing ways to inform and engage potential industrial users of SMRs, such as energy-intensive industries, hydrogen producers, and urban districts
- facilitating and coordinating projects to address future research and innovation needs, particularly for Advanced Modular Reactors under the Euratom Research and Training Programme, and by establishing a nuclear skills academy with the support of the Net-Zero Industry Platform
- promoting public engagement about SMRs by working with relevant civil society organisations and NGOs, as well as cooperating with relevant international bodies to help European SMR projects reach international markets

Later in 2024, the Alliance started to elaborate a Strategic Action Plan **to identify**, for example:

- **Technically mature and commercially viable SMR technologies that could be supported under the Alliance.**
- Potential gaps and solutions in the European supply chain for SMRs (including fuel and raw materials).
- Investment barriers, funding opportunities and new financial blending options to support SMR development.
- Future needs for research on SMRs and AMRs and identify existing skills gaps along the supply chain to be addressed under the Euratom Research and Training Programme, and at national level.

In October 2024, the Alliance announced the first set of Project Working Groups (PWGs). These nine projects were selected from a competitive pool of 22 proposals with another opportunity to apply for selection in Q2 2025. The nine projects groups are:

1. **EU-SMR-LFR** (Ansaldo Nucleare, SCK CEN, ENEA, RATEN)
2. **CityHeat** (Calogena, Steady Energy)
3. **Project Quantum** (Last Energy)
4. **European LFR AS** (newcleo)
5. **NUWARD** (EDF)
6. **European BWRX-300 SMR** (ORLEN Synthos Green Energy)
7. **Rolls-Royce SMR** (Rolls-Royce SMR)
8. **NuScale VOYGR™ SMR** (RoPower Nuclear)
9. **Thorizon One** (THORIZON)

The Number 1 in the list above is now managed by a consortium agreement and is very well integrated into the European Union's Energy Strategy. The consortium, since June 2025 known as **Eagles 300** is working on a 350MWe LFR (Lead cooled Fast Neutron Reactor). This work is supported by the European Union through the SNETP (Sustainable Nuclear Energy Technology Platform) and ESNII (European Sustainable Nuclear Industrial Initiative) initiatives, for the development of Generation IV nuclear systems.

Great British Nuclear (GBN)

Great British Nuclear is a UK state owned company. It is designed to boost UK energy security, reduce dependence on volatile fossil fuel imports and deliver the government's priority to grow the economy. Its aim is speed-up the development and commissioning of SMRs and AMRs. As well as launching GBN, the government has launched a competition to provide grant funding totalling up to £157 million. Up to £77.1mn of funding will be for companies to enable the acceleration of advanced nuclear development in the UK and support advanced nuclear designs to enter UK regulation to maximise the chance of small and advanced modular reactors being built during the next Parliament.

International Bank for Nuclear Infrastructure (IBNI) – the geopolitical angle

Of the approximately 50 reactors currently being constructed in the world, the majority are funded by the Chinese or Russian governments. Russia, through Rosatom, is currently financing projects in Bangladesh, Egypt and Turkey. Russian reactors are said to cost less than half of the selling price that American or Europeans companies would have to charge and are around 60% of the price of South Korean reactors. Rosatom also offers a 'turnkey' service providing technology, construction, maintenance and fuel as well as financing so, not surprisingly, most emerging countries choose Rosatom products even if Rosatom's business model is to offer the reactor at a low price and recover capital cost later from high priced fuel sales (as with printers and cartridges) and from service and repair costs. If European countries, the US, South Korea, Japan and other states with an interest in nuclear power wish to compete in global markets against the current market leaders, they could benefit from a new multilateral institution similar in structure to the World Bank but exclusively for nuclear. Its role would be to offer risk reduction mechanisms as well as funding for the purchase of reactors. The IBNI is intended to be just such a bank. The IBNI Implementation Organisation has now been set up and the targeted establishment date is 2026. Some of the types of financing and support tools they would intend to offer are:

- Emerging technologies supplemental grant and concessionary financing
- Committed equity financing of projects and companies through supply chains
- Committed senior and/or subordinate debt financing
- Hedging contracts: interest rates, inflation, currency risks
- Completion risk insurance and committed contingent debt and/or equity financing
- Commercial guarantees and sovereign risk guarantees

15.7 Nuclear Fusion: **Iter**, **Tokamak Energy** and **Helion**

Although nothing to do with Gen IV reactors or indeed with fission reactors, it would be remiss to conclude this chapter on nuclear without providing a brief discussion of nuclear fusion technology. There is currently no fusion reactor producing power on a commercial scale. Most forecasts say commercial operation is unlikely before 2035. It is a nuclear technology which does not involve highly radioactive materials. Fusion is the process which is continually taking place in the sun and generating all the energy we use in solar, wind and concentrating solar power (CSP) and now mankind is trying to re-create this process on Earth. Fusion could generate four times more energy per kilogram of fuel than fission (used in current nuclear power plants) and nearly four million times more energy per kg than burning oil or coal.

Fusion (as distinct from fission) involves two lighter atomic nuclei combining to form a heavier nucleus. This releases some of the binding energy in the atoms which thereby releases energy in the form of heat. The first attempt to generate power from fusion was a UK experiment in 1957 under the name Zero Energy Thermonuclear Assembly (ZETA). The scientists working on the project believed that they had achieved nuclear fusion but sadly, it was shown shortly after that the measuring instruments were at fault and there in fact been no energy output produced in excess of input energy. **ITER** ("The Way" in Latin) is one of the most ambitious energy projects in the today. It involves a collaboration amongst 35 countries. **Tokamak** is a magnetic fusion device which it is hoped can demonstrate the feasibility of generating power from the atom but without the problem of nuclear waste.

Today, there are several companies active in attempting to achieve commercial success with fusion. One of these is a joint UK/ US venture **Tokamak Energy** which is a private sector spin-off from UK Atomic Energy Authority (UKAEA). Another such venture is **Helion** is expected to be online by 2028 and generating of 50 MW of electric power. **Helion** claims that its **Trenta** fusion prototypes will be the first privately funded fusion device to reach commerciality. The company has already raised more than $1bn from private investors including Sam Altman of OpenAI and the steel company Nucor. Its current valuation based on its last fundraising is over £5bn. Helion already has a contract to supply power to Microsoft by 2028 (we assume a non-binding contract) and to supply a fusion reactor to Nucor (a steel company) at a future date.

Another approach to generating energy from fusion rather than fission is Inertial Fusion Energy (IFE), which has been achieved on an experimental scale. This has accelerated momentum towards expanding national programmes and strengthening long-term support for commercialisation. One UK company would appear to be one of the leaders in this field. **First Light's** FLARE Fusion via Low-power Assembly and Rapid Excitation) programmes is aiming to enable commercial reactors based on much lower power requirements than other fusion technologies.

Part VI: Innovation: Heat and Power from Concentrating Solar Power & Geothermal Heat from Decay of Natural Radioactivity

Chapter 16: Solar Heat from Concentrating Solar Power (CSP) for Industrial Heat, Electricity and Energy Storage – DEWA, Midelt, NREL, Rondo

Heat as an energy storage medium – the Carnot battery

While electric power itself cannot be stored, heat energy can be. Electric power has traditionally been generated from heat energy from the combustion of fossil fuels – oil, gas, coal – and from the fission reaction in nuclear. Thus heat is a source of energy that can be converted to electric power by heating water in a steam generator which then drives a turbogenerator set. This process can also operate in reverse as electric power once generated, can be passed through heating elements, and thereby converted back to heat.

If that heat can be stored, then it can be produced at one point in time and used to produce power at another point in time (separation in time between production and use of the use of the heat to generate power). This is an extremely valuable attribute since it means that while variable sources of direct power generation (solar and wind) are not producing power, if adequate heat storage is available then that stored heat can be used later to generate power when the variable power source is not generating. It is thus one alternative to battery storage and long duration energy storage which we cover in Chapter 11. Heat storage is also a key feature of the TerraPower Natrium reactor which we cover in Chapter 13.

Radiation from the sun is the source of all natural forms of energy on earth (apart from nuclear). Radiation striking the earth's surface has three main components – visible light (around 45% of the sun's radiation) infrared (around 50/55% of radiation and ultraviolet (which gives our skin a tan) about 5 – 10%. *Photovoltaic solar*, generally known as PV converts visible light into electric power. This can subsequently indirectly generate heat by passing the electrical power being generated through an electric resistance material as in an electric fire or underfloor heating mat. The other type of solar is *solar thermal* where longer wavelength solar electro-magnetic radiation (infra-red) turns into heat directly when it impacts a surface which absorbs it.

The DEWA project in Dubai

A good example of a project using the heat from the sun's infrared radiation is ACWA Power's 700 MW Concentrating Solar Power (CSP) project for DEWA in Dubai which was the world's first facility to offer a 35-year long, day and night, round the clock (RTC) solar contract. It also had a record-low price for solar with storage – of just 7.3 US cents per kWh. It is an example of an increasingly common request for solar combined with storage to enable the provision of power through the 24 hours of a day. This facility uses parabolic mirrors shining on a collector tube which carries a heat transfer fluid (HTF) which transfers the heat to a storage facility or directly to a generating unit. The ability to offer a long (35 year) contract at a fixed price arises from the fact that 'fuel' cost, i.e. sunshine, is zero compared with the highly variable cost of natural gas or coal which, in the case of natural gas, rose by more than 5X in 2021 and 2022 as a result of the Russian invasion of Ukraine.

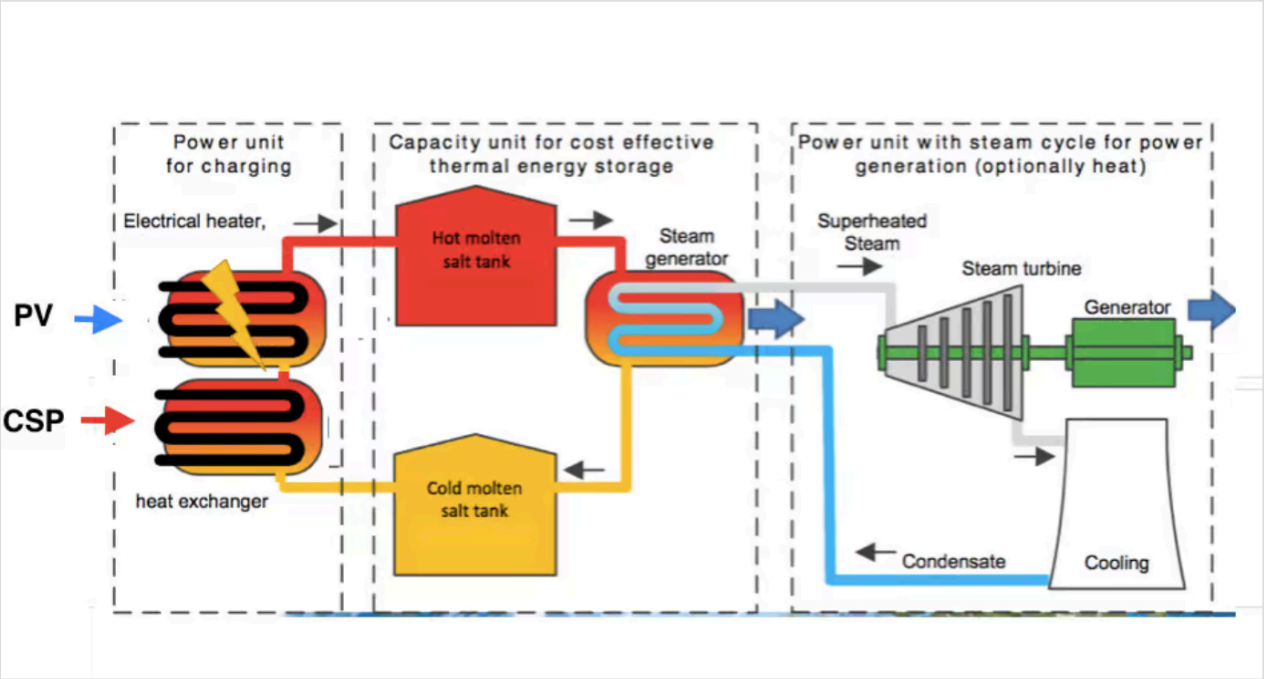
Thus a contract of this type can offer a longer power purchase agreement (PPA) compared with fossil generation as the key risk of natural gas generation is the volatility of the price of gas. This enables the developer of the facility to amortise its cost over a longer period which enables them to offer a lower cost of power in any PPA contract. ACWA has built a similar storage-equipped plant in Dubai, with one of the lowest tariffs in the industry.

An alternative design of CSP is to use a tower surrounded by mirrors focussed on the top of the tower where the molten salt storage medium passes through and is heated.

Midelt Project in Morocco: Hybrid generation + storage: CSP + PV + molten salts

Until recently, thermal storage tanks were connected only to CSP. But in the graphic below which shows the operation of the hybrid solar project at Midelt in Morocco, it can be seen that this plant is *also* connected to solar PV! So while the CSP works by converting sunlight into heat energy (but at a temperature of only 400 °C), PV works by converting sunlight into power. This power can then be passed through resistance heaters (just like in an electric kettle) to convert the power into heat which then heats the molten salt to a higher temperature than with CSP

alone. Using PV in this way for generating energy to store has only become economically viable recently with the plunge in solar PV costs. It is less likely that CSP costs will fall to anything like the same extent. But by adding solar PV it becomes possible to raise the storage temperature to 565°C which means greater heat storage and hence greater power output and also higher thermodynamic efficiency. Figure 14.1 and the text that follows[63] explain this system further:



How colocated PV and CSP would store solar energy in the molten salts thermal energy storage of a CSP plant. Illustration @DLR/Bauer (ADAPTED): from Experimental and Numerical Investigation of a 4 MWh High Temperature Molten Salt Thermocline Storage System with Filler

Figure 14.1 The Midelt project using both solar PV and solar CSP

Morocco's clean energy agency MASEN is executing a national renewable policy with an eye on how a future grid can operate reliably with dispatchable firm electricity from 100% renewables. Midelt in Morocco will be the first solar project in the world with both heat storage from PV and from CSP. Midelt's first-of-a-kind hybrid solar and shared storage project will deliver dispatchable solar at 7 cents per kWh. Morocco was paying up to 30 cents per kWh for fuel oil electricity. The nation of 36 million needs five hours of power after dark and is on track to meet its target to have 52% renewable electricity by 2030.

To date, when PV solar projects have included storage, they have only been paired with batteries. But at Midelt the solar energy from not just the CSP plant, but also from the PV plant will be, for the first time, stored in the thermal energy storage of the CSP portion of the project. CSP projects built today routinely include 10 or more hours of thermal energy storage in tanks of low-cost molten salts. Originally the plan for Midelt was to include PV for daytime with a little battery backup for transient daytime needs and for the CSP to include five hours of thermal storage to cover the evening peak. But as PV prices dropped, the storage plan changed. "If we limit the number of storage hours, batteries can win. But in our studies we find we also need CSP to complete the mix, as we need five hours after dark," Abderrahim Jamrani, (MASEN Technical Director) said.

This idea of co-locating PV and CSP and sharing the CSP thermal storage is one that Schmitz (TSK Flagsol technical director) believes will be widely applicable as energy grids become more saturated with renewables, not just Morocco's, and as a result more regulators move from lowest cost to "best fit" procurement. "If you are operating a grid which is still mostly based on carbon, then merely replacing as many kilowatt hours as you can with green energy is what you do; you just dump the PV in as much as you can. But at a certain point you will reach a level where there is just too much PV in the grid during sunny hours. So just feed it into the already existing tanks with the CSP."[64]

[64] Susan Kraemer Solar Paces April *2020 Morocco Pioneers PV with Thermal Storage at 800 MW Midelt CSP Project* https://www.solarpaces.org/morocco-pioneers-pv-to-thermal-storage-at-800-mw-midelt-csp-project/

NREL

The US National Renewable Energy Laboratory (NREL) is in the late stages of prototype testing a new thermal energy storage technology that uses inexpensive silica sand as a storage medium. It is a low cost, build-anywhere particle thermal energy storage technology in which silica particles are fed through an array of electric resistive heating elements to heat them to 1,200°C. The heated particles are then gravity-fed into insulated concrete silos for thermal energy storage. The baseline system is designed for economical storage of up to 26,000 MWh of thermal energy! As it is a modular design, storage capacity can be scaled up or down relatively easily. This heating process takes place when power prices are low, normally when there is an excess of solar power available to the grid. Note that this system is taking power from the grid, not from its own generating capacity. Thus when the imbalance of total supply and demand (excess supply) would be overcome by selling solar to heat storage systems at a low cost rather paying curtailment fees to solar producers.

When power is needed and therefore power prices are high, the hot particles are gravity-fed through a heat exchanger, heating and pressurizing a working gas (which could be helium or steam) to drive the generators that create electricity for the grid. The system discharges during periods of high electricity demand and when limited solar photovoltaic or wind power are available, such as early in the morning and evening, during dinner preparation, and when TVs are on. Once discharged, the spent, cold particles are once again fed into insulated silos for storage until conditions (and economics) are appropriate again for charging.

See also Chapter 17.2 on the use of CSP for the manufacture of synthetic liquid hydrocarbon fuels (or how to power aeroplanes on sunlight).

Chapter 17: Geothermal Heat: Earth's heat as a source of carbon free heat & dispatchable CFE power

The most comprehensive study of geological heat has been produced by the US Dept. of Energy in a report entitled "Geovision – harnessing the heat beneath our feet"[65] which concludes that:

> "Geothermal is America's untapped energy giant. Geothermal energy is a renewable and diverse solution for the United States—providing reliable and flexible electricity generation and delivering unique technology solutions to America's heating and cooling demands. Geothermal resources can be found nationwide, are "always on," [firm power] and represent vast domestic energy potential. We could increase geothermal power generation nearly 26-fold from today, representing 60 gigawatts-electrical (GWe) of always-on, flexible electricity-generation capacity by 2050. This capacity would make up 3.7% of total U.S. installed capacity in 2050 and would produce 8.5% of all U.S. electricity generation.

The earth is essentially a fireball with a central temperature of around 6,000 degrees (comparable to the sun) which loses heat gradually at the surface but whose heat is replenished continually by natural nuclear processes (like those in nuclear reactors) in the core of the Earth. The Earth's temperature rises with increasing depth and at the core it is like that of the sun.

To evaluate opportunities for the success of geothermal energy, the U.S. Department of Energy's Geothermal Technologies Office (GTO) initiated the GeoVision analysis. This technical analysis evaluated future geothermal deployment opportunities based on three core objectives:

- Increased access to geothermal resources
- Reduced costs and improved economics for geothermal projects
- Improved education and outreach about geothermal energy

It should be noted that for power generation deep drilling is generally required and also high temperature water to generate steam whereas district heating can be provided by shallower wells (but at sufficient depth to tap geothermal heat) and lower temperature water. In addition, domestic heating can be provided by ground source heat pumps which are normally no more than a metre underground. They are in fact operating mainly through incident solar heat on the ground (not geothermal heat) which penetrates to about this depth and keeps the ground at this depth at a relatively constant temperature.

The International Geothermal Association (IGA)

The role of the Association is to encourage, facilitate and promote the development of geothermal resources, as well as innovative research in geothermal technologies, through visible and integrated position and representation of geothermal power, heat, geo-exchange, cooling and other uses of geothermal resources. The IGA connects the global geothermal community, serving as a hub for networking opportunities aimed at promoting and supporting global geothermal development. With industry partners the IGA sets standards, matures the technology agenda and nurture entrepreneurs engaged in clean technology.

The European Geothermal Energy Council (EGEC)

EGEC is a non-profit international organisation founded in 1998 to promote the European geothermal industry and enable its development both in Europe and worldwide, by shaping policy, improving business condition, and driving more research and development.

Based in Brussels, it provides a link between the industry and European institutions. More than 120 members from 28 countries, including developers, equipment manufacturers, electricity providers, national associations, consultants, research centres, geological surveys, and public authorities, make EGEC a unique network, uniting and representing the entire geothermal sector.

Innovative technologies in the field of geothermal energy include new drilling technologies, new exploration methods, closed loops, smart systems, and 3D models.

[65] http://www.osti.gov/scitech & https://www.energy.gov/sites/default/files/2019/06/f63/2-GeoVision-Chap2-opt.pdf

Advanced plants and technologies include:

> **Ultra-deep geothermal systems enhanced geothermal systems and supercritical geothermal systems.** These enable access to resources, which would otherwise remain unexploited.
>
> **Enhanced geothermal systems (EGSs)** are an emerging energy technology which could expand geothermal power generation.

An example of a company that is advanced in using Enhanced Geothermal Systems is the US company Fervo.

> **Fervo Energy** is commercializing a geothermal technology that could expand the regions of the world that could tap into underground heat by 'fracking' i.e. creating or widening cracks under the surface to allow water to more easily circulate underground.

EGS uses techniques not dissimilar to those used in fracking for oil and natural gas. There are therefore risks of setting off small 'earthquakes'. But this may be a price to pay for the benefits of dispatchable power and an increase in the portfolio of technologies available to achieve the energy transition. The temperature of the water coming to the surface will depend on geological conditions.

Part VII: Innovation: Other Technologies to Enable the Energy Transition

A number of the technologies covered in this Part of the text require the use of innovative materials to enable their effective application. These include catalysts which are materials which increase the rate of a chemical reaction without undergoing any permanent chemical change i.e. they are not used up in a process. It also includes ceramic materials which have a number of uses. Several companies, such as Johnson Matthey (JM) provide catalysts. It describes itself as a British multinational speciality chemicals and sustainable technologies company. Its 'strapline' is "Our purpose is to catalyse the net zero transition." By doing this, the company is moving away from its previous key product which was platinum as the catalyst in catalytic converters for internal combustion engines (ICE) as battery and hydrogen energy sources take over in the production of vehicles.

Another company which specialises in products which enable several of the technologies below is NGK, a Japanese company which is active in Europe (NGK Europe) and uses the strapline "Unique ceramic technology for a future of coexistence between environment and humanity." It is active in Energy & Climate tech, H2-Economy, IoT, Microelectronics & Semiconductors and Sustainability. Its 'New Value' business is transforming from automotive industry business to CN (Carbon Neutral) and DS (Digital Society) business as the production of ceramic components such as spark plugs for ICE engines will diminish over time as the energy transition advances.

One use of ceramic-based membranes is in carbon capture and storage including for direct air capture. NGK is also developing materials which are enabling more efficient production and transportation of hydrogen. The energy transition also requires new processes which enable the manufacture of materials but with a lower need for the use of heat and electricity (to reduce costs). In Europe, the importance of this arises from the fact that many constituent countries have amongst the highest energy costs in the world which can make a high energy input company uneconomic.

Chapter 18: Green Hydrogen (H_2): Manufacture and Applications:

18.1: Green hydrogen value chain

The World Economic Forum (WEF) identifies green hydrogen as one of the top technologies to have emerged in 2020 and as a technology with the greatest potential to transform industry and society. Green hydrogen is not something new, but its use is increasing and at an increasing rate, hence it is regarded as a top emerging technology.

Before we explore the importance of green hydrogen, we note that in the process of decarbonising the economy, direct electrification of industrial processes, transport and heating rather using hydrogen is the most technically efficient way to meet most energy needs. Interestingly, only 20% of total final energy demand is generated by electricity. However, this could reach 60–70% by 2050. For some sectors direct electrification will likely remain impossible or uneconomic for many decades. Unfortunately, the 2020's have not been kind to many hydrogen investors. This is in contrast to a number of Advanced Nuclear Technology reactor companies whose share prices have risen substantially through 2024 and 2025.

It is in those uses and others where hydrogen may offer considerable advantages over current battery technology, that hydrogen or its derivatives are likely to become sources of energy in a decarbonised economy. As well as being a source of power generation from energy storage (see Chapter 5) it can also be used as to provide of heat for domestic and commercial space heating and for industrial processes. But in practice it should be noted that given that carbon-free power to manufacture 'green' hydrogen will be in short supply for more than a decade, initially the most valuable uses of H2 are likely to be in decarbonising fertiliser production (from natural gas) and also decarbonising the H2 used in fossil fuel manufacture.

Hydrogen along with electricity, may be a solution in some 'hard to decarbonise' sectors i.e. not easily directly electrified. For example it can used directly or in the form of derived fuels such as ammonia and synthetic liquid hydrocarbon fuels (also called synfuels, e-fuels or drop-in-fuels). In steel and long-distance shipping, for instance, hydrogen's (or ammonia's) potential new role is increasingly certain. For the manufacture of fertiliser, it is hoped

that green hydrogen will displace hydrogen made from natural gas with its consequential emissions of CO2. In a number of other sectors it is the leading decarbonisation option.

Today is not the first time we have heard that hydrogen is the fuel of the future and it has certainly come 'into vogue' in the last few years as one of the elements of any successful decarbonisation of a country. But the idea behind the 'hydrogen economy' is in fact an old one developed back in the 1970s when it involved hydrogen being used mainly to power all vehicles in cities in order to reduce pollution i.e., not for reasons of climate change mitigation. While in theory this was achievable, the energy majors and the major automobile companies, certainly in the US, were not in in favour of such a change. And without their support, it was always unlikely the idea could succeed. But in practice it was the low cost of carbon-based fuels along with the inertia of car manufacturers which ensured that the internal combustion engine (ICE).

While hydrogen is back 'in vogue' due to the need for decarbonisation of the economy in fact it has been for many decades, and is continuing to be, manufactured in large volumes. However, the starting point for such manufacture is fossil fuel, in particular natural gas (methane) though in China mostly coal, which can be converted to hydrogen. The hydrogen produced by this process is known as 'grey'. It uses a process called Steam Methane Reforming (SMR). This results in a high level of emissions of CO2 because natural gas (methane) has the chemical formula CH4 and therefore the carbon atom has to be broken away from the hydrogen atoms (creating CO2) which is then released into the atmosphere. Producing natural gas 'at the well head' also can create large 'fugitive' methane emissions.

Today there are many so-called 'colours' of hydrogen. Grey hydrogen is hydrogen made from natural gas without capturing the CO2 that the process generates. Green hydrogen is manufactured from water (H_2O) as the 'raw material' through which an electric current is passed to separate the two elements although heat can also be used as we noted later. Blue hydrogen is made from natural gas but with carbon capture and storage. While the cost of green hydrogen has, until very recently been higher than grey, when the price of natural gas rose sharply due to Russia's invasion of Ukraine in some cases it was lower. In addition, technological developments currently under way and the likelihood of carbon taxes being extended considerably in scope, could make green hydrogen very economic

The production of green hydrogen requires a large amount of electric power (or heat energy) to break the molecular bonds in water in order to separate the two gases – hydrogen and oxygen – which in chemical combination produce water (H2O). Thus when we are considering the cost of manufacturing green hydrogen and attempting to lower it substantially, the cost of power is a central factor.

Figure 16.1[66] below provides a useful representation of the links in the value chain for H2 from green power sources of energy to applications of H2. This value chain involves firstly, the provision of the large amounts of green power required. The majority of the operating cost (OPEX) for hydrogen production arises from the cost of the power input. The capital investment (CAPEX) required both for power generation and electrolysis are also high. Reducing the cost of H2 requires both a lower OPEX and CAPEX.

The graphic shows solar, wind and nuclear as the three sources of clean power. The price at which the power generated can be delivered to the hydrogen production facility depends in part on the distance between the power source and the electrolyser as this gives rise to transmission charges. Also whether it is 'dispatchable' or variable renewable power and if variable, whether there is energy storage to enable 24/7 operation of the electrolysers. These are much more efficient if operating 24/7.

[66] Zixuan Luo et.al. *Cost-Economic analysis of Hydrogen for China's Fuel Cell Transportation Field.* Energy, December 2020 https://www.mdpi.com/1996-1073/13/24/6522

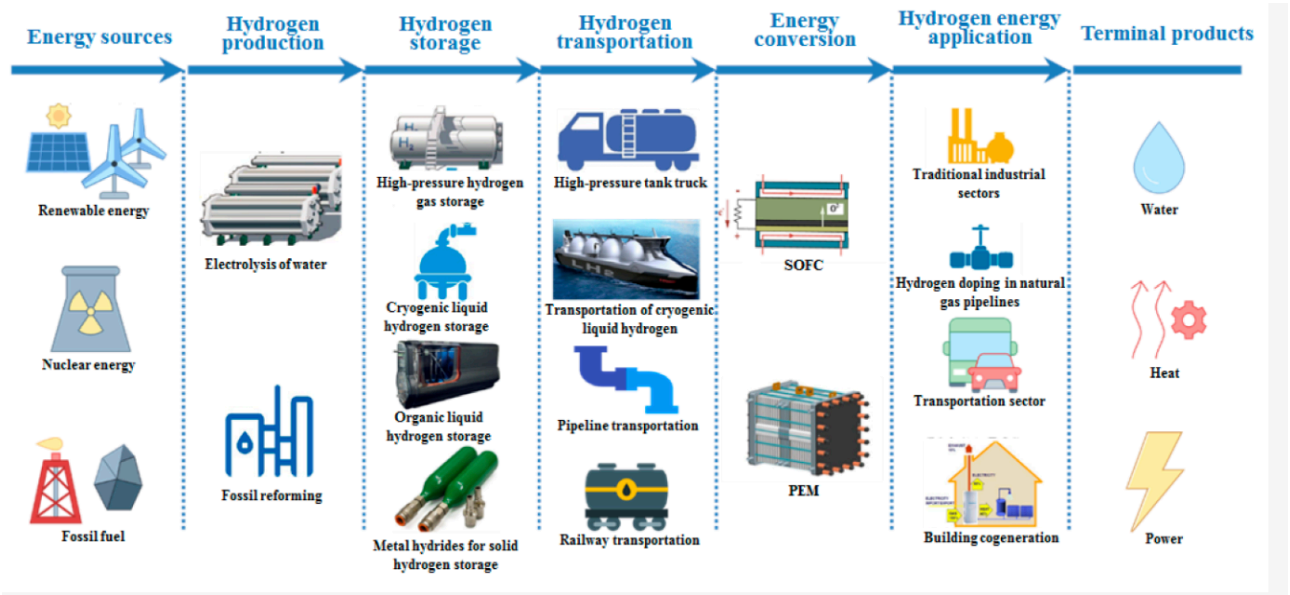


Figure 16.1 Links from power generation to applications for hydrogen

Columns 1 & 2 above: Energy sources (generation of power) and connection to hydrogen production facility

In respect of transmission costs, both wind and solar power have the drawback that good sunshine and strong wind are often found at a considerable distance from users including those who might wish to install electrolysers. This is quite different from traditional power which was once generated within each city and today is still often located near users. Thus the 'delivered' price of VRE will depend on how the two facilities – power generation and electrolysis – are connected to each other. If power goes through the grid there will be transmission charges but if the two facilities are directly connected it will not.

The construction of a wind or solar project which can supply the energy necessary for the manufacture of hydrogen is now relatively straightforward. Storage facilities are relatively straightforward but there is still not as much experience of constructing large scale storage. Traditionally, an investment in renewable power is undertaken with the intention of supplying power to the grid. This is a well-established business model. If, however, it is to finance hydrogen production, there is no such well-established business model and therefore project financing might be harder.

The fact that there are few established business models for large scale independents in this highly capital intensive business (developing a power project alongside an electrolysis plant) explains why such projects are generally undertaken on the balance sheet of a large corporate and not as independent project financings. The logical type of corporate would be an energy or power company/ utility (such as Ørsted) or a company which uses hydrogen in large quantities such as oil refineries. This should greatly reduce the cost of capital compared with project financing where lenders and investors would have a single, undiversified risk (including some technology risk) but instead puts the risk on the corporate's larger and more diversified balance sheet.

Columns 2 and 5: Hydrogen production from power (electrolysis) and power production from energy (fuel cell)

The first key technology is the electrolyser. This produces hydrogen from water and electric power/heat. The original technology is the alkaline electrolyser. Somewhat more recently the proton exchange membrane (PEM) technology has been developed and is a very successful alternative. The most recent is known as the High Temperature Solid Oxide Electrolyser Cell (SOEC) which we examine in more detail in the section on 'electrofuels'. For those uses which require hydrogen to be converted to power, the other key technology is the fuel cell which involves the opposite (reverse) process to that of the electrolyser. It takes in hydrogen (or natural gas) and converts it to heat and power.

The technology of both the alkaline and PEM electrolysers is well established. The Solid Oxide Cell (SOC) is a newer technology which is the basis of the technology for both the SOEC electrolyser and solid oxide fuel cell (SOFC). Although available on the market, these are relatively recent technologies and like most technologies are likely to

be continually improved. They have higher efficiency in the conversion process (around 30% more efficient) and some models may be operated reversibly, that is, switching between operating as an electrolyser (SOEC) and operating as a fuel cell (SOFC). Such cells, known as reversible solid oxide fuel cells (RSOFC), can perform a valuable function on a grid by arbitraging between use as an electrolyser when there is surplus (low cost) power available then switching to generation for the grid when power prices are high.

By alternating between SOFC mode and SOEC modes, RSOFCs can provide ''load shifting'' services for the energy system. Solid oxide devices also operate at between 600°C and 850°C, a much higher temperature than the other types of electrolyser which operate at between 60° and 160°C. This results in considerable efficiency gains compared with the other two main types of electrolyser. They can also be manufactured without the use of scarce and expensive chemical materials which some other technologies require.

Columns 3 & 4: Hydrogen Storage and Transportation

An important part of the infrastructure for hydrogen if it is to become a major contributor to a country's energy needs is storage and transport. Companies in this sector of the value chain include Snam, Lindt and Air Products. While SNAM (originally part of ENI) was, and still is, a major natural gas distributer with the largest natural gas transport pipeline network in Europe, Air Products and Linde are industrial gas suppliers of oxygen, nitrogen, argon, helium, and hydrogen as well as medical and specialty gases for welding and cutting..

Financing hydrogen electrolyser and fuel cell companies

A feature of many electrolyser and fuel cell companies is that they are relatively capital-light and try to avoid debt which is how larger companies obtain much of their financing. One way in which they achieve this is by licensing their technology to much larger companies. They are very different to utilities which are the most highly indebted of all corporates along with infrastructure companies (such as airports). Utilities are highly leveraged with up to 80% debt financing. This high debt load is possible because such companies are expected to provide their investors with a steady low-risk flow of income from a low volatility asset. They are not expected to be 'growth stocks'.

The types of investor in companies manufacturing electrolysers and fuel cells are quite different from those investing in utilities. Investors in electrolysers and fuel cells are not seeking regular, steady income such as utilities generally offer but large capital gains at the point when a dramatic improvement in technology and scale of operation which will lead to much lower costs and selling prices and consequently a dramatic increase in sales and, correspondingly, in profit and share price. In this characteristic they are more like 'tech' companies such as Amazon, Google, etc. However, this same investor expectation can lead to sharp price declines when market conditions change as we note later. Also, because many of them are relatively new with relatively new technologies many of which are still being developed and hence relatively risky, they cannot easily support debt financing. Hence in their early stages at least they will be almost 100% equity financed.

Sparc Technologies: Hydrogen and Photocatalytic Water Splitting (PWS)[67]

This is an alternative method of producing renewable green hydrogen which avoids the need to convert solar energy into electricity to split water via electrolysis (direct solar to hydrogen technology). In a similar approach to that which is used in concentrating solar power (CSP) to produce heat, PWS employs a highly specialised photocatalyst which triggers the dissociation of water into hydrogen and oxygen. Sparc Hydrogen aims to commercialise PWS to produce green hydrogen at large scale.

The potential advantage of Sparc technology is that it obviates the need to generate electrical power from sunlight and instead uses the whole sunlight spectrum to provide the energy required to break the molecular bonds in water. The consequence is potentially lower capital investment requirements and higher efficiency by supplying the required energy via a concentrating solar power (CSP) source rather than by using electricity directly.

Sparc Hydrogen is a joint venture between Sparc Technologies, the University of Adelaide and Fortescue Limited. It has now commenced front-end engineering and design (FEED) for a global, first-of its-kind, pilot plant planned to be located at the University of Adelaide. The decision to commence FEED, obtain site approvals and procure long lead equipment is on the back of several key milestones achieved over recent months with respect to the

[67] See company website at https://www.sparchydrogen.com/blog

pilot plant, including securing an in principle agreement from the University of Adelaide, progressing pilot scale reactor design and engineering and the signing of a Collaboration Framework Agreement with Shinshu University in Japan, a leading developer of photocatalyst materials. In parallel, work continues in the laboratory to test and optimise Sparc Hydrogen's photocatalytic water splitting reactor under a range of conditions, using various high-efficiency photocatalyst materials.

Sparc Technologies is a good example of a company which is developing an idea which works in theory and in practice. However, the first question is whether the technology and scale and whether the unit cost and thus selling price of its hydrogen at the optimal scale of production will be below that of electrolyser H2 production. The second question is whether a company of this type will suffer from the 'missing middle' phenomenon leading to the 'valley of death' or will it be able to raise the necessary growth funding to enable it to move to scale. We covered this problem in the Preface and also cover it in Chapter 2.3.

Repsol & Photo electrocatalysis

Repsol is a 'global multi-energy' company 'striving to become a net zero emissions' company. It is a Spanish based multinational. Repsol is the leading producer and the main consumer of hydrogen in Spain as hydrogen is a key component in the crude oil refining process. It is used in desulfurisation and hydrocracking treatments that improve the performance and the environmental quality of the refined fuels.

Repsol is also developing photo electrocatalysis to produce hydrogen directly from solar energy. The project also involves Enagás, and several leading research centres including the Catalan Institute for Energy Research, the University of Alicante, and the Aragon Hydrogen Foundation. In photo electrocatalysis "the device receives solar radiation directly and with a photoactive material it generates the electrical charges that cause this separation" into hydrogen and oxygen.

18.2 Applications of Hydrogen: energy storage, transport, hard to decarbonise sectors

There are four categories of use for H2 – as a **fuel**, as a provider of **heat**, as an **energy storage medium** for 'time-shifting electricity production and as a **feedstock** for manufacturing industry. In the first two sets of uses, hydrogen is acting as an 'energy carrier' i.e. it has to be converted to usable energy in some way either by combustion or by passing through a fuel cell. On the other hand, in the case of its use as a feedstock, it is the hydrogen molecule itself which is being used to create products or, in the case of steel, to 'reduce' iron ore (iron oxide) by combining with the oxygen molecule in iron oxide to produce water (H2O) in contrast to reducing iron ore using the carbon content of coal whereby the carbon molecule combines with oxygen to produce CO2 which is dissipated into the atmosphere.

Direct electrification is always likely to be a lower cost means of achieving an objective than indirect electrification using hydrogen for the simple reason that conversion of power to hydrogen and back to power is a low efficiency process. Hydrogen has a very low volumetric energy density meaning that storing and transporting it is difficult. But it does have a very high energy gravimetric density (energy for a given weight of fuel). However, for long distance aeroplanes, hydrogen would probably not be used directly but be used to manufacture liquid fuels such as ammonia or e-fuel.

Batteries are the worst possible source of energy in terms of the weight of the 'fuel' to power a plane. Jet fuel has by far the highest energy content at around 45MJ/kg whereas batteries are only around 0.6 – 0.8MJ/kg. in consequence even a short-haul aeroplane would have to carry some hundreds of tonnes of batteries which would be more than the weight of the plane itself.

The *current* major uses of hydrogen (mainly grey hydrogen) are in the manufacture of liquid fuels by the oil majors e.g. petrol from crude oil (hydrogenation to improve quality) and the manufacture of nitrogen-based fertilisers. Thus the initial markets for green hydrogen are likely to be in these two industries but only once the cost of manufacture falls to levels comparable to grey hydrogen or once governments legislate to prohibit grey hydrogen use (or its price rises dramatically as it has) or impose a carbon tax which makes it less economic in use than green hydrogen.

Energy storage for dispatchable power

Wind power and photovoltaic (PV) solar are, as we noted earlier, variable power generators. Thus there also needs to be provision of 'dispatchable power' in a power system with a high proportion of VRE (as we examined earlier), to 'fill in the gaps' and provide Round the Clock (RTC) power.

One approach is to use hydrogen which is in fact stored chemical energy used as an energy carrier. Hydrogen produced by electrolysis using green power can be converted back to power in two ways. The first is to convert it using fuel-cells. The other is to combust it in a conventional gas turbine or internal combustion engine which turns a generator to produce electric power.

Once the cost of green hydrogen falls from present day levels of around $6 per kg to nearer $2 which some electrolyser manufacturers believe is achievable, it should become highly competitive for longer-term storage. However, it must still compete with alternative long-term storage such as heat energy and the iron-based flow batteries we covered earlier.

Transport

In the European Union, transport is the largest emitter of GHGs (in part because other sectors have reduced their emissions). Electrifying transport directly or using H2 is a key requirement for achieving the decarbonisation of the EU and in many other countries.

Most electric vehicles (EVs) currently use batteries as the source of energy rather than hydrogen. Hydrogen vehicles also have quite substantial batteries in the electric drive train. Power is not fed to the motors directly from the electrolyser but via the battery. The battery is also used to store power produced by regenerative braking. There are also electric vehicles (hybrids) where all the power is produced by a fossil fuel internal combustion engine (ICE) and then passes to a battery from which it is delivered as required to the wheels. An example of this in passenger vehicles is the Toyota hybrid Prius (now also available as a plug-in hybrid or PHEV). Daimler Trucks which is the largest European truck manufacturer has said that while diesel will dominate sales up to at least 2025, hydrogen could take-off sometime between 2027 and 2030 and then rise steeply. In the UK, the government is planning to ban the sale of new diesel and petrol heavy goods vehicles (HGVs) by 2040. At this point it is possible that heavy trucks will be an important source of demand for hydrogen and for H2 re-fuelling stations around the country.

The benefits of using hydrogen as a source of energy for a fuel cell to power an electric vehicle (FCEV) rather than batteries to power an electric vehicle (BEV) come to the fore once large vehicles, heavy trucks and buses are considered or vehicles where a source of electric power to recharge them is not readily available (earth movers on building sites or fork-lift trucks in warehouses). When time is at a premium, hydrogen vehicles can also be refuelled as quickly as traditional vehicles whereas re-charging a large vehicle's batteries is a lengthy process or, alternatively, requires very high rates of current flow if adequate power is available from the grid.

Hard to de-carbonise sectors - heat from H2 for iron, steel & glass etc.

Some industrial sectors are much harder to de-carbonise than others yet decarbonisation is critical to achieving climate objectives. These include iron & steel, petrochemicals, aluminium, cement, glass, paper and fertilizers, that in total are said to be responsible for around 22 percent of global CO2 emissions.

Iron & Steel Sector

This has long been a sector where emission-free manufacture has been an aim of producers but has been hard to achieve. The Swedish company H2Green Steel (H2GS) now known as Stegra, is trying to make steel without emitting CO2. It has stated that "existing steel production technology leaves the steel industry as Sweden's single largest emitter of carbon dioxide. At European level, the steel industry accounts for six percent of carbon dioxide emissions, and globally the figure is seven to nine percent".

Stegra and Hybrit both based in Sweden, are working towards producing iron and steel with virtually no emissions of greenhouse gases. In both cases the technology involves 'reducing' i.e. removing the oxygen from iron ore (Fe_2O_3) whereby the oxygen in the ore combines with hydrogen to produce water (H2O) as a bi-product and produces pure iron (Fe). In traditional iron making using coking coal, the ore is reduced by combining the oxygen in the ore with carbon from the coking coal which produces CO2 as a by-product. For both startups, the key to being able to manufacture 'green' steel at a reasonably economic price is to have 'green' power available at the

lowest possible cost. Sweden is fortunate in having a surplus of electricity produced mainly from hydropower. But this power is also wanted by datacentres (Meta currently has a datacentre in Sweden), and battery producers such as Northvolt (now insolvent) and fertiliser manufacturers such as Power2Earth. This latter company is set to establish Sweden's first fossil-free ammonia and mineral fertiliser plant in Luleå. It is hoped to commence production in 2028.

In France, in February 2022, the Prime Minister announced support from the French Government for ArcelorMittal's decarbonisation programme in France, which involves a €1.7 billion investment in its Fos-sur-Mer and Dunkirk sites in France. This investment will enable a profound transformation of steelmaking in France and a total reduction of close to 40% or 7.8 million tonnes per year in ArcelorMittal's CO2 emissions in France by 2030. This transformation will represent a 10% reduction in greenhouse gas emissions from manufacturing industry in France and put France's steelmaking industry on the path of the Paris Agreement.

In Dunkirk, ArcelorMittal will build a 2.5 million tonne Direct Reduction of Iron (DRI) unit to transform iron ore using hydrogen instead of coal. This DRI will be coupled with an innovative technology electric furnace and completed by an additional Electric Arc Furnace (EAF). Other investments are already under way to continue to increase the proportion of scrap steel used.

Cement

Cement manufacture is particularly hard to decarbonise since it is not just the fossil fuel currently used to manufacture it which emits CO2 but the inherent nature of the manufacturing process itself which results in further CO2 emissions from the chemical process. The cement industry alone is responsible for about at least 25% of all industry CO2 emissions. About two-thirds of these result from what is known as calcination which is the chemical reaction that occurs when limestone is exposed to high temperatures. While the use of hydrogen would eliminate CO2 emissions from the heating process, it would not eliminate the natural emissions from manufacture.

As cement production accounts for about 8 percent of all CO2 emissions, it is a critically important industry to decarbonise. It has been said that if cement production were a country, it would be the world's third-largest emitter. While cement could be de-carbonised if carbon capture and storage became available at a much larger scale and at a more economic price, other experiments are focussed on absorbing carbon dioxide into cement at the time of making concrete. Carbon Cure is one such company. It has been backed by Microsoft, Amazon (Climate Pledge Fund) and Mitsubishi but that does not guarantee that it will succeed commercially.

Float sheet glass

Sheet glass has been produced using hydrogen fuel at a plant near Liverpool in the UK, in an achievement that has been touted by developers as a world first and a major milestone in industrial manufacturing's transition towards net zero emissions. Progressive Energy, one of a number of companies involved in the HyNet project to decarbonise industrial sites in the North-West, announced that hydrogen fuel had successfully replaced natural gas in the manufacture of float glass at a facility owned by Pilkington UK in the town of St Helen's in Merseyside. The achievement is believed to be the first full-scale, 100 per cent hydrogen trial on a float glass line. In October 2021, Pilkington was in danger of having to shut to reduce losses as UK (and global) gas prices rose to unprecedentedly high levels. This shows an advantage of using green hydrogen which can be supplied on very long-term contracts with no risk of price increases.

Ammonia

Ammonia is used extensively in the manufacture of fertilisers. It is made from hydrogen plus nitrogen (NH3). Currently, natural gas is used by most companies in the fertiliser industry as the source of the necessary hydrogen atoms. However, when natural gas is used (CH4) the carbon atoms are released into the atmosphere as carbon dioxide (CO2).

Clean ammonia can be made using hydrogen produced through any electrolysis process. One company, Supercritical Solutions, has developed a new type of electrolyser which not only produces hydrogen at a lower cost than other technologies but also generates it at the high pressure (200 bar) required in the process of converting it to ammonia thus reducing the need for, and cost of, further compression. This company also sees the manufacture of synthetic methanol and synthetic liquid hydrocarbon fuels, as potentially large and expanding markets for their high pressure hydrogen.

Chapter 19: Hydrogen for the manufacture of Synthetic Liquid Hydrocarbon Fuels: Biofuels, electrofuels (e-fuels).

Transport is a large user of fossil fuels. To date, direct electrification (as in electric railways or tramways) has been one way of avoiding direct use of fossil fuels though clearly the power grid is currently not providing carbon-free electricity. But for ships and aircraft, direct electrification is not possible. Alternatives include compressed hydrogen, liquid hydrogen, methanol, ammonia and batteries. Increasing attention is, however, now focussing on producing liquid synthetic hydrocarbon fuels.

The manufacture of synthetic liquid hydrocarbon fuels such as petrol and diesel for road vehicle and other uses and Sustainable Aircraft Fuel (SAF) as a substitute for kerosene **may (or may not) become one of the major uses of hydrogen.** This is because the hydrogen molecule (H2) is one of the two components of all hydrocarbon fuels the other, obviously, being the carbon atom C. H2 molecules and C atoms can be sourced in two different ways. The first is to use biological material which is comprised on both or, alternatively, to use what is known as power-to-liquids technology involving the manufacture of hydrogen using electrolysis of water and the collection of CO2 either from smokestacks or from the air.

Hydrogen-based synthetic fuels (e-fuels) are in the EU's plans to decarbonise transport. In particular, for aviation and maritime transport, the EU has set minimum quotas for "renewable fuels of non-biological origin" (RFNBOs) i.e. electrofuels, which includes both hydrogen and e-fuels. Indeed, the Renewable Energy Directive (RED) requires 1.2% of all kerosene (aircraft fuel) demand to be met by 2030 by the use of renewable e-kerosene. This will rise to 35% by 2050. EU countries are also required to set a minimum quota for RFNBOs, of at least 1% of all fuels used across all transport modes including road transport.

19.1 Biogenic fuels – hydrogen + carbon from biological materials.

Biofuel is a contentious area. First, there needs to be certainty that their use is at least carbon neutral. Second, if they come from agricultural crops, there needs to be an understanding of the extent to which they may damage food security by taking land out of agricultural production. The starting point for understanding the use of biological materials for the production of energy and the manufacture of gaseous and liquid fuels is to consider the process through which biomass is created (photosynthesis) and the different sources of biomass that are available from this natural process. Plants produce biomass through photosynthesis which involves absorbing CO2 from the air and taking in sunlight and water which produces biomass such as trees. Biomass can be burned directly for heat or converted to renewable liquid and gaseous fuels through various processes as we consider below but in either case, **the CO2 taken in by the plants that have made the biomass is then released back into the atmosphere**. Figure 19.1[68] shows the process of photosynthesis and the sources of biomass available for conversion to fuels:

[68] US Energy Information Administration *Biomass explained* https://www.eia.gov/energyexplained/biomass/

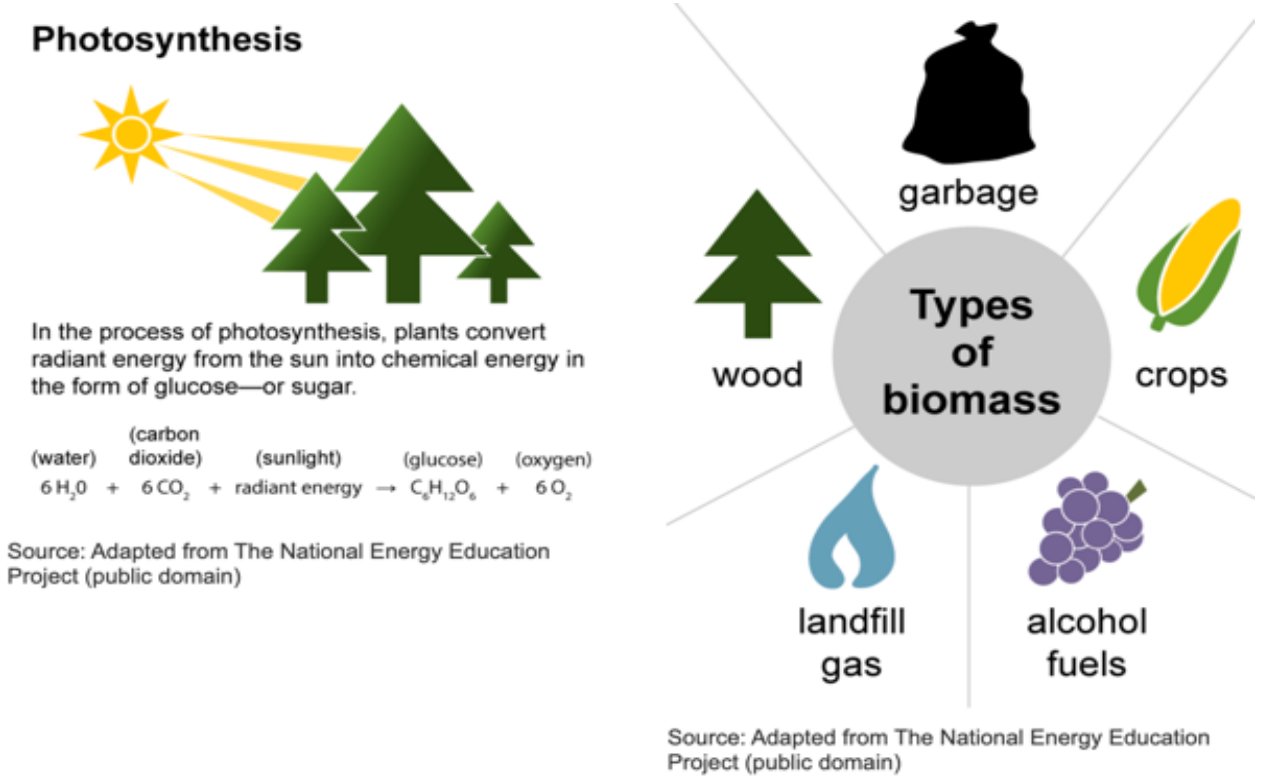
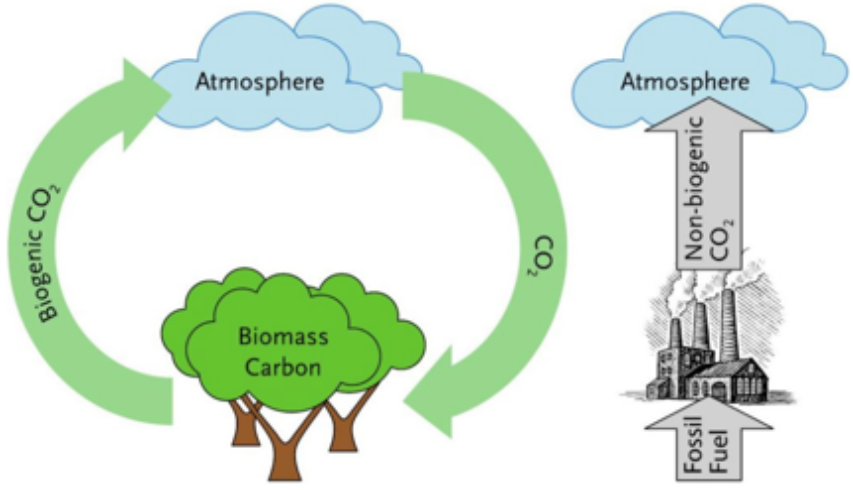


Figure 19.1 Photosynthesis and sources of biomass

A key question is whether the use of biomass to generate carbon for manufacturing liquid fuels is compatible with the need for 'carbon neutrality'. The answer come down to the 'length of the carbon cycle' and to 'what happens to the emissions of CO2 in each case. Figure 19.2[69] tries to answer this question.

On the left, the graphic shows the 'biogenic carbon cycle' – in this case a tree absorbing energy from sunlight and using this energy to convert CO2 and water into plant growth. If a tree takes 20 years to grow to a point where it becomes economic to cut it down and burn it for heat, then all the benefit it has provided by absorbing CO2 from the atmosphere over that time is lost immediately and will not be recovered. Even if it is replaced one-for-one with a newly planted tree it will take another 20 years to provide the same benefit. If the tree is not replaced then burning it will be carbon additive in many people's remaining lifetime and hence will add to atmospheric concentration. Thus CO2 capture from trees may not be the answer to removing CO2 permanently. Another important reason why trees are unlikely to be the answer is that forest fires covering huge acreages of land, would seem to be becoming more frequent. At the point in time when a tree is destroyed by the action of fire, the benefit of planting it has been lost.



On the other hand, other types of biological materials such as sweetcorn, have a much shorter growth period (sweetcorn/ corn on the cob only 3 months) and the cob (core) of the corn can be used as biomass. If replanted each year, then the carbon in the cob when burned either by the farmer or when burned in the form of a liquid fuel, is just recycled over the space of a year as shown in the first diagram. In the case of the liquid hydrocarbon fuel,

Figure 19.2 Understanding the biogenic carbon cycle

[69] IEA *Fossil vs biogenic CO2 emissions* https://www.ieabioenergy.com/iea-publications/faq/woodybiomass/biogenic-co2/

the carbon and hydrogen atoms/molecules it contains can be converted to synthetic hydrocarbon fuels (biogenic fuels) as these two elements are the only ones required. When a biogenic fuel is burned in an internal combustion engine (ICE) or jet engine, the carbon is returned to the atmosphere making the biogenic cycle carbon neutral over a period of less than a year. Carbon in the form of CO2 is then re-absorbed by the replacement crop over the crop growing season.

The graphic on the right (above) shows something quite different. In this case, CO2 which has been locked-up underground for millennia is being released straight into the atmosphere. Even if this were captured and used as a component of synthetic fuel, when that fuel is burned, there is no mechanism to re-cycle the emissions from combustion as there is no room on earth to plant the number of trees that would be required to absorb all the carbon released by vehicles and the time taken to re-absorb the carbon would, anyway, be very many years. Thus this process is creating non-biogenic CO2 emissions as it allows continuing use of fossil fuels in factories and elsewhere.

Such fossil fuel combustion releases carbon that has been locked up in the ground for millions of years. Thus at the point of combustion, atmospheric carbon concentration is increased. Burning biomass on the other hand emits carbon that is part of the biogenic carbon cycle which is much shorter and thus not increasing carbon concentration over the biogenic carbon cycle period though not reducing it either.

Factory chimney emissions as a source of carbon for liquid hydrocarbon electrofuels rather from direct air capture (DAC) is controversial. But it may not necessarily be bad in the short run. Iron and steel works are an example of industry assets that cannot be converted overnight to using hydrogen as their source of energy and thus are likely to be emitting CO2 for many years into the future. Thus, when considering any possible benefit to decarbonisation in the case of an electrofuel manufacturer purchasing carbon atoms from a steel company's emissions (so that the cost to the steel company of CCS is either zero or relatively small), we need to consider the alternative – namely that the emissions are simply put straight into the atmosphere where they clearly add to GHG concentration.

When factory emissions (from the use of recently produced fossil fuels) are used as the source of carbon atoms in the manufacture of an electrofuel, the vehicle using that fuel (for example, an aircraft) is not carbon neutral – it is still adding to atmospheric GHG concentration. However, the steelworks (if all the CO2 were removed from its processes and the carbon content re-used) would in fact be carbon neutral. Thus all that is happening is that the original CO2 from the steelworks is being recycled through being used twice. So which process is carbon neutral? The steel factory or the vehicle, such as an aircraft which uses synthetic fuel sourced from the steel factory's chimney? The answer is that only one (it hardly matters which) can be considered carbon neutral but certainly not both as is sometimes claimed.

By using such an electrofuel, atmospheric CO2 concentration does increase though in a slightly delayed timeframe from its first round of use in the factory. What has happened, however, is that this roundabout process has saved the additional fossil fuel emissions that the plane would have produced had it not been converted to an electrofuel. Thus, although the whole process is not carbon neutral, it reduces what would have been a greater amount of emissions if this re-use of carbon had not taken place. However, this only applies in the short run.

In the longer-run, to make the whole process carbon neutral in this and other industries, steel producers and others must eliminate their emissions by converting to non-fossil fuels in particular hydrogen. Electrofuel manufacturers would then be forced to source their carbon from the air (DAC). At this point we have overall carbon neutrality assuming the electric power producing the hydrogen now used by steelworks and the hydrogen used to manufacture the synthetic fuel were both 100% green.

19.2 Electrofuels (e-fuels) – synthetic liquid hydrocarbon fuels from power-to-liquids (PtL)

CO2 should be captured from the atmosphere in order to reduce atmospheric concentration of this greenhouse gas. However, the current markets for CO2 (such as in sparkling drinks) is not very large. But the other reason for wishing to capture CO2 is to obtain carbon atoms to use in combination with hydrogen in the manufacture of liquid hydrocarbon fuels.

The first thing we can say about electrofuels is that hydrogen itself is an electrofuel but in gaseous form. The key input in its manufacture is electric power. However making hydrogen, which is a gas, from water (hydrogen was traditionally known as water-gas) involves 'forcing' a lot of energy in the form of electricity or heat or both into the water. To break the atomic bonds between the hydrogen and the oxygen which originally converted these two gases into water (H2O), a huge amount of energy is required. After electrolysis, a proportion of the energy used in the electrolysis of water is then available as a source of energy in the form of hydrogen. Unfortunately, the hydrogen produced has considerably less energy content than the energy used to create it resulting in an efficiency loss.

The processes by which we extract the CO2 for manufacturing e-fuels from chimney stacks and even more so from the air to produce carbon atoms for an electrofuel is also very energy intensive. There are substantial losses in each link in the chain of manufacture – both in electrolysis of H2 and CO2 capture. It is for this reason that it is always said that if an activity can be electrified directly (such as by using overhead power lines for railways), then it should be. In the case of passenger vehicles it means using battery power even though this is slightly less efficient than a direct supply of power (such as electric trains with overhead power cables use). If neither means of direct electrification is possible then the activity might use hydrogen or ammonia and if they are not suitable then, as a last resort, the activity or process should use electrofuels.

Haldor Topsoe (now just called 'Topsoe')

Topsoe is a Danish chemicals company founded in 1940. It specialises in carbon reduction technologies. Topsoe is focussed on carbon emission reduction technologies for the chemical and refining industries. It therefore has a particular focus on liquid electrofuels.

Topsoe has been selected by Zhejiang Jianglan Bio-Energy Technology Co. Ltd (Zhejiang Jianglan) to deliver technology and services for Sustainable Aviation Fuels (SAF) production. Once fully operational, the plant is set to produce 300,000 tons of SAF annually which will initially be exported to European markets. Construction is expected to start in September 2025, with operations expected to commence in December 2026.

Porsche/ Volkswagen Group

One vehicle manufacturer which has invested substantially in the manufacture and use of e-fuels is Porsche This is initially for the Porsche racing team. The e-fuel is manufactured in Chile by HIF. It will use wind power (wind in this part of Chile is strong, relatively constant and from the same direction) to make hydrogen and will extract CO2 from the air using DAC. The project will integrate a wind powered electrolyser to produce green hydrogen, which will then be combined with CO2 captured from the atmosphere to produce synthetic methanol. The methanol will be converted into e-fuel via a methanol-to-gasoline technology. Porsche has been testing a 911 running on synthetic liquid hydrocarbon fuel made by HIF from electricity, air and water rather than crude oil. This points up one of the many possible directions of the energy transition. May 5th, 2025**,** HIF Global, signed an agreement with Porsche AG and Shell for the delivery of e-Fuels produced by its Haru Oni e-Fuels facility in Punta Arenas, Chile. HIF is delivering e-Gasoline produced at the HIF Haru Oni e-Fuels facility in Chile to Shell's Technology Centre in Hamburg, Germany. The e-Gasoline will be blended by Shell and then used by Porsche at different events to showcase their compatibility with existing combustion engines.

Norsk e-fuel: Aviation

A good European example of the production of e-fuel for aviation (Sustainable Aircraft Fuel or SAF) is provided by Norsk e-fuel. This uses CO2 captured from the air which is converted to what is called 'syngas' also called synthesis gas, a mixture of hydrogen and carbon monoxide (CO). This syngas can then be converted to liquid fuels in a 'reactor'. The production of this type of fuel involves a closed carbon cycle in which carbon is first extracted from the air (DAC) and then returned to the air as an aircraft burns it in a jet engine. The process is, therefore, carbon neutral.

19.3 **Thermochemical Processes:** for manufacture of H2 and Synthetic Liquid Hydrocarbon Fuels: *Case Studies: Synhelion, Heliogen*

Synhelion: (or how to enable jets to fly on sunlight)[70]

Synhelion (now incorporating Heliokon) is focussed on synthetic liquid hydrocarbon fuels produced by turning sunlight into fuels but without the need for an electrolyser and on producing the heat required for manufacturing cement and energy storage to enable round-the-clock (RTC) manufacture. The CO2 is sourced from chimney stacks or from Direct Air Capture or from Methane. The H2 comes from water which, as well as being available from a water main, may also be obtained by direct air capture.

It is the first company able to produce syngas on an industrial scale using only solar heat as an energy source. It has developed a solar thermal process to produce synthetic fuels that does not require electricity by instead using high-temperature solar heat to produce syngas which can then be used in standard industrial processes to synthesize liquid fuels, such as kerosene, gasoline, or diesel that are compatible with conventional jet engines and internal combustion engines. The 'formula' is CO2 + H2O + sunlight energy = fuel (in the form of a hydrocarbon with the H coming from water and the C from carbon dioxide). As with other synthetic hydrocarbon fuels, the process is carbon neutral (defined in Chapter 18.1) as the process releases an equal amount of CO_2 during its combustion as was previously used for its production.

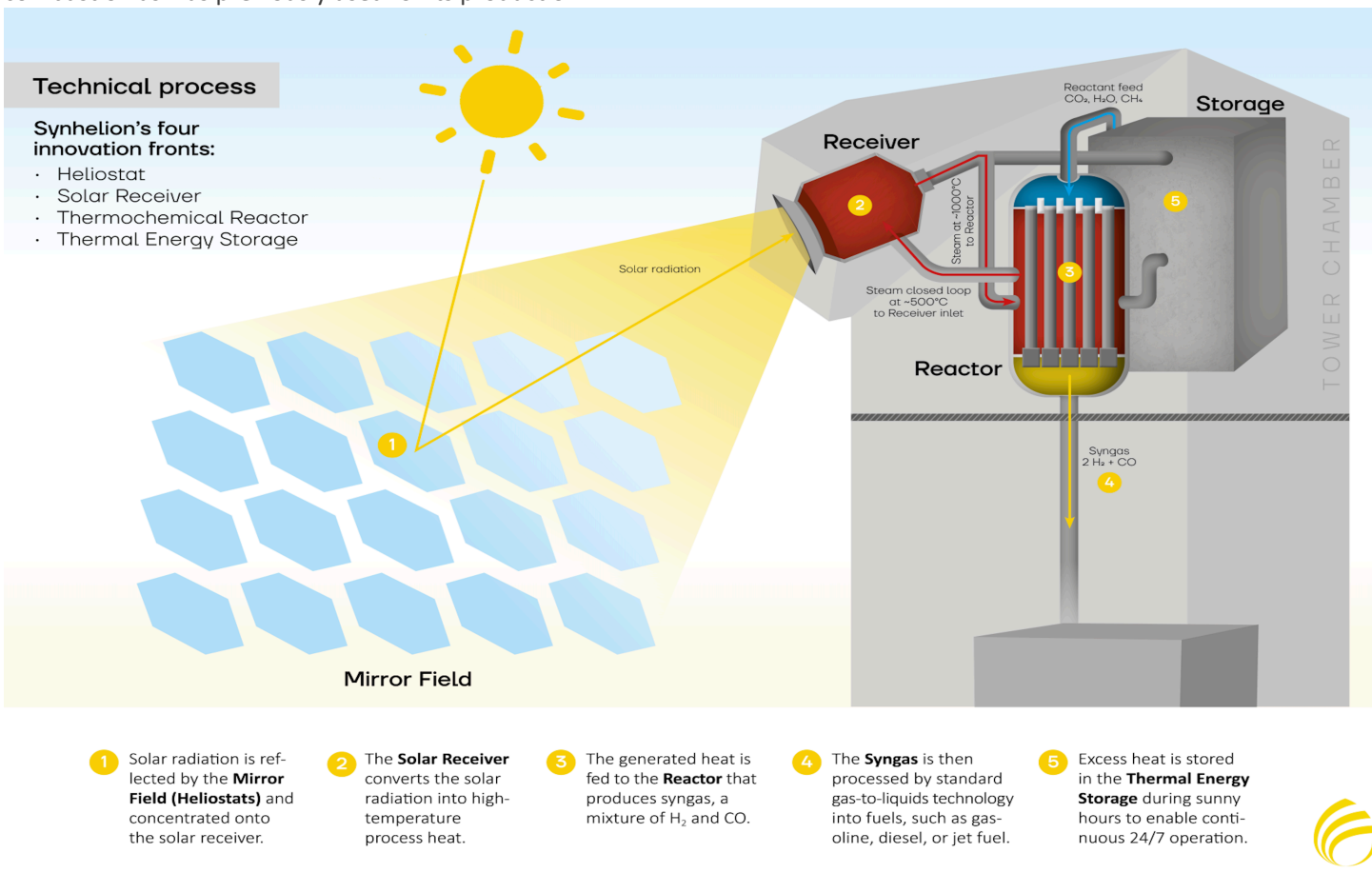


Figure 19.3 Synhelion's technical process

Making liquid fuels requires in the first instance heliostats (which are the mirrors in Figure 17.2), a solar receiver (in the tower), a thermochemical reactor (to enable the production of syngas from CO2 and water), and thermal energy storage (to enable round the clock [RTC] operation of the 'refinery' or industrial process such as cement manufacture. A great advantage of the technology is that it can sited anywhere. The company already works with international partners such as Lufthansa Group, Wood, Eni, CEMEX, and Zurich Airport.

[70] This section based on information from company website at https://synhelion.com/

Direct air capture (DAC) could also be a source of water (for making hydrogen) as well as CO2 as the air is full of moisture. Differences in the chemistry of CO2 and water adsorption may mean that it will be possible to simultaneously remove water from the atmosphere alongside CO2.

Heliogen[71]

This company was founded in 2013. It was trying to integrate related aspects of energy – heat, power and hydrogen (which is one of what they call HelioFuels). It developed a technology using heat from concentrating solar power (CSP) to manufacture clean hydrogen which can then be used for the manufacture of liquid fuels. The company's technology produces primary energy from the sun in the form of heat rather than electric power by using concentrating solar power (CSP) to produce heat rather than using photovoltaic panels (PV) which use sunlight to produce power. They call it a Sunlight Refinery.

Heliogen is focussed on Industrial Decarbonisation. As we have noted before, electrolysers and synthetic fuel plants operate more efficiently and hence at a lower unit cost of production if round-the-clock (RTC) clean power is available. As the energy produced by CSP is heat, not power, it can be stored economically in the form of heat (heat storage is covered in Chapter 14) and thus a manufacturing plant can operate 24/7. Clearly, however, the addition of storage raises the total cost of RTC clean power above that of intermittent heat or power that the CSP system would produce without heat storage.

The first thing which Heliogen emphasise on their website is that their CSP technology is best employed where solar 'insolation' is high. Although they say that sunlight is 'one of the most evenly distributed natural resources', in fact the economic value of that energy is determined by the 'insolation' of sunlight at any particular location. Most parts of Europe apart from Spain and Southern Italy are not obvious places to use CSP or even PV whereas California, the northern part of South America, Africa, parts of Asia and Australasia are all good locations for solar, either CSP or PV or both. Indeed, Hydrogen Europe in its "Compass for the Hydrogen World" publication noted that a photovoltaic plant in Chile, Namibia or Morocco produces around **double the yield of a plant in central Europe**. In North Africa, a one kWp (p = peak nameplate output) plant generates 2,190 kilowatt-hours while in European Union countries the average is only 1,051 kilowatt-hours.

In the solar industry, the capacity factor and dispatchability are key. If they are low as in Europe, then it is hard to compete with North Africa. The graphic below, also from Hydrogen Europe, shows that with the same solar panel 'nameplate' power the actual power generated is so much less in Europe (apart from Spain and southern Italy). As a result, even though at present the efficiency of fuel cell electric vehicles (FCEVs) is much lower than that of batteries, the result 'at the wheel' is the same.

Heliogen sees a competitive advantage they have as being the ease with which they can provide 'dispatchable' power by means of heat storage despite using variable sunlight. In this way, local power in parts of the US may become nearly as economically efficient as distant sources of energy that have to be transported over long distances. Such a combined heat and storage system for power generation is ideal for those operating power distribution networks as well as for those manufacturing hydrogen or synfuels.

[71] This section based on company website which also provides a video explanation of this particular CSP design, see https://heliogen.com/

Figure 19.4 shows a CSP system without storage (in yellow) which can produce 600 MW of power output when the sun is at its peak. However, that power is wasted or can only be sold at a low price compared with the demand at other times of day. By storing that heat, the system can make truly dispatchable power available for industries that require 24 hour availability.

The other key competitive advantage is the higher temperatures, up to 1000 °C at some point which is necessary for many industrial processes (Heliogen's main target market).

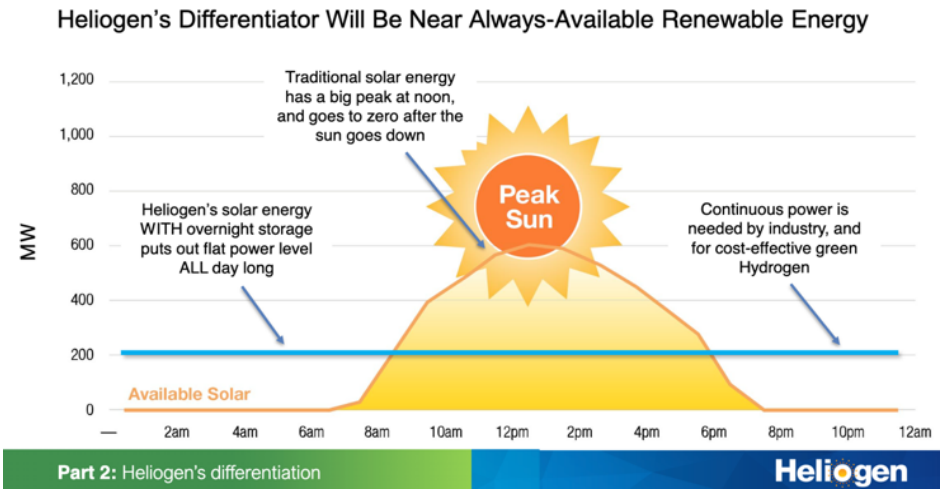


Figure 19.4 Smoothing out peaks and troughs of electricity production

What they have achieved is a system using cameras that focus on the mirrors continuously and which provide a signal to a micromotor which can make continuous slight adjustments in the positioning of each mirror to ensure it is working at full efficiency. This is in contrast to previous technologies which used 'tables' of sun position at every time of day in all locations but which is inevitably not so accurate as Heliogen's positioning.

Heliogen's storage system enables all the peak power to be easily stored as heat which can then be used to provide continuous power supply to industry, hydrogen production or for the production of synthetic liquid fuels. While solar PV installations can construct special storage facilities using battery, hydrogen or heat storage with CSP the energy starts as heat and does not have to be converted to heat. Thus adding a storage facility to a CSP heat generator, should be much simpler (and hopefully lower cost) than adding it to a PV facility. By this means, Heliogen can provide round-the-clock (RTC) heat direct to a steam generator-set for the production of power. Alternatively heat and power can be supplied to electrolysers to make hydrogen 24 hours per day. In this latter use, the added value is from the much higher 'capacity factor' than with just a PV power supply. Also, of course, it can provide heat for industrial processes to replace natural gas.

Heliogen make claims for several innovations that will sharply increase the efficiency of their system. First, in the size of the mirrors (the heliostats). They are very much smaller units compared with earlier generations of CSP technology and therefore easier to install. Because they are so much lighter, they are also easier to continually adjust to ensure that they always point directly at the sun.

Heliogen's so-called closed-loop system (rather than just using tables of sun positions linked at time of day and day of the year) uses computer vision (four cameras which can focus on each heliostat to identify exactly how to move it to keep it exactly centred on the sun) to identify the precise position of every single mirror, then make micro-adjustments in real time to achieve a perfect focus. Closed- loop AI ensures that the mirrors are pointed more accurately by using optical feedback.

They have also developed an 'Installation & Cleaning Autonomous Robot & Utility Solution' (ICARUS). By bringing the same advanced technologies used by its AI-enabled concentrated solar power systems to the task of installing and maintaining its systems, the Company's latest innovation is expected to significantly reduce the time to deploy its concentrated solar facilities, as well as the costs associated with construction and ongoing maintenance such as heliostat cleaning. Their 20 module 100MW plant is forecast to: produce power at a cost of 5 US cents per kW-hour and use much less ground space than comparable PV or other CSP.

Plant estimated to produce:

- > 85% capacity factor renewable power
- < 5 cents per kW-hour power cost
- < 1/6th square mile footprint (650m x 650m)
- > 850,000 kilograms of hydrogen/year

In July 2021, it was announced that Heliogen would merge with Athena Technology Acquisition Corp, a so-called Special Purpose Acquisition Company (SPAC) which would enable Heliogen to raise funds as well as to become a public company on the New York Stock Exchange with an expected value of approximately $2 billion. The SPAC is a company led by two women with extensive experience in venture capital who have also been board members of public and private companies. Sadly, it de-listed from the New York Stock Exchange in November 2023. It is still operating, however. But now the business model is "commodities-as-a-service". It provides steam, power, or green hydrogen to municipalities and heavy industry, with charging based on quantity supplied (e.g. steam-as-a-service).

Price of SAF, eSAF and Kerosene.

The European Union Aviation Safety Agency (EASA) has calculated 2024 average prices for SAF from biofuels, advanced biofuels, and recycled carbon biofuels at €2,085/t ($2,169/t). The reference price for synthetic aviation fuels (eSAF) is estimated at €7,695/t ($8,007/t), while conventional aviation fuel is €734/t ($763.8/t). This creates price differentials of €1,351/t ($1,406/t) for SAF and €6,961/t ($7,244/t) for eSAF compared with conventional fuel (kerosine).

Under Article 12 of ReFuelEU Aviation, penalties for non-compliance must be at least twice the price difference between SAF and fossil kerosene. Based on 2024 average prices, SAF non-compliance penalties would be around €2,700/t ($2,809/t), while eSAF penalties would be significantly higher at €13,992/t ($14,560/t), though the eSAF sub-mandate doesn't begin until 2030.

Significantly, jet fuel suppliers who fail to meet EU mandates must pay these penalties but aren't exempted from meeting quotas. Unfulfilled quotas carry over to subsequent years, making non-compliance increasingly costly over time.

Chapter 20: Carbon Credits, Carbon Capture, Usage and Storage, Direct Air Capture of CO2 [72]

Until recently, it was common practice for companies which were large emitters of CO2 to claim that they were carbon neutral or carbon negative through using voluntary carbon markets. However, the real challenge for those buying such credits is knowing how to separate the real, impactful credits available on the market from those that won't deliver the promised emission reduction, removal, or avoidance. These latter are a form of greenwashing.

Reducing, removing or avoiding emissions.

A carbon credit is a mechanism for one party to compensate another for the action of reducing, avoiding or removing carbon. Based on their net emissions impact, there are three types of carbon credits: carbon reduction, carbon avoidance and carbon removal. Carbon avoidance is an action that prevents a carbon-emitting activity from happening. Carbon reduction is an action that decreases the amount of greenhouse gas emissions, compared to prior practices. Carbon removal is the process of removing carbon dioxide from the atmosphere and locking it away for decades, centuries, or millennia. For all three types, credits are measured and issued by assessing or estimating the climate impact from carbon credit purchases—specifically, how much carbon is avoided, reduced, or removed based on the purchase of the credit.

Avoidance, reduction, and removal are three very different actions. But without accurate baselines, the climate impact cannot be determined. According to Carbon Direct:

> **Carbon Avoidance Credits**
> Examples of carbon avoidance include avoiding deforestation that would result in the release of carbon dioxide into the atmosphere. **Roughly 75% of all certified credits available are avoidance credits.**
>
> **Carbon Reduction Credits**
> While carbon reduction must drive most of the push to net zero, the transmission mechanism of carbon reduction to a purchase of an external credit is challenging. Examples of carbon reduction credits include reducing fossil-fuel use by improving fuel efficiency, or programs that reduce the methane that is generated from farms or municipal waste processing. **Reduction credits make up roughly 22% of certified credits on the voluntary carbon market.**
>
> **Carbon Removal Credits**
> Projects that remove carbon come from a diverse set of solutions, from nature-based solutions like reforestation, to engineered solutions such as direct air capture and storage. **Roughly 3% of credits on the carbon market today are classified as removals.**

While at present only 3% of credits are from carbon removal activities, we focus on the technologies for achieving this as removal provides greater certainty of the actual impact on global warming. However, to date, there is little sign of actual removals taking place on more than a tiny fraction of emissions. A key reason is the high cost of removals and the absence of revenue from most removals. One market, however, is that for synthetic hydrocarbon fuels which, by combining green hydrogen with the carbon atoms extracted from CO2, can create fuels for planes and road vehicles. Today, however, the cost of such fuels is substantially higher than that of natural hydrocarbon fuels.

In February 2024, the European Commission welcomed the provisional agreement between the European Parliament and the Council on the first EU-wide voluntary framework for the certification of high-quality carbon removals. The agreed criteria will ensure that carbon removals are: correctly quantified; store carbon for an agreed long-term period (a minimum of 35 years for carbon stored in products); go beyond existing practices and don't just reward the status quo; and contribute to broader sustainability goals, for instance by providing positive impacts on biodiversity.

[72] This introduction is based on an abbreviated and edited version of information on the website of Carbon Direct https://www.carbon-direct.com/ which offers services to measure, reduce, and remove carbon emissions with Carbon Direct science-backed carbon management solutions.

20.1 Capturing CO2

When CO2 is removed from 'smokestacks' or from the air, it can either be 'stored' or 'used'. It may be removed for two reasons. The first is to reduce the concentration of CO2 in the atmosphere and hence limit global warming or, second, as we noted above, to provide the carbon atoms to enable the production of synthetic hydrocarbon fuels by combining the carbon atoms with hydrogen molecules.

Reducing the concentration of greenhouse gas emissions in the global atmosphere is the critically important need. In 2022 emissions amounted to 53.8bn tonnes (CO2 equivalent). According to the IPCC, any pathway that could limit global warming to 1.5°C requires carbon removal of between 100bn and 1000bn tons over the remainder of the century. In effect, the IPCC is saying that we will not achieve the Paris Agreement temperature target without a massive investment in carbon removal and storage while at the same time, eliminating most new carbon emissions by moving towards the rapid phasing out of most fossil fuel use.

CO2 can be 'captured' in two ways.

The first is from industrial processes that emit a high tunnage of CO2 from their 'chimney stacks/flues'. It thus comes directly from hydrocarbon fuels being burned shortly after they have been extracted from the Earth. If not captured these emissions would immediately be increasing the concentration of CO2 in the atmosphere. When this CO2 is stored or sequestered, the process is known as carbon capture and storage (CCS)

The second way is to capture/ extract it from the air which is known as Direct Air Carbon Capture (DACC). Given that the concentration of CO2 in the air is only 422 parts per million (PPM) this is very much lower than from flue emissions. It has thus been much more costly, at least up to now, to extract directly from the air. When this CO2 is stored or sequestered, the process is known as direct air carbon capture and storage (DACCS).

Capturing flue CO2 or atmospheric CO2 to obtain carbon atoms to use as the carbon component in the manufacture of synthetic liquid hydrocarbon fuels such as aircraft kerosine (see Chapter 17 on synthetic fuels) only enables carbon in the air or from a flue to be 're-used'. If recycled this way, then CO2 extracted from flue gases will still increase atmospheric concentration of CO2 though with a slight delay as carbon is still being combusted and released into the atmosphere in its second time of use for example as aircraft fuel. **Thus this process is carbon additive as it still involves combusting newly extracted fossil fuel**. **In the case of DACC capture, initially the CO2 concentration in the air will be reduced but when the synthetic fuel is used, the concentration returns to what it was before. This process is carbon neutral**. We cover these concepts in more detail below using aircraft synthetic kerosine as an example:

Carbon neutral

The use of carbon captured by using DACC is carbon neutral because it is recycling existing CO2 in the atmosphere and therefore re-using CO2 that has already been emitted into the atmosphere from the combustion of fossil fuel sometime since the start of the industrial revolution. When combusted a second time, it is simply returning that CO2 to the atmosphere. There is thus no net impact of the two processes – DAC (CO2 removal) and an aircraft flight (CO2 additive) provided the capture process uses carbon free energy. But one company, Tree Energy Solutions (TES), has developed a system whereby carbon removed from smoke stacks to collect carbon atoms to manufacture methane (CH4) is in fact carbon neutral.

Carbon additive

If the carbon atoms that are being used for the manufacture of electrofuel are sourced from a factory or power station exhaust, the combined processes from the two linked manufacturing processes (for example making aircraft fuel from the emissions of a factory making steel) is carbon additive.

The double use of CO2 can be considered as

carbon neutral steel-manufacture and carbon additive flying *or*
carbon neutral flying and carbon additive steel making *or*
the two activities each being only half as carbon additive as the alternative.

But certainly not both being carbon neutral as is often claimed.

In the longer run, however, there should be almost no industrial processes (apart perhaps for the manufacture of cement) which emit CO2. As we noted above, at that point, the manufacture of synfuels would have to rely entirely on DAC. These fuels are likely always to be expensive.

Carbon negative

If rather than re-using the carbon obtained from DACC which returns it to the atmosphere, we 'sequestrate' it i.e. put it somewhere for very long-term storage normally underground in caverns, then the concentration of CO2 in the atmosphere actually declines (or rises less than it otherwise would).

Microsoft recently announced that it aims to become carbon negative by 2030 and by 2050 to have removed from the atmosphere all the carbon that it has emitted since it was founded in 1975. This is a significant commitment by the company but it underscores the potential for approaches using negative emissions to play an important role in meeting international climate goals.

We will cover below some of the companies which already provide, or are developing, methods of carbon capture. We will then consider how carbon can be sequestered. We will also cover companies with different methods of sequestering or (re)use. The first DAC company we cover (Climeworks) is one which has teamed up with another company (Carbfix) which provides an example of an interesting method of sequestering.

Other methods of sequestration involve converting CO2 to solid carbon. This could be black carbon (with the oxygen being released from the carbon bond). An example of this technology in use is provided below.

20.2 Direct Air Carbon Capture (DACC) Companies – some examples: Climeworks and Phlair

Climeworks[73]

Rather than supplying carbon dioxide to electrofuel manufacturers (which remains carbon additive), Climeworks main business is using DACC to permanently remove CO2 from the atmosphere and mineralising it underground. As they note, "air-captured carbon dioxide can either be recycled and used as a raw material, or completely removed from the air by safely storing it. Direct air capture, therefore, is one of the key ways to stop global warming by reducing the amount of potentially harmful CO_2 in the Earth's atmosphere."

One of the ways the ways they generate revenue is by 'selling' subscriptions to individuals as well as companies to help finance an expansion of their DAC facilities.

"Why should I subscribe or what should I use the subscription for?

> This is of course up to you! Do you want to enable the removal of some carbon dioxide to reduce your personal footprint? Good! Are you planning a long flight and want to build up some carbon dioxide "allowance" first? Even better! Do you believe that the Climeworks technology is surprisingly cool and deserves some more publicity? Great! Join us on the climate positive journey.

Those who contribute are called Pioneers and these are the people who finance Climeworks (it's a bit like crowdfunding). There are now over 20,000 Pioneers."

It would not be possible to scale up to the CO2 removal capacity that would make a large difference to the current CO2 concentration in the atmosphere (around 430 ppm) using only crowdfunding. On 6 April 2022 Climeworks announced that it had signed an equity round of CHF 600 million (USD 650 million), which marks a major milestone in the history of Climeworks and an even bigger step for the carbon removal industry.

[73] https://climeworks.com/news/climeworks-builds-first-commercial-scale-direct-air-capture-plant

The financing is co-led by Partners Group (acting on behalf of its clients) and GIC, with further participation from (in alphabetical order) Baillie Gifford, Carbon Removal Partners, Global Founders Capital, John Doerr, M&G, Swiss Re, as well as other new and existing shareholders including long-term investor and anchor shareholder BigPoint Holding AG. J.P. Morgan Securities LLC served as sole placement agent for Climeworks in connection with the transaction.

Since its foundation in 2009, Climeworks has pioneered direct air capture technology and is the most advanced industry player, with the world's largest direct air capture and storage plant launched in September 2021. The new investor base includes several of the most renowned and largest institutional technology and infrastructure investors globally. The funding will unlock the next phase of Climeworks' growth, scaling direct air capture up to multi-million-ton capacity and implementing large-scale facilities as carbon removal becomes a trillion-dollar market.

In terms of what has happened on the ground (and below ground), in September 2021 Climeworks launched its flagship facility Orca, in Iceland. It is the world's largest direct air capture and storage plant. As carbon capture is just the first stage of CCS, they joined forces with the Icelandic company Carbfix (we cover it below) which has a strong scientific backbone and is one of the world's experts in rapid underground mineralisation of carbon dioxide. Orca removes 4,000 tons of carbon dioxide per year while Carbfix mixes the carbon dioxide with water and pumps it deep underground. Through natural mineralization, the carbon dioxide reacts with the basalt rock and turns into stone within a few years.

It is probably fair to say that carbon capture and storage has not yet taken off at a sufficiently large scale to make more than a minimal contribution to reducing CO2 concentration in the atmosphere. However, over time the contribution will increase particularly if carbon taxes become more widespread.

Its latest (Generation 3) technology is said to use novel structured sorbent materials replacing the packed filter beds used in previous technology generations. The new structures increase surface contact with CO_2, reducing the time to capture and release CO_2 by a factor of at least two, thus capturing more than twice as much CO_2 as previous filters. The new filter materials consume half the energy and are designed to last three times longer than prior materials. The company's cost reduction strategy aims to achieve costs of 250-350 USD per ton captured and total costs of 400-600 USD per ton net removal by 2030. This represents an overall cost reduction of up to 50 percent compared to today.

Phlair[74]

This is a relatively new company which has developed a carbon removal process from the air which is less energy intensive than other methods. It uses what is known as a hydrolyser. It claims that its system has 3x lower energy capture costs than thermal DAC approaches, yielding lower removal costs. It can also operate intermittently to take advantage of periods when electricity prices are low or negative. Phlair already has contracts with companies that will benefit from being able to use this new DAC technology.

20.3 Sequestration, mineralisation and Reduction of CO2 by removing the O2. Storegga, Carbfix, CarbonBuilt, Converting CO2 to Solid Carbon

Most CCS projects inject CO2 into sedimentary basins and require an impermeable cap rock to prevent the CO2 from migrating to the surface. A good example of a company using this technique is Storegga.

Storegga[75]

Storegga uses proven geological stores many kilometres below the earth's surface, to safely and permanently lock away harmful CO2 which has been captured from industrial emissions processes or drawn out of the atmosphere.

CCS has been proven in industrial-scale projects globally, with the world's first industrial CCS project in continuous operation since 1996. Now, CCS projects are being developed in industrial regions across the globe

[74] Source: Company website https://phlair.com/technology
[75] Source: Company website https://storegga.earth/co2-storage

because it is the most effective way to prevent or remove the CO2 emissions from our atmosphere in volumes significant enough to slow the pace of climate change. These emissions have been generated from modern society through things like power generation and industrial processes like manufacturing.CO2 storage in deep geological formations is also known as 'geo-sequestration'. It has a big part to play in a sustainable future.

Storegga is also developing both electrolytic and CCS-enabled hydrogen projects in Scotland.

Carbfix

Captured carbon can, alternatively, be stored through injection into reactive rocks provoking CO2 mineralization and thereby, permanently fixing carbon.

Mineral carbonation extends the applicability of carbon capture and storage by enabling storage in areas previously not considered feasible. The rapid mineralization of CO2 through injection into reactive rock formations also increases storage security. Carbon mineralization in basaltic rocks offers a global storage potential that exceeds anthropogenic emissions.

CarbonBuilt Inc

"CarbonBuilt's Reversa™ process includes innovations to both the concrete mix design and its curing process. On the formulation side, it introduced portlandite (a widely available commodity chemical also known as calcium hydroxide or hydrated lime), reduce the usage of traditional Portland cement and increase the use of low-cost and low-carbon supplementary cementitious and filler materials. Concrete products are then formed using the same processes and equipment that are used today.

After forming, the concrete products are cured with CO_2 using a process that does not require expensive capture, compression or purification of the CO_2, high temperature or elevated pressure. This flexibility enables use of a wide range of CO2 containing streams ranging from point-source or distributed emissions that can be used directly, to merchant gas sourced from such sources, to emerging direct air capture technologies."

Chapter 21: Connecting up the Energy Value Chain – the Sector-Coupled Economy and Clusters: H2, Electric Power, Synthetic Fuels, CCUS and Fertilisers
Case studies: Iberdrola, HyDeal Espana

Hydrogen will be the driving force of our energy system of the future, but the bloc will need to rely on imports. 'Europe is never going to be capable of producing its own hydrogen in sufficient quantities.'

Frans Timmermans European Commissioner for Climate Action

The statement above is one of hope. But that hope will not be realised unless it becomes possible to provide hydrogen to the market at a much lower cost than today, whether manufactured in the EU or abroad in locations with higher capacity factors for wind and solar. Such locations would be parts of Morocco or Saudi Arabia. Alternatively, legislation could be introduced to require manufacturers of ammonia-based fertilisers and oil refiners (who are currently the largest users of hydrogen and others, to switch to green hydrogen.

Sector Coupling

Synergies exist on a wider scale in the energy sector than those explored above and if exploited, can result in a considerable reduction in unit costs. The European Parliament Committee which covers the energy sector has produced a report entitled *Sector coupling: how can it be enhanced in the EU to foster grid stability and decarbonise*, the Committee made it clear that the energy system will have to function without coal, oil *or gas*.

The question then is what will replace these carbon-based *primary energy sources* in the future provision of power (currently ~20%), heat (currently ~40% including industrial heat) and transport (currently ~40%). The fact that electricity represents a lower proportion of energy use than either of the other uses, points up the illogic of considering power de-carbonisation on its own. Hydrogen can provide energy for power, heat and transport and is therefore one integrating element of the equation for all three main energy uses. There are, however, technology issues in transporting hydrogen over large distances.

While there remain some differences in usage across sectors essentially sector coupling means interconnecting and integrating the energy consuming sectors – buildings (heating and cooling), transport and industrial heat - with a diverse energy sector which includes electricity, hydrogen and the associated transport, storage and grid systems for these energy carriers. It might also involve the use of two-way fast charging points for electric transport vehicles to provide power storage (vehicle-to-grid) and also hydrogen refuelling points and increased contractual automatic disconnection of some power users at times of high power prices (demand management) by the use of interruptible contracts.

It includes the conversion of power to hydrogen, ammonia and fertiliser and to liquid fuels and the use of nuclear heat and power to produce hydrogen. But also, for the nuclear power station owner, when spot prices of power are high, to arbitrage between the revenue generated from hydrogen production and the revenue generated from selling power to the grid at times of such high prices (normally when wind and solar are making little or no contribution). In addition for some nuclear installations will be selling both industrial heat and heat for district heating systems which will provide them with an important source of revenue.

Figure 18.1[76] below indicates examples of sector coupling and integration based around:

- the conversion of power to hydrogen (through electrolysis) and;
- the reverse conversion of hydrogen to power (though fuel-cell regeneration) and;
- the conversion of hydrogen to liquid fuels and ammonia for use in decarbonising shipping and aviation and;
- hydrogen in use for space heating and combined heat and power and;
- hydrogen in the transport sector and;
- hydrogen as a source of heat for industrial processes such as steel manufacture.

For investors, such integration potentially provides investment opportunities. In this vision of the very near future, hydrogen is a pivotal technology in sector integration.

[76] US Dept. of Energy (DOE) National Renewable Energy Lab. (NREL) *Hydrogen in Future Integrated Energy System* https://www.nrel.gov/news/program/2020/study-shows-abundant-opportunities-for-hydrogen-in-a-future-integrated-energy-system.html

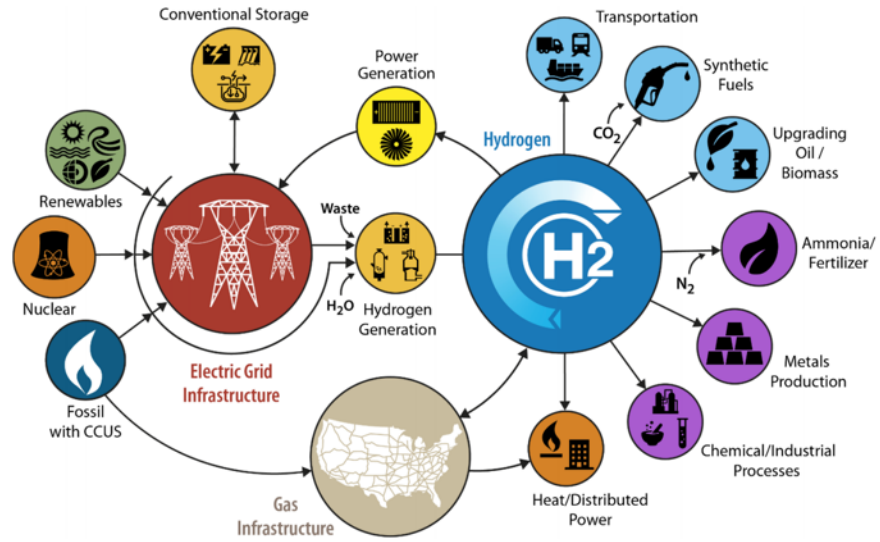


Figure 18.1 Sector coupling

And educating politicians, civil servants, school children and others on why we need to go the nuclear route. We also need countries to enable the required training for constructing this new generation of assets.

Clusters

We have already noted that there are considerable advantages in siting renewable or clean power (small nuclear & wind, solar) in close proximity to hydrogen manufacture and other users of hydrogen. This is designed to give economies of scale and scope and such agglomerations are known as industrial clusters. They are characterized as geographic areas that co-locate companies across multiple related industries. Concentrating manufacturers and users in a relatively small geographical area enables the scaling of low-carbon technologies By sharing risk and resources among multiple companies working in partnership with each other, industrial clusters enable economies of scale, scope and learning. A good example of a cluster is the WESTKÜSTE100 project in Germany:

WESTKÜSTE100 is a project which is trying to achieve sector coupling in a region of Germany. It is trying to achieve both decarbonisation and the use of green hydrogen across a range of industries such as air transport, construction and space and industrial heating. It has the benefit of strong winds in the region as well as geological storage possibilities for CO2. It also plans to use the CO2 emissions from cement manufacture to provide the required CO2 for the production of synthetic hydrocarbons. What it is doing is in line with the EU's 'Fit for 55' Hydrogen Strategy and in particular the EUs plans in its ReFuelEU Aviation initiative which is trying to achieve an increase in the use of sustainable aviation fuel in Europe from 0.05% in 2020 to 5% by 2030. The 2050 target is 63% which may be achievable as investors are encouraged by the fact that synthetic liquid can use existing infrastructure for storage, delivery etc.

It is also aimed at obtaining the efficiencies gained from a cluster and from an integrated energy/ industrial system. The objective in this case is to build the regional infrastructure to enable sustainably produced steel, ammonia, ethylene and fuels to the Netherlands and Belgium accelerate carbon reductions in a world where wind is the new North Sea oil and gas.

Case Study: Iberdrola

Iberdrola, a Spanish utility, provides a good example of clusters/ sector coupling. It is a global energy corporate which describes itself as "Utility of the Future" and is the number one producer of wind power in the world, and one of the world's largest electricity utilities in terms of market capitalisation. 90 % of the group's long-term investment plan is aligned with the green taxonomy of the EU. It has a global reach with assets in the UK (Scottish Power), across Europe, Australia and Japan and in Mexico, Brazil and the US.

Iberdrola SA and fertilizer maker Fertiberia SA have proposed a €1.8bn ($2.1 billion) plan to turn the country into an exporter of ammonia made with renewable energy. Ammonia is a key component in fertilizers.

It is all very well building large electrolyser manufacturing plants or operating a large green hydrogen plant **but without end customers guaranteeing offtake of the product (as in any project financing) to mitigate volume and**

price risks and a means of transporting the end-product to customers, a project will not be bankable. To overcome the issues involved in successfully scaling-up the hydrogen market, in February 2021 thirty energy players in Europe announced that they had developed a plan for an integrated value chain known as HyDeal to deliver green hydrogen across Europe at the price of fossil fuels. The idea is to create an industry platform covering the complete green hydrogen value chain, from upstream to midstream, downstream and finance: solar power generation, electrolyser manufacturing, project engineering, gas transmission and storage, industrial applications in steel, chemicals and electric power.

HyDeal España is the first industrial implementation of the agreement between the parties. It will manage the development, funding and construction of renewable hydrogen production and transmission infrastructures. It should deliver price competitive renewable hydrogen to an industrial complex in Asturias from facilities based in Northern Spain. Anchor sponsors include ArcelorMittal, Enagás, Grupo Fertiberia and DH2 Energy. Production is planned to start in 2025 and total installed capacity expected to reach 9.5 GW of solar power and 7.4 GW of electrolysers by 2030.

HyDeal España has developed an industrial and financial model based on value chain integration, captive solar power, electrolysers operating on an industrial scale, hydrogen pipelines and demand aggregation to give it a cost advantage. Mass-scale renewable hydrogen will sustainably replace fossil fuels in industry, energy and mobility and will provide Spain with a domestically produced, secure, competitive and zero-carbon energy, steel and food supply alternative. HyDeal España aims to provide the equivalent of 5% of Spain's natural gas imports, contributing to the country's energy independence. ArcelorMittal and Grupo Fertiberia – together with other large hydrogen clients set to join the project - plan to purchase 6.6 million tons of renewable hydrogen over 20 years (avoiding annually the equivalent of 4 % of current Spanish CO2 emissions) to produce green steel, green ammonia, green fertilisers, and other low-carbon industrial and energy products, positioning the companies as European leaders on their respective markets. The direct connection of mass-scale renewable hydrogen production and long-term bankable consumption create enormous value for the system.

Potential Future Energy Systems

The US Idaho National Laboratory (INL) [77]. Has as its objective to advance integrated energy generation, storage and delivery technologies needed for a net-zero future. The integrated systems approach is a marked change from traditional energy system designs typically focused on single generation sources to support a single energy demand. The footnote gives useful information and graphics on how the future may look.

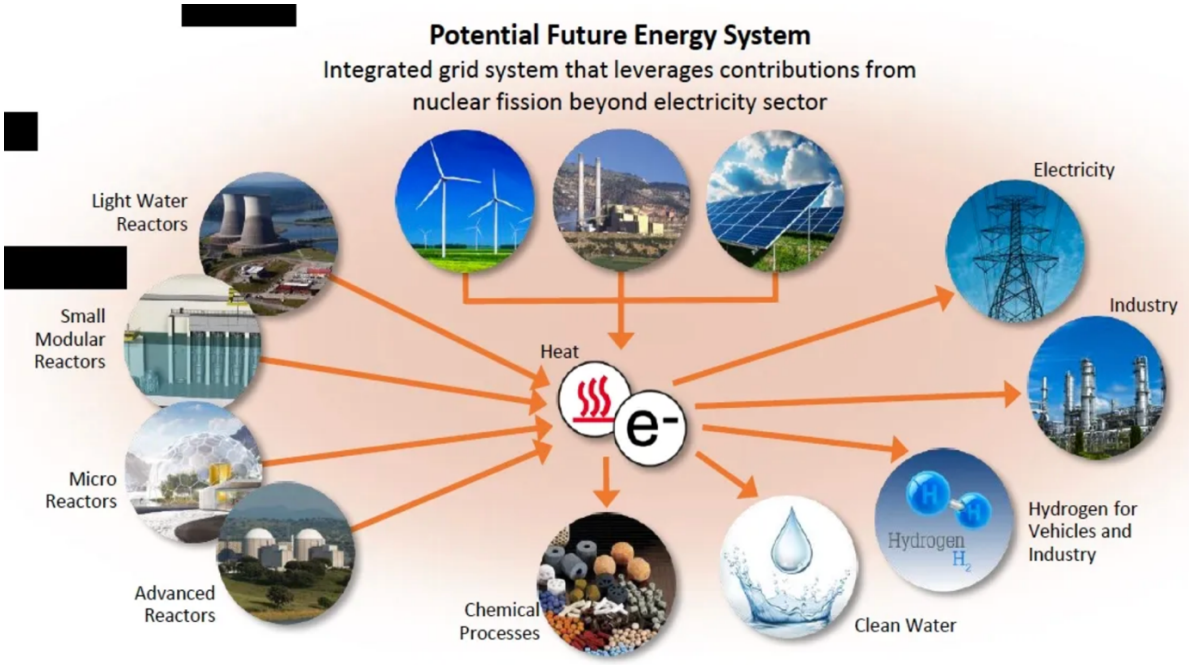


Image Source: Idaho National Laboratory

[77] Sourced from *The role of technology in the energy transition* Power Electronics News January 2024 https://www.powerelectronicsnews.com/the-role-of-technology-in-the-energy-transition-2/

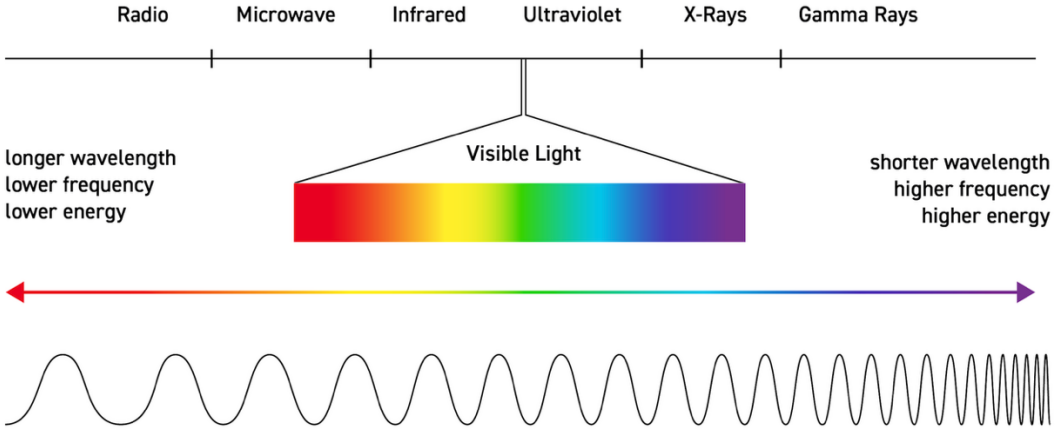
Appendix 1: *The Science of Climate Change and its Relevance to Lenders, Investors and Insurers: Climate tipping points and Global Cascades*

The Earth's Carbon Cycle

The science starts with the fact that:

> *The Sun delivers more energy to Earth in an hour than we use in a year from fossil, nuclear and all renewable sources combined.*

Fortunately, we can now convert *a little* of that sunlight (photons) into electricity (electrons) at low cost using photovoltaic (PV) cells (at least when the sun is shining). Other electromagnetic energy frequencies from the sun (infra-red) can convert directly into heat which happens when infra-red rays strike matter such as the air (or, indeed, you or me).



The energy in the air can then generate powerful air currents (winds) which can be converted into electric power through wind turbines. Infra-red (IR) rays can also be captured by what is known as Concentrating Solar Power (CSP) mirror collectors and then focussed on a collector, normally in the form of molten salts which then reach a high temperature suitable for industrial heat use, for the manufacture of hydrogen by breaking down water molecules into oxygen and hydrogen or for raising steam to generate power.

Some of the sunlight not used by our solar or wind farms or CSP is used by plants for photosynthesis as plants absorb CO2 from the air which 'locks in' carbon in all plant life at least until a plant is used for animal feedstock or burned or simply decays.

Clearly after these uses of sunlight, there is still a huge amount of energy from the sun, incident on the earth. It is this energy that keeps the earth at an average global temperature (across the whole globe and averaged over day and night) of around 15˚C - 17˚C which is the equilibrium temperature of the Earth. But there is a problem! Basic thermodynamics tells us that if all the solar energy that arrives on earth while the sun is out, is not dissipated back into space at night, the earth will heat up day-by-day as it tries to restore a temperature equilibrium which will be higher than before.

The problem we are now facing (and its manifestations in higher temperatures, more violent and more frequent storms and hurricanes and rising sea levels) arises from the fact that the Earth's energy systems are *not* in equilibrium. The only way in which Earth can achieve a new equilibrium rather than suffering ongoing rising temperatures is for us to cut greenhouse gas (GHG) emissions to close to zero. The deadline for doing so to have a good chance of keeping the global temperature rise to 'less than 1.5˚C or, alternatively 2˚C, is 2050. However, equally critical is the pathway to 2050. Cutting emissions only from, say, 2040 onwards, would not achieve the same objective as halving emissions between 2022 and 2030 and continuing the reduction at that rate until 2050, something which has now become necessary.

All life forms on Earth, including humans, are carbon life forms. Without carbon, humans and animals as well as plants and animals, would not exist. Carbon is a key element for life (as we know it) to exist. However, CO2

generated by coal, oil and gas is emitted into the atmosphere where, unless it was stored in some way, would cause global warming. It is partly stored in the atmosphere, partly in the oceans. Much is also stored in calcium carbonate ($CaCO_3$) deposits which is a product of the remains of sea creatures from millennia past. This CO2 is released in large quantities when calcium carbonate (limestone) is converted to cement for building purposes.

The CO2 emissions absorbed by the ocean (around 25% of fossil fuel emissions) results in acidification of the ocean which is now 30% more acidic than before the industrial revolution. This has dramatic effects on ocean life including damage to corals and to creatures with shells such as oysters. But it also affects fish and while the full impact of acidification (a change in the Ph measure of the water) is not yet fully researched, it is one of the major issues arising from fossil fuel emissions. The rising temperature of the ocean also results in the extinction of some forms of life including corals. Already the Great Barrier Reef off Australia, has suffered considerable destruction as a consequence.

Earth's Energy Budget[78]*, The Natural Greenhouse Effect and the Anthropogenic (human-related) Contribution*

The climate's heat engine (an engine in the sense that heat provides the energy that causes weather such as storms and tornadoes) must not only redistribute solar heat from the equator toward the poles, but also from the Earth's surface and lower atmosphere back to space. Otherwise, Earth would heat up endlessly as it has since the start of the industrial revolution in 1750 when anthropogenic emissions started to grow.

Fortunately, when sunlight strikes the earth, land and ocean surfaces and the atmosphere are simultaneously radiating heat back into space. The heat is generated from the infra-red component of sunlight incident on the Earth. Earth does not suffer the temperature extremes of the Moon (-173˚ at night to 127˚ during the day) because the blanket of atmosphere around it creates the Natural Greenhouse Effect. This is the process by which radiation from the Earth's atmosphere warms the planet's surface to a temperature above what it would be without this natural 'blanket' This is due to the greater transparency of the atmosphere to visible radiation from the sun (light) than to infrared radiation (heat) emitted from the Earth's surface. This net flow of energy into and out of the Earth system during the day and night, is Earth's energy budget as explained in the following section.

As is shown at the top of Figure A1.1[79], the incoming solar energy on average across the earth's surface is 340 watts per square metre or 1/3rd of a kilowatt – a very substantial amount of power when measured over the whole area of earth! Some is reflected back into space (around 29%) leaving around 240 watts of solar power per square metre. About 23 percent of incoming solar energy is absorbed in the *atmosphere* by water vapor, dust, and ozone, and 48 percent passes through the atmosphere and is absorbed by the surface.

On the surface, absorption of this energy drives photosynthesis (plant life), photovoltaic (PV) power generation, wind turbine generated power and Concentrating Solar Power (CSP). It also provides the energy that causes evaporation of water from the oceans and rivers (and when this falls back to earth as rain, enables hydropower), melts snow and ice, and warms the Earth which, on average provides an equable temperature which enabled the development of animal life (including human life). This is in contrast to everywhere else in the universe (at least as far as we know). The natural greenhouse effect from which earth benefits has enabled humans to develop on earth and to survive and generally to prosper. However, mankind's footprint on earth is now giving rise to concern for future generations.

[78] The material following is based on a NASA publication: https://earthobservatory.nasa.gov/features/EnergyBalance/page1.php
[79] NASA Earth Observatory https://earthobservatory.nasa.gov/features/EnergyBalance/page4.php

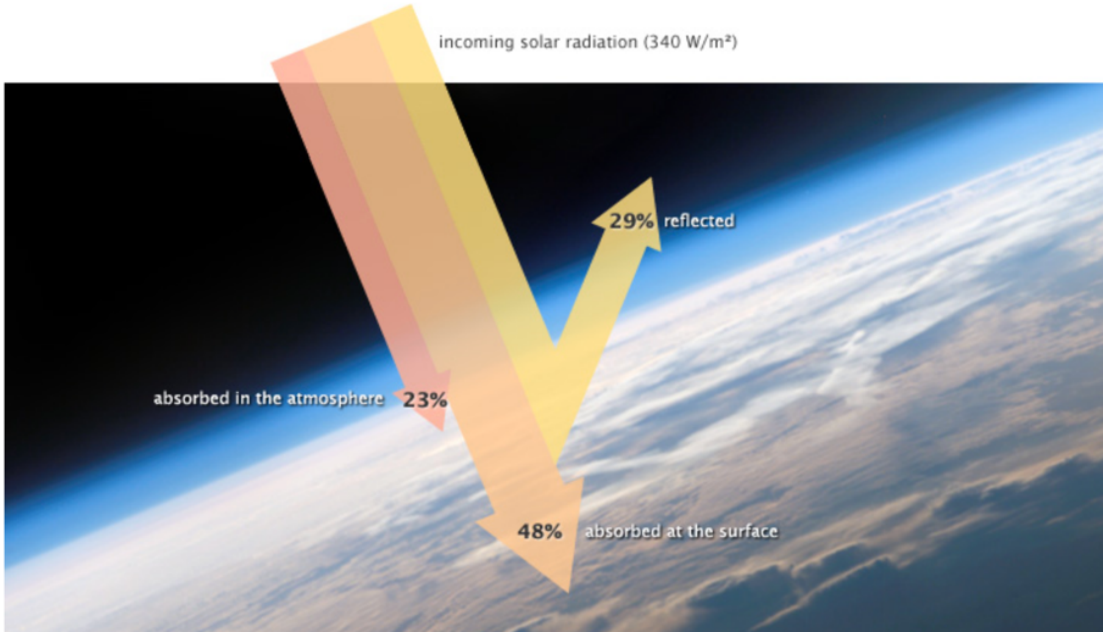


Figure A1.1 Incoming solar radiation must balance outgoing solar radiation to keep Earth's temperature constant

Earth's natural greenhouse effect was critical in enabling life on Earth to come into existence and thrive. It is the atmosphere above Earth that enables it to retain sufficient heat rather than being a frozen planet like the moon. So in an (imperfect) analogy to a garden greenhouse, the atmosphere enables Earth to have a much more equable temperature range than otherwise. During the day, light from the sun falls on (is incident on) the Earth and as a result of striking the solid body of the Earth, changes from light energy (shorter electromagnetic wavelengths) to heat energy (longer wavelength infrared radiation) which warms up the land and the sea. The land and sea then radiate some of this heat (infrared radiation) back to the atmosphere and in turn the atmosphere absorbs some of it and radiates some of it back to Earth.

This balance between Earth's heat being radiated back into space and or being radiated back to Earth is particularly evident at night in the dessert. During the day a cloudless dessert region is extremely hot. At night it can cool down dramatically not just because the sun is no longer incident on that part of the earth but also because the absence of cloud cover enables more infrared energy to be radiated back out to space. In contrast, in a country with frequent cloud cover, it will be warmer at night when it is cloudy compared with when the sky is clear when it may be frosty.

The key question is whether or not the *balance* between incoming and outgoing energy i.e. an equilibrium is achieved without a change in average Earth temperature. This energy balance is most meaningfully evaluated at the top of the atmosphere, where the net outgoing energy encapsulates all the complexity of the greenhouse effect and climate feedbacks. If more heat is trapped within the atmosphere, equilibrium can only be achieved through the Earth (land and oceans and atmosphere) warming in order to radiate more heat out into space. It is the factors that lead to more or less heat being trapped which determine whether the Earth's temperature remains constant, rises or falls.

Of course the earth has had temperature cycles throughout time including a number of ice ages. We have also had temperatures very much higher than today. Thus climate change has been a fact of life even before life existed on Earth. Temperature changes can be caused by changes in the sun's energy emission which can be related to 'sunspots' or to changes in the amount of snow cover or cloud cover resulting in a higher albedo (whiteness or reflective coefficient). This can lead to feedback effects. As the brightness of the earth increases for whatever reason, the earth will cool more which leads to snow covering greater areas which then leads to an even higher albedo. In contrast, as the earth heats up for example as a result of human caused (anthropogenic) GHG emissions, snow melts, the albedo decreases and the feedback effect then leads to the earth heating up more and the snow receding further leading to even higher temperatures depending on the so-called 'climate sensitivity' i.e. an estimate of how much Earth's surface will warm.

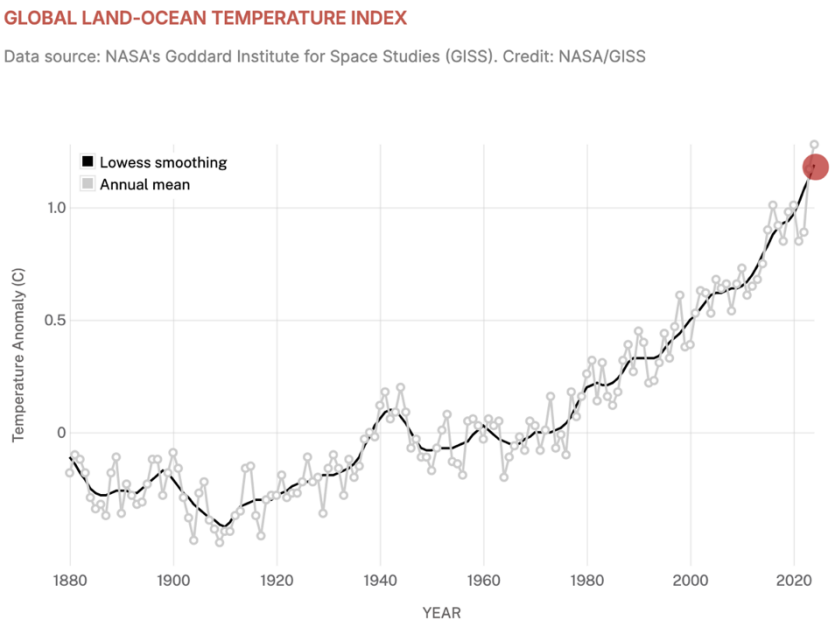


Figure A1.2 Change in Earth's temperature over time

A good up-to-date source of information on the impact of rising temperatures leading to climate change is summarised below by Carbon Brief on 10 January 2025. It can be accessed at https://www.carbonbrief.org/state-of-the-climate-2024-sets-a-new-record-as-the-first-year-above-1-5c/

Why CO2 Causes Global Warming

In the atmosphere, the major atmospheric gases (oxygen and nitrogen) are transparent to incoming sunlight (solar light energy) and also to infrared radiation (heat). Thus infrared radiation (heat) would be able to pass in or out through the atmosphere if these were the only constituents of the atmosphere. The key to the greenhouse effect is that water vapor, carbon dioxide and methane are opaque to thermal infrared energy.

While water vapour comprises the major part of greenhouse gases it is not the direct cause of warming. It is carbon dioxide (CO2) and methane (CH4) which cause what is known by scientists as "Radiative Forcing" or Climate Forcing. This is the difference between sunlight absorbed by the Earth and energy radiated back to space. Changes to Earth's radiative equilibrium that cause temperatures to rise or fall over decadal periods, are called *climate forcings*. Positive radiative forcing means Earth receives more incoming energy from sunlight than it radiates to space. This net gain of energy will cause warming. Conversely, negative radiative forcing means that Earth loses more energy to space than it receives from the sun, which produces cooling.

"Natural climate forcings include changes in the Sun's brightness, Milankovitch cycles (small variations in the shape of Earth's orbit and its axis of rotation that occur over thousands of years), and large volcanic eruptions that can inject light-reflecting particles into the stratosphere. Man-made forcings include particle pollution (aerosols), which absorb and reflect incoming sunlight; deforestation, which changes how the surface reflects and absorbs sunlight; the rising concentration of atmospheric carbon dioxide and other greenhouse gases, which decreases heat radiated back into space. A forcing can trigger feedbacks that then intensify or weaken the original forcing. The loss of ice at the poles, which makes them less reflective, is an example of a positive feedback. A system in thermal equilibrium has zero radiative forcing"[80] which is the stability that needs to be restored to Earth quickly before excessive damage to life on Earth is the outcome.

Greenhouse gas molecules absorb thermal (infrared) energy resulting in a rise in their temperature. Because greenhouse gas molecules radiate heat in all directions, some of it spreads downward and ultimately comes back into contact with the Earth's surface, where it is absorbed. The temperature of the surface becomes higher than

[80] Sources: Wikipedia https://en.wikipedia.org/wiki/Radiative_forcing and NASA Earth Observatory, Op.Cit. https://earthobservatory.nasa.gov/features/EnergyBalance

it would be if it were heated only by direct solar heating (sunshine). This supplemental heating of the Earth's surface by the atmosphere is the natural greenhouse effect. So it is important to note that while much of the ability of Earth to support life depends on the water vapour in the atmosphere providing a 'blanket' around the Earth – the greenhouse effect – it is the additional molecules in the atmosphere added by mankind (principally CO2 and CH4) which strengthen this greenhouse effect and lead the Earth's energy system's attempt to return to equilibrium.

However, as the concentration of greenhouse gases in the atmosphere increases this equilibrium can only be achieved at an increasingly higher temperature. If the proportion of GHGs in the atmosphere is continually increasing as a result of the annual release of CO2 and CH4 into the atmosphere, then over time the Earth's temperature will rise and continue to rise and remain high possibly for centuries. However, estimates of climate sensitivity is hugely important to climate policy and has been a key focus of the IPCC. However, the exact number has been difficult to determine due to the complexity of the processes involved and the relatively short historical records. The AR5 reported a likely range of 1.5C-4.7C and AR6 reported 2.5C-4C per doubling of CO2.

Radiative Forcing and its Impact on Earth

In accordance with the basic laws of thermodynamics, as Earth absorbs energy from the sun, it must eventually emit an equal amount of energy to space if its temperature is to remain constant. The difference between incoming and outgoing radiation is a planet's radiative forcing (RF). Thus Incoming Energy – Outgoing Energy = Radiative Forcing. So if more of the energy that reaches Earth stays on Earth this will force a change in the Earth's temperature. **Figure A1.3[81] shows how much radiative forcing there has been up to 2011. Thus while the energy beaming onto the Earth is, on average, 340 watts per m^2 by 2011, an additional 2.29 watts per m^2 of Earth's surface was being absorbed by land and ocean rather than being sent back out into space thus raising Earth's temperature.**

Year	Radiative Forcing (RF) Relative to 1750 ($W\ m^{-2}$)
1750	0.0
1950	0.57
1980	1.25
2011	2.29

Prior to 1750, before the Industrial Revolution, Earth's average RF remained relatively stable. To document how the atmosphere has changed since then, scientists calculate current RF levels as if it were zero in 1750. Source: IPCC AR5 WG1 Figure SPM.5.

Figure A1.3 Radiative forcing over time

Human influence causes heating of the climate system

In figure A 1.4 Earth's energy budget compares the flows of incoming and outgoing energy that are relevant for the climate system. Since at least the 1970, less energy is glowing out than is flowing in which leads to excess energy being absorbed by the ocean, land, ice and atmosphere with the ocean absorbing 91%. Graphic from from IPCC AR6

[81] NOAA Climate.gov https://www.climate.gov/maps-data/primer/climate-forcing

Since at least 1970, there has been a persistent imbalance in the energy flows that has led to excess energy being absorbed by different components of the climate system.

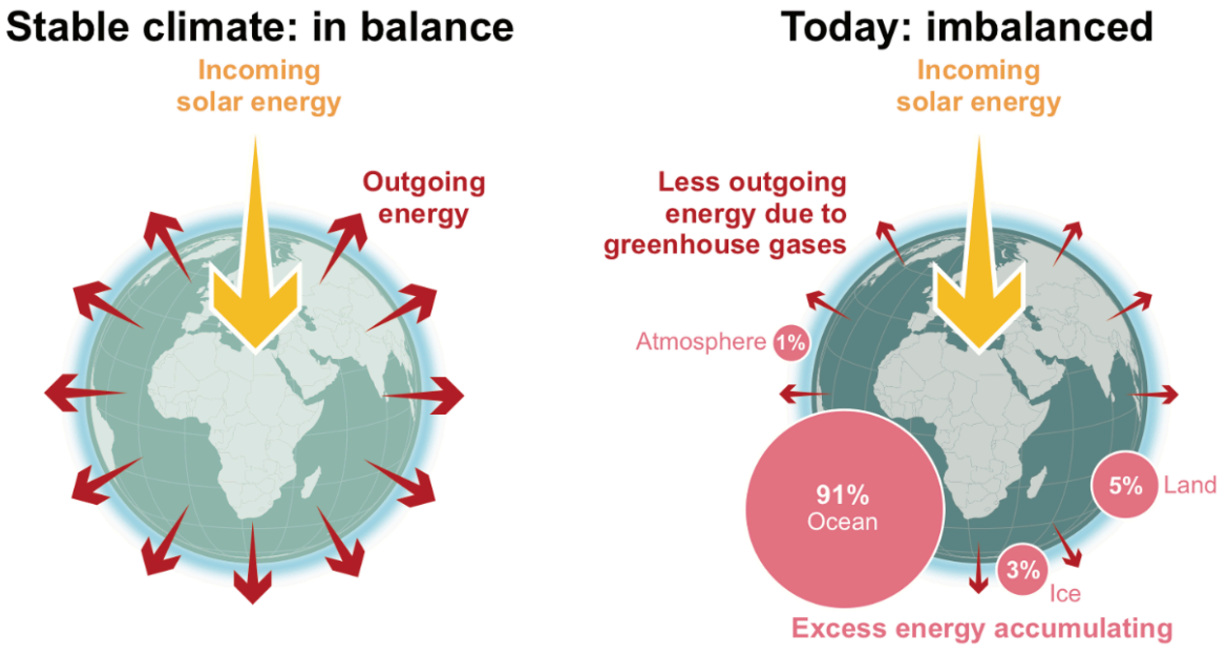


Figure A 1.4 Explaining a stable or unstable climate on Earth

Climate scientists[82] have defined four possible *scenarios for our future* based on possible amounts of radiative forcing. Which one will come to pass depends on the actions that are taken now to limit the causes of such forcing. Each scenario is based on a plausible future *pathway* regarding global emissions of greenhouse gases. The scenarios, known as *Representative Concentration Pathways*, or RCPs in Figure A1.5, specify the amount of radiative forcing (RF) in 2100 relative to 1750 and are determined by the parts per million of CO2 in the atmosphere. Thus the first physical connection is between how much CO2 (and other GHGs) we put into the atmosphere and the extent to which this disturbs the balance between the energy landing on earth in the form of sunlight and the heat leaving the earth as infrared radiation – hence radiative forcing. The final column shows the impact of each pathway in terms of temperature rise. Note: Atmospheric carbon dioxide (CO2) is around 426.78 to 432 parts per million (ppm), with recent daily and monthly readings averaging near **427 to 432 ppm** according to recent tracking data

Representative Concentration Pathway (RCP)	Forcing compared to 1750 (Wm^{-2})	Climate policy associated with scenario	CO_2 Equivalent (ppm)	Projected global average temperature increase from 1986-2005 (°C)
2.6	2.6	Mitigation	475	1.0
4.5	4.5	Stabilization	630	1.8
6.0	6.0	Stabilization	800	2.2
8.5	8.5	None	1313	3.7

Source: IPCC Fifth Assessment Report Summary for Policymakers, Figure SPM.5 and Box SPM.1.

Figure A 1.5 Four scenarios for our future

[82] IPCC Fifth Assessment Report Summary for Policymakers: IPCC, 2014: *Climate Change 2014: Synthesis Report. Contribution of Working Groups I, II and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change* [Core Writing Team, R.K. Pachauri and L.A. Meyer (eds.)]. IPCC, Geneva, Switzerland, 151 pp.
https://www.ipcc.ch/site/assets/uploads/2018/02/SYR_AR5_FINAL_full.pdf

The Representative Concentration Pathways (RCPs) shown above were developed in 2014 by the Intergovernmental Panel on Climate Change (IPCC) and are a key part of their Fifth Assessment Report (AR5). The pathways show four possible scenarios for CO2 output leading to a higher concentration in the atmosphere and hence the resultant climate forcing in watts per square metre. The Bank of England uses the explanation below in its Report on the Impact of Climate Change on the UK Insurance Sector

> The scenarios of human influence underlying the AR5 projections are known as RCPs, because they are expressed in terms of greenhouse gas *concentrations* (the result of emissions) rather than emission levels. Each RCP implies a different amount of human-driven climate change (i.e. each RCP results in a different amount of extra heat energy being stored in the Earth system because of greenhouse gas emissions). The scenarios are developed using assumptions concerning economic growth, choices of technology and land-use and reflect a wide range of possible mitigation actions.[83]

The graphic[84] (below) shows There are clear linkages between the factors we have been discussing from radiative forcing through surface temperature change, to economic and other impacts and ultimately to policy actions such as the likelihood of the so-called 'inevitable policy response' such as a high carbon tax. CO2 emissions in the atmosphere are already changing the Ph value of the oceans making them more acidic which gives ocean life survival problems. That is a direct effect of CO2 emissions. But it is the indirect effect from CO2 leading to higher Earth surface temperatures that is critical in relation to life on land.

Land-use change (LUC) is an important source of GHGs. LUC emissions account for around 11 per cent of the global total GHG emissions. The British Ecological Society has recently focussed on an area of land use change often not considered and that is changes in emissions from peat bogs (which, like coal, comes from decayed vegetation laid down over centuries). Peat is a fuel in many countries such as India and in developed countries it is used to improve the soil in gardens and in horticulture. Globally, according to Wikipedia, peat stores up to 550 gigatons of carbon, representing 42% of all soil carbon which exceeds the carbon stored in all other vegetation types, including the world's forests. This is more than 50 times global GHG emissions in 2019.

The problem that peat presents is that if peat bogs are drained as they are when peat is being harvested or a peat bog catches fire as they often do, then the CO2 is released. As growing human populations expand peat harvesting might be expected to increase and peatlands may be stripped. As global warming raises fire risk, peat bogs may dimmish substantially or disappear as a result of fire, thereby putting large volumes of stored carbon into the atmosphere.

Impact of climate change on life on Earth

The Paris Agreement (look it up) aim was to hold the increase in the global average temperature to well below 2°C The extent of temperature rise is linked to impacts on humans, animals and plants on Earth. But the impact of climate change does not scale linearly with temperature increase (some impacts scale much harder than others; every half a degree counts). A major result of the post-Paris 1.5C special report from the IPCC is an evaluation of the impacts of climate change on life on Earth at different warming levels. The text below is from the Carbon Brief website[85].

> 'At 1.5C, risks will increase for "health, livelihoods, food security, water supply, human security and economic growth", the report says. At this level of global warming, many low-elevation and small glaciers around the world would lose most of their mass or disappear, the report says with high confidence. Coral reefs are expected to decline by a further 70–90%, it adds with high confidence.
>
> At 2C, risks associated with extreme weather events will transition to "very high", the report says with medium confidence. At this level of warming, changes in food availability and diet quality could increase nutrition-related diseases and undernourishment for up to "hundreds of millions of people", particularly among low-income households in sub-Saharan Africa, south Asia and central America, the report says with high confidence.
>
> At 3C, "risks in many sectors and regions reach high or very high levels, implying widespread systemic impacts", the report says. The number of endemic species in biodiversity hotspots at a very high risk of extinction is expected to be 10 times higher than at 1.5C, it says with medium confidence.

[83] Source: University of Cambridge (2013), 'Climate Change: Action, trends and implications for business'.

[84] (NAP 2005) The National Academies Press, Radiative Forcing of Climate Change https://www.nap.edu/read/11175/chapter/1#vi

[85] https://www.carbonbrief.org/qa-ipcc-wraps-up-its-most-in-depth-assessment-of-climate-change/#warming

> At 4C and above, around half of tropical marine species could face local extinction, the report says with medium confidence. Around four billion people could face water scarcity, it says with medium confidence. It adds that the global area burned by wildfires could increase by 50-70% (medium confidence).

Climate tipping points (too risky to bet against) and Global Cascades

"Tipping point" refers to a threshold at which a tiny disturbance (sometimes jokingly described as a butterfly flapping its wings) can alter the state of, or the development of, a system.. In the case of the Earth system, a "tipping element" is something that causes a component of the Earth's climate system to pass a tipping point. As explained in an article in Nature[86]:

> If current national pledges to reduce greenhouse-gas emissions are implemented — and that's a big 'if' — they are likely to result in at least 3 °C of global warming. Some economists, assuming that climate tipping points are of very low probability (even if they would be catastrophic), have suggested that 3 °C warming is optimal from a cost–benefit perspective. However, if tipping points are looking more likely, then the 'optimal policy' recommendation of simple cost–benefit climate-economy models align with those of the recent IPCC report. In other words, warming must be limited to 1.5 °C. This requires an emergency response.
>
> In our view, the clearest emergency would be if we were approaching a global cascade of tipping points that led to a new, less habitable, 'hothouse' climate state. Interactions could happen through ocean and atmospheric circulation or through feedback that increase greenhouse-gas levels and global temperature. Alternatively, strong cloud feedback could cause a global tipping point. We argue that cascading effects might be common.

Cascading effects (multiple tipping points triggering one after another) suggests the need to protect the world from such possible climate "outliers". The normal mechanism used by society for such risks is insurance. But in this case our societal response needs to be through regulation and other mechanisms. Such mechanisms would include relative price changes between fossil fuels and clean fuels by, for example ceasing to subsidise fossil fuels and putting a tax (carbon tax) on fossil fuels. Another would be subsidising clean energy technologies until they reached the scale of manufacture at which they were highly economic in use. Although Integrated Assessment Models (IAMs) which we discuss below are widely used as a basis for policy making, they are unable to account for the *tail risks* i.e. the risks resulting from tipping points and cascades which lead to much more dramatic climate change that is envisaged in IAMs.

Integrated Assessment Models (IAMs) and IPPC AR6

General Circulation Models (GCMs) that we looked at above are driven by the fundamental physics of the atmosphere, the oceans and the Earth's land surface. They were later enhanced in so-called the Earth System Models (ESMs) which considered in greater detail, the complex interactions between the atmosphere, the oceans and the land surface. Later still, so-called Integrated Assessment Models (IAMs) were developed.

IAMs are an interdisciplinary research platform which combine a climate science module in which greenhouse gas emissions are linked to temperature increases with an economic module which links such increases in temperature to economic and policy outcomes.

The interaction of human and earth systems requires that these models analyse not only physical interactions between atmosphere, oceans and land surface but the human aspects which in this case means incorporating economic and social data. This should enable these models to project future greenhouse gas emissions and climate impacts. Most models use the Representative Concentration Pathways (RCPs) that we discussed earlier as the foundation for various temperature pathways that the world may take. As decision-making is what we are concerned with, IAM's enable decision makers to move towards the necessary requirement of understanding the potential economic impact of climate change on industries, countries and regions and corporate real assets. This then allows those in the finance industry to estimate the likely impact on the financial assets in their portfolios.

Carbon Brief have produced an excellent 'Explainer'[87] on IAMs.

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[86] https://www.nature.com/articles/d41586-019-03595-0#ref-CR4

[87] Q&A: How integrated assessment models are used to study climate change 2018 https://www.carbonbrief.org/qa-how-integrated-assessment-models-are-used-to-study-climate-change

Depending on the IAM, researchers must answer myriad questions before they can start their work. The answers to these questions are the assumptions that regulate how the model behaves. For example:

> What is the relationship between income and energy use?
>
> How does demand for private transport change over time, as incomes increase and as populations move away from rural areas towards urban centres?
>
> What can we say about average diets in relation to wealth and in particular demand for beef and other ruminant protein sources relative to previous diet and the likelihood of the populations of developed countries shifting dramatically from animal protein to other protein sources including plant and insect sources.

Researchers have worked on modelling possible changes in socioeconomic factors over time. These changes include population growth, economic growth, education levels, the rate of urbanisation and the rate of technological change. These groups of researchers have agreed on what are known as "Shared Socioeconomic Pathways" (SSPs).

Although IAMs are used by the IPCC to explore some of the relationships between society and the natural world, their modelling limitations must also be acknowledged. They use what are known as 'general equilibrium models' based on 'neoclassical' economics assumptions. Many would consider these assumptions about how capital markets function as unrealistic and in particular the use of the rational expectations hypothesis. Newer models also allow for 'cascade effects':

Models which allow for non-linearities may in future be able to provide better predictions. Such models may also, of course, demonstrate the high probability of much more severe outcomes than general equilibrium models. In addition, because of different assumptions built into different models, their outputs differ, sometimes substantially. Carbon Brief also point out that those models that suggest there is less need for urgent action on reducing carbon emissions are likely to be those that rely on a 'deus ex machina' being developed and available on a large scale, sometime in the future that will enable carbon capture and storage and direct air capture of CO2, to 'solve' the problem at some point[88]. They may also use a high discount rate which weight near-term costs much more than future costs. This means there is a tendency in such models to delay action today in favour of hoped for technology in the future particularly given the very high cost of carbon capture, transport and storage.

Although they could cause governments and regulators to limit their climate intervention, they do provide a guide and a platform to enable rational debate on how to move ahead with the additional actions urgently required to give the world a chance of minimising adverse outcomes for Earth and its human, animal and plant life. The fact of uncertainty about the severity of outcomes from greenhouse gas emissions, rather than being a reason to slow down action, is a reason to speed up inter-governmental action given the high probability of 'tail' outcomes resulting from, for example, further loss of Amazon forest or Antarctic ice sheet.

It must also be accepted that whilst climate models are based on physical laws which don't change, IAMs involve human behaviour which does change in response to incentives such as taxes, subsidies, new technologies such as plant-based foods and regulation. Socio economic models cannot easily predict human behaviour in response to price changes, diets, need for transportation etc. Indeed, some would argue that we are in a situation of "Radical Uncertainty" (see Chapter 18.6) where it is not possible to 'scientifically' calculate probabilities of damage from physical or transition risk. In such a world, they would argue that the only answer is what is known as Precautionary Policy where the focus must be on avoiding *tipping points* and building resilience into systems.

Recently, there has been greater concern in the United States about the implications of climate change – in particular on the implications of migrations from south of the globe to the north of the globe as a result of starvation in the global south. But also about a newly resurgent Russia on the back of its increasingly cultivatable land area as global warming leads to the tundra melting and increasing crop yields over larger areas of land at the same time as US crop yields are declining due to the same warming climate. In addition, Russia would benefit from an ice-free Northeast passage for ships between Europe and Asia along the north coast of Russia. More recently,

[88] Translated as "God from a Machine". Deus ex Machina is a phrase used to describe any situation where something unexpected or **implausible** is brought in to a story line to resolve a situation, in this case an unwillingness to do anything about climate change.

the high temperatures and forest fires in the north-west US and in western Canada in July 2021 and even more in the years since have made people ever more aware of the likely impact of climate change.

Limitations and assumptions of commonly used climate-change scenarios in financial services[89]

> History is replete with incidents in which we have placed too much faith in models, including the global financial crisis, the collapse of Long-term Capital Management, and, more recently, the UK pension scheme LDI crisis. A common theme is that models under-estimate the level of risk because it is mathematically tractable to do so, in some cases due to underpinning assumptions that do not hold in the real world.

Consequently, many climate models are severely underestimating the economic impact of climate change because:

> 1. Modelled scenarios do not incorporate our experience on climate change and may not be realistic when compared to experience. There is also limited consideration of higher warming scenarios.
>
> 2. There is considerable uncertainty in key climate-system modelling assumptions, including how rapidly the climate will warm for a given level of emissions, which are likely to be prudent meaning the carbon budget for limiting global warming to 1.5°C may now be exhausted. However, no margins are currently included to represent this uncertainty.
>
> 3. Damage functions that are used to estimate the economic impacts of climate change exclude many of the risks we expect to face, such as those impacts from tipping points, or societal consequences such as involuntary mass migration.
>
> 4. General equilibrium models that are widely used to estimate economic impacts contain a number of simplifying assumptions that do not hold in the real world.

We conclude that it is essential for model users to be climate literate in order to understand the context in which models are used and the limitations that apply to climate models. They should be used alongside narrative scenarios to provide direction and rank on risks – but users must not interpret results as accurate. In particular, the observed benign results for the hot-house world are deeply flawed and underestimate the impact of the risks we expect to face.

[89] The text provides excerpts from a paper (July 2023) by the Institute and Faculty of Actuaries and the University of Exeter written by Sandy Trust, Sanjay Joshi, Tim Lenton and Jack Oliver entitled *Limitations and assumptions of commonly used climate-change scenarios in financial services.* https://actuaries.org.uk/media/qeydewmk/the-emperor-s-new-climate-scenarios.pdf

Appendix 2 *Frameworks: The EU and the European Commission as Exemplars in the Creation of Sustainable Assets and Achieving the Paris Agreement*

The EU Regulatory Framework – Shaping markets, not just fixing market failures

It is certainly the case that the European Union is well ahead of other countries and regions in taking necessary measures to mitigate climate change risk. But it is possible that much more is needed in terms of phased-in regulation that prohibits more activities rather than simply trying to 'nudge' investors, industrialists, and the public, to act 'responsibly'.

In conditions of Deep Uncertainty where there are threats of serious or irreversible damage from climate change, governments and regulators need to employ the Precautionary Principle which in practice means radical regulatory action rather than market pricing alone. We strongly believe that governments (including the EU) and regulators should *shape* markets not just think in terms of fixing market *failures*. It has to be active government policy rather than simply relying on market forces through, for example, small changes such as to capital ratios for "brown" or "green" assets. The *Inevitable Policy Response* is an example of the possible likely radical action. Such a large sudden crackdown on emitting industries including farming would result in large falls in values of affected assets.

The European approach to sustainable finance and 'Paris compatibility' is still dominated by what old (and many young) bankers and others were taught in university finance classes. As a result, policy intervention has essentially been devoted to stimulating market efficiency based on what many consider to be the erroneous efficient market hypothesis (EMH) model. But market forces simply work too slowly.

If these policies are to be successful, they must help ensure that carbon emissions fall as shown in the first graphic in Figure A2.1 as a result of a transition shown in the second graphic (both from the Rocky Mountain Institute)[90]. This shows the path that emissions must take for investors to achieve 'climate aligned portfolios' i.e. what needs to be done to align investor portfolios with Paris.

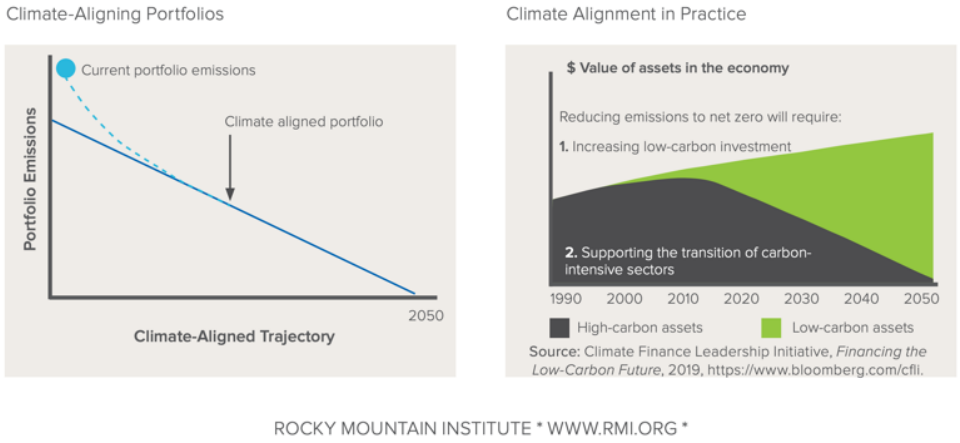

Figure A2.1 How to achieve climate aligned portfolios

The second graphic in Figure AB1 shows that the transition involves the phasing out of carbon intensive activities and the investment required to replace this through investment in low-carbon producing assets. Thus we are looking here at how to achieve Climate Aligned Finance. At present, world CO2 emissions continue to increase annually and, record temperatures have been felt across the world in summer 2022 and wildfires and other manifestations of climate change continue to cause huge losses to society.

We may now have to wait for the impact of a sudden climate crisis even greater than the heat and drought in Europe and China and other countries in 2022 to drive governments to meet their Paris commitments. It must be remembered, however, that it is not currently possible to 'reverse' what has already happened as climate impact depends on atmospheric concentrations of GHGs, not annual emissions (which just add to concentrations). It can

[90] https://rmi.org/insight/navigating-five-barriers-to-climate-aligned-finance/

take 100 years or very much longer (depending on which GHG we are considering) for the impact of GHGs emitted today, to dissipate and thus we cannot reverse what has been happening over past decades as it is unlikely that carbon removal technology (DACC) will be able to remove more than a small proportion of the existing concentration. All we can do with today's and tomorrow's technology is to stop emitting GHGs as soon as possible – much sooner than 2050, if we are to have any chance of meeting 1.5°C or even 2°C.

The EU Action Plan and the European Green Deal for Financing Sustainable Growth

The Action Plan [https://www.unpri.org/download?ac=5173] outlines ten reforms in three areas:

REORIENT CAPITAL FLOWS TOWARDS SUSTAINABLE INVESTMENT, IN ORDER TO ACHIEVE SUSTAINABLE AND INCLUSIVE GROWTH BY:

1) Establishing an EU classification system for sustainability activities
2) Creating standards and labels for green financial products
3) Fostering investment in sustainable projects
4) Incorporating sustainability when providing investment advice
5) Developing sustainability benchmarks

MAINSTREAMING SUSTAINABILITY INTO RISK MANAGEMENT

6) Better integrating sustainability in ratings and research
7) Clarifying institutional investors and asset managers' duties
8) Incorporating sustainability in prudential requirements

FOSTER TRANSPARENCY AND LONG-TERMISM IN FINANCIAL AND ECONOMIC ACTIVITY

9) Strengthening sustainability disclosure and accounting rulemaking
10) Fostering sustainable corporate governance and attenuating short-termism in capital markets

The four legislative proposals adopted in May 2018 address several areas of the Action Plan:

1. Taxonomy
A proposal for a regulation to establish a framework to facilitate sustainable investment. The proposed regulation establishes the conditions and the framework to create, over time, a unified classification system (or taxonomy) on what can be considered environmentally sustainable economic activities. This is widely seen as a first and essential enabling step in the overall effort to channel investments into sustainable activities.

2. Disclosure and duties
A proposal for a regulation on disclosures relating to sustainable investment and sustainability risks, amending Directive (EU) 2016/2341(IORP 2 - Pensions). This regulation will introduce obligations on institutional investors and asset managers to disclose how they integrate ESG factors in their risk processes. Requirements to integrate ESG factors in investment decision-making processes, as part of their duties towards investors and beneficiaries, will be further specified through delegated acts.

3. Benchmarks
A proposal for a regulation amending the benchmark regulation. The proposed amendment will create a new category of benchmarks, comprising low-carbon and positive carbon impact benchmarks, to help investors better understand the relative carbon impact of their investments.

4. Sustainability Preferences (consultation)
In addition, the Commission has solicited feedback on amendments to delegated acts under the Markets in Financial Instruments Directive (MiFID II) and the Insurance Distribution Directive to include ESG considerations into the advice that investment firms and insurance distributors offer to their clients.

The European Green Deal which was approved in 2020 is a roadmap is the much wider plan than the Action Plan for Financing, for making the EU's economy sustainable by turning climate and environmental challenges into opportunities across all policy areas and making the transition just and inclusive for all. The European Green Deal is the EUs new growth strategy. It shows how to transform ways of living and working, of producing and consuming

so that people live more healthily and make businesses innovative. As can be seen below at the bottom left, Financing the Transition is a key aspect of the Green Deal.

The Green Deal provides a roadmap with actions to boost the efficient use of resources by moving to a clean, circular economy and stop climate change, revert biodiversity loss and cut pollution. It outlines investments needed and financing tools available and explains *how to ensure a just and inclusive transition.* The Green Deal covers all sectors of the economy, notably transport, energy, agriculture, buildings, and industries such as steel, cement, ICT, textiles and chemicals.

Achieving the 2030 climate and energy targets is estimated to require €260bn of additional annual investment, representing about 1.5% of 2018 GDP. This will need the mobilisation of both the public and private sectors. The Green Deal Investment Plan will mobilise EU funding and create an enabling framework to facilitate and stimulate the public and private investments needed for the transition.

Complementing other initiatives announced under the Green Deal, the Plan is based on three dimensions: financing, enabling, and practical support. It will mobilise at least €1 trillion of sustainable investments over the next decade, provide incentives to unlock and redirect public and private investment, and provide support to public authorities and project promoters in planning, designing and executing sustainable projects. The Commission has committed to a least 25% of the EU's long-term budget being for climate action and beyond this, the European Investment Bank, Europe's climate bank, will also provide further support.

The Role of the European Investment Bank

At the end of 2020, EU member states approve EIB Group Climate Bank Roadmap 2021-2025. In summary, the key points in the roadmap are:

- Support €1 trillion in investment for climate action and environmental sustainability in the decade to 2030
- EIB Group is now aligning all financing activities from end of 2020, with goals of Paris agreement
- More than 50% of annual financing dedicated to green investment by 2025
- More Green advisory services and financing of innovative low carbon technologies
- Support for green capital markets, climate change adaptation, Just Transition projects

Announcing the Roadmap, EIB Vice President Ambroise Fayolle responsible for climate action and environment said in a speech:

> "2021-2030 is the critical decade to mobilise the trillions of investment that is required to limit global warming to 1.5 degrees and protect the environment and biodiversity. As requested by the European Council and the EU member states, I am proud to announce that the EIB Group has achieved an important milestone in its transformation to become Europe's fully fledged climate bank."

The Technical Expert Group (TEG) Report[91] *and the Green Taxonomy*

The EU has been active in a wide range of ways to help Europe achieve sustainability. 'Sustainable finance' in the view of the EU refers to the process of *taking due account of environmental and social considerations when making investment decisions, leading to increased investment in longer-term, sustainable activities.* In July 2018, the European Commission set up a Technical Expert Group on sustainable finance (TEG) to assist it in four key areas of the Action Plan through the development of the following:

Mission:
The group's tasks shall be to assist the Commission, notably in the development of:

[91] TEG [2020] European Commission Report on Climate-related Disclosures
https://ec.europa.eu/info/sites/info/files/business_economy_euro/banking_and_finance/documents/200309-sustainable-finance-teg-final-report-taxonomy_en.pdf

- an EU *taxonomy* or classification system of climate change mitigation, climate change adaptation and other environmental activities;
- an EU *Green Bond Standard*;
- *benchmarks* for low-carbon investment strategies; and
- *metrics* for climate-related disclosures.

The *taxonomy* is the key output of the TEG and is the foundation for the other aspects of the Action Plan – the Green Bond Standard (GBS), benchmarks and metrics. We cover below the Taxonomy, the Green Bond Standard and EU Paris Aligned Benchmarks and EU Climate Transition Benchmarks. Metrics have not yet been tackled by the TEG.

The EU Green Taxonomy Regulation (which resulted from the TEG Report) notes that the Taxonomy is "the most important and urgent action envisaged by the EU Action Plan" and an important aspect of the EU Green Deal package.

> Until now, there has been no common classification system at EU or global level that provides a definition of "environmentally sustainable" economic activities. The green list[92] aims to:
>
> - Reduce fragmentation resulting from market-based initiatives and national practices;
> - Reduce "greenwashing", i.e. the practice of marketing financial products as "green" or "sustainable", when in fact they do not meet basic environmental standards

There are two more criteria that a project must meet. First, the project must not do any significant harm to any of the other environmental objectives and second, it must meet the legal minimum safeguards.

Arguments have however arisen in particular over nuclear power (and also natural gas as a 'transition fuel'). Nuclear power is non-polluting and can provide 'clean' power, heat and hydrogen from electrolysis and contributes to at least two objectives. The question is whether it could cause any harm in terms of the other environmental objectives.

On this question, there are strong views on both sides. The European Council decision to support the Finnish Presidency's wording in favour of "renewable and climate-neutral energy sources" left the door open for nuclear energy to be classified as green in the new EU Sustainable Finance Taxonomy which it was not originally. Indeed, the exclusion of nuclear energy on the basis of concerns about waste management may be quite invalid since all commercial nuclear activities in the EU (including spent fuel management) are already regulated to a 'Do No Significant Harm' (DNSH) standard through the Laws, Regulations, and Procedures of the EU. On the other hand, the world's scientific consensus is that nuclear energy is necessary to achieve climate change mitigation within the necessary timescale.

In March 2021 there was some resolution of this issue in a technical assessment[93] of nuclear energy with respect to the 'do no significant harm' criterion of EU regulation:

Technical assessment of nuclear energy with respect to the '**do no significant harm' criteria** of Regulation (EU) 2020/852 ('Taxonomy Regulation') by the EU JRC[94]

Inclusion or exclusion of nuclear energy in the EU taxonomy was a debated subject throughout the negotiations on the EU Taxonomy Regulation. While there are indirect references in the regulation to the issue of nuclear

[92] Source: https://ec.europa.eu/commission/presscorner/detail/en/ip_19_6793

[93] Source: World Nuclear News 29 March 2021 https://www.world-nuclear-news.org/Articles/EU-Taxonomy-can-move-forward-with-nuclear-JRC-find

[94] The Joint Research Centre (JRC) is the European Commission's science and knowledge service. It aims to provide evidence-based scientific support to the European policymaking process. The text in this section is from a 387 page *JRC Science for Policy Report* published by the European Commission in 2021 under the title of the Chapter heading above. Available at https://publications.jrc.ec.europa.eu/repository/handle/JRC125953

energy (including on radioactive waste), co-legislators ultimately left the assessment of nuclear energy to the Commission as part of its work on the delegated acts establishing the technical screening criteria.

As the in-house science and knowledge service of the Commission with extensive technical expertise on nuclear energy and technology, the JRC was invited to carry out such analysis and to draft a technical assessment report on the 'do no significant harm' (DNSH) aspects of nuclear energy including aspects related to the long-term management of high-level radioactive waste and spent nuclear fuel, consistent with the specifications of Articles 17 and 19 of the Taxonomy Regulation. This report is the result of that JRC analysis.

- The detailed assessment of the impacts of nuclear energy in its various lifecycle phases shows that all non-radiological effects and potential impact indicators are dominated by the mining & milling phase, except the greenhouse gas (GHG) emissions, where Nuclear Power Plant (NPP) operation gives the largest contribution (see Figure 3.3.1-12 of Part A and Tables A.2-1 and A.2-2 in Annex 2).

- The analyses did not reveal any science-based evidence that nuclear energy does more harm to human health or to the environment than other electricity production technologies already included in the Taxonomy as activities supporting climate change mitigation.

- The comparison of impacts of various electricity generation technologies (e.g. oil, gas, renewables and nuclear energy) on human health and the environment, based on recent Life Cycle Analyses (LCA) presented in Chapter 3.2 of Part A, shows that the impacts of nuclear energy are mostly comparable with hydropower and the renewables, with regard to non-radiological effects.

- Management of radioactive waste and its safe and secure disposal is a necessary step in the lifecycle of all applications of nuclear science and technology (nuclear energy, research, industry, education, medical, and other). Radioactive waste is therefore generated in practically every country, the largest contribution coming from the nuclear energy lifecycle in countries operating nuclear power plants. Presently, there is broad scientific and technical consensus that disposal of high-level, long-lived radioactive waste in deep geologic formations is, at the state of today's knowledge, considered as an appropriate and safe means of isolating it from the biosphere for very long time scales

- Measures to ensure that radioactive waste does not harm the public and the environment include a combination of technical solutions and an appropriate administrative, legal and regulatory framework. Although there remain contrasting views, it is generally acknowledged, that the necessary technologies for geological disposal are now available and can be deployed when public and political conditions are favourable. No long-term operational experience is presently available as technologies and solutions are still in demonstration and testing phase moving towards the first stage of operational implementation. Finland, Sweden and France are in an advanced stage of implementation of their national deep geological disposal facilities, which are expected to start operation within the present decade.

The comparison of environmental impacts of various electricity generation technologies on human health and the environment, leads to the following main findings:

— Average lifecycle GHG emissions determined for electricity production from nuclear energy are comparable to the values characteristic to hydropower and wind;

— Nuclear energy has very low NOx (nitrous oxides), SO2 (sulphur dioxide), PM (particulate matter) and NMVOC (non-methane volatile organic compounds) emissions. The values are comparable to or better than the corresponding emissions from the solar PV and wind energy chains (see Figure;

— Land occupation of nuclear energy generation is about the same as for an equivalent capacity gas-fired power plant, but significantly smaller than wind or solar PV.

Resolution of Taxonomy issues relating to natural gas and nuclear, 01 January 2022
On 01 January 2022, the European Commission issued a press release headed *EU Taxonomy: Commission begins expert consultations on Complementary Delegated Act covering certain nuclear and gas activities:*

> Taking account of scientific advice and current technological progress, as well as varying transition challenges across Member States, the Commission considers there is a role for natural gas and nuclear as a means to facilitate the transition towards a predominantly renewable-based future. Within the Taxonomy framework, this would mean classifying these energy sources under clear and tight conditions (for example, gas must come from renewable sources or have low emissions by 2035), in particular as they contribute to the transition to climate neutrality.

The Commission notes that "The activities covered in this complementary Delegated Act would accelerate the phase out of more harmful sources, such as coal, and in moving us towards a lower-carbon energy mix."

The EU Taxonomy or "unified classification system" *bridges the gap between international goals and investment practice and signals the types of activities that are consistent with the low-carbon transition, adaptation and other environmental objectives*. It puts environmental data in context and helps investors understand which companies are contributing to the low-carbon transition. In the security selection process undertaken by investors, the Taxonomy should enable them to identify companies whose carbon footprints and, very importantly, *transition measures* are actively contributing to environmental sustainability and are compatible with EU objectives. The EU's performance standards under the Taxonomy, or technical screening criteria as they are known, enable identification of an eligible economic activity which truly sustainable and thus "taxonomy-aligned".

Reform of the EU Electricity Market Design

2019 – Clean Energy Package for All Europeans [https://energy.ec.europa.eu/topics/energy-strategy/clean-energy-all-europeans-package_en]

In May 2019, the Council of ministers of the EU formally adopted four new pieces of EU legislation that redesign the EU electricity market to make it fit for the future. This concludes the remaining elements of the Clean energy for all Europeans package and represents a major step towards completing the Energy Union, delivering on the priorities of the Juncker Commission. The four new pieces of legislation are in the areas of Efficient Energy Use, More Renewables, Better Governance of the Energy Union, More Rights for Consumers and a Smarter and More Efficient Energy Market.

The EU Green Bond Standard[95]

The ICMA Green Bond Principles (ICMA GBP) which we covered above were produced well before the TEG issued its report on the question of 'what is green'. Under these Principles (ICMA GBP), the cornerstone is that the proceeds of a Green Bond issue can only be for Green Projects, which should be appropriately described in the legal documentation for the security. All designated Green Projects should provide clear environmental benefits, which will be assessed and, where feasible, quantified by the issuer.

One objective of the EU Green Taxonomy was to provide definitions of what is a Green Project and what is not – something that has been lacking up to now. The benefit of having the EU Green Bond Standard (EUGBS) is to help achieve the aims of the Action Plan to reorient capital flows towards sustainable investment. EU Green Bonds are a capital market instrument that is fully aligned with that aim. The EU GBS should not only aid in overcoming the biggest barriers to further grow this market but also create a strong foundation to accelerate the flow of capital towards the EU's environmental objectives. Indeed the TEG Report goes on to say that "green bonds have provided bond investors with the capacity to become involved in corporate environmental strategies. It has also enabled bond markets to become a powerful force in green finance, notably including climate mitigation finance."
Once the taxonomy had been developed by ICMA's Principles Community it became much easier to propose an EU Green Bond Standard, building on ICMA's existing Green Bond Principles. It is therefore an example of an industry association leading by developing standards but, given that it cannot impose these on its Members nor create a taxonomy, it is appropriate for a governmental body to enhance the standards and at some point to make them mandatory for a 'Green Bond'. While currently voluntary, (like TCFD), it is not likely to remain voluntary for any entity wishing to issue a Green Bond.

[95] EU Technical Expert Group (TEG) on Sustainable Finance: *Usability Guide: EU Green Bond Standard* published in March 2020 https://ec.europa.eu/info/sites/info/files/business_economy_euro/banking_and_finance/documents/200309-sustainable-finance-teg-green-bond-standard-usability-guide_en.pdf

The new EUGBS can be applied by any issuer in any country including those outside of the EU. There are four key requirements under the framework:

1. The funds raised by the bond should be allocated fully to projects aligned with the EU Taxonomy;
2. There must be full transparency on how bond proceeds are allocated through detailed reporting requirements;
3. All EU green bonds must be checked by an external reviewer to ensure compliance with the Regulation and that funded projects are aligned with the Taxonomy. Specific, limited flexibility is foreseen here for sovereign issuers;
4. External reviewers providing services to issuers of EU green bonds must be registered with and supervised by the European Securities Markets Authority. This will ensure the quality and reliability of their services and reviews to protect investors and ensure market integrity. Specific, limited flexibility is foreseen here for sovereign issuers

The EU/EC as an Exemplar on Sustainability (including the ETS)

The EU has been ahead of central banks and the BIS in taking action on disclosure of 'non-financial' information. But as of the present date, the TCFD framework is essentially the one that it has adopted for future disclosure, at least on climate change. A key early document (and accompanying legislation) was the "*Single Market Act — Twelve levers to boost growth and strengthen confidence*", published by the Commission in 2011 which identified the need to improve financial and non-financial companies' *disclosure of social and environmental information*, by presenting a legislative proposal in this field. Thus the European Commission sees a positive connection between non-financial information disclosure and environmentally sound growth.

The European Parliament also acknowledged the importance of businesses divulging information on sustainability such as social and environmental matters with a view to identifying sustainability risks and increasing both institutional investor and retail consumer trust. Indeed, the European Parliament acknowledged that the disclosure of non-financial (sustainability) information was vital for managing the transition towards a sustainable global economy by combining long-term profitability with social justice and environmental protection – the profits, people, planet (PPP) perspective.

The concept of each country setting its own Nationally Determined Contribution (NDC) to the goals of the Paris Agreement is a key starting point for understanding the role of the finance industry and its corporate customers in enabling a government to achieve its Paris Agreement Commitments. It is based on the fact that at the present time, the countries of the world combined are emitting 55bn tons (CO2e) of GHG each year resulting in the concentration of long-lived GHGs in the atmosphere increasing year by year. In order to achieve the 1.5°C or well under than 2°C degree target to which all governments have committed, there is a need to achieve 'net zero' emissions and hence no increase in concentration. There is a limited number of years available to reach net zero emissions if the world chooses to limit the temperature rise to 'well under 2 °C'. At a maximum, this is 30 years i.e. by 2050.

Transitioning the financial system in financial firms to maximise the likelihood of a country's financial system being climate aligned requires understanding the current financial market structure and how it must change towards meeting society's new needs. Given that the world in 2020 was estimated to be 1.25 – 1.4 °C warmer than in pre-industrial times, there is only a small, *remaining carbon budget.* To meet the goal of limiting global warming to 1.5°C, the whole world now needs to reduce emissions by 7% per year every single year in the next decade between 2020 and 2030! However, that 7% is a global average. Some countries would need to de-carbonise at perhaps twice that rate yet at present global emissions are still rising.

Earlier we looked at why it is critical to the success of the Paris Agreement process that the public sector and the private sector (market approach) work together to achieve the Paris goals. Figure A2.3 from the Paris-based *Institute for Climate Economics* (IC4E)[96] is a useful summary of the main reasons why public intervention is necessary on the corporate front (numbers 1,2 & 3 below) and why the same public intervention is necessary in the case of the financial sector (numbers 4,5 & 6, below):

[96] https://www.i4ce.org/home/

Figure A2.2 The limitations of the market approach to green its activities

The European Commission has seen the need for legislation and regulation to minimise climate damage and sees the finance industry as the key player in achieving its goal of having no net GHG emissions across the EU by 2050 and decoupling economic growth from resource use. It sees the finance industry as being key because of its linkages to the highly polluting sectors of the economy i.e. the lending, investing, capital raising and insuring services that facilitate the growth of an economy. This viewpoint would also apply to most other governmental and regulatory agencies worldwide.

As the finance sector currently facilitates the growth of high GHG emitting companies ('financed emissions' when a bank itself provides financing or 'facilitated emissions' when a bank or investment bank underwrites a bond issue), the question the Commission always asks, as a representative of society's concerns in this area, is how it can put pressure on financiers, and thereby on the highly GHG-emitting corporate sectors as well as those sectors that are destroying natural capital, in order to achieve its 2050 goal. The Commission cannot overcome all the limitations in the graphic above, but its regulatory intervention can certainly contribute. While there are many other issues that are pertinent to its objectives, it believes that, as shown below, a combination of the first five points plus (No. 6) can make a substantial impact:

1. a higher level of carbon tax or a revised cap and trade system
2. improvement in non-financial data disclosure (ESG disclosures);
3. providing incentives for the finance industry to provide additional financing for green projects;
4. the provision of *risk mitigation services* by the European Investment Bank;
5. regulations on infrastructure project design, construction and operation;
6. self-regulation for example through trade associations such as ICMA and through stock exchanges

No.1 above refers to the EU Emissions Trading System (ETS) which we cover below:

If we look now at the 'effective carbon tax rate' in the EU, it has been much lower than the actual rate (US$ 126 per metric ton) in Sweden or the IMFs estimate of the rate necessary to achieve the desired objective. The level of the tax in the EU is determined through the EU emissions trading system (EU ETS) which they call "a cornerstone of the EU's policy to combat climate change and it is a key tool for reducing greenhouse gas emissions cost-effectively". It is the world's first major carbon market and remains the largest. For some years the price was very low and indeed it declined to around €5 as recently as 2016. It then rose to nearly €30 a decade ago before falling again to below €20 in March and April 2020 as economic activity fell across the union as a result of the coronavirus outbreak. By January 2021 it had risen again to over €30 for the first time in over a decade and hit nearly EUR 100 twice in 2022 and then again in February 2023.

It is not a straightforward carbon tax but a system whereby companies are given EU Emission Allowances (EUAs). Some companies have allowances they don't need and others have a need for more allowances. Companies are then permitted to trade allowances.

In Germany the majority of MPs agreed in 2020 to setting the price for one tonne of CO_2 at 25 euros as of 1 January 2021 for companies that put petrol, diesel, heating oil and natural gas into circulation. Unlike in the past, they will now have to buy emission allowances. The price will increase to 55 euros per tonne by 2025 resulting in a litre of petrol becoming 7 cents more expensive and natural gas 0.6 cents per kilowatt-hour. China also recently introduced a carbon trading system.

We cover the EU Corporate Sustainability Reporting Directive (CSRD) and European Sustainability Reporting Standards (ESRS) in Chapter 7.2

EU Corporate Sustainability Due Diligence Directive [97] *(CSDDD)*

Undertaking "due diligence" means making enquiries in respect of a company to uncover material information in a careful manner by putting in the "requisite effort" to be thorough in that investigation.

In December 2022, the Council and the European Parliament today reached a *provisional* deal on the corporate sustainability due diligence directive (CSDDD), which aims to enhance the protection of the environment and human rights in the EU and globally. The due diligence directive will set obligations for large companies regarding actual and potential adverse impacts on human rights and the environment, with respect to their own operations, those of their subsidiaries, and those carried out by their business partners.

In February 2022, the European Commission issued draft regulations on both human rights and environmental due diligence. Requiring due diligence is an attempt to ensure that in these two important areas for achieving sustainability, directors of companies do not simply cast a 'blind eye' at what is happening in their company and in this case also in their whole value chain. This proposed Directive is part of an attempt by the Commission to improve corporate governance in respect of sustainability. EU companies and, importantly, non-EU companies doing more than a minimum amount of business in the EU, will be required to assess both environmental impacts and human rights matters not only in their own operations but equally in their supply chains. They will be required to take action to either prevent, or if not to mitigate, any environmental or human rights harms. In respect of both the social and the environmental category, this is potentially a huge step-up in requirements and is likely to lead to more civil cases of the type we considered above. The damages involved could be considerable and are thus highly relevant to financial firms which are considered to have provided the supply chain input (cash) to enable 'financed emissions' or their underwriting skills to provide 'facilitation of bond issues'

The need for this Directive seems to have arisen from the fact that voluntary approaches in both areas – environmental and human rights (E&S) have failed to stop abuses or to limit GHG emissions. The Directive complements NFRD (and thus CSRD when it comes into effect). It also complements the Taxonomy. There will thus be a substantive requirement to perform due diligence to identify, prevent, mitigate and account for external harm resulting from adverse human rights and environmental impacts in a company's own operations, its subsidiaries and their value chains.

In December 2022, the Council and the European Parliament reached a *provisional* deal on the corporate sustainability due diligence directive (CSDDD), which aims to enhance the protection of the environment and human rights in the EU and globally. The due diligence directive will set obligations for large companies regarding actual and potential adverse impacts on human rights and the environment, with respect to their own operations, those of their subsidiaries, and those carried out by their business partners.

CSDDD will enhance information flow by mandating companies to set up additional risk management systems to those which most already have in place to report on their findings in the areas relevant to this new due diligence system whereas up until now or at least when CSDDD is implemented, due diligence systems have been optional and are little more than a by-product from an obligation (i.e. not easily enforceable) to report under the NFRD.

[97] Much of the information in this chapter is based on the European Commission document on this proposed Directive. Available at https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52022PC0071

The Directive could make litigation against corporates and possibly also financial firms more likely to succeed than in the past. According to Lara Wolters, a Dutch MEP

> "This new law on corporate due diligence will set the standard for responsible business conduct in Europe and beyond. We refuse to accept that deforestation or forced labour are part of global supply chains. Companies will have to avoid and address harm done to people and planet in their supply chains. The new rules will give victims a legal right to access support and to seek reparations, and will ensure fairness, a level playing field and legal clarity for all businesses, workers and consumers".

The purpose of this new directive is to improve environmental and social governance by making the absence of effective due diligence, either on environmental or social issues, grounds for a claim for reparation. The word 'reparations' is derived from the word 'repair, i.e. to return something to how it was in the past before it was damaged. In the case of climate the problem all that can be done is to try to compensate for damage to property

The Directive is designed to help deliver the Green Deal which requires the EU to:

- reach net-zero greenhouse-gas (GHG) emissions by 2050;
- decouple economic growth from resource use;
- leave no person and no place behind

However, achieving this will require a rapid doubling or tripling of the flow of capital into existing types of sustainable investment and new innovative energy technologies. The OECD estimates that further investments of $630 billion a year worldwide for the next decade will be needed to have even a 66% chance of limiting the temperature increase on the earth's surface to below 2 degrees.

The Directive is also designed to help achieve the *United Nations Sustainable Development Goals (UN SDGs)* which we cover in Appendix A6.1 in both the environmental and social fields in the European Union. Achieving these objectives:

> "requires implementing comprehensive mitigation processes for adverse human rights and environmental impacts in their value chains;
>
> integrating sustainability into corporate governance and management systems, and
>
> framing business decisions in terms of human rights, climate and environmental impact, as well as in terms of the company's resilience in the longer term".

The Commission notes that "Identifying these adverse impacts in value chains will become easier if more companies exercise due diligence and thus more data is available on human rights and environmental adverse impacts." This is in line with the objective of trying to ensure that companies do take the required effort to undertake due diligence not only on their own operations but those of suppliers and, we presume, users of their products.

A key reason for introducing this Directive is that attempts to achieve the same outcome by voluntary means have been unsuccessful as noted here: "Voluntary action does not appear to have resulted in large scale improvement across sectors and, as a consequence, negative externalities from EU production and consumption are being observed both inside and outside the Union. Certain EU companies have been associated with adverse human rights and environmental impacts, including in their value chains. Adverse impacts include, in particular, human rights issues such as forced labour, child labour, inadequate workplace health and safety, exploitation of workers, and environmental impacts such as greenhouse gas emissions, pollution, or biodiversity loss and ecosystem degradation."

The Commission believes that the Directive will:

> (1) improve corporate governance practices to better integrate risk management and mitigation processes of human rights and environmental risks and impacts, including those stemming from value chains, into corporate strategies;

2) avoid fragmentation of due diligence requirements in the single market and create legal certainty for businesses and stakeholders as regards expected behaviour and liability;

(3) increase corporate accountability for adverse impacts, and ensure coherence for companies regarding obligations under existing and proposed EU initiatives on responsible business conduct;

(4) improve access to remedies for those affected by adverse human rights and environmental impacts of corporate behaviour. This latter objective on access to remedies is one that may have considerable.

"The Directive introduces directors' duties to set-up and oversee the implementation of due diligence and to integrate it into the corporate strategy. In addition, when fulfilling their duty to act in the best interest of the company, directors must take into account the consequences of their decisions for sustainability matters, including human rights, climate change and environmental consequences. In general, Member States shall ensure that national provisions on breach of directors' duties also apply to these provisions, *e.g. by adding personal liability of directors for non-compliance with sustainability.* Furthermore, Group 1 companies (large companies as defined in this Proposal) need to adopt a plan to ensure that their business strategy is compatible with limiting global warming to 1.5 °C in line with the Paris Agreement." Below is a case study of a company which according to an NGO, failed to properly manage climate risk which means that it is breaching its legal duties

Case study: ClientEarth v Shell
ClientEarth is an NGO which take up cases and defines itself as a non-profit organisation that uses the law to create systemic change that protects the Earth for – and with – its inhabitants. Its approach in the Shell case is somewhat unexpected. It has a small ownership of Shell shares and is taking action against Shell as a shareholder on the grounds that its behaviour is harming shareholders in general. It alleges that Shell's directors are mismanaging the risks of climate change through an inadequate plan for the energy transition and that this leaves Shell's long-term commercial viability and competitiveness in danger due to the UK government's Net Zero Strategy.
ClientEarth's[98] lawyer argues that the Board's failure to properly manage climate risk means that it is breaching its legal duties. The Board has failed to adopt and implement a climate strategy that truly aligns with the Paris Agreement goal to keep global temperature rises to below 1.5°C by 2050. We believe the Board is breaching its duties under sections 172 and 174 of the UK Companies Act, which legally requires it to act in a way that promotes the company's success, and to exercise reasonable care, skill and diligence.

This will be the first time ever that a company's board has been challenged on its failure to properly prepare for the net zero transition.

ClientEarth is pursuing shareholder litigation to compel Shell's Board to act in the best long-term interests of the company by strengthening Shell's climate plans. Its current strategy and insufficient targets put the enduring commercial success of the company and employees' jobs at risk and is not good for people or the planet. We want to make sure the Board's 'wait and see' approach to the energy transition does not come at the expense of long-term viability for the company's stakeholders, including shareholders and employees. Putting in place sufficient targets to reduce its emissions over the next 3, 5 and 10 years in order to meet net zero will secure the company's long-term value, while protecting investors' capital and the climate.

It is taking 'derivative action' against Shell in the UK. A derivative action is a claim brought by a shareholder of the company, ultimately on behalf of the company, in this case to argue alleged breaches of duty by the Board. That means the shareholder bringing the claim is effectively seeking to step into the company's shoes, to pursue the Board for wrongs allegedly committed against the company.

Institutional investors are increasingly calling on all companies to align their business plans with the goals of the Paris Agreement. ClientEarth believes Shell is not responding to this objective of shareholders. As an example of this, in 2021, Shell was ordered by a Dutch Court to reduce its overall emissions – including those from the fossil fuel products it sells – by net 45% by the end of 2030. However, its Board has since rebuffed parts of the verdict, indicating that it is unreasonable and essentially incompatible with its business. The company has appealed the Court's decision.

Scientists say that to avoid the worst impacts of climate change, we need to halve the world's greenhouse gas emissions by 2030. Currently more than 70% of the world's emissions come from fossil fuels, and fossil fuel producers must focus on cutting their production every year if we are to avoid the worst climate impacts. On the other hand, Shell publicly maintains that its strategy is consistent with the 1.5°C temperature goal of the Paris

[98] This section based on material on ClientEarth's website at https://www.clientearth.org/latest/latest-updates/news/we-re-taking-legal-action-against-shell-s-board-for-mismanaging-climate-risk/

Agreement and has set a target to become a net zero emission energy business by 2050, in step with society. However, its interim targets to get there seem quite inadequate and have indeed by challenged in court as we note in the case study above.

The Just Transition – Inter-generational and North/ South issues: who bears the cost?

Simply applying a tax to all uses of carbon by consumers such as on gas heating, petrol, beef etc. would have a disproportionate effect on the cost of living of many people, increasing it beyond the price rises already seen in 2022 which have impacted those on low incomes much more than those on high incomes. Clearly a carbon tax would increase the financial stress on those on lower incomes. Equally the ensuing transition from fossil energy to clean energy would hit the employment prospects of many of those working in, for example, the coal mining industry and some types of farming. Not only individuals would be affected but regions of Europe and particular towns which have focussed on industries based on fossil fuel.

An example of a previous industrial transition can be seen in the City of Glasgow in Scotland (the City in which the main author of this book was born). When he was growing up, the choice of employment was mainly in the iron and steel companies, coal mining, shipbuilding and dock work on the River Clyde. Long before COP 26 in Glasgow in 2021, all those industries had disappeared. They were eventually replaced after decades of transition by jobs in the service industries such as investment and commercial bank operations centres, call centres, research and development and consulting. Service industries now employ around 800,000 people while manufacturing employs only 52,000. That transition to a restructured local economy was much too slow. The objective of the EU Just Transition is to ensure that in future, those individuals, towns and regions impacted by the energy transition should not suffer the long drawn out transition seen in previous eras in cities such as Glasgow. This was a city built on coal, steel and shipbuilding. Now there are none of these industries but over many decades, the city has managed to transition to a service industry and SME city.

Many of those involved in the debate on carbon taxes have emphasised the need for a 'Just Transition'[99] that is one that does not result in low-income groups regions or towns, suffering disproportionately from a rise in their costs and a potential loss of income, due to measures taken to mitigate climate change. Such mitigation measures would include re-training, investment grants, expansion of universities etc.

Imposition of a carbon tax which hit all consumers without corresponding measures to mitigate its impact on individuals and communities could lead to an outcome such as the 'gilet jaune' protests in France when the President of France initiated higher taxes on fossil fuels on the grounds of it being one of the potentially most effective climate-change measure. The protestors said that it hit lower income people and those living in the countryside, disproportionately.

The European Union is a region that is very focussed on the need for a just transition as without measures to help those disadvantaged by the transition, social solidarity could be severely weakened. Frans Timmermans, Executive Vice-President of the European Commission has said "We must show solidarity with the most affected regions in Europe, such as coal mining regions and others, to make sure the Green Deal gets everyone's full support and has a chance to become a reality." The Just Transition Mechanism (JTM) in the European Union is a key tool to ensure that the transition towards a climate-neutral economy happens in a fair way, leaving no one behind. It provides targeted support to help mobilise at least €150 billion over the period 2021-2027 in the most affected regions, to alleviate the socio-economic impact of the transition.

Inter-generational Issues

The Just Transition is also relevant to inter-generational equity (fairness) within a country. The question is whether it is fair, and hence in line with the Just Transition, or not, to put more than a 'fair' share of the burden on to future generations than on the current generation? In 2021, Germany's highest court agreed that it was unfair to push so much of the burden on to young people in what may become a landmark case. Among the plaintiffs in court was Sophie Backsen, 22, daughter of a farming family on the North Sea island of Pellworm, who fears that rising sea levels would engulf her low-lying island, leaving her with no inheritance. In the case, the judge ruled that it

[99] See, for example, https://climatejusticealliance.org/just-transition/ and https://www.gov.scot/groups/just-transition-commission/

would be wrong to use up large parts of the remaining CO2 budget by requiring only small reductions now if that simultaneously meant that following generations were left with a radical reduction burden and their lives are exposed to extensive limits to freedom.

The North South Divide

The argument here is not about fairness (Climate Justice) between different regions or cities in one country or different age groups in a country but between countries. In particular it's about which countries have been responsible for emitting GHGs in the largest amounts since the industrial revolution which set off the worldwide use of fossil fuels. The usual date for the starting point is taken to be 1750. Clearly, a large country will have much greater cumulative emissions than a small country. But as shown in the graphic below, two countries of approximately the same size, India and China, have very different total emissions. They also have very different annual emissions as China is more heavily industrialised than India.

The problem arises from the fact that all countries in the world suffer from these cumulative emissions and thus developing countries with low cumulative emissions (and annual emissions per capita) suffer as much, and possibly more, than developed countries. Some will even disappear under the sea. The graphic below would give a rough indication of the 'fair' contributions to a global climate adaptation fund if all major historical emitters contributed on the basis of their cumulative emissions to date.

Sustainable Finance Disclosure Regulation (SFDR): Financial firms

The extent of 'greenwashing' in the finance industry meaning, for example, that on closer analysis 'green funds' prove not to be green, is a matter of considerable concern to the European Commission. To try to limit the extent of greenwashing and to provide a high degree of comparability.

SFDR is one of the mechanisms whereby the European Commission is attempting to reorient capital towards sustainable businesses. It is designed to try to ensure that those providing capital to corporates are financing them in a sustainable manner and avoiding 'greenwashing:' It also requires managers to explain how they take adverse impacts into account. The regulation is very focussed on adverse sustainability factors. This refers not just to climate issues but social and employee issues which the Commission considers to be material to society as a whole.

EIOPA: European Insurance and Occupational Pensions Authority

EIOPA is an independent advisory body to the European Commission, the European Parliament and the Council of the European Union. It is one of the EU agencies carrying out specific legal, technical or scientific tasks and giving evidence-based advice. In this way, it helps shape informed policies and laws at EU and national levels. EIOPA is one of three European Supervisory Authorities. The other two are the European Banking Authority (EBA) and the European Securities and Markets Authority (ESMA).

EIOPA published its 'Opinion'[100] on the supervision of the use of climate change risk scenarios in ORSA (Own Risk and Solvency Assessment). The Opinion sets out supervisory expectations on the integration of the use of climate change scenarios by insurance undertakings in their ORSA. Given that undertakings will be impacted by climate change-related physical and transition risks, EIOPA believes it is important to encourage a forward-looking management of these risks, also in the long term. Currently, only a small minority of undertakings assess climate change risk using scenario analysis in the ORSA. Moreover, where undertakings perform a quantitative analysis of climate change risk, most assessments take a short-term perspective. The graphic[101] below from KPMG (Holland) indicates the areas that EIOPA has focussed on in 2022.

[100] https://www.eiopa.europa.eu/document-library/other-documents/application-guidance-climate-change-materiality-assessments-and_en

[101] https://home.kpmg/nl/en/blogs/home/posts/2021/10/eiopa-announces-key-focus-on-esg-in-2022-and-beyond.html

The EU Innovation Fund

The EU is concerned not only with regulation but with providing funding for projects that otherwise might not proceed or would proceed more slowly. With revenue of more than €38 billon until 2030 from the EU Emissions Trading System (ETS), the Innovation Fund aims to create the right financial incentives for companies and public authorities to invest in the next generation of low-carbon technologies and give EU companies a first-mover advantage to become global technology leaders. The text below is from an EU press release[102].

> Today, the EU is investing over €1.8 billion in 17 large-scale innovative clean-tech projects with a third round of awards under the Innovation Fund. Grants will be disbursed from the Innovation Fund to help bring breakthrough technologies to the market in energy-intensive industries, hydrogen, renewable energy, carbon capture and storage infrastructure, and manufacturing of key components for energy storage and renewables. The selected projects are located in Bulgaria, Finland, France, Germany, Iceland, the Netherlands, Norway, Poland and Sweden.
>
> Executive Vice-President Frans Timmermans said: "Today's grants support innovative businesses across Europe to develop the cutting-edge technologies we need to drive the green transition. The Innovation Fund is an important tool to scale up innovations in renewable hydrogen and other solutions for European industry. Compared to the first disbursement round, the funds available have increased by 60%, enabling us to double the number of projects supported. This is a big boost for the decarbonisation of energy-intensive industry in the European Union."
>
> The selected projects cover a wide range of sectors contributing to the EU's decarbonisation efforts such as production, distribution and use of green hydrogen, waste-to-hydrogen, offshore wind, manufacturing of photovoltaic (PV) modules, battery storage and recycling, carbon capture and storage, sustainable aviation fuels, and advanced biofuels. Together, they have a potential to save 136 million tonnes of CO2eq over their first 10 years of operation.
>
> In addition, up to 20 projects that are promising but not yet sufficiently mature for a grant will be pre-selected for project development assistance by the European Investment Bank. These will be announced in the fourth quarter of 2022.

A New EIB Risk sharing Initiative[103]

The European Investment Bank (EIB) is launching an initiative to support wind turbine parts manufacturers in Europe. It is doing so by extending a EUR 500m counter-guarantee to Deutsche Bank AG in Germany. According to the announcement, this will allow Deutsche Bank to create a portfolio of up to EUR 1bn in bank guarantees for new investments in wind farms in the EU.

The agreement is the first under the EIB's December 2023 plan to make EUR 5bn available to strengthen the provision of commercial bank guarantees to companies across the European wind industry. The initiative is part of an EU wind energy package presented by the European Commission in October 2023 to maintain a healthy and competitive wind energy supply chain across the Union. It includes counter-guarantees to the sector's main lenders, including Deutsche Bank.

"A rapid build-up of wind energy is essential for the decarbonization of the economy," Nicola Beer, EIB vice president, said in the announcement and continued:

> "Together with Deutsche Bank, we are promoting the deployment of renewable energy in Europe, bringing down the price of sustainable energy. This project will also help secure and create jobs in a sustainable and competitive European industry."

The EIB counter-guarantees under the EUR 5bn initiative are designed to share the credit risks commercial banks face when dealing with wind industry players. It will facilitate access to prepayment and performance guarantees, benefiting both banks and the industry. In this first project, Deutsche Bank will use the EIB's backing to provide bank guarantees to wind energy producers. This will allow manufacturers to receive advance payments and provide performance guarantees when taking on new wind projects.

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[102] Press Release. European Commission 12 July 2022 https://ec.europa.eu/commission/presscorner/detail/en/IP_22_4402

[103] This is an article published in EnergyWatch on 2 August 2024. EnergyWatch is a subscription publication. https://energywatch.com/EnergyNews/Renewables/article17321467.ece

The guarantee scheme also allows manufacturers to pay their suppliers in advance for the delivery of wind farms and the associated components of the wind value chain, which includes turbines, grid connection infrastructure, cables and substations. It is estimated that the EUR 1bn EIB-Deutsche Bank guarantee facility will trigger private investment of up to EUR 8bn.

Carbon Border Adjustment Mechanism (CBAM)

Some countries have reported that they have reduced their carbon emissions and claimed that this shows the success of their regulatory system. However, in practice, what has happened in many instances is that countries outside Europe have taken market share from European countries in industries with high emission levels. The CO2 emissions have simply moved from a European country to another country outside Europe. If European legislation then requires European companies to make large reductions in their CO2 emissions or pay a high carbon tax, but countries outside Europe do not, then European companies are being disadvantaged.

CBAM is designed to prevent "carbon leakage" and ensure that non-EU producers pay a comparable carbon price to EU manufacturers. The particular sectors which are carbon-intensive are steel, aluminium, cement, fertilisers, electricity, and hydrogen. Companies importing such products to Europe will have to purchase CBAM certificates based on the emissions "embedded" in their products.

EU Omnibus Simplification Package

In February 2025, the Commission published the EU Omnibus Simplification Package which scales back or delays key sustainability reporting legislation. At the start of its new five-year term the Commission focussed on EU industrial competitiveness a top priority. As a result, in November 2024, EU leaders adopted the Budapest declaration, which lays out a new European Competitiveness Deal (see under the Clean Industrial Deal below). The Omnibus proposal seeks to reduce reporting burdens by 25% for large companies and 35% for SMEs.

The Commission outline five activities aimed at increasing competitiveness:

Simplification
Lowering barriers to the Single Market
Financing competitiveness
Promoting skills and quality jobs
Better coordination of EU and national policies

The key pieces of legislation impacted by the Omnibus Package are:

Corporate Sustainability Reporting Directive (CSRD)
Corporate Sustainability Due Diligence Directive (CSDDD)
EU Taxonomy
Carbon Border Adjustment Mechanism (CBAM)

The Clean Industrial Deal (CID) [104]

In February 2025, the EU published details of the Clean Industrial Deal. It describes the CID as **A plan for EU competitiveness and decarbonisation.** It goes on to explain the CID thus:

> Faced with high energy costs and fierce global competition, European industries need urgent support. The Clean Industrial Deal outlines concrete actions to turn decarbonisation into a driver of growth for European industries. This includes lowering energy prices, creating quality jobs and the right conditions for companies to thrive.
>
> The Deal presents measures to boost every stage of production, with a focus on:

[104] Source: https://commission.europa.eu/topics/eu-competitiveness/clean-industrial-deal_en

- energy-intensive industries such as steel, metals, and chemicals, that urgently need support to decarbonise, switch to clean energy, and tackle high costs, unfair global competition, and complex regulations.

- the clean-tech sector which is at the heart of future competitiveness and necessary for industrial transformation, circularity, and decarbonisation.

The two foci are clearly complementary in that it is the clean-tech sector (renewable energy, nuclear energy, energy storage, carbon capture and storage, clean hydrogen, clean heat etc.) that will enable the energy intensive sectors to decarbonise.

It is hoped that the CID will boost demand for clean products. It is expected that the industrial decarbonisation accelerator act will increase demand in the EU for EU-made clean products. It will achieve this by introducing sustainability, resilience, and 'made in Europe' criteria in **public and private procurements**. In addition the Commission will review the public procurement framework to introduce **sustainability, resilience and European preference criteria in public procurement for strategic sectors**.

In relation to how to finance the necessary actions that are required under the CID, the Commission believes that the CID will mobilise over €100 billion to support EU-made clean manufacturing. The Commission will:

- adopt a new Clean Industrial Deal state aid framework to accelerate the approval of state aid to roll out renewable energy, decarbonise industry and ensure sufficient manufacturing capacity of clean tech

- strengthen the Innovation Fund and propose an industrial decarbonisation bank, aiming for €100 billion in funding, based on available funds in the Innovation Fund, additional revenues resulting from parts of the ETS as well as the revision of InvestEU

- launch a dedicated call under Horizon Europe to stimulate research and innovation in these areas

- amend the InvestEU Regulation to increase the amount of financial guarantees that InvestEU can provide to support investments. This will in turn mobilise up to €50 billion for the deployment of clean tech, clean mobility and waste reduction.

Appendix 3 *Pexapark on Cannibalisation Risk*

This term and others are well explained in a publication by Pexapark on cannibalisation[105]. An edited excerpt from this excellent publication is provided below:

> What is cannibalisation?
> Because of the marginal pricing method, when renewable energy injects to the grid, spot prices tend to go down. If a low-marginal cost technology such as wind and solar is available, the system operator will give priority dispatch to that technology and will need less of the more expensive energy generated by other (typically fossil-fuel based) technologies such as gas. Therefore, the average price for this particular time slot will fall. Cannibalization is often referred to as 'profile risk'.
>
> In the long run this poses a serious threat to the viability of renewables operating on the free market. When a lot of renewable energy is injected at the same time. Nowadays, renewables increasingly participate in the electricity markets on a merchant basis. The amount of renewables in the grid is set to increase very substantially, posing a significant risk for their long-term remuneration.
>
> Baseload price: It's the average spot price over a selected timeframe. Remember that spot prices usually change on a half-hourly (hourly) basis. So when we refer to baseload prices, we either refer to them on a daily, monthly, or yearly basis accumulating all the different prices throughout the day.
>
> Baseload production/power: This usually refers to the ability of a power plant to meet the average daily demand continuously. If needed, a natural gas power plant can be on 24/7. Therefore, it is able to provide baseload production which is the opposite of what VRE renewables can provide which is intermittent production.
>
> Capture Price:
> Capture prices are the average prices that each generator will manage to realise over a period of time hence they are also referred to as 'realised prices'. This is where seasonality of production, and the ability to provide baseload power comes into play. Suppose a solar power plant produces during specific hours of the day but its production lowers the spot price. In that case, the capture price could be lower than a natural gas plant will achieve because the latter will be able to sell during peak hours.
>
> Cannibalization is the phenomenon of capture prices for renewables dropping as solar peak production is at times when prices are low. With more and more solar with the same generation profile producing at these hours prices would drop further and the capture price would be lower.
>
> Capture prices are calculated as an asset's volume-weighted average price. So, the price at which it sold, based on the volume and the time.
>
> Capture Factors: The capture factor illustrates the ratio between baseload prices and capture prices i.e. capture price divided by the baseload price. Capture factors = capture price/baseload price.

[105] Pexapark *Cannibalisation: Behind the Renewables Silent Risk* https://pexapark.com/cannibalization-renewables/
About Pexapark: We empower renewable energy players to thrive in post subsidy markets. Keeping renewable investments attractive is our contribution to a more sustainable world. Through the phasing out of subsidy schemes, renewable energy investors become exposed to the electricity market and face entirely new challenges. How to replace guaranteed subsidy revenue with Power Purchase Agreements (PPAs)? How to sell and hedge renewable energy and actively manage energy market risks?